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Revision History

The following table lists the revision history of the PHY from release to release. For a detailed description of PHY component updates, refer to the PHY release notes.

Note

- Links and references to section, table, figure, and page numbers in this table are only assured to be valid for the version in which the change is made.
- In some instances, documentation-only updates occur. The DesignWare IP product information (http://www.designware.com) has the latest documentation.
- Documentation-only updates are designated using the following numbering structure: <phy_version>_d<x>.

PHY Version	Date	Description
2.02a	May 17, 2022	Updated version and date
2.01a	April 13, 2022	Updated: ■ "Product Overview" on page 23 ■ "Signal Descriptions" on page 37 ■ "PHY Usage and Configuration" on page 79 ■ "Physical-Level Implementation" on page 131 ■ "Package-Level Implementation" on page 157 ■ "Register Descriptions" on page 177
2.01a_pre2	February 25, 2022	Updated: ■ "Signal Descriptions" ■ "PHY Usage and Configuration" ■ "Test and Diagnostics" ■ "Package-Level Implementation" ■ "Register Descriptions" ■ "PHY Configuration"
1.01a_sup1_d1	August 30, 2021	Updated: ■ "PHY Usage and Configuration"

PHY Version	Date	Description
1.01a_sup1	August 16, 2021	Updated: ■ “Deliverables” ■ “Functional Description” ■ “PHY Usage and Configuration” ■ “Package-Level Implementation” ■ “PHY Configuration” Added: ■ “Test and Diagnostics”
1.01a	June 7, 2021	Updated: ■ “Quick Reference” ■ “Electrical Specifications” ■ “PHY Supply Power-Up”
1.00a	April 9, 2021	Updated: ■ “Quick Reference” ■ “Features” ■ “Electrical Specifications” ■ “Signal Descriptions” ■ “Register Descriptions”

PHY Version	Date	Description
1.00a	March 31, 2021	Updated: ■ “Quick Reference” ■ “General Product Description” ■ “General Product Description” ■ “Electrical Specifications” ■ “Electrical Specifications” ■ “Signal-Integrity Views” ■ “Signal Descriptions” ■ Figure 4-1 ■ “PMA” ■ “Microcontroller” ■ “Raw PCS Initialization Sequence” ■ “PHY Initialization Sequence” ■ “Datapath Latency” ■ “Datarates” ■ “clk_mode[2:0] Settings” ■ “Forwarded Clock Mode” ■ Figure 5-7 ■ Figure 5-9 ■ “Address Map” ■ “CPU JTAG Interface” ■ “Firmware” ■ “Scan Mode (Stuck-at, At-speed)” ■ “RX IDDQ Test Sequence” ■ “TX IDDQ Test Sequence” ■ “Loopback Modes” ■ “On-Chip Supply Decoupling” ■ “ESD Back-to-Back Diode Connection to Core VSS” ■ “Reference Resistor” ■ “Reference Resistor Contention Bus” ■ “Synthesis Database” ■ “Raw PCS Constraints” ■ “Clock Constraints” ■ “Power Integrity – Supplies Requirements” ■ “Register Descriptions” ■ “PHY Configuration”

PHY Version	Date	Description
1.00a	March 31, 2021	(continued) Added: ■ “Power Conditions” ■ “TX Datapath” ■ “RX Datapath” ■ “JTAG 1149.1 Port” ■ “JTAG 1149.1 Programmer's Model” ■ “JTAG 1149.1 TAP Controller” ■ “Raw PCS Level Loopback” ■ “Internal Per-Lane Loopback” ■ “Die-to-Die Loopback” ■ “Syncronizers” ■ “txX_clk Constraints”

PHY Version	Date	Description
1.00a_pre2	January 29, 2021	Updated: ■ “Product Overview” ■ “Features” ■ “Electrical Specifications” ■ “Test Views” ■ “Signal Descriptions” ■ “Support Module” ■ “PHY Supply Power-Up” ■ “PHY Initialization Sequence” ■ “Datapath” ■ “Datarates” ■ “Clock Modes” ■ “PMA Block Size” ■ “Reference Resistor” ■ “Synthesis Database” ■ “RX Bump Map” ■ “TX Bump Map” ■ “Power Supply Connectivity” ■ “ESD Protection Strategy” ■ “ESD Protection Strategy for Die-to-Die Pins” ■ “ESD Back-to-Back Diode Connection to Core VSS” ■ “Repeat Clock Output Specification” ■ “Reference Clock Implementation” ■ “Register Descriptions” Added: ■ “PHY Initialization Sequence” ■ “Raw PCS JTAG Interface” ■ “CPU JTAG Interface” ■ “TX and RX Power States” ■ “Firmware” ■ “Test Modes” ■ “Clock Constraints” ■ “Power Integrity – Supplies Requirements” ■ “Clocking Architecture” ■ “PHY Configuration”

PHY Version	Date	Description
1.00a_pre1	November 8, 2020	Updated: ■ “Revision History ■ “Features” ■ “Firmware Based” ■ “Datapath” ■ “Datarates” ■ “Verilog Model Files and Simulators” ■ “Signal Descriptions” ■ “Register Descriptions” ■ Table 9-2 Added: ■ Figure 5-10
1.00a_pre1	October 30, 2020	Updated: ■ “Quick Reference” ■ “Raw PCS” ■ “Raw PCS Initialization Sequence” ■ “Datarates” ■ “Signal Descriptions” ■ “Register Descriptions” Added: ■ “Electrical Specifications” ■ “Datapath” ■ “Clock Modes” ■ “Control Register and Memory Interface” ■ “Raw PCS Synthesis” ■ “Physical-Level Implementation” ■ “Package-Level Implementation” ■ “Reference Clock Implementation”
0.00a_pre1	August, 2020	Initial preliminary release

Preface

This databook describes the Synopsys DesignWare® Cores Die-to-Die 112G USR/XSR PHY.

Databook Organization

This databook is organized as follows:

- Chapter 1, "Product Overview", provides an introduction to the DesignWare Cores Die-to-Die 112G USR/XSR PHY and its features.
- Chapter 2, "Deliverables", describes the deliverables included in the PHY full-package release.
- Chapter 3, "Signal Descriptions", describes the PHY top-level signals.
- Chapter 4, "Functional Description", provides a functional overview of the PHY.
- Chapter 5, "PHY Usage and Configuration", explains the usage of PHY features and lists the PHY configuration settings for the protocols and data rates that are supported.
- Chapter 6, "Test and Diagnostics", details supported test procedures and diagnostic features.
- Chapter 7, "Verilog Model Description", provides an overview of simulating the PHY Verilog model.
- Chapter 8, "Physical-Level Implementation", describes the physical implementation of the PHY.
- Chapter 9, "Package-Level Implementation", describes requirements and guidelines for implementing the PHY at the package level.
- Chapter 10, "Reference Clock Implementation", describes the PHY reference clock.
- Chapter 11, "Register Descriptions", details all possible registers in the PHY.
- Appendix A, "PHY Configuration", provides configuration values for configuring the PHY for your application.

Web Resources

For more information, check the following Synopsys online resources:

- DesignWare IP product information: https://www.synopsys.com/designware-ip.html
- Your custom DesignWare IP page: https://www.synopsys.com/dw/mydesignware.php
- Documentation through SolvNetPlus: https://solvnetplus.synopsys.com (Synopsys password required)
- Synopsys Common Licensing (SCL): https://www.synopsys.com/keys

STAR on the Web (SotW)

You must review all STARs on the Web (SotWs) associated with your product. SotWs are considered a part of the Synopsys documentation suite, and show critical information related to your product. To review product SotWs, refer to the DesignWare IP product information:

https://www.synopsys.com/designware-ip.html

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

Customer Support

To obtain support for your product, prepare the required files and contact the support center using one of the methods described:

- Prepare the following debug information, if applicable:
  - For environment set-up problems or failures with configuration, simulation, or synthesis that occur within coreConsultant or coreAssembler, select the following menu:
    - **File > Build Debug Tar-file**

    Check all the boxes in the dialog box that apply to your issue. This option gathers all the Synopsys product data needed to begin debugging an issue and writes it to the <*core tool startup directory*>/debug.tar.gz file.
  - For simulation issues outside of coreConsultant or coreAssembler:
    - Create a waveform file (such as VPD or VCD).
    - Identify the hierarchy path to the DesignWare instance.
    - Identify the timestamp of any signals or locations in the waveforms that are not understood.
- For the fastest response, enter a case through SolvNetPlus:

  a. https://solvnetplus.synopsys.com

  
  Note SolvNetPlus does not support Internet Explorer. Use a supported browser such as Microsoft Edge, Google Chrome, Mozilla Firefox, or Apple Safari.

  b. Click the **Cases** menu and then click **Create a New Case** (below the list of cases).

  c. Complete the mandatory fields that are marked with an asterisk and click **Save**.

  Make sure to include the following:

  - **Product L1:** DesignWare Cores

- **Product L2:** Die-to-Die 112G USR/XSR PHY

d. After creating the case, attach any debug files you created.

For more information about general usage information, refer to the following article in SolvNet-Plus:

https://solvnetplus.synopsys.com/s/article/SolvNetPlus-Usage-Help-Resources

- Or, send an e-mail message to mailto:support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
  - Include the Product L1 and Product L2 names, and Version number in your e-mail so it can be routed correctly.
  - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.
  - Attach any debug files you created.
- Or, telephone your local support center:
  - North America:

    Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
  - All other countries:

    https://www.synopsys.com/support/global-support-centers.html

1

Product Overview

This chapter includes the following sections:

- "Quick Reference" on page 24
- "General Product Description" on page 25
- "Features" on page 26
- "Applications" on page 27
- "Process Information" on page 27
- "Electrical Specifications" on page 29

1.1 Quick Reference

This section provides the most relevant details of the DesignWare Cores Die-to-Die 112G USR/XSR PHY (dwc_d2d_sr112_phy_tsmc12ffc_x8ns), which are provided along with other details throughout this databook.

Features
Two independent macros: ■ x8 TX macro ■ x8 RX macro **Receiver** AFE containing ATT and CTLE with programmable boost. **Transmitter** 3-tap digital FFE, low-power driver **PLL** Two cascaded PLLs, one LC-tank based and the other ring-oscillator based which enable gapless data rate support **Lane Configurations:** x8

Electromigration (EM)	NRZ Data Rates [Gbps]	PAM4 Data Rate [Gbps]
88KPOH@105°C	■ 2.5 to 56	■ 56 to 112

Process Technology	Metal Stack
TSMC 12-nm FinFET Flip-Chip150 um pitch	11M_2Xa1Xd_h_3Xe_vhv_2Y2R_HDMIM

Protocol/Specification Target	Electrostatic Discharge (ESD)
■ CEI-112G-XSR (pending official specification publication) ■ CEI-56G-USR	■ CDM 1 A, HBM 100 V for all high-speed pins on the die-to-die link (TX/RX) ■ CDM 6 A, HBM 2 kV for all other pins

TX PHY Drawn Area [um^2]	RX PHY Drawn Area [um^2]
x8: 1.289339566848	x8: 1.289339566848

Power Supply	Junction Temperature
■ Core Supply: 0.90 V ±5% ■ I/O Supply: 1.2 V-3% to 1.8 V +10% ■ Digital Core Supply: 0.8 V ±10% (SoC voltage is used for level shifting in the PHY)	■ Functional Range: -40 to 125°C ■ Performance Range: -20 to 105°C

Power Consumption x8 @ 112 Gbps PAM4 Typical (preliminary)
■ TX Total Power per Lane: 103.70 mW per lane (0.93 pJ/b unidirectional power per lane) ■ RX Total Power per Lane: 95.58 mW per lane (0.85 pJ/b unidirectional power per lane)

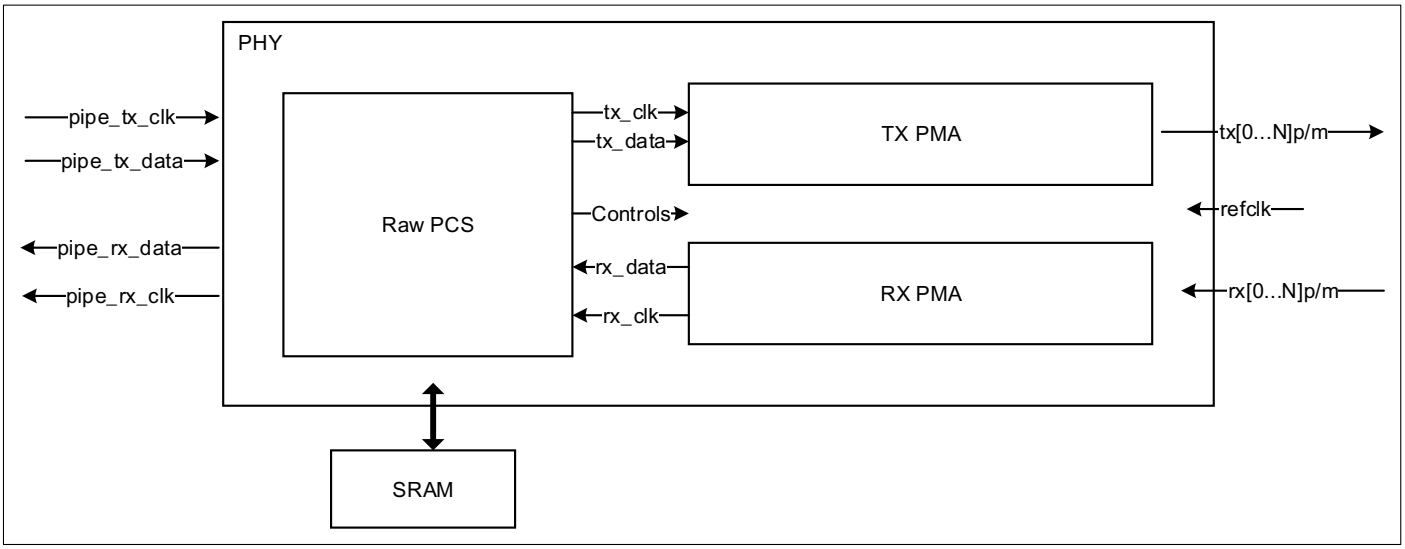
1.2 General Product Description

The DesignWare Cores Die-to-Die 112G USR/XSR PHY meets the growing high bandwidth and low latency needs of high-performance data center applications. The PHY delivers signal integrity and jitter performance that targets to align with the CEI-112G-XSR and CEI-56G-USR standards electrical specifications. The PHY is small in area, low in power consumption, and high in performance, meeting the needs of die-to-die interconnects with channel losses up to 10dB.

Figure 1-1 shows the Die-to-Die 112G USR/XSR PHY usage model, considering the following:

- PHY: dwc_d2d_sr112_phy_tsmc12ffc_x8ns
- TX PMA: dwc_d2d_sr112_tx_top_x8
- RX PMA: dwc_d2d_sr112_rx_top_x8

Figure 1-1 PHY Usage Model



1.3 Features

The PHY supports the following features:

- Two independent macros. Unless stated otherwise, the information for the TX PMA and RX PMA is the same. The term PMA is used for both TX and RX PMAs.
  - x8 TX macro
  - x8 RX macro
- Receiver: AFE containing ATT and CTLE with programmable boost
- Transmitter: 3-tap digital FFE, low-power driver
- PAM-4 and NRZ signaling supported
- Supports the following baudrates (1.75 GSps to 112 GSps):
  - Quarter Rate
  - Half Rate
  - Full Rate
  - 2x oversampling
  - 4x oversampling
- PLL: Two cascaded PLLs, one LC-tank based and the other ring-oscillator based which enable gapless data rate support
- Low jitter Phase-Locked Loops (PLLs)
- Visibility and controllability of hard macro functions through programmable registers in the design:
  - Overrides on all ASIC side functional control inputs for easy debug
  - Access register space through a parallel interface
  - Access register space through a JTAG port
- Both forwarded clock and embedded clocking modes supported
- Extensive PMA debug capability through read/write and read-only registers in Raw_PCS
- Configurable Raw_PCS with extensive debug options:
  - Independent TX and RX (control per lane)
  - Pseudo-random bit sequence (PRBS) generation and checker PRBS31, PRBS15, PRBS7
- Register-based control of all Raw_PCS–to–PMA signals
- Advanced, built-in diagnostics including on-chip eye scope
- Die-to-Die channels for intra-package links
- Programmable 128-bit pattern generation with error injection capability
- TX Total Power Consumption per Lane x8 @ 112Gbps PAM4 (preliminary)
  - 103.70 mW per lane
  - 0.93 pJ/b per lane (unidirectional)
- RX Total Power Consumption per Lane x8 @ 112Gbps PAM4 (preliminary)
  - 95.58 mW per lane
  - 0.85 pJ/b per lane (unidirectional)

> **Note** Specification compliance exception list is under review.
> Contact Synopsys for more information about the feature list.

1.4 Applications

The PHY targets hyperscale data centers, enterprise and campus networks, cloud computing and networking, service provider networks, artificial intelligence, and machine learning.

The PHY enables designers to introduce competitive products using the latest standards because it is designed for low power, efficient area, low test costs, and process centering for optimal yield and interoperability.

1.5 Process Information

Table 1-1 describes the process information for the Die-to-Die 112G USR/XSR PHY. For availability in alternate process nodes or variants, contact Synopsys Customer Support.

Table 1-1 Process Information

Foundry	Process	Core Voltage	I/O Voltage	Digital Core Supply Voltage[a]	Metal Layers
TSMC	12-nm	0.90 V ±5%	1.2 V /1.5 V / 1.8 V	0.8 V ±10%	11M_2Xa1Xd_h_3Xe_vhv_2Y2R_HDMIM

a. SoC voltage is used for level shifting in the PHY.

> **Note** SoC voltage is used for level shifting in the PHY. Digital logic in the PHY is not using the SoC supply voltage.

Table 1-2 Primitive Devices Used

Devices	Description
crtmom_2t	2T RTMOM
mimcap_sin	HD-MIM decap 2T
moscap_rf_nw	CORE RF MOSCAP Varactor wo DNW
nch_18_mac	I/O 1.8V VT NMOS
nch_18ud12_mac	Thick Oxide UnderDrive 1.2V NMOS
nch_18ud15_mac	Thick Oxide UnderDrive 1.5V NMOS
nch_lvt_mac	CORE Low VT NMOS

Devices	Description
nch_svt_mac	CORE Standards VT NMOS
nch_ulvt_mac	CORE Ultra Low VT NMOS
ndio_gated18_mac	2T IO 1.8V gated N+/PW DIODE
ndio_hia18_mac	CORE 2T ESD diode N+/PW
ndio_mac	N+/PW Junction DIODE
nmoscap	NMOS in N-Well Varactor
nmoscap_18	1.8V MOSCAP Varactor
pch_18_mac	I/O 1.8V VT PMOS
pch_18ud12_mac	Thick Oxide UnderDrive 1.2V PMOS
pch_18ud15_mac	Thick Oxide UnderDrive 1.5V PMOS
pch_lvt_mac	CORE Low VT PMOS
pch_svt_mac	CORE Standard VT PMOS
pch_ulvt_mac	CORE Ultra Low VT PMOS
pdio_gated18_mac	2T IO 1.8V gated P+/NW DIODE
pdio_hia18_mac	CORE 2T ESD diode P+/NW
pnp_i2_mac	Cascade PNP (P+/NW/Psub)
rhim_m	P+ Poly Non-Salicided resistor 3T
rm10w	M10 resistor
rm11w	M11 resistor
rm1w	M1 resistor
rm2w	M2 resistor
rm3w	M3 resistor
rm4w	M4 resistor
rm5w	M5 resistor
rm6w	M6 resistor
rm7w	M7 resistor
rm8w	M8 resistor
rm9w	M9 resistor
rmap	AP metal resistor

Note This IP implementation uses CPODE, High R resistor, VTL_N, VTL_P, and HDMIM devices which may incur additional mask cost.

1.6 Electrical Specifications

Note The information in the next tables assumes the power conditions in Table 1-7 on page 31.

Table 1-3 shows the TX x8 power consumption.

Table 1-3 TX Power Consumption

Configuration	Current vp [mA]	Current vph [mA]	Current vpdig [mA]	Total Current [mA]	Total Power per Lane [mW/lane]	Unidirectional Power per Lane [pJ/b]
Power On 112G PAM4 (Typical)	821.6057255	31.37400316	0.363287	853.3430157	97.17080275	0.867596453
Power On 112G PAM4 (Max)	1168.948002	38.42304418	0.705314	1208.076361	144.4945206	1.290129648
Power On 56G NRZ (Typical)	721.1969708	31.37400134	0.103318	752.6742901	85.85044548	1.533043669
Power On 56G NRZ (Max)	1035.352402	38.42304588	0.360895	1074.136343	128.6780224	2.297821828
Power On 56G PAM4 (Typical)	519.183205	31.37400048	0.183389	550.7405945	63.13140335	1.127346488
Power On 56G PAM4 (Max)	779.2269349	38.42304574	0.47373	818.1237107	98.43483764	1.757764958
Power On 28G NRZ (Typical)	458.0760149	31.37400273	0.203826	489.6538436	56.25876077	2.009241456
Power On 28G NRZ (Max)	711.818205	38.42304666	0.495312	750.7365636	90.47440721	3.231228829

Table 1-4 shows the RX x8 power consumption.

Table 1-4 RX Power Consumption

Configuration	Current vp [mA]	Current vph [mA]	Current vpdig [mA]	Total Current [mA]	Total Power per Lane [mW/lane]	Unidirectional Power per Lane [pJ/b]
Power On 112G PAM4 (Typical)	744.6454	28.7141	2.20116	775.56066	88.28608125	0.788268583

Configuration	Current vp [mA]	Current vph [mA]	Current vpdig [mA]	Total Current [mA]	Total Power per Lane [mW/lane]	Unidirectional Power per Lane [pJ/b]
Power On 112G PAM4 (Max)	1122.5887	34.0322	4.38347	1161.00437	138.6731485	1.238153112
Power On 56G NRZ (Typical)	570.244	28.7141	0.566438	599.524538	68.51266856	1.22344051
Power On 56G NRZ (Max)	900.5683	34.0322	1.70842	936.30892	112.1711243	2.00305579
Power On 56G PAM4 (Typical)	468.8053	28.7141	1.10191	498.62131	57.15101531	1.020553845
Power On 56G PAM4 (Max)	769.4767	34.0242	2.52795	806.02885	96.76912303	1.728020054
Power On 28G NRZ (Typical)	400.3521	28.7221	1.12495	430.19915	49.45339031	1.766192511
Power On 28G NRZ (Max)	689.2451	34.0242	2.65543	725.92473	87.30491166	3.118032559

Table 1-5 shows the TX x8 leakage.

Table 1-5 TX Leakage

	Current vp [mA]	Current vph [mA]	Current vpdig [mA]	Total Current [mA]	Total Power per Lane [mW/lane]
TX x8 power down leakage (Typical)	0.992410573	7.1649E-05	0.00238684	0.994869062	0.111880703
TX x8 power down leakage (Max)	87.9971861	0.202598985	0.243359	88.44314409	10.45319284

Table 1-6 shows the RX x8 leakage.

Table 1-6 RX Leakage

	Current vp [mA]	Current vph [mA]	Current vpdig [mA]	Total Current [mA]	Total Power per Lane [mW/lane]
RX x8 power down leakage (Typical)	2.28652903	6.99995E-05	0.00919	2.295789029	0.258106578
RX x8 power down leakage (Max)	134.9264045	0.21398801	0.98463	136.1250225	16.07502952

1.6.1 Power Conditions

Table 1-7 shows the power conditions.

Table 1-7 Power Conditions

	Process	Temperature	Analog Core Supply (vp)	Digital Core Supply (vpdig)	Analog I/O Supply (vph)
Typical Corner	TT	25 °C	0.90 V	0.80 V	1.20 V
Maximum Corner	FF	125 °C	0.945 V	0.88 V	1.98 V

2

Deliverables

This chapter includes the following sections:

- "Introduction" on page 34
- "Front-End Views" on page 34
- "Back-End Views" on page 35
- "Signal-Integrity Views" on page 35
- "Test Views" on page 35

Note For more information about the deliverables included in the package, contact Synopsys.

2.1 Introduction

The full-package for each DesignWare Cores Die-to-Die 112G USR/XSR PHY release contains all of the design views and information required to instantiate the PHY in a top-level chip design. It also allows for the simulation of both the digital functionality of the PHY and the signal integrity of the link, and provides the information to enable production testing of the PHY.

Since the Die-to-Die 112G USR/XSR PHY consists of two independent TX and RX Macros, the PHY package includes front-end and back-end views for TX, as well as front-end and back-end views for RX.

For information about the Die-to-Die 112G USR/XSR PHY release directory structure, see the top-level Readme file included in the product deliverables.

The deliverables are grouped into four basic categories based on usage:

- Front-end view—Includes deliverables for system simulation and floor planning
- Back-end view—Includes deliverables for physical design and finishing
- Signal-integrity view—Includes deliverables for simulation of link performance at the analog level
- Test view—Includes deliverables for testing

2.2 Front-End Views

Front-end views are provided for both the hard and soft macro portions of the XSR PHY.

A front-end view includes the following:

- Verilog Testbench (VTB) examples

> **Note** The packaged testbench is a sample testbench for you to use for reference and pipe-cleaning purposes. The testbench does not exhaustively include all validation tests for all protocols.

- Digital simulation environment
- RTL for soft macro
- GTECH netlist for hard macros
- Synthesis timing models in .db and .lib formats
- Behavioral models for analog blocks
- Library Exchange Format (LEF) file with pin sizes and locations
- Protocol-specific testbench examples
- Synthesis constraints for the RAW PCS soft macro portion
- IP-XACT XML file with register details
- Spyglass views
- UPF views
- DesignWare Cores Die-to-Die 112G USR/XSR x8 PHY for TSMC 12-nm FFC/1.8V Databook

2.3 Back-End Views

Note For more information about the deliverables, contact Synopsys.

Back-end views are provided for the hard macro portions of the TX and RX PHY. These deliverables tend to be metal stack dependent, therefore, multiple files for each hard macro are provided. The top-level cell name inside all of the files is the same, but the content is not—you must choose the appropriate file that matches your metal stack-up.

A back-end view includes the following:

- GDSII layout and layer map files
- Layout-Versus-Schematic (LVS) netlist in SPICE format and LVS report
- LVS samples
- Design Rule Check (DRC) report
- DRC samples
- Generic technology (GTECH) netlists and models for simulation
- Gate-level Verilog
- Gate-level netlist and Standard Delay Format (SDF) timing file

2.4 Signal-Integrity Views

A signal-integrity view includes the following:

- S-Parameter models of RX and TX to facilitate package design
- IBIS-AMI models of RX and TX for signal integrity simulations
- Current waveforms, along with on-die power grid equivalent capacitance and resistance (Cdie and Rdie, respectively) for power delivery network simulations in the package

2.5 Test Views

Test view includes the following:

- Scan test environment with Automatic Test Pattern Generation (ATPG)
- Scan Verilog netlist

3

Signal Descriptions

This chapter details all I/O signals in the IP. Inputs are on the left of the signal diagrams; outputs are on the right. In addition to describing the function of each signal, the signal descriptions in this chapter might include the following information:

- **Active State:** Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).
- **Registered:** Indicates whether or not the signal is registered directly inside the IP boundary without intervening logic (excluding simple buffers). A value of *No* does not imply that the signal is not synchronous, only that there is some combinatorial logic between the signal's origin or destination register and the boundary of the IP. A value of N/A indicates that this information is not provided for this IP title.
- **Synchronous to:** Indicates which clocks in the IP sample this input (drive for an output). This clock might not be the same as the clock that your application logic should use to clock (sample/drive) this pin. For more details, consult the clock section in the databook.

The I/O signals are grouped as follows:

- "Reset and Clock Signals" on page 39
- "APB Interface Signals" on page 41
- "Miscellaneous Configuration and Status Signals" on page 43
- "JTAG Signals" on page 44
- "Interrupt Signals" on page 45
- "CPU Signals" on page 46
- "Closely-Coupled Memories AHB-Lite Slave Interface Signals" on page 49
- "Data Interface Signals" on page 51
- "Analog Signals" on page 53
- "Pad Signals" on page 54
- "Resistor Tune Signals (SUP block)" on page 56
- "TX - EQ Signals" on page 57
- "DFT - Scan Signals" on page 58

- "PIPE Lite interface Signals" on page 63
- "Power Supply Signals" on page 64

3.1 Reset and Clock Signals

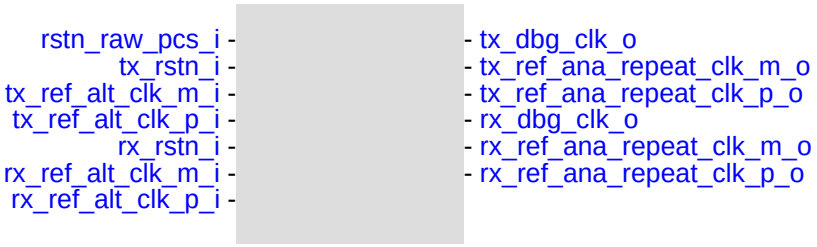


Table 3-1 Reset and Clock Signals

Port Name	I/O	Description
rstn_raw_pcs_i	I	Asynchronous reset. Active low. This reset is just for the logic on the Raw PCS. ■ Asserted as soon as possible during or after supplies are enabled. ■ De-asserted after all supplies are enabled. **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** Low
tx_rstn_i	I	Asynchronous reset. Active low. Asserted during power-up. De-asserted after all supplies are up. Reset for the TX lanes **Synchronous To:** Asynchronous **Active State:** low
tx_dbg_clk_o	O	Output debug clock. **Synchronous To:** Clock **Active State:** N/A
tx_ref_alt_clk_m_i	I	Transmitter differential negative input ref_alt_clk. **Synchronous To:** N/A **Active State:** N/A
tx_ref_alt_clk_p_i	I	Transmitter differential positive input ref_alt_clk. **Synchronous To:** N/A **Active State:** N/A
tx_ref_ana_repeat_clk_m_o	O	TX SUP buffered version of input reference clock (ref_repeat_clk_m) This is a SUP-buffered version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m, depending on the settings of refa/b_clk_sel register. This pair of clocks can be used as reference clocks for other PHYs on chip. **Synchronous To:** N/A **Active State:** N/A

Port Name	I/O	Description
tx_ref_ana_repeat_clk_p_o	O	TX SUP buffered version of input reference clock (ref_repeat_clk_p) This is a SUP-buffered version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m, depending on the settings of refa/b_clk_sel register. This pair of clocks can **Synchronous To:** N/A **Active State:** N/A
rx_rstn_i	I	Asynchronous reset. Active low. Asserted during power-up. De-asserted after all supplies are up. Reset for the RX lanes. **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** low
rx_dbg_clk_o	O	Debug clock. **Synchronous To:** Clock **Registered:** N/A **Active State:** N/A
rx_ref_alt_clk_m_i	I	Receiver differential negative input ref_alt_clk. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A
rx_ref_alt_clk_p_i	I	Receiver differential positive input ref_alt_clk. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A
rx_ref_ana_repeat_clk_m_o	O	RX SUP buffered version of input reference clock (ref_repeat_clk_m) This is a SUP-buffered version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m, depending on the settings of refa/b_clk_sel register. This pair of clocks can be used as reference clocks for other PHYs on chip. **Synchronous To:** N/A **Active State:** N/A
rx_ref_ana_repeat_clk_p_o	O	RX SUP buffered version of input reference clock (ref_repeat_clk_p) This is a SUP-buffered version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m, depending on the settings of refa/b_clk_sel register. This pair of clocks can **Synchronous To:** N/A **Active State:** N/A

3.2 APB Interface Signals

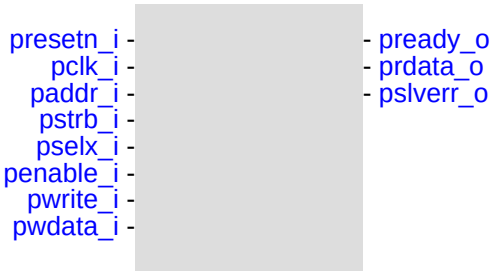


Table 3-2 APB Interface Signals

Port Name	I/O	Description
presetn_i	I	APB reset. Asserted as soon as possible during or after supplies are enabled. De-asserted after all supplies are enabled. **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** low
pclk_i	I	APB clock. **Synchronous To:** Clock **Registered:** N/A **Active State:** N/A
paddr_i[17:0]	I	APB byte address. Two least significant bits are always 0b00. **Synchronous To:** pclk_i **Registered:** N/A **Active State:** N/A
pstrb_i[3:0]	I	APB byte strobe. **Synchronous To:** pclk_i **Registered:** N/A **Active State:** N/A
pselx_i	I	APB slave select. **Synchronous To:** pclk_i **Registered:** N/A **Active State:** N/A
penable_i	I	APB transaction enable. **Synchronous To:** pclk_i **Registered:** N/A **Active State:** N/A

Port Name	I/O	Description
pwrite_i	I	APB write/read transaction control signal. ■ 0b1: Write ■ 0b0: Read **Synchronous To:** pclk_i **Registered:** N/A **Active State:** N/A
pready_o	O	APB slave ready. **Synchronous To:** pclk_i **Registered:** N/A **Active State:** N/A
pwdata_i[31:0]	I	APB write data. **Synchronous To:** pclk_i **Registered:** N/A **Active State:** N/A
prdata_o[31:0]	O	APB read data. **Synchronous To:** pclk_i **Registered:** N/A **Active State:** N/A
pslverr_o	O	APB slave error. **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** N/A

3.3 Miscellaneous Configuration and Status Signals

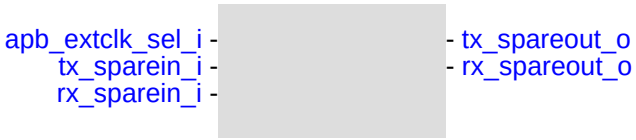


Table 3-3 Miscellaneous Configuration and Status Signals

Port Name	I/O	Description
apb_extclk_sel_i	I	Selects the clock source for the internal configurations and status registers bus. Change only when clocks are disabled. ■ 0b0: (cpu_clk_i) ■ 0b1: (pclk_i) or (rawpcs_jtag_tck_i). This is based on the value of (jtag_apb_sel_i) **Synchronous To:** Asynchronous
tx_sparein_i	I	Transmit spare inputs. Reserved. **Synchronous To:** Asynchronous **Active State:** N/A
tx_spareout_o	O	Transmit spare outputs. Reserved. **Synchronous To:** Asynchronous **Active State:** N/A
rx_sparein_i	I	Spare inputs. Reserved. **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** N/A
rx_spareout_o	O	Spare outputs. Reserved. **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** N/A

3.4 JTAG Signals

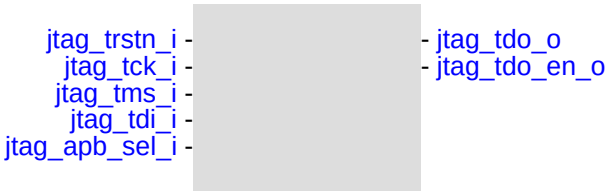


Table 3-4 JTAG Signals

Port Name	I/O	Description
jtag_trstn_i	I	JTAG reset. Raw PCS JTAG state machine reset signal. While this signal is held low, the state machine is held in test logic reset state. **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** low
jtag_tck_i	I	JTAG clock. Raw PCS JTAG state machine clock signal. **Synchronous To:** clock **Registered:** N/A **Active State:** N/A
jtag_tms_i	I	JTAG state machine control. Raw PCS JTAG test mode select signal. **Synchronous To:** jtag_tck_i **Registered:** N/A **Active State:** High
jtag_tdi_i	I	JTAG input port. Raw PCS JTAG test data input signal. **Synchronous To:** jtag_tck_i **Registered:** N/A **Active State:** N/A
jtag_tdo_o	O	JTAG output port. Raw PCS JTAG test data output signal. **Synchronous To:** jtag_tck_i **Registered:** N/A **Active State:** N/A
jtag_tdo_en_o	O	JTAG output enable. Raw PCS JTAG test data output enable signal. **Synchronous To:** jtag_tck_i **Registered:** N/A **Active State:** High
jtag_apb_sel_i	I	Controls the selection between Raw PCS JTAG interface or APB interface for the configurations and status registers access. Change only when both (jtag_tck_i) and (pclk_i) are disabled. ■ 0b0: APB interface ■ 0b1: Raw PCS JTAG interface **Synchronous To:** Asynchronous

3.5 Interrupt Signals

Table 3-5 Interrupt Signals

Port Name	I/O	Description
intr_o	O	Reserved **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** N/A

3.6 CPU Signals

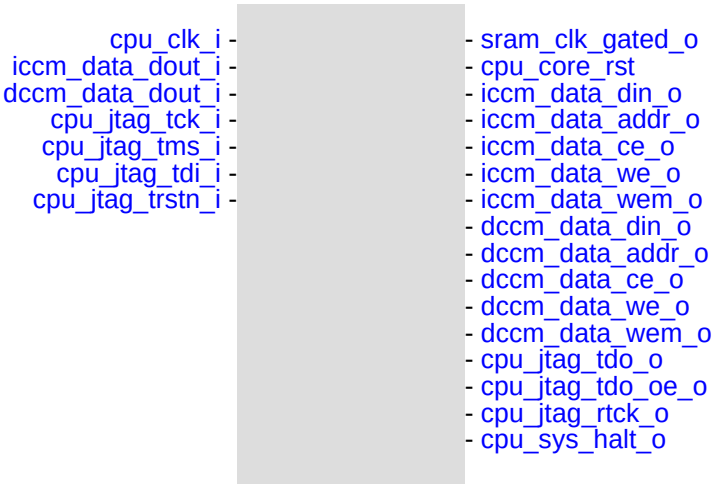


Table 3-6 CPU Signals

Port Name	I/O	Description
cpu_clk_i	I	CPU core clock. **Synchronous To:** Clock **Registered:** N/A **Active State:** N/A
sram_clk_gated_o	O	SRAMs clocks. Used for instruction Closely-Coupled Memory (ICCM) and the Data Closely-Coupled Memory (DCCM). Gates off when CPU is halted or memory is not being accessed for boot loading. **Synchronous To:** Clock **Registered:** N/A **Active State:** N/A
cpu_core_rst	O	Reset that is the input for the CPU. Is also outputted in case need to connect it to memories. **Synchronous To:** Asynchronous
iccm_data_dout_i[31:0]	I	Instruction Closely-Coupled Memory (ICCM) read data. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** N/A
iccm_data_din_o[31:0]	O	Instruction Closely-Coupled Memory (ICCM) write data. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** N/A

Port Name	I/O	Description
iccm_data_addr_o[13:0]	O	Instruction Closely-Coupled Memory (ICCM) address. This is word/32-bit aligned address. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** N/A
iccm_data_ce_o	O	Instruction Closely-Coupled Memory (ICCM) chip enable. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** High
iccm_data_we_o	O	Instruction Closely-Coupled Memory (ICCM) write enable. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** High
iccm_data_wem_o[3:0]	O	Instruction Closely-Coupled Memory (ICCM) byte write enable. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** High
dccm_data_dout_i[31:0]	I	Data Closely-Coupled Memory (DCCM) read data. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** N/A
dccm_data_din_o[31:0]	O	Data Closely-Coupled Memory (DCCM) write data. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** N/A
dccm_data_addr_o[13:0]	O	Data Closely-Coupled Memory (DCCM) address. This is word/32-bit aligned address. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** N/A
dccm_data_ce_o	O	Data Closely-Coupled Memory (DCCM) chip enable. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** High
dccm_data_we_o	O	Data Closely-Coupled Memory (DCCM) write enable. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** High

Port Name	I/O	Description
dccm_data_wem_o[3:0]	O	Data Closely-Coupled Memory (DCCM) byte write enable. **Synchronous To:** sram_clk_gated_o **Registered:** N/A **Active State:** High
cpu_jtag_tck_i	I	CPU JTAG TCK signal. **Synchronous To:** clock **Registered:** N/A **Active State:** N/A
cpu_jtag_tms_i	I	CPU JTAG TMS signal. **Synchronous To:** cpu_jtag_tck_i **Registered:** N/A **Active State:** N/A
cpu_jtag_tdi_i	I	CPU JTAG TDI signal. **Synchronous To:** cpu_jtag_tck_i **Registered:** N/A **Active State:** N/A
cpu_jtag_tdo_o	O	CPU JTAG TDO signal. **Synchronous To:** cpu_jtag_tck_i **Registered:** N/A **Active State:** N/A
cpu_jtag_tdo_oe_o	O	CPU JTAG TDO_OE signal. **Synchronous To:** cpu_jtag_tck_i **Registered:** N/A **Active State:** N/A
cpu_jtag_rtck_o	O	CPU JTAG RTCK signal. **Synchronous To:** Clock **Registered:** N/A **Active State:** N/A
cpu_jtag_trstn_i	I	CPU JTAG TRSTN signal. **Synchronous To:** Asynchronous **Registered:** N/A **Active State:** High
cpu_sys_halt_o	O	CPU halt status signal. Indicates that the CPU core is halted. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** High

3.7 Closely-Coupled Memories AHB-Lite Slave Interface Signals

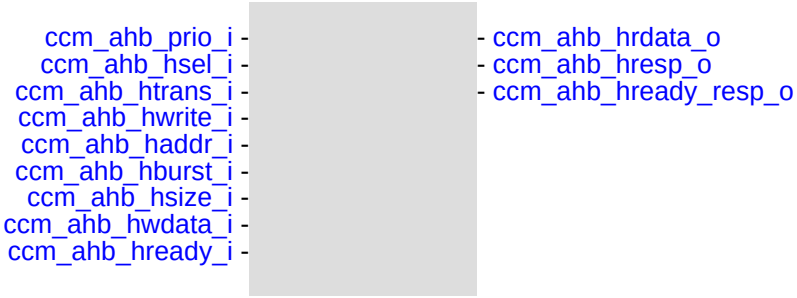


Table 3-7 Closely-Coupled Memories AHB-Lite Slave Interface Signals

Port Name	I/O	Description
ccm_ahb_prio_i	I	Priority signal. If true then high priority transfer, else low priority. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** High
ccm_ahb_hsel_i	I	AHB-Lite slave select. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** High
ccm_ahb_htrans_i[1:0]	I	AHB-Lite transfer type of the current transfer. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A
ccm_ahb_hwrite_i	I	AHB-Lite write/read transaction control. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A
ccm_ahb_haddr_i[19:0]	I	AHB-Lite Address bus to access ICCM and the DCCM. This is byte address. ICCM is at base address 0h10000. DCCM is at base address 0h18000. Note: PHY registers can not be accessed using this interface. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A
ccm_ahb_hburst_i[2:0]	I	AHB-Lite burst type. Note: master must not attempt to start an incrementing burst that crosses a 1KB address boundary. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A

Port Name	I/O	Description
ccm_ahb_hsize_i[1:0]	I	AHB-Lite size of the transfer. Only supports 32-bit access. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A
ccm_ahb_hwdata_i[31:0]	I	AHB-Lite write data. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A
ccm_ahb_hready_i	I	AHB-Lite ready input to indicate that previous transfer in the AHB bus has completed. This should be controlled by the AHB multiplexer, to allow this slave to know when other slaves has completed their transfers and that the master is ready to receive data from this slave. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A
ccm_ahb_hrdata_o[31:0]	O	AHB-Lite read data. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A
ccm_ahb_hresp_o	O	AHB-Lite response. ■ 0b0: indicates that the transfer status is OKAY. ■ 0b1: indicates that the transfer status is ERROR. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A
ccm_ahb_hready_resp_o	O	AHB-Lite slave ready. **Synchronous To:** cpu_clk_i **Registered:** N/A **Active State:** N/A

3.8 Data Interface Signals

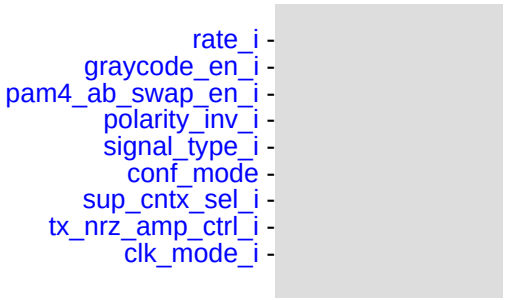


Table 3-8 Data Interface Signals

Port Name	I/O	Description
rate_i[2:0]	I	Defines the phy rate for the phy: ■ 3b000 Quarter rate ■ 3b001 Half rate ■ 3b010 Full rate ■ 3b011 2xoversampling ■ 3b100 4xoversampling ■ 3b1xx Not used **Synchronous To:** Asynchronous **Active State:** N/A
graycode_en_i	I	This bit is valid only when (tx_signal_type_i) indicates PAM4 signaling. ■ 0b0: Gray encoding is disabled ■ 0b1: Gray encoding is enable **Synchronous To:** Asynchronous **Active State:** N/A
pam4_ab_swap_en_i	I	This bit is valid only when (tx_signal_type_i) indicates PAM4 signaling. ■ 0b0: AB bit swapping is disabled ■ 0b1: AB bit swapping is enabled **Synchronous To:** Asynchronous **Active State:** N/A
polarity_inv_i	I	Inverts the datapath polarity. ■ 0b0: Disabled ■ 0b1: Enabled **Synchronous To:** Asynchronous **Active State:** N/A

Port Name	I/O	Description
signal_type_i	I	Signal type: ■ 0b0: NRZ mode ■ 0b1: PAM4 mode **Synchronous To:** Asynchronous **Active State:** N/A
conf_mode[1:0]	I	Reserved - No functionality associated Should be set to zero **Synchronous To:** Asynchronous **Active State:** N/A
sup_cntx_sel_i[6:0]	I	Controls the selection of the Clock frequencies (refclk and ropll_clk) on a LUT table. This functionality is not yet implemented. Will be on a future release. **Synchronous To:** Asynchronous **Active State:** N/A
tx_nrz_amp_ctrl_i[1:0]	I	Controls the amplitude of the TX in NRZ mode 11: = +/-3, 10=+/-2, 01=+/-1, 00=reserverd **Synchronous To:** Asynchronous
clk_mode_i[2:0]	I	Selects the clock mode of operation of the phy: ■ 0b00 - forwarded clock ■ 0b01 - embedded clock ■ 0b10 - common clock ■ 0b11 - reserved **Synchronous To:** Asynchronous

3.9 Analog Signals

Table 3-9 Analog Signals

Port Name	I/O	Description
tx_ana_atb_s_m_io	IO	Analog test bus for observe and control of critical internal signals. Connected by abutment to associated macro and eventually connected to chip pin via associated (CMS) macro. Negative. **Synchronous To:** N/A **Active State:** N/A
tx_ana_atb_s_p_io	IO	Analog test bus for observe and control of critical internal signals. Connected by abutment to associated macro and eventually connected to chip pin via associated (CMS) macro. Positive. **Synchronous To:** N/A **Active State:** N/A
rx_ana_atb_s_m_io	IO	Analog test bus for observe and control of critical internal signals. Connected by abutment to associated macro and eventually connected to chip pin via associated CMS macro negative. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A
rx_ana_atb_s_p_io	IO	Analog test bus for observe and control of critical internal signals. Connected by abutment to associated macro and eventually connected to chip pin via associated CMS macro positive. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A

3.10 Pad Signals

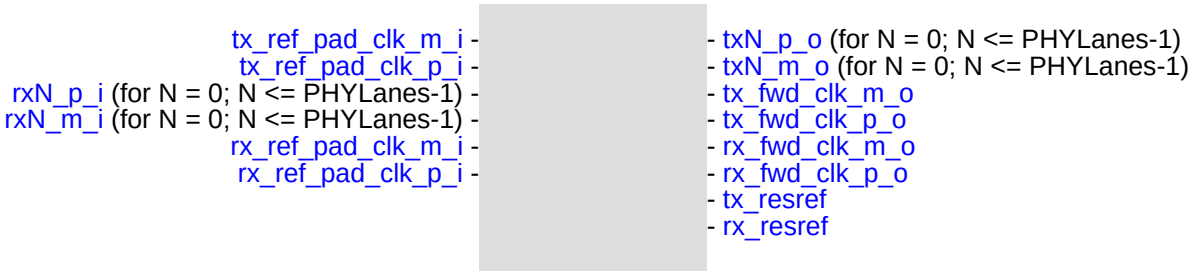


Table 3-10 Pad Signals

Port Name	I/O	Description
txN_p_o (for N = 0; N <= PHYLanes-1)	O	Transmit differential output positive terminal lane N. **Synchronous To:** N/A **Active State:** N/A
txN_m_o (for N = 0; N <= PHYLanes-1)	O	Transmit differential output negative terminal lane N. **Synchronous To:** N/A **Active State:** N/A
tx_ref_pad_clk_m_i	I	Transmitter differential negative input for ref_pad_clk. **Synchronous To:** N/A **Active State:** N/A
tx_ref_pad_clk_p_i	I	Transmitter differential positive input ref_pad_clk. **Synchronous To:** N/A **Active State:** N/A
tx_fwd_clk_m_o	O	Transmitter differential negative output diffbuf_clk. **Synchronous To:** N/A **Active State:** N/A
tx_fwd_clk_p_o	O	Transmitter differential positive output diffbuf_clk. **Synchronous To:** N/A **Active State:** N/A
rxN_p_i (for N = 0; N <= PHYLanes-1)	I	Receive differential input positive terminal lane N. **Synchronous To:** N/A **Active State:** N/A
rxN_m_i (for N = 0; N <= PHYLanes-1)	I	Receive differential input negative terminal lane N. **Synchronous To:** N/A **Active State:** N/A

Port Name	I/O	Description
rx_ref_pad_clk_m_i	I	Receiver differential negative input for ref_pad_clk. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A
rx_ref_pad_clk_p_i	I	Receiver differential positive input ref_pad_clk. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A
rx_fwd_clk_m_o	O	Receiver differential negative output diffbuf_clk. **Synchronous To:** N/A **Active State:** N/A
rx_fwd_clk_p_o	O	Receiver differential positive output diffbuf_clk. **Synchronous To:** N/A **Active State:** N/A
tx_resref	IO	Reference resistor connection to the Support module of the transmitter. Attach a 200 ohm 1% 100-ppm/C precision resistor-to- ground on the board. For information about resref, see the "Physical-Level Implementation" and "Package-Level Implementation" chapters. **Synchronous To:** N/A **Active State:** N/A
rx_resref	IO	Reference resistor connection to the Support module of the receiver. Attach a 200 ohm 1% 100-ppm/C precision resistor-to- ground on the board. For information about resref, see the "Physical-Level Implementation" and "Package-Level Implementation" chapters. **Synchronous To:** N/A **Active State:** N/A

3.11 Resistor Tune Signals (SUP block)

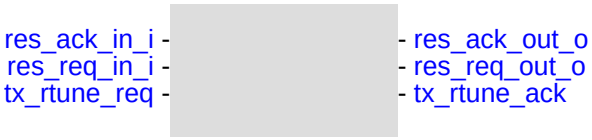


Table 3-11 Resistor Tune Signals (SUP block)

Port Name	I/O	Description
res_ack_in_i	I	Arbitration lines for sharing single reference resistor. Arbitration lines used for sharing a single-reference resistor among multiple cores. **Synchronous To:** Asynchronous
res_ack_out_o	O	Arbitration lines for sharing single reference resistor. Arbitration lines used for sharing a single reference resistor among multiple cores. **Synchronous To:** Asynchronous
res_req_in_i	I	Arbitration lines for sharing single reference resistor (res_req_in). Arbitration lines used for sharing a single-reference resistor among multiple cores. **Synchronous To:** Asynchronous
res_req_out_o	O	Arbitration lines for sharing single reference resistor (res_req_out). Arbitration lines used for sharing a single reference resistor among multiple cores. **Synchronous To:** Asynchronous
tx_rtune_ack	O	Resistor tune acknowledge. Indicates that a resistor tune has completed. **Synchronous To:** Asynchronous
tx_rtune_req	I	Resistor tune request. Assertion triggers a resistor tune request (if one is not already in progress). Resistor tuning is done automatically by PHY during POR and results are stored in Raw PCS AON registers. So external rtune_req is not needed for normal operation. In normal operation, if rtune_req is triggered, then customer will need to copy the results in Raw PCS AON registers. **Synchronous To:** Asynchronous

3.12 TX - EQ Signals

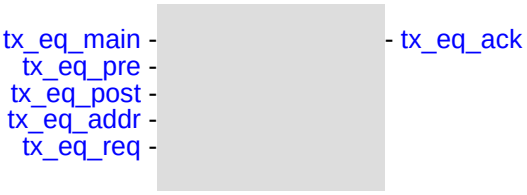


Table 3-12 TX - EQ Signals

Port Name	I/O	Description
tx_eq_main[6:0]	I	Transmitter Equalizer signal for main bits - This function is not yet present on the phy, Future improvement **Synchronous To:** Asynchronous
tx_eq_pre[5:0]	I	Transmitter Equalizer signal for main pre This function is not yet present on the phy, Future improvement **Synchronous To:** Asynchronous
tx_eq_post[5:0]	I	Transmitter Equalizer signal for main post This function is not yet present on the phy, Future improvement **Synchronous To:** Asynchronous
tx_eq_addr[4:0]	I	Transmitter Equalizer signal for address the lane This function is not yet present on the phy, Future improvement **Synchronous To:** Asynchronous
tx_eq_req	I	Transmitter Equalizer signal for start equalization on the lane This function is not yet present on the phy, Future improvement **Synchronous To:** Asynchronous
tx_eq_ack	O	Transmitter Equalizer signal for ack, will indicate when is eq is completed. This function is not yet present on the phy, Future improvement **Synchronous To:** Asynchronous

3.13 DFT - Scan Signals

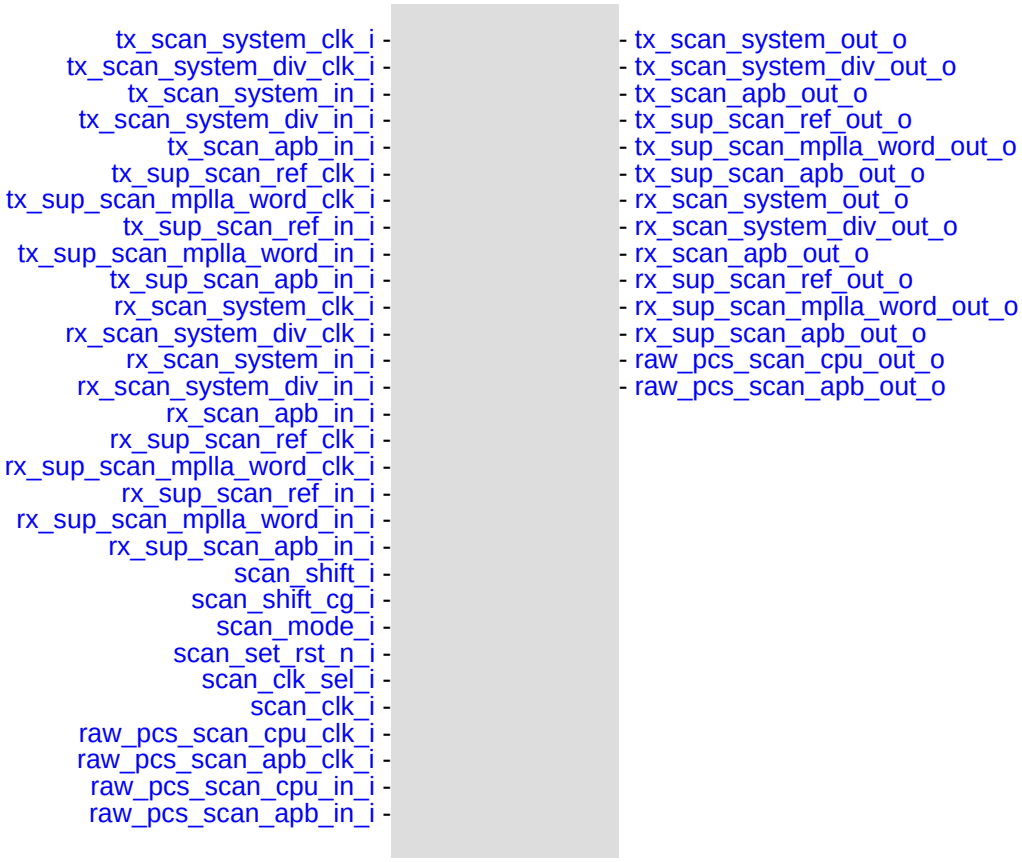


Table 3-13 DFT - Scan Signals

Port Name	I/O	Description
tx_scan_system_clk_i	I	Scan clock ASST system clock. **Synchronous To:** clock **Active State:** N/A
tx_scan_system_div_clk_i	I	Scan clock ASST system divided clock. **Synchronous To:** clock **Active State:** N/A
tx_scan_system_in_i[39:0]	I	Scan chain inputs. Serial data stream for the system input when macro is in scan mode. **Synchronous To:** scan_clk_i **Active State:** N/A
tx_scan_system_div_in_i[7:0]	I	Scan chain inputs. Serial data stream for the system divided clock input when macro is in scan mode. **Synchronous To:** scan_clk_i **Active State:** N/A

Port Name	I/O	Description
tx_scan_apb_in_i[31:0]	I	Scan chain inputs. Serial data stream for the ref input when macro is in scan mode. **Synchronous To:** scan_clk_i **Active State:** N/A
tx_scan_system_out_o[39:0]	O	Scan chain outputs. Serial data stream for the system output when macro is in scan mode. **Synchronous To:** scan_clk_i **Active State:** N/A
tx_scan_system_div_out_o[7:0]	O	Scan chain outputs. Serial data stream for the system divided clock output when macro is in scan mode. **Synchronous To:** scan_clk_i **Active State:** N/A
tx_scan_apb_out_o[31:0]	O	Scan chain outputs. Serial data stream for the apb output when macro is in scan mode. **Synchronous To:** scan_clk_i **Active State:** N/A
tx_sup_scan_ref_clk_i	I	Scan clock ASST ref_clk on SUP TX. **Synchronous To:** clock
tx_sup_scan_mplla_word_clk_i	I	Scan clock ASST mplla_word_clk on SUP TX. **Synchronous To:** clock
tx_sup_scan_ref_in_i[1:0]	I	Scan chain inputs. Serial data stream for the ref sup tx output when macro is in scan mode. **Synchronous To:** scan_clk_i
tx_sup_scan_mplla_word_in_i[6:0]	I	Transmit scan chain inputs for the mplla clock SUP TX. **Synchronous To:** scan_clk_i
tx_sup_scan_apb_in_i[2:0]	I	Scan chain inputs. Serial data stream for the apb sup tx input when macro is in scan mode. **Synchronous To:** scan_clk_i
tx_sup_scan_ref_out_o[1:0]	O	Scan chain outputs. for the reference clock SUP TX. **Synchronous To:** scan_clk_i
tx_sup_scan_mplla_word_out_o[6:0]	O	Transmit scan chain outputs for the mplla clock SUP TX. **Synchronous To:** scan_clk_i
tx_sup_scan_apb_out_o[2:0]	O	Scan chain inputs. Serial data stream for the apb sup tx output when macro is in scan mode. **Synchronous To:** scan_clk_i
rx_scan_system_clk_i	I	Scan clock ASST system clock RX. **Synchronous To:** clock **Active State:** N/A

Port Name	I/O	Description
rx_scan_system_div_clk_i	I	Scan clock ASST system divided clock RX. **Synchronous To:** clock **Active State:** N/A
rx_scan_system_in_i[55:0]	I	Scan chain inputs. Serial data stream for the system input when macro is in scan mode RX. **Synchronous To:** scan_clk_i **Active State:** N/A
rx_scan_system_div_in_i[7:0]	I	Scan chain inputs. Serial data stream for the system divided clock input when macro is in scan mode RX. **Synchronous To:** scan_clk_i **Active State:** N/A
rx_scan_apb_in_i[39:0]	I	Scan chain inputs. Serial data stream for the apb input when macro is in scan mode RX. **Synchronous To:** scan_clk_i **Active State:** N/A
rx_scan_system_out_o[55:0]	O	Scan chain outputs. Serial data stream for the system output when macro is in scan mode RX. **Synchronous To:** scan_clk_i **Active State:** N/A
rx_scan_system_div_out_o[7:0]	O	Scan chain outputs. Serial data stream for the system divided clock output when macro is in scan mode RX. **Synchronous To:** scan_clk_i **Active State:** N/A
rx_scan_apb_out_o[39:0]	O	Scan chain outputs. Serial data stream for the apb output when macro is in scan mode RX. **Synchronous To:** scan_clk_i **Active State:** N/A
rx_sup_scan_ref_clk_i	I	Scan clock ASST ref_clk on SUP RX. **Synchronous To:** clock
rx_sup_scan_mplla_word_clk_i	I	Scan clock ASST mplla_word_clk on SUP RX. **Synchronous To:** clock
rx_sup_scan_ref_in_i[1:0]	I	Scan chain inputs. Serial data stream for the ref SUP RX output when macro is in scan mode. **Synchronous To:** scan_clk_i
rx_sup_scan_mplla_word_in_i[6:0]	I	Scan chain inputs. Serial data stream for the mplla SUP RX output when macro is in scan mode. **Synchronous To:** scan_clk_i

Port Name	I/O	Description
rx_sup_scan_apb_in_i[2:0]	I	Scan chain inputs. Serial data stream for the APB SUP RX input when macro is in scan mode. **Synchronous To:** scan_clk_i
rx_sup_scan_ref_out_o[1:0]	O	Scan chain outputs. for the reference clock on SUP RX. **Synchronous To:** scan_clk_i
rx_sup_scan_mplla_word_out_o[6:0]	O	Scan chain inputs. Serial data stream for the mplla SUP TX output when macro is in scan mode. **Synchronous To:** scan_clk_i
rx_sup_scan_apb_out_o[2:0]	O	Scan chain inputs. Serial data stream for the APB SUP RX output when macro is in scan mode. **Synchronous To:** scan_clk_i
scan_shift_i	I	Scan shift. ■ 1b1: PHY scan shift enable ■ 1b0: PHY capture mode if scan_mode =1 **Synchronous To:** scan_clk_i **Active State:** High
scan_shift_cg_i	I	■ 1b1: PHY scan shift for clock gates enable ■ 1b0: PHY capture mode if scan_mode =1 **Synchronous To:** Asynchronous **Active State:** High
scan_mode_i	I	Scan mode. ■ 1b1: phy in scan mode ■ 1b0: phy not in scan mode **Synchronous To:** Asynchronous **Active State:** High
scan_set_rst_n_i	I	Scan reset signal PHY. **Synchronous To:** Asynchronous **Active State:** low
scan_clk_sel_i	I	Select scan_clk or at-speed clock. **Synchronous To:** scan_clk_i
scan_clk_i	I	Scan shift and stuck-at capture clock. **Synchronous To:** clock
raw_pcs_scan_cpu_clk_i	I	Scan Clock ASST CPU clock. **Synchronous To:** Clock
raw_pcs_scan_apb_clk_i	I	Scan Clock ASST APB clock. **Synchronous To:** Clock

Port Name	I/O	Description
raw_pcs_scan_cpu_in_i[7:0]	I	Scan chain inputs. Serial data stream for the CPU input when Raw PCS is in scan mode. **Synchronous To:** scan_clk_i
raw_pcs_scan_cpu_out_o[7:0]	O	Scan chain inputs. Serial data stream for the cpu output when Raw PCS is in scan mode. **Synchronous To:** scan_clk_i
raw_pcs_scan_apb_in_i[1:0]	I	Scan chain inputs. Serial data stream for the apb input when Raw PCS is in scan mode. **Synchronous To:** scan_clk_i
raw_pcs_scan_apb_out_o[1:0]	O	Scan chain inputs. Serial data stream for the apb output when Raw PCS is in scan mode. **Synchronous To:** scan_clk_i

3.14 PIPE Lite interface Signals

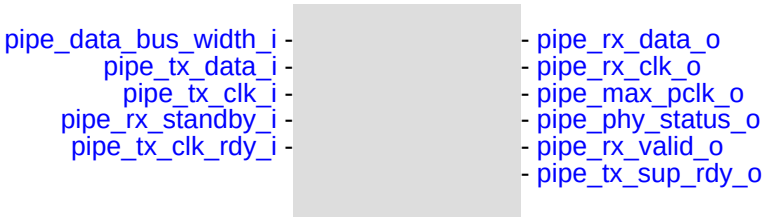


Table 3-14 PIPE Lite interface Signals

Port Name	I/O	Description
pipe_data_bus_width_i[1:0]	I	Reports the PHY data bus width. 1'b0 is 64bit per lane (only option valid at this point. **Synchronous To:** Asynchronous
pipe_tx_data_i[511:0]	I	Input data of the TX. **Synchronous To:** pipe_tx_clk_i
pipe_tx_clk_i	I	Input clock of the TX. **Synchronous To:** clock
pipe_rx_data_o[511:0]	O	Output data from receiver that will start after the rx_valid is asserted. **Synchronous To:** pipe_rx_clk_o
pipe_rx_clk_o	O	Output clock from receiver. **Synchronous To:** clock
pipe_max_pclk_o	O	Output clock of the PLL, this should be used to feed in back the pipe_tx_clk_i. **Synchronous To:** Clock
pipe_phy_status_o[7:0]	O	This signal will be de-asserted when the TX is calibrated and ready for transmission. TX data will be ignored until this signal is de-asserted. **Synchronous To:** Asynchronous
pipe_rx_valid_o[7:0]	O	This signal will indicate that All calibrations are done and RX output data is valid **Synchronous To:** Asynchronous
pipe_rx_standby_i[7:0]	I	This signal is asserted when phy_status is de-asserted. **Synchronous To:** Asynchronous
pipe_tx_sup_rdy_o	O	This signal is asserted when the SUP of the tX is calibrated and clk on the pipe_max_pclk_o is running . **Synchronous To:** Asynchronous
pipe_tx_clk_rdy_i[7:0]	I	Signal is asserted when the input clock of the TX is running. **Synchronous To:** Asynchronous

3.15 Power Supply Signals

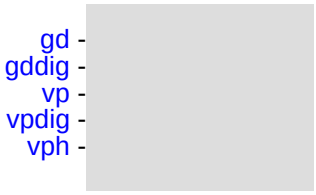


Table 3-15 Power Supply Signals

Port Name	I/O	Description
gd	I	Analog ground. This is a bump. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A
gddig	I	Digital ground. **Synchronous To:** N/A **Active State:** N/A
vp	I	Analog supply at core level (See Quick Reference). This is a bump. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A
vpdig	I	Digital supply at core level (See Quick Reference). **Synchronous To:** N/A **Registered:** Active Low **Active State:** N/A
vph	I	Analog supply at IO level (See Quick Reference). This is a bump. **Synchronous To:** N/A **Registered:** N/A **Active State:** N/A

4

Functional Description

This chapter includes the following sections:

- "PHY" on page 66
- "Raw PCS" on page 68
- "PMA" on page 69
- "Microcontroller" on page 77

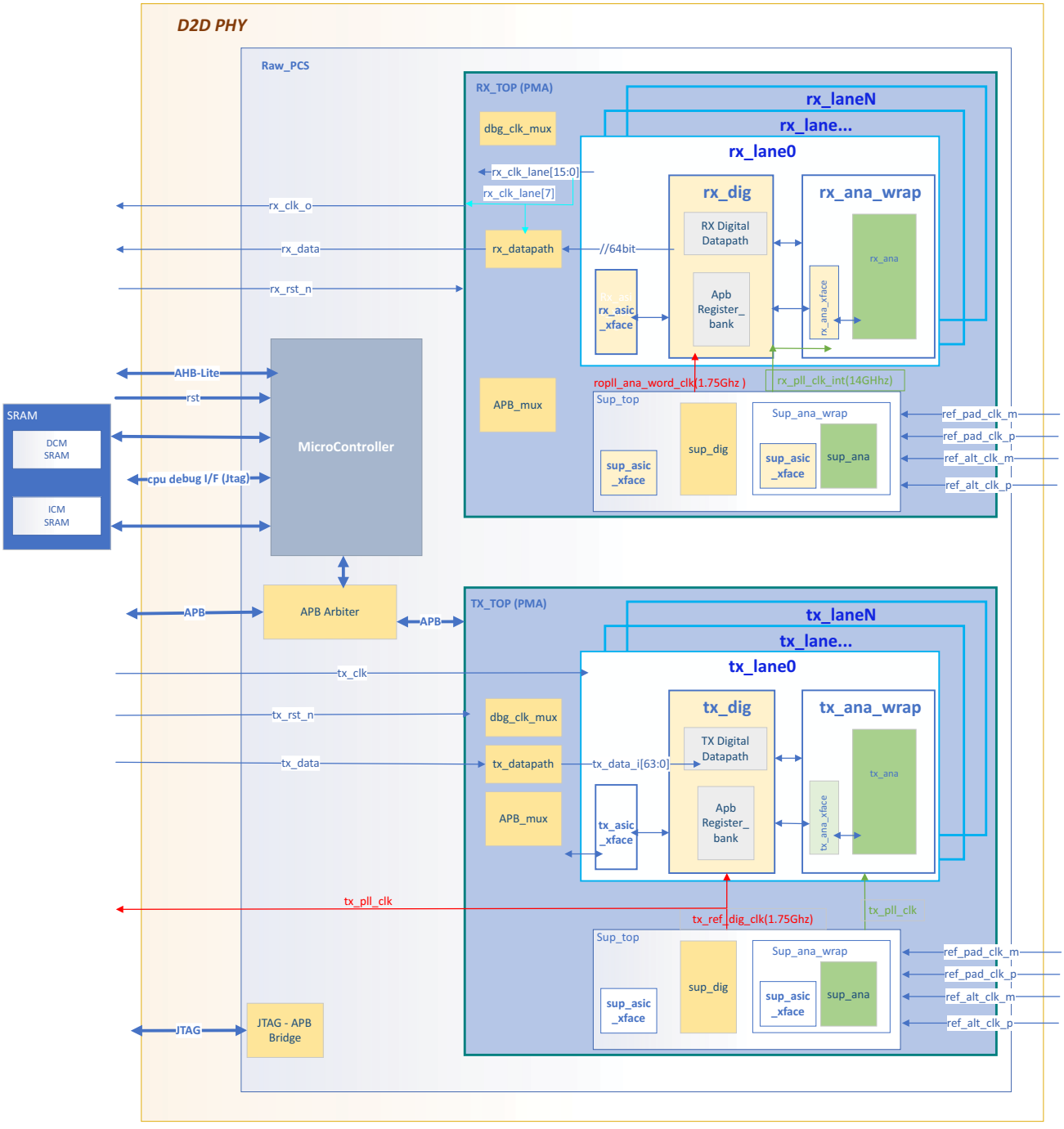
4.1 PHY

The DesignWare Cores Die-to-Die 112G USR/XSR PHY, also referred to in this document as raw PCS, consists of two hard macro (TX and RX)—Physical Medium Attachment Sublayer (PMA)—and a soft macro, referred to as Raw PCS soft macro.

The following sections describe the Raw PCS soft macro, the PMA, and their sub-blocks.

Figure 4-1 shows the PHY top-level architecture, including RX_TOP, TX_TOP, and raw PCS.

Figure 4-1 Die-to-Die PHY Architecture

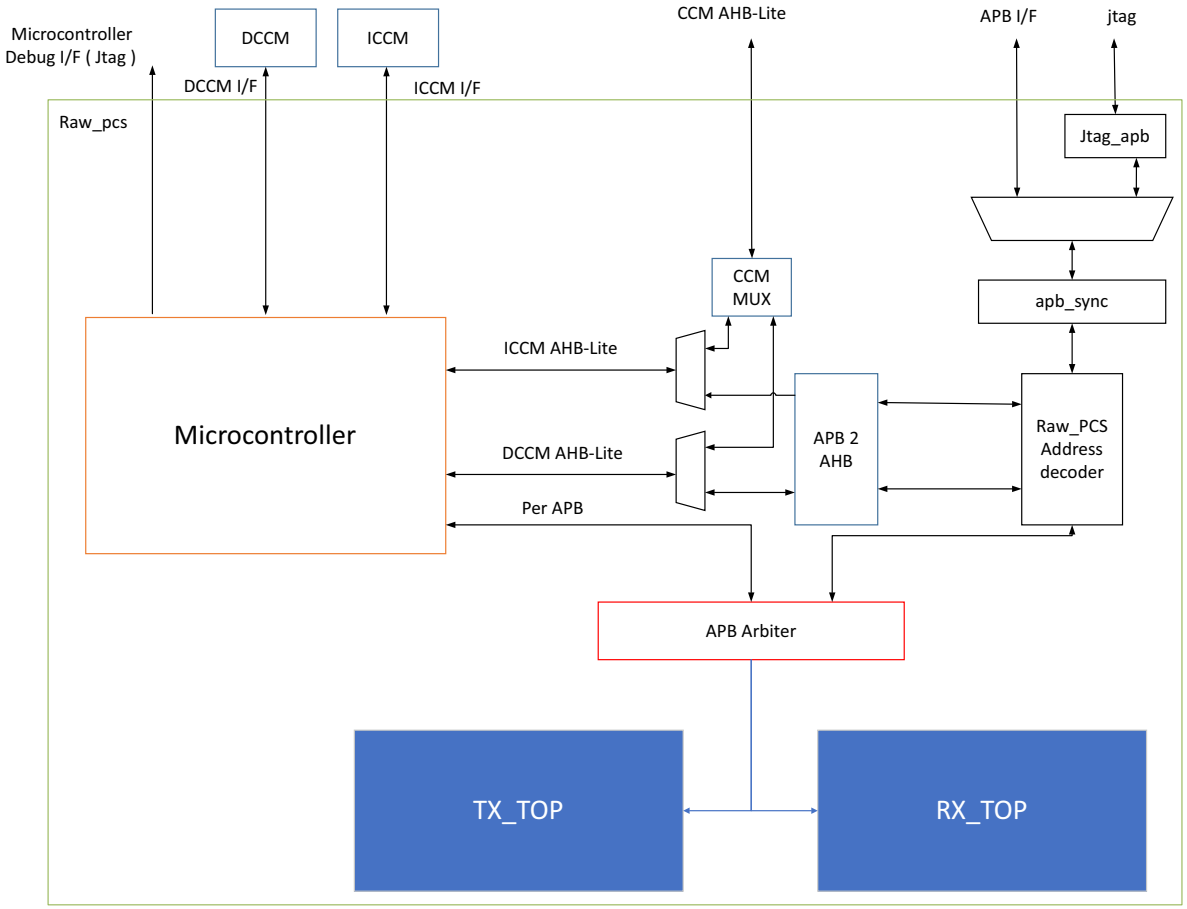


4.2 Raw PCS

The raw PCS is a soft macro provided as Verilog RTL.

Figure 4-2 shows the raw PCS block diagram.

Figure 4-2 Raw PCS Block Diagram



The soft macro portion of the raw PCS includes the following features:

- TX and RX datapaths
- Microcontroller

  Requires external SRAM, not part of IP deliverable. For the SRAM requirements see "Microcontroller" on page 77.
- External Register access
  - APB
  - JTAG
- Arbiter

  Arbitrates register access between external register access interface (JTAG/APB) and internal microcontroller register access.

4.3 PMA

The PHY includes two PMAs. Unless stated otherwise, information for the TX PMA and RX PMA is the same. The term PMA is used for both TX and RX PMAs.

- PMA_TX
  - 8 lanes TX
  - Support Block (SUP)
- PMA_RX
  - 8 lanes RX
  - Support Block (SUP)

Figure 4-3 shows the RX PMA architecture.

Figure 4-3 RX PMA Architecture

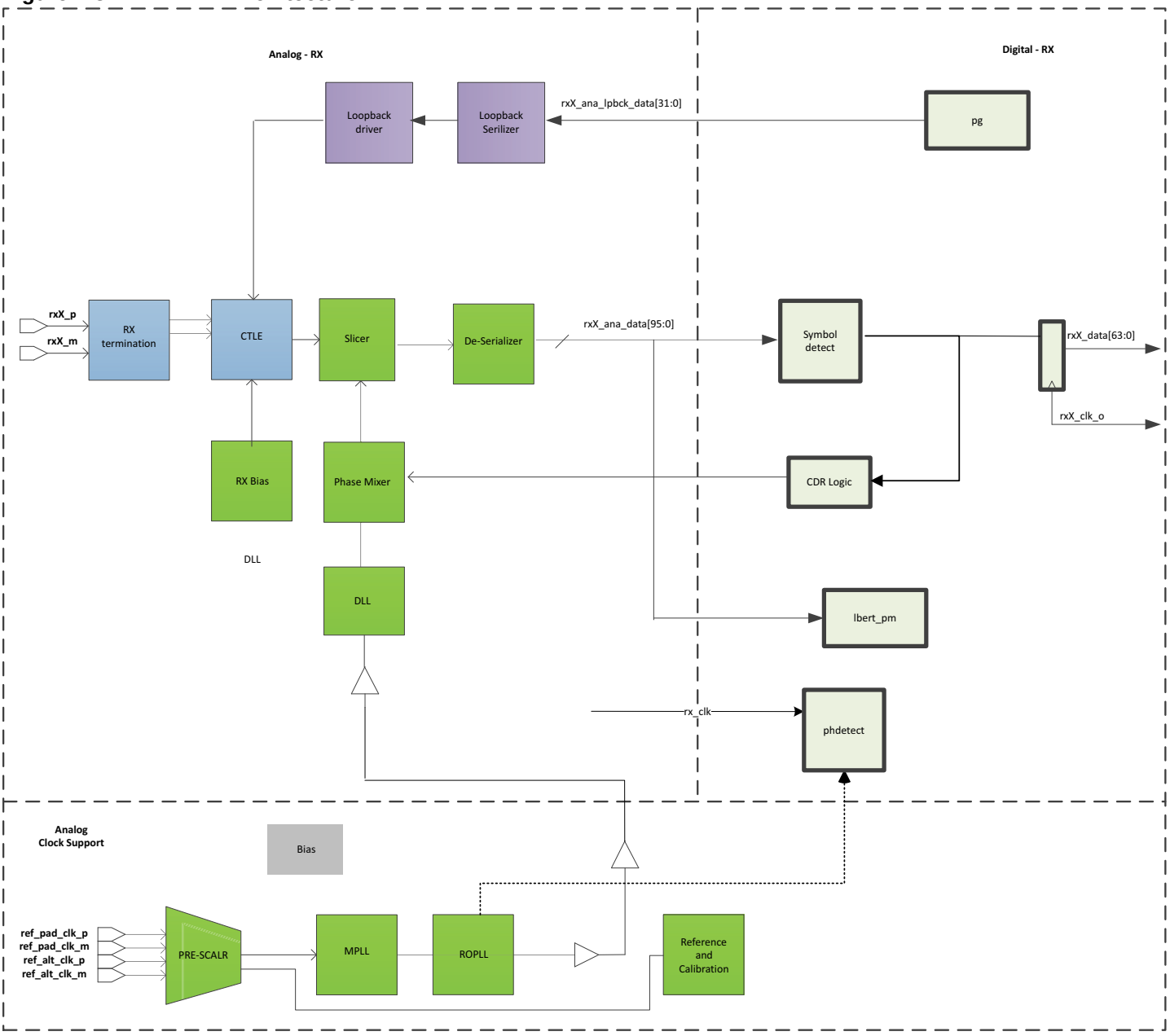
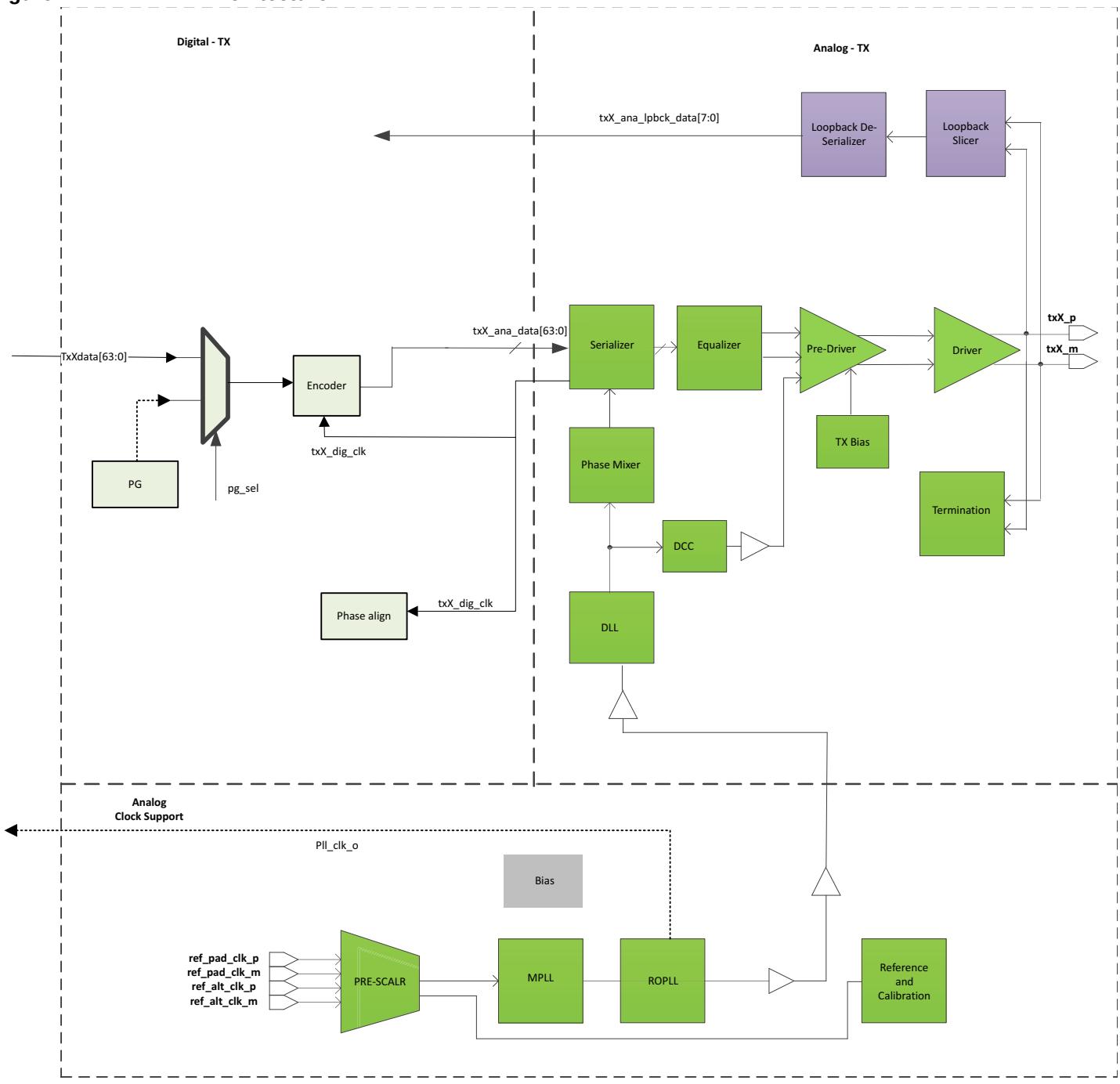


Figure 4-4 shows the TX PMA architecture.

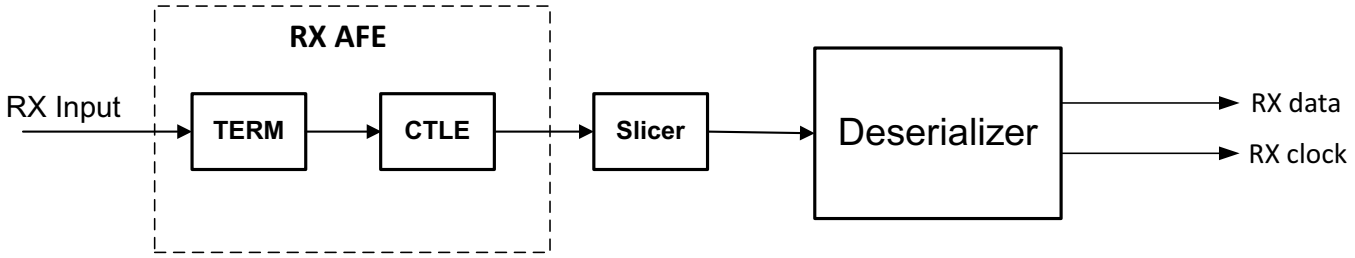
Figure 4-4 TX PMA Architecture



4.3.1 Receiver

Figure 4-5 provides the block diagram of the receiver (RX).

Figure 4-5 RX Block Diagram



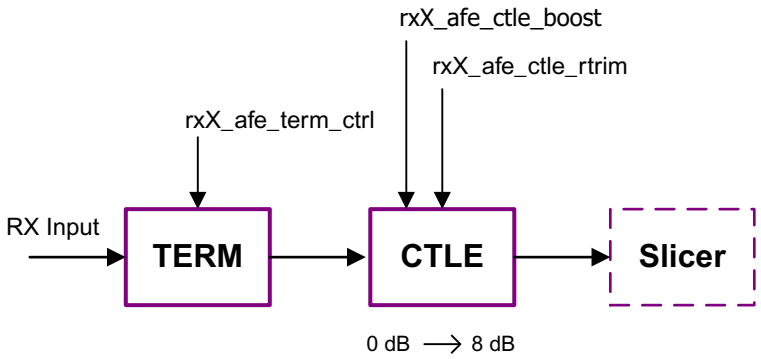
The receiver functions as follows:

1. The PMA receives differential serial data from a pair of external pads. Receiver characteristics, such as equalization and termination, are user-controllable.
2. A continuous-time linear equalizer (CTLE) is first used to compensate for channel loss.
3. A Slicer is implemented after the CTLE to sample the serial data.
4. The deserializer produces parallel data and a parallel data clock for the relevant lane.

4.3.1.1 RX Analog Front End

The analog front end (AFE) circuitry receives off-chip data and boosts high-frequency components to compensate part of the channel loss with programmable or adaptive CTLE.

Figure 4-6 RX AFE Diagram



As shown in Figure 4-6, the AFE consists of the following blocks:

- Termination (TERM)

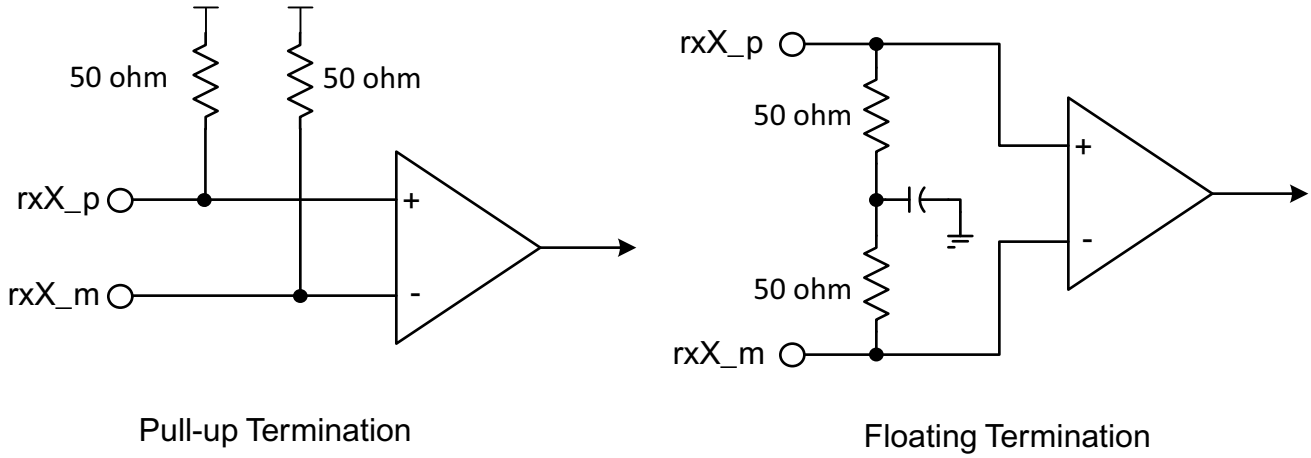
  The TERM block is tuned by the PHY using an off-chip reference resistor.

  To configure the RX termination in pull-up or floating mode (see Figure 4-7), use rxX_afe_term_ctrl_fix_vp:

- 0=floating mode termination
- 1=mode termination to supply

Note The PHY supports DC coupled link into RX floating termination.

Figure 4-7 RX Termination



- Continuous-time linear equalizer (CTLE)

  The CTLE block provides high-frequency boost range from 0–8 dB. The CTLE block has programmable boost level and boost bandwidth (BW) control (provided by rxX_afe_ctle_boost and rxX_afe_ctle_rtrim, respectively).

To achieve optimal link performance, the receiver equalization parameters are automatically tuned to their optimal values for protocols that are adapted.

4.3.1.2 RX Slicer and Deserializer

The Slicer and Deserializer convert serial data from the rx_afe and send the data to the raw PCS as parallel data. The Slicer samples the serial data stream, converting it to the 4UI data domain. The Deserializer then down-converts the data further, before sending the parallel data to the receiver's PCS channel.

4.3.2 Transmitter

The Transmitter functions as follows:

1. The PMA receives parallel data from the relevant lane of the raw PCS.
2. The data is serialized and sent to the TX driver, which is a high-speed, current mode, differential-output driver. The TX driver characteristics, such as amplitude, pre-emphasis, and termination, are user-controllable.
3. The TX driver output is sent to a pair of external pads.

4.3.2.1 TX Serializer

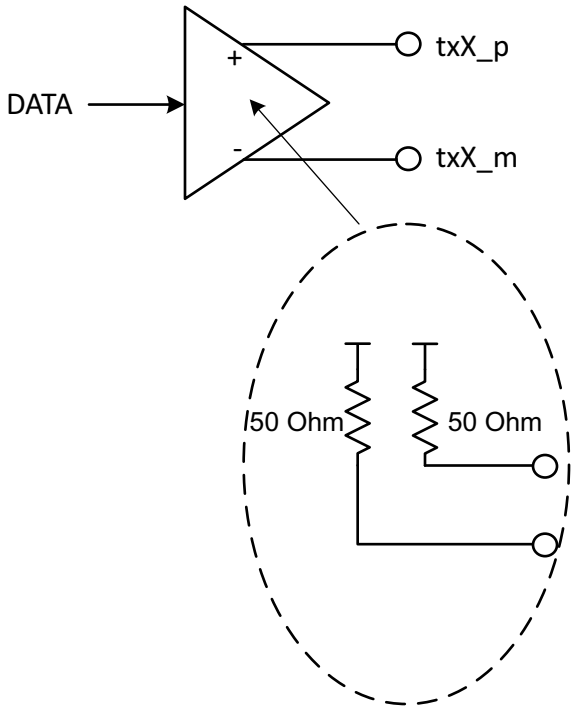
Parallel data from the raw PCS is latched-in using a clock that is synchronous to the lane's TX clock, and then serialized by the Parallel-to-Serial (P2S) converter.

4.3.2.2 TX Termination

The transmit driver is a current-mode driver, with nominally 50 ohm termination on both the TXP and TXN outputs, as shown in Figure 4-8. The termination is bandwidth extended through on-die t-coils and output matched to target return-loss specifications.

Note The PHY supports DC coupled link into far end floating termination.

Figure 4-8 TX Termination



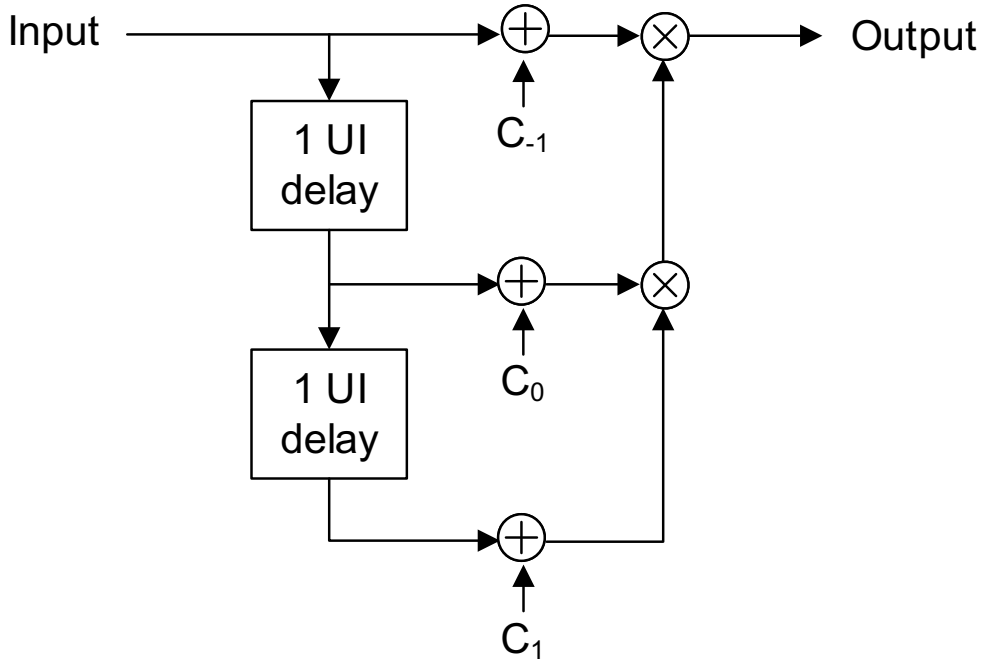
4.3.2.3 TX Amplitude

The transmit driver is a current-mode driver, and the DC launch amplitude is determined by the current through the main tap. You can control the amplitude by adjusting the C_0 tap weight.

4.3.2.4 TX Equalization

The TX Digital implements a 3-tap feed-forward equalizer (FFE) with one main tap (C_0), one post-cursor tap (C_{+1}) and one pre-cursor taps (C_{-1}) as shown in Figure 4-9. These tap values are controlled by tx_eq_{pre/main/post} primary input signals.

Figure 4-9 TX Driver Architecture



Where:

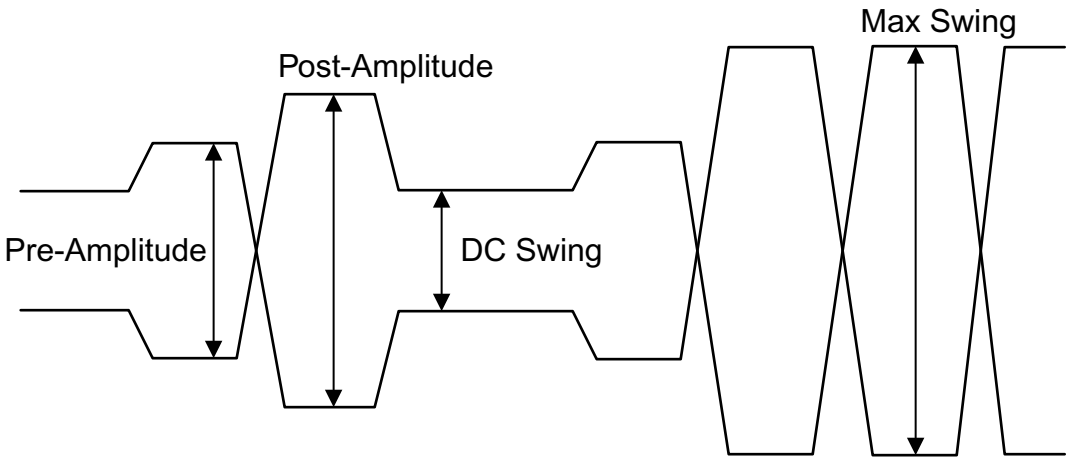
- C_0, C_{-1}, and C_1 are TX driver main tap, pre-cursor tap and post-cursor tap coefficients
- C_{-1} and C_1 are defined as negative values

The FFE coefficients C_0, C_{-1}, C_1 are set by three control signals (tx_eq_main, tx_eq_pre, and tx_eq_post) by the following equations:

- C_0 = tx_eq_main
- $|C_{-1}|$ = tx_eq_pre
- $|C_1|$ = tx_eq_post

TX Equalization with pre-amplitude, post-amplitude, and DC swing is illustrated in Figure 4-10 on page 76.

Figure 4-10 TX Equalization



To calculate the pre-cursor and post-cursor equalization, use the following equations:

- PRE_EQ_dB = $20*LOG10((C_0 - C_{-1} - C_1) / (C_0 + C_{-1} - C_1))$
- POST_EQ_dB = $20*LOG10((C_0 - C_{-1} - C_1) / (C_0 - C_{-1} + C_1))$

4.3.3 Support Module

The PHY includes two support blocks, one for TX and one for RX, working independently.

The support block (see Figure 4-1 on page 67) provides the common functions for:

- TX and RX clock generation
- TX and RX termination calibration
- Biasing voltage
- Reference clock buffering

4.3.3.1 Reference Clock Input

The PHY requires a reference clock. It can be provided by one of the following sources:

- External off-chip reference clocks using ref_pad_clk_p and ref_pad_clk_m
- On-chip reference clocks using ref_alt_clk_p and ref_alt_clk_m

The external or core-side input is selected by refa_clk_sel. When refa_clk_sel is set to 1 is asserted, the external reference clock is used; otherwise, the on-chip reference source is used.

For more information about implementation, see "Reference Clock Implementation" on page 165.

4.3.3.2 Reference Clock Output

The support block includes a reference clock repeat function. A Current-Mode Logic (CML) differential reference clock is driven on ref_repeat_clk_{p,m} when ref_repeat_clk_en is asserted. This clock can be used as the reference clock source for other PHYs.

For more information about reference clock output, see "Repeat Clock Specification" on page 168.

4.3.3.3 PLL

Each Support module has a PLL that generate the high-frequency clocks used by the transmitter lane (TX) and the receiver (RX).

The PLL has the following key features:

- PLL has a frequency range of 7 GHz - 14 GHz
- With a 14 GHz clock and quarter-rate clocking, a symbol rate of 56 Gbps is achieved
- With a 7 GHz clock and quarter-rate clocking, a symbol rate of 28 Gbps is achieved
- With a 7 GHz clock and 4x oversampling, a symbol rate of 1.75 Gbps is achieved
- The PLL is an integer PLL, and the simultaneously supported data rates are a function of supplied reference clock frequency

4.4 Microcontroller

The raw PCS contains a microcontroller to assist with execution of the raw PCS initialization, configuration, and power-state transitions.

The microcontroller firmware uses the APB interface to read and write the raw PCS /PMA internal registers. It requires two external memories (not included in the raw PCS).

The external memories connect to the raw PCS through the top-level microcontroller memory interface, see "Control Register and Memory Interface" on page 99.

4.4.1 Features

The microcontroller includes the following features:

- 32-bit Harvard architecture core to execute the firmware
- Maximum CPU clock frequency is 100 MHz
- CPU JTAG maximum clock frequency 100 MHz
- ICCM instruction closely-coupled memory populated to 64 kB (must be implemented externally)
- DCCM data closely-coupled memory populated to 64 kB (must be implemented externally)
- An IO port with APB interface mapping raw PCS/PMA Registers into microcontroller memory space
- A pair of AHB-Lite interfaces for ICCM and DCCM loading and for general access for debug purposes
- Big Endian memory orientation
- JTAG debug unit for runtime firmware debugging
- Real time counter to implement firmware delay
- Interrupt controller to handle critical events
- Watchdog timer to protect the system from CPU core and firmware faults

5

PHY Usage and Configuration

This chapter explains the usage of PHY features and lists the PHY configuration settings for the supported protocols and data rates. It includes the following sections:

- "PHY Supply Power-Up" on page 80
- "Raw PCS Initialization Sequence" on page 80
- "PHY Initialization Sequence" on page 83
- "Datapath" on page 89
- "Datarates" on page 90
- "Clock Modes" on page 92
- "Control Register and Memory Interface" on page 99
- "TX and RX Power States" on page 116
- "Firmware" on page 118

5.1 PHY Supply Power-Up

Power up the vp* and vph power supplies and then proceed with the PHY reset and initialization sequence (vp* indicates the grouping of all core supplies as well as vp and vpdig).

For a complete power supply pin list, including analog and digital grounds, see "Power Supply Signals" on page 64.

Note

- The PHY does not support power modes. The PHY is either On or Off.
- The state of the PHY is unknown after power up but before assertion of reset.
- While the unknown state is undesirable, the PHY has been designed to prevent damage if it is left in this state.
- Supply ramp should be monotonic with a ramp time of no faster than 10 us and a rate no faster than (supply voltage)/10 µs.
- vpdig ramping up before vp is supported; vp ramping down before vpdig is supported.
- vp ramping up before vpdig is not supported; vpdig ramping down before vp is not supported.
- vpdig and vp ramping up/down together is supported.

5.2 Raw PCS Initialization Sequence

You can initialize the raw PCS using a hardware or firmware based option. This impacts many of the raw PCS configuration settings. In the following sections, make sure you follow the procedures according to the raw PCS initialization option you choose (hardware or firmware based).

The hardware based solution should be only used during the initial phase of the PHY model integration, and for debug purposes. The PHY initialization must be done by the Firmware. The Raw PCS has a built-in microcontroller core that needs to be initialized. Once the microcontroller core boots up, the IP Firmware helps with the initialization sequence based on commands described in "Firmware Based" on page 81.

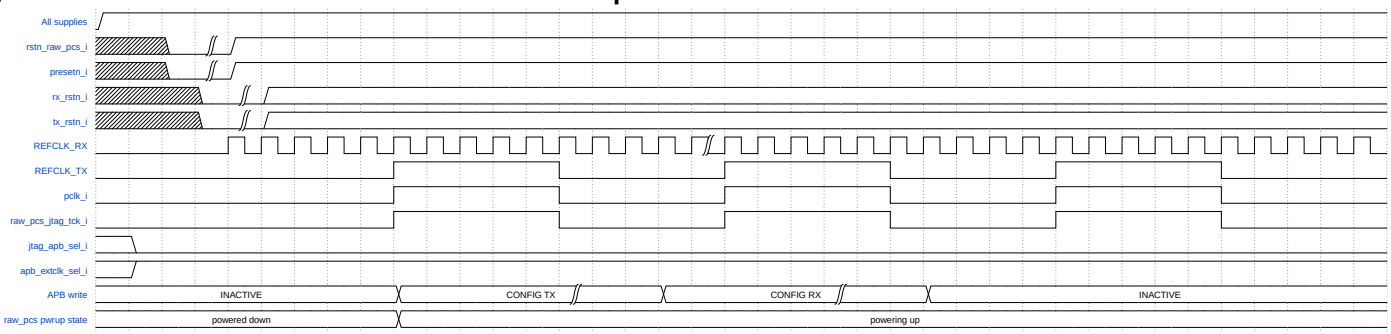
5.2.1 Hardware Based

The hardware based test setup, and initialization must be done during the debug, and/or initial phase of the PHY model integration.

This mode gets deprecated by the usage of the Firmware mode (see "Firmware Based" on page 81), that is the default mode of usage of the PHY.

Figure 5-1 shows the PHY hardware based initialization sequence. Note that the hardware based sequence cannot see full functional PIPE interface behavior.

Figure 5-1 Hardware Based PHY Initialization Sequence



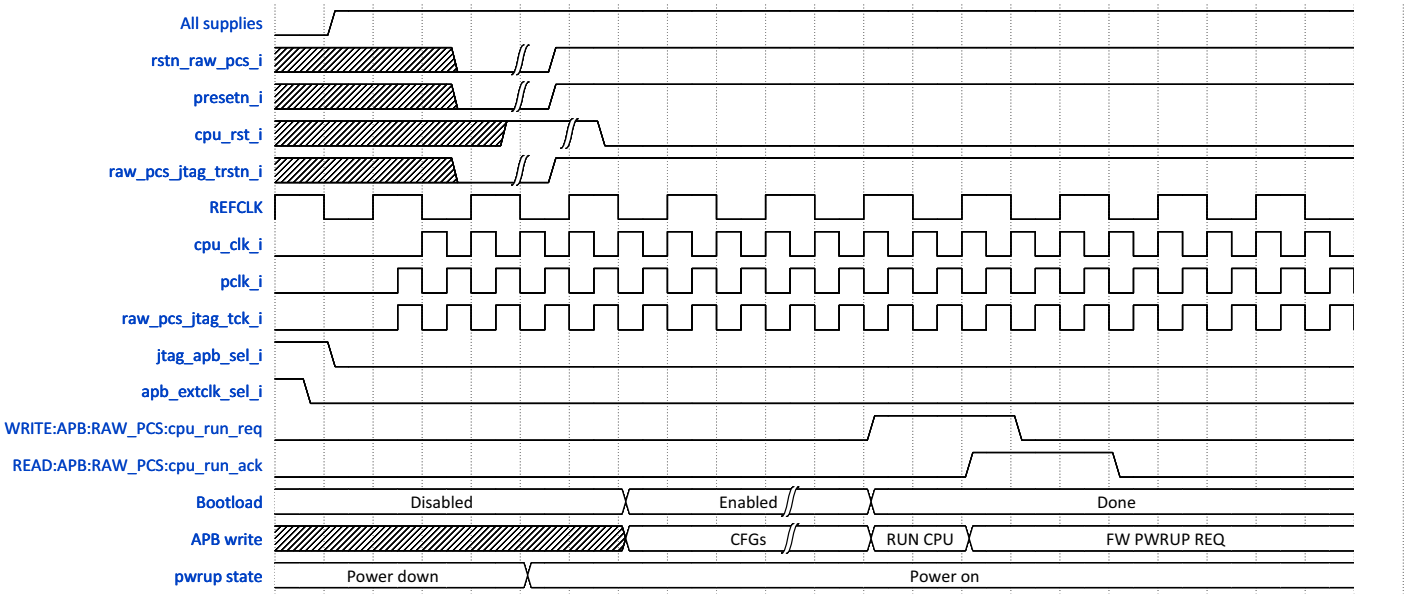
To initialize or reset the raw PCS through hardware, follow the next steps:

1. Power-up power supplies and make sure they are stable.
2. Assert resets.
3. De-assert the resets. PHY is ready for APB programming.
4. Program PHY register to specify the bring up mode as hardware based (current programing is done on the provided testbench).
5. Program the PHY registers for the desired configuration, such as data rate (see "Datarates" on page 90).

5.2.2 Firmware Based

Figure 5-2 shows the PHY firmware based initialization sequence.

Figure 5-2 Firmware Based PHY Initialization Sequence



To initialize or reset the raw PCS through firmware, follow the next steps:

1. Power-up power supplies and make sure they are stable.
2. Assert resets.

   Resets need to be asserted for a minimum of 25 us after power supplies are ramped-up (power-on). For any subsequent reset assertions (warm-reset) the resets need to be asserted for a minimum of 10 ns.
3. De-assert the resets.

   Ensure the APB clock and CPU clocks are stable. Raw PCS is now ready for APB programming.
4. Load the ICCM SRAM with the provided firmware image using any of the available methods.

   This can be done through one of the AHB or APB interfaces, see "Control Register and Memory Interface" on page 99.
5. Program RAW_PCS register to run the CPU. Write `APB::RAW_PCS::CPU_CFG::cpu_run_req=1b1`.

   The CPU is halted by default upon removal of reset. This step allows the processor to start fetching the firmware from the SRAM and executing it.

6. Wait for `APB::RAW_PCS::CPU_STAT::cpu_run_ack=1b1`.
   Make `APB::RAW_PCS::CPU_CFG::cpu_run_req=0b0`.
   The microcontroller core is fully functional and the IP Firmware can take over.

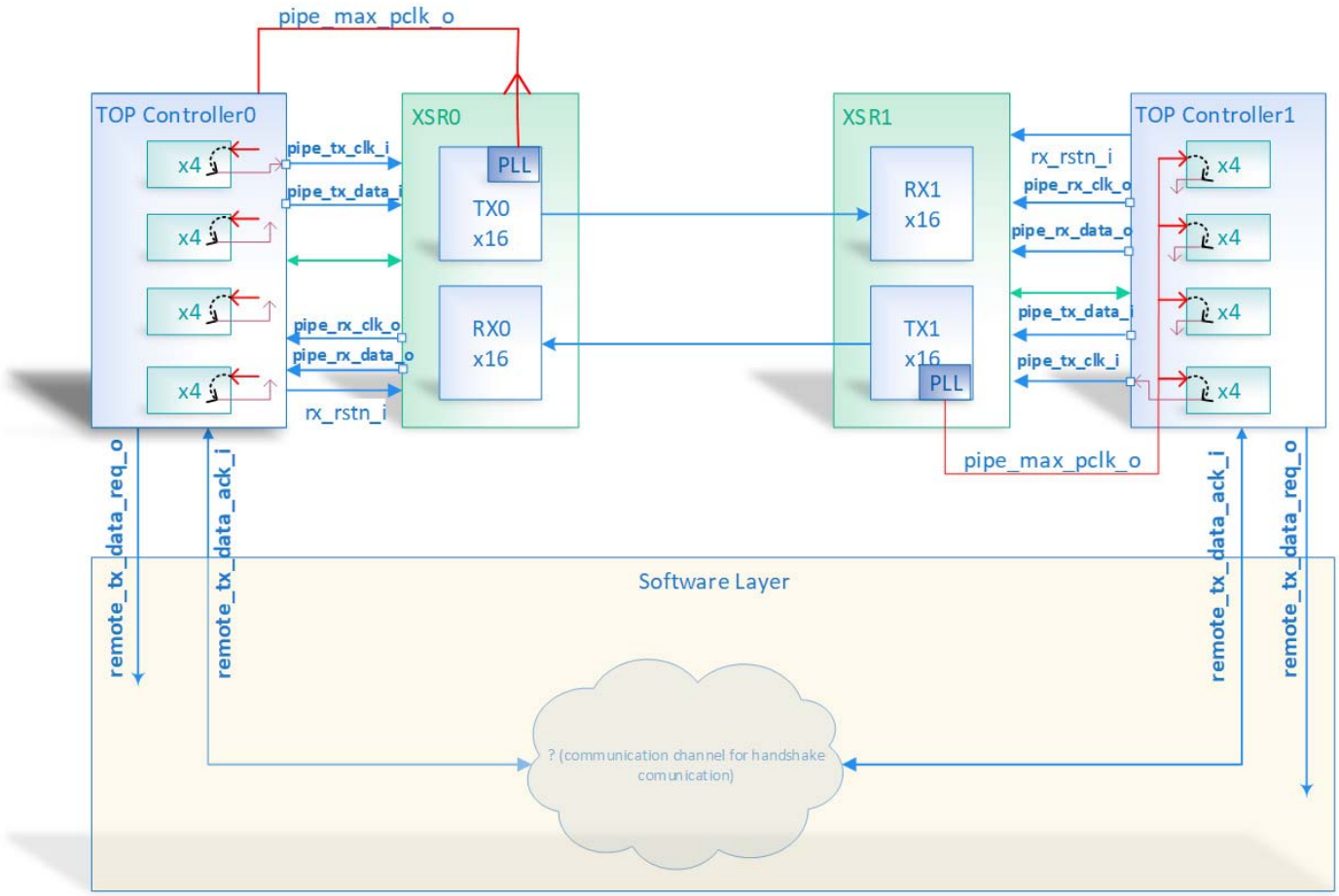
5.3 PHY Initialization Sequence

This section describes the PIPE lite interface for the D2D connections. Since 112G XSR PHY is a product protocol independent, it uses a lite version of the Pipe CxL 5.2.

The PIPE lite interface is handled by firmware.

Figure 5-3 shows an example implementation of the PIPE interface, having near-end and a far-end connection, assuming x16 connection PHY, and forwarded clock mode connection. For more information about forwarded clock connection, see "Forwarded Clock Mode" on page 93.

Figure 5-3 PIPE Interface PHY and Controller Example



The software layer implements the protocol for communication and handshaking between the far-end and near-end sides of the PHY. Communication is done through the controller, each side of the PHY has a controller. The handshaking uses sideband communication.

Although the controller side supports several links (on the example there are four), you only need to perform one synthesis, to respect the clock tree distribution.

The PHY interface uses only one clock and one data stream.

The startup sequence depends on the selected clock mode configuration. This mode is selected prior to the initialization.

5.3.1 PIPE Lite Interface Usage Model with Controller

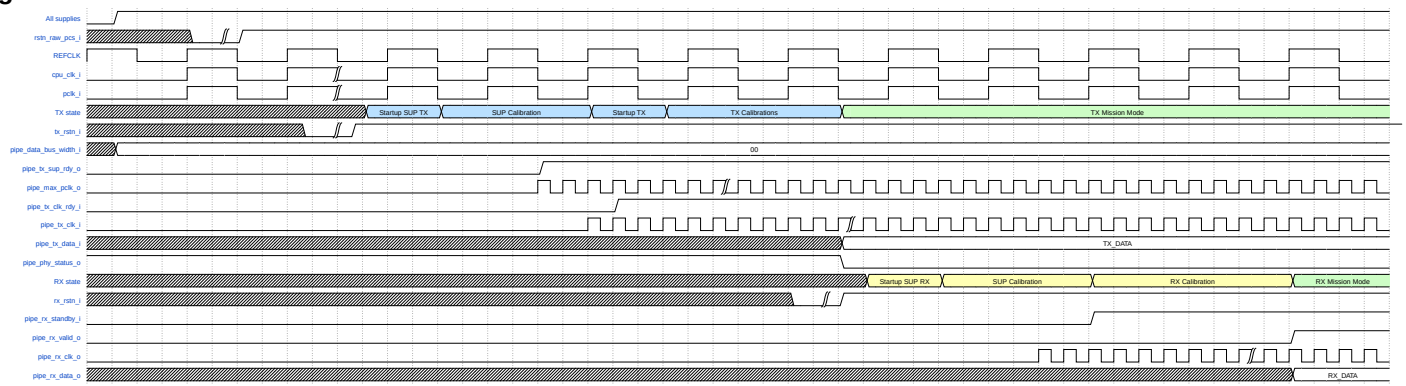
You should respect constraints of data transmission on the PHY. Even if several links exist on the controller side (four in the next example) only one synthesis must be done to respect the clock tree distribution.

For PHY interface, you need to have one clock and one data stream.

5.3.2 Signal Interface and Pinout Dedicated to PIPE

Figure 5-4 shows the PIPE interface initialization timing diagram.

Figure 5-4 PIPE Interface



5.3.3 PIPE Data Organization

Lane ordering the data on the PIPE interface on pipe_tx_data and pipe_rx_data should have the following organization.

Table 5-1 Lane Ordering

pipe_tx_data_i[1023:0]	Lane
pipe_tx_data_i[63:0]	Tx_lane0
pipe_tx_data_i[127:64]	Tx_lane1
pipe_tx_data_i[191:128]	Tx_lane2
pipe_tx_data_i[255:192]	Tx_lane3
….	…..
pipe_tx_data_i[1023:960]	Tx_lane15

Table 5-2 shows the lane data bit organization within lane for PAM4.

Table 5-2 PAM4 Bit Ordering

txX/rxX_data_bit	Order	MSB/LSB	Symbol #
0	oldest	LSB	0
1	oldest	MSB	0

txX/rxX_data_bit	Order	MSB/LSB	Symbol #
2		LSB	1
3		MSB	1
4		LSB	2
5		MSB	2
6		LSB	3
7		MSB	3
8		LSB	4
9		MSB	4
10		LSB	5
11		MSB	5
12		LSB	6
13		MSB	6
14		LSB	7
15		MSB	7
16		LSB	8
17		MSB	8
18		LSB	9
19		MSB	9
20		LSB	10
21		MSB	10
22		LSB	11
23		MSB	11
24		LSB	12
25		MSB	12
26		LSB	13
27		MSB	13
28		LSB	14
29		MSB	14
30		LSB	15

txX/rxX_data_bit	Order	MSB/LSB	Symbol #
31		MSB	15
32		LSB	16
33		MSB	16
34		LSB	17
35		MSB	17
36		LSB	18
37		MSB	18
38		LSB	19
39		MSB	19
40		LSB	20
41		MSB	20
42		LSB	21
43		MSB	21
44		LSB	22
45		MSB	22
46		LSB	23
47		MSB	23
48		LSB	24
49		MSB	24
50		LSB	25
51		MSB	25
52		LSB	26
53		MSB	26
54		LSB	27
55		MSB	27
56		LSB	28
57		MSB	28
58		LSB	29
59		MSB	29

txX/rxX_data_bit	Order	MSB/LSB	Symbol #
60		LSB	30
61		MSB	30
62	newest	LSB	31
63	newest	MSB	31

Table 5-3 shows the data organization within lane for NRZ.

Table 5-3 NRZ Bit Ordering

txX/rxX_data_bit	Order	MSB/LSB	Symbol #
0	older		0
1			1
2			2
3			3
4			4
5			5
6			6
7			7
8			8
9			9
10			10
11			11
12			12
13			13
14			14
15			15
16			16
17			17
18			18
19			19
20			20
21			21

txX/rxX_data_bit	Order	MSB/LSB	Symbol #
22			22
23			23
24			24
25			25
26			26
27			27
28			28
29			29
30			30
31	Newer		31
32…63		…	Zero value

5.4 Datapath

TX and RX datapaths are activated after PIPE Lite interface is initialized. For details, see "PHY Initialization Sequence" on page 83.

5.4.1 TX Datapath

Make sure that the pipe_tx_clk is synchronous with the full bus of data on the pipe_tx_data[N_lanes*64-1:0] for the TX datapath data symbols start to be captured by the PHY on the rising edge of pipe_tx_clk, when pipe_phy_status[N_Lane] is de-asserted.

For lanes belonging to the same link ensure that the pipe_tx_data_ is provided from the same link using the pipe_tx_clk.

The PMA contains a circuit that aligns the incoming txX_clk with the internal transmit clock used in the Analog. This alignment is done internally by the Firmware on the calibrations phase.

This alignment results in a maximum TX skew on txX_p/m across different lanes in the same link to within 2 unit-intervals (UI) + 500 ps.

This procedure requires (mandatory) that Lane 0 is always On.

5.4.2 RX Datapath

The RX datapath data symbols are valid and outputted on the rising edge of the pipe_rx_clk, when pipe_rx_valid[N_lane] is set to one.

Each lane of the PHY has a correspondent pipe_status and rx_valid signal, meaning that lanes are being powered up individually.

The PMA contains a circuit that aligns the internal clock of each lane to the top clock This alignment is done internally by the Firmware on the calibrations phase.

This procedure requires (mandatory) that Lane 0 is always On.

5.4.3 Datapath Latency

The TX and RX datapath latencies for the PMA are measured as per Figure 5-5 and Figure 5-6.

On TX there is a clock alignment mechanism in the PHY to align the internal transmit clocks of each lane to the common pipe_tx_clk inputs. This alignment results in a maximum TX skew on txX_p and txX_m across different lanes of 500 ps.

Figure 5-5 Transmit Datapath Latency Measurement

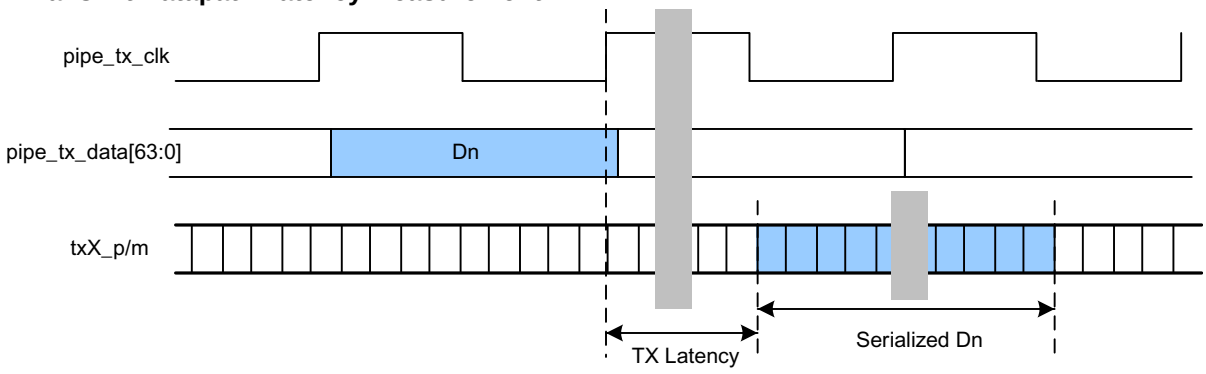


Figure 5-6 Receive Datapath Latency Measurement

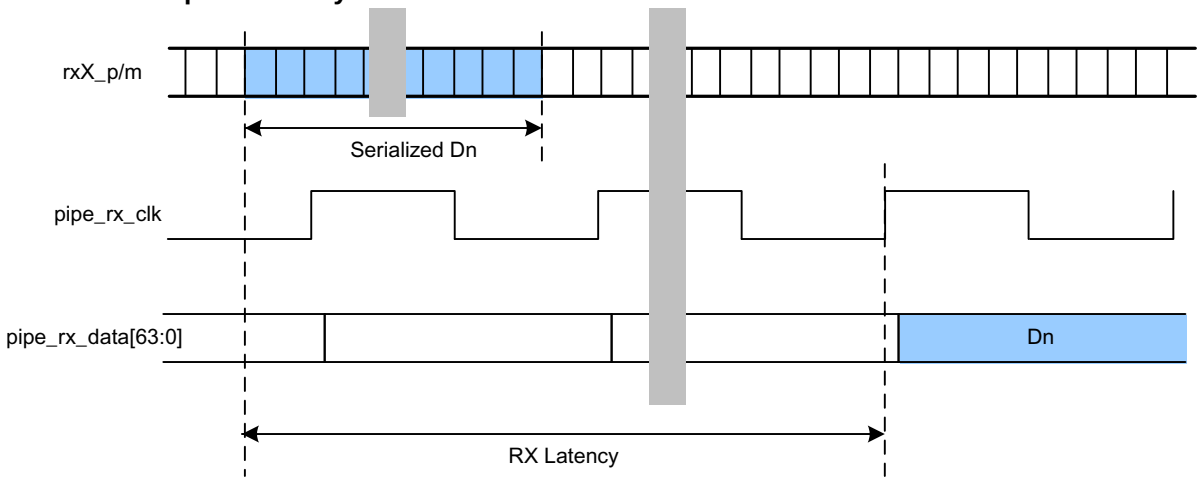


Table 5-4 provides the PHY latency numbers in nano-seconds (ns) and Unit Interval (UI) for TX, RX, and total for TX plus RX datapaths. Minimum and maximum variations are due to pvt corners variations.

Table 5-4 PHY Datapath Latency (Preliminary)

	RX [UI]	RX [ns]	TX [UI]	TX [ns]	Total [UI]	Total [ns]
Minimum	164	2.93	146	2.62	310	5.55
Maximum	NA	NA	NA	NA	NA	NA

Note Latency numbers are under review.

5.5 Datarates

The TX and RX datapaths support different datarates. The datarate configuration affects both TX and RX datatpaths.

TX and RX do not support different datarate configurations, rate needs to always be aligned.

The data rate pin `rate[2:0]` only affects the amount of data per clock cycle. Changing the rate does not affect either the PLL frequencies, or the pipe_tx/rx_clk frequency.

The clock frequency of the PHY is only affected by the `sup_cntx_sel` information, for more information see Appendix A, “PHY Configuration”.

Table 5-5 shows the PHY operation datarates.

Attention Rate signal cannot be changed dynamically. The PHY does not support rate changes on demand.

To change the rate of the PHY, you need to restart the PHY by asserting and resets and re-doing the power-up sequences. Not doing this may result in compromised data.

Table 5-5 Datarates

rate[2:0]	Clocking Rate	PLL Clock Frequency [GHz]	NRZ Datarate [Gbps]	PAM4 Datarate [Gbps]
0	quarter-rate (1/4)	7 – 14	28 – 56	56 – 112
1	half-rate (1/2)	7 – 14	14 – 28	28 – 56
2	full-rate (1)	7 – 14	7 – 14	14 – 28
3	2x oversample	7 – 14	3.5 – 7	7 – 14
4	4x oversample	7 – 14	1.75 – 3.5	3.5 – 7
5,7	reserved	N/A	N/A	N/A

5.6 Clock Modes

The PHY supports the following clock modes configurations.

- forwarded clock
- common clock
- embedded clock

Use pin `clk_mode_i[2:0]` to set the clock mode. You can set the clock source and the type of configuration. This pin must be set before startup.

Table 5-6 is coded in the firmware, no other configuration is needed for control the clock mode.

See Appendix A, "PHY Configuration" for more information about the PHY configuration.

Table 5-6 clk_mode[2:0] Settings

clk_mode[2:0]	'd0	'd1	'd2	'd3	'd4	'd5	'd6	'd7
Clock Source Driven On	(internal) --> tx_ref_alt0_clk_p/m (external) --> rx_ref_pad_clk_p/m	(internal) --> tx_ref_alt0_clk_p/m (internal) --> rx_ref_alt0_clk_p/m			(external) --> tx_ref_pad_clk_p/m (external) --> rx_ref_pad_clk_p/m			
Clock Mode	Forward Clock Mode	Common Clock	CDR Assist	Reserved	Forward Clock Mode	Common Clock	CDR Assist	Reserved

Each configuration type allows to have internal clock sources from the SoC into the PHY or external clock sources from the SoC into the PHY. Table 5-7 shows clock source connection.

Table 5-7 Clock Source Connection

External Clock Source	Internal Clock Source
Connect the external clock to the following bumps. ■ Tx/Rx_ref_pad_clk_p ■ Tx/Rx_ref_pad_clk_m	Connect the internal clock to the following pins. ■ Tx/Rx_ref_alt_clk_m ■ Tx/Rx_ref_alt_clk_p
The following pins should be tight to zero on the inputs. ■ Tx/Rx_ref_alt_clk_p ■ Tx/Rx_ref_alt_clk_m	The following pins should be tight to zero on the inputs. ■ Tx/Rx_ref_pad_clk_p ■ Tx/Rx_ref_pad_clk_m

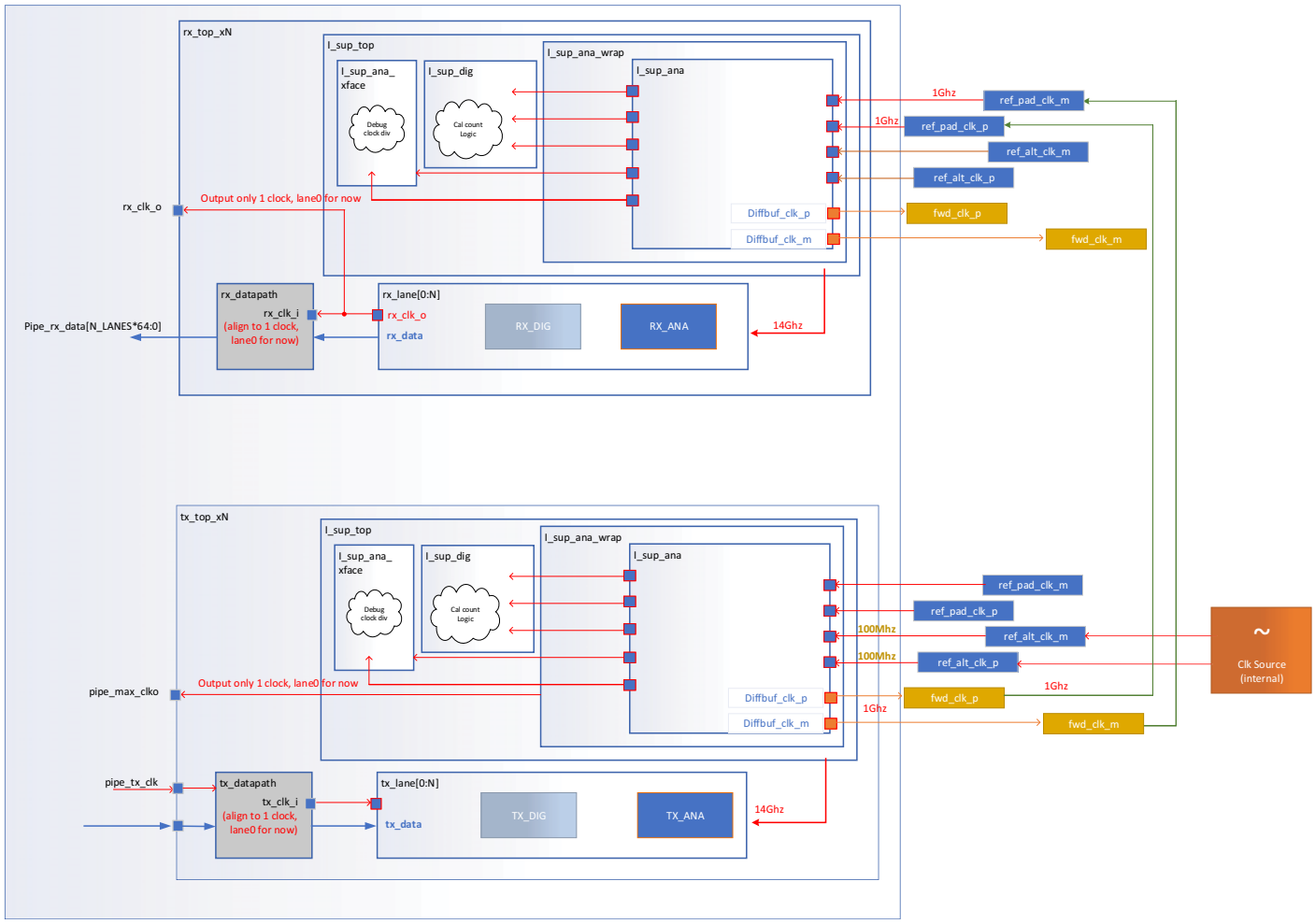
5.6.1 Forwarded Clock Mode

In the forwarded clock mode, TX forwards the clock to the RX. This is the default and the recommended operation mode for the XSR PHY.

Figure 5-7 illustrates a forward clock mode usage example.

For this example configuration make `clk_mode_i[2:0]=3'b000`.

Figure 5-7 Forwarded Clock Mode



5.6.2 Common Clock Mode

The common clock mode uses the same (internal or external) clock source for TX and RX.

Figure 5-8 shows the common clock mode using an external clock source.

For this example configuration make `clk_mode_i=2'b101`.

Figure 5-8 Common Clock Mode Using External Source

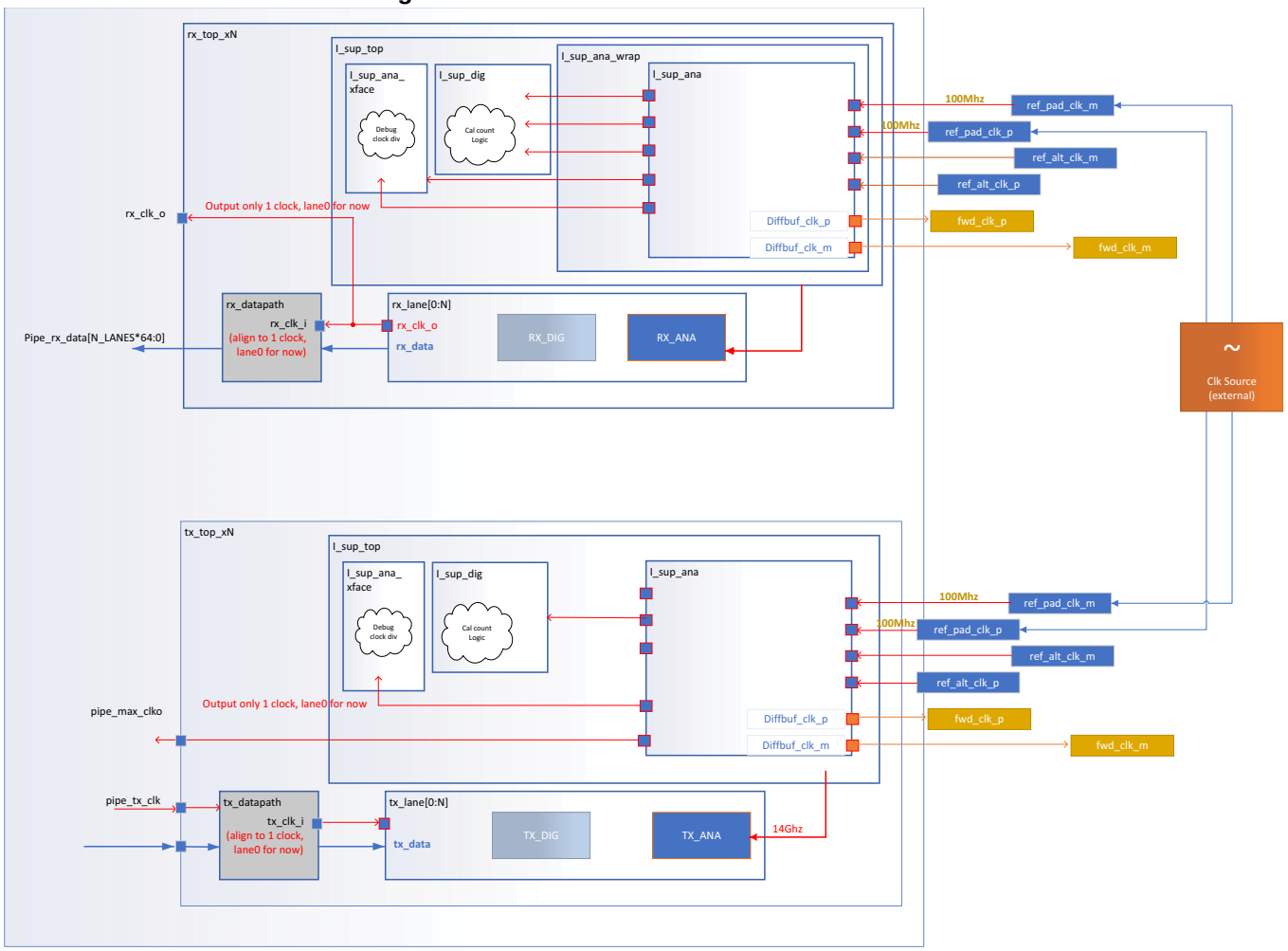
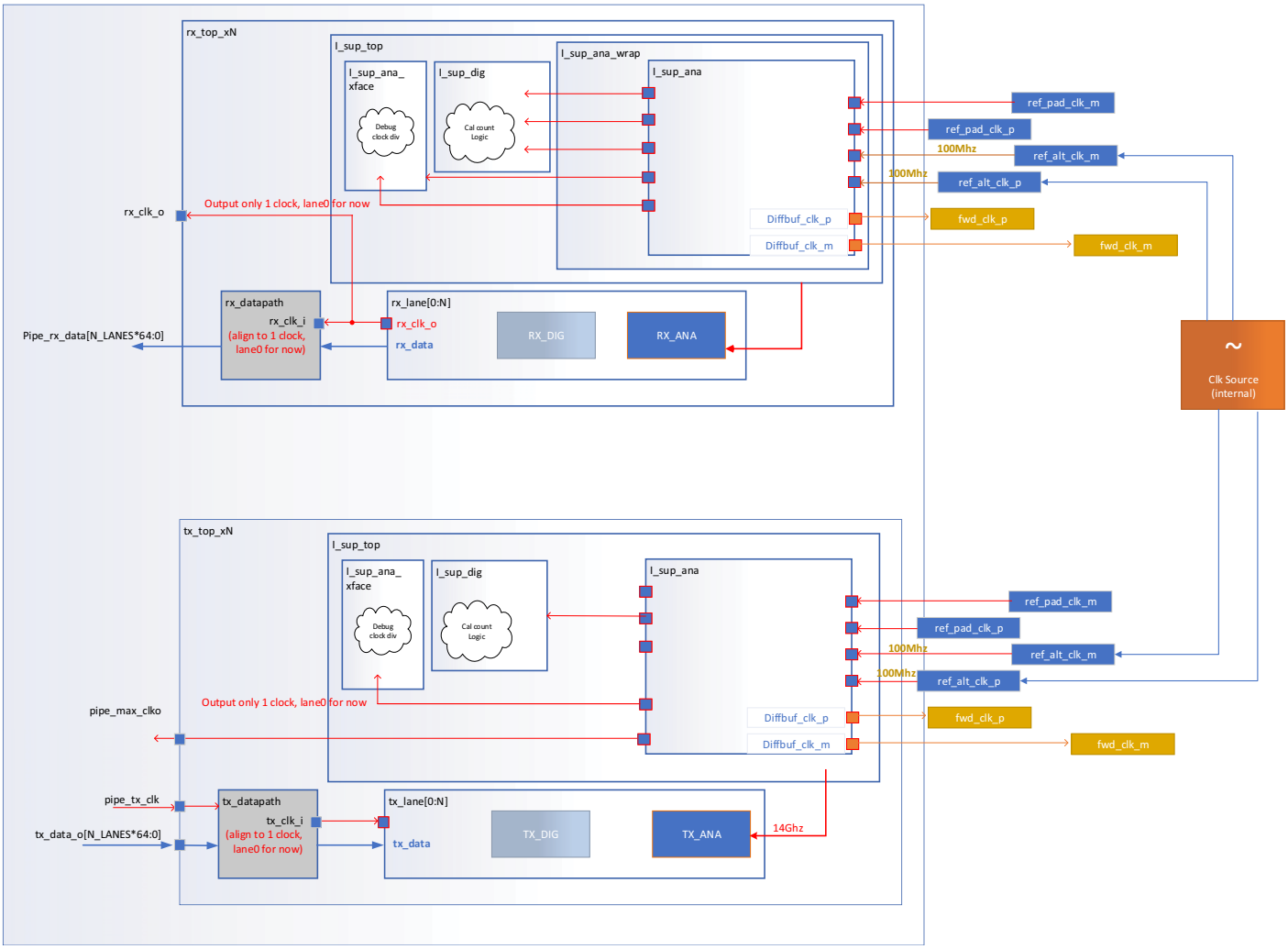


Figure 5-9 shows the common clock mode using an internal clock source.

For this example configuration make `clk_mode_i=2'b001`.

Figure 5-9 Common Clock Mode Using Internal Source



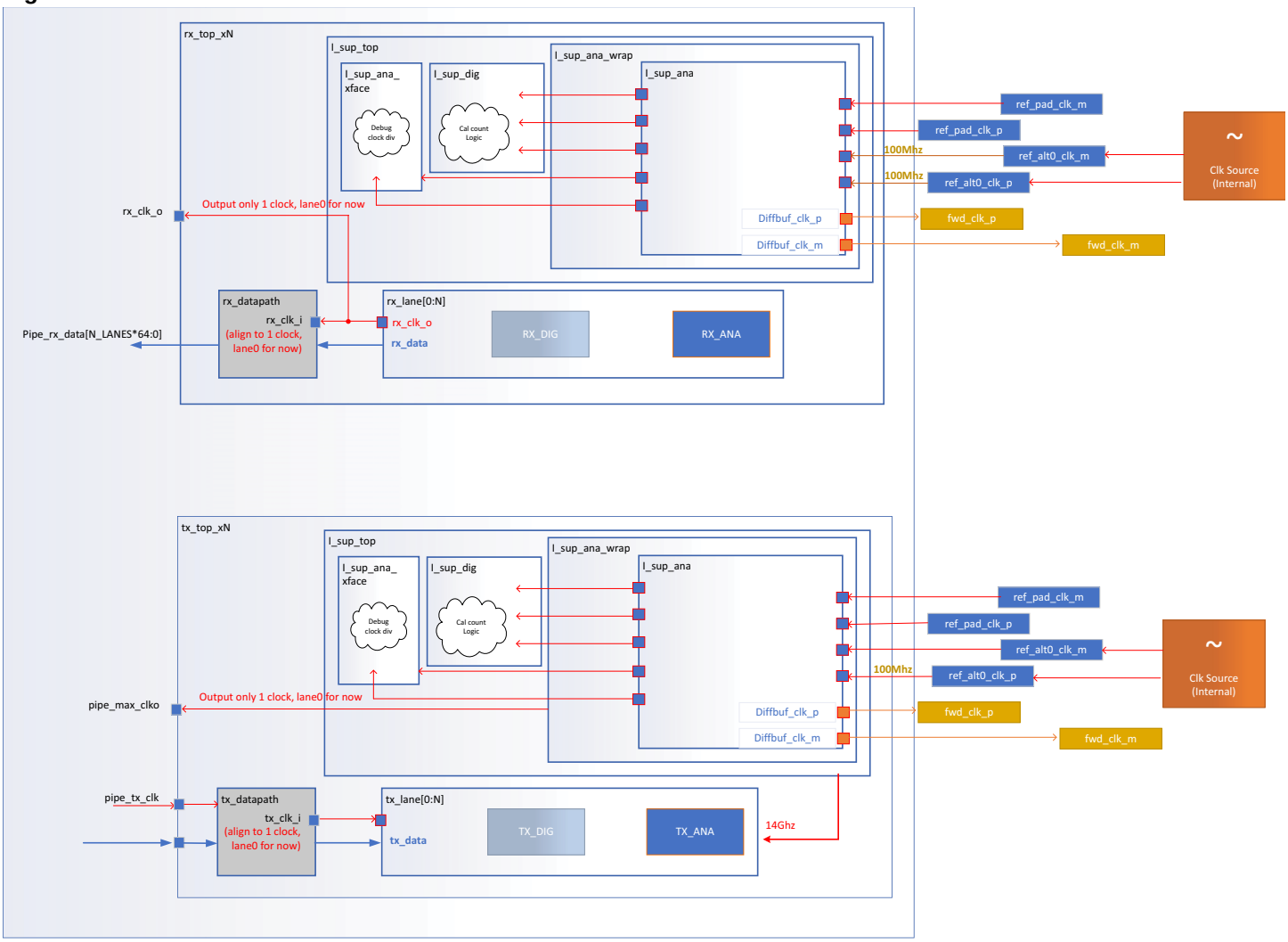
5.6.3 CDR Assist (Embedded Clock)

In the CDR Assist, or embedded clock mode, the PHY has two different clock sources for TX and RX. In this mode, the internal CDR on the Receiver is On.

Figure 5-10 shows the embedded clock mode.

For this example configuration make `clk_mode_i=2'b010`.

Figure 5-10 Embedded Clock Mode



5.6.4 Clock Modes Summary

Each clock mode presented in this section has its own advantages and disadvantages. Table 5-8 outlines the differences between the clock modes as well as summarizes their advantages and disadvantages.

Table 5-8 Clock Modes Usage

Clock Mode	Clock Source	Notes
Forward Clock	One and forwarded to 2nd PHY	■ Lowest Jitter ■ Additional high-speed clock needs to be routed
Common Clock	One Common	■ Slightly higher jitter but relaxed clock routing requirements to PHYs

Clock Mode	Clock Source	Notes
CDR Assist (Embedded clock)	Two Separated	■ Required when PHYs have separate clock references ■ Non-linearities of the additional circuits needed to recover the clock can reduce margin vs. other clocking schemes

5.7 Control Register and Memory Interface

The raw PCS has two control register interfaces and three memory interfaces.

The control register interfaces are used for register access and for the CPU SRAMs slow access; they are listed as follows:

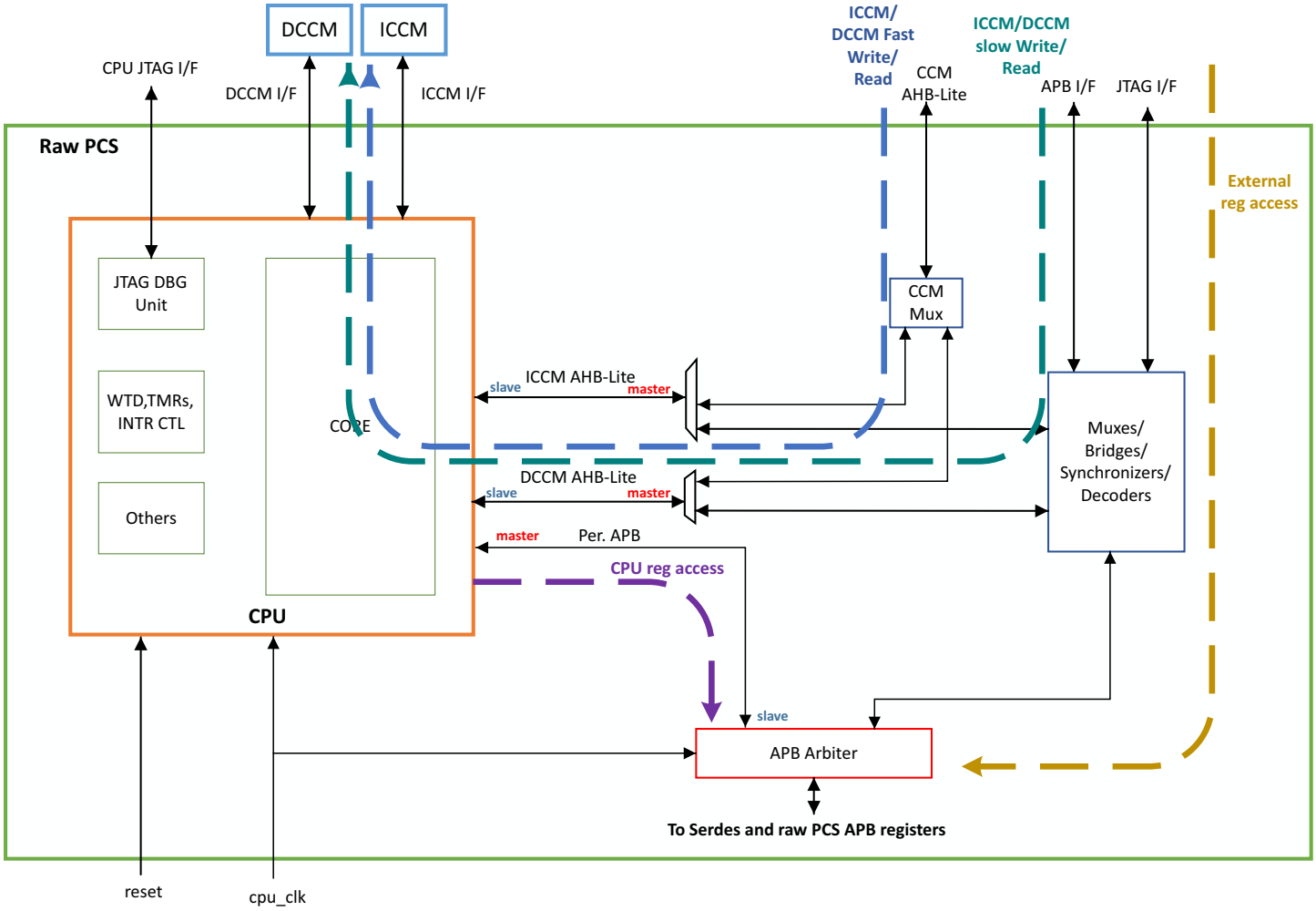
- APB external interface, is the main interface for accessing raw PCS registers. It can also be used to access ICCM and DCCM memories.
- Raw PCS JTAG interface, is an alternative interface to the APB, it has the same access capability.

The memory interfaces are listed as follows:

- CCM AHB-Lite interface, provides a fast loading and access to the CPU SRAMs.
- SRAM interfaces:
  - ICCM interface, used by the CPU to access the external instruction memory.
  - DCCM interface, used by the CPU to access the external data memory.

Figure 5-11 shows the raw PCS control register and memory interfaces.

Figure 5-11 Raw PCS Control Register and Memory Interfaces



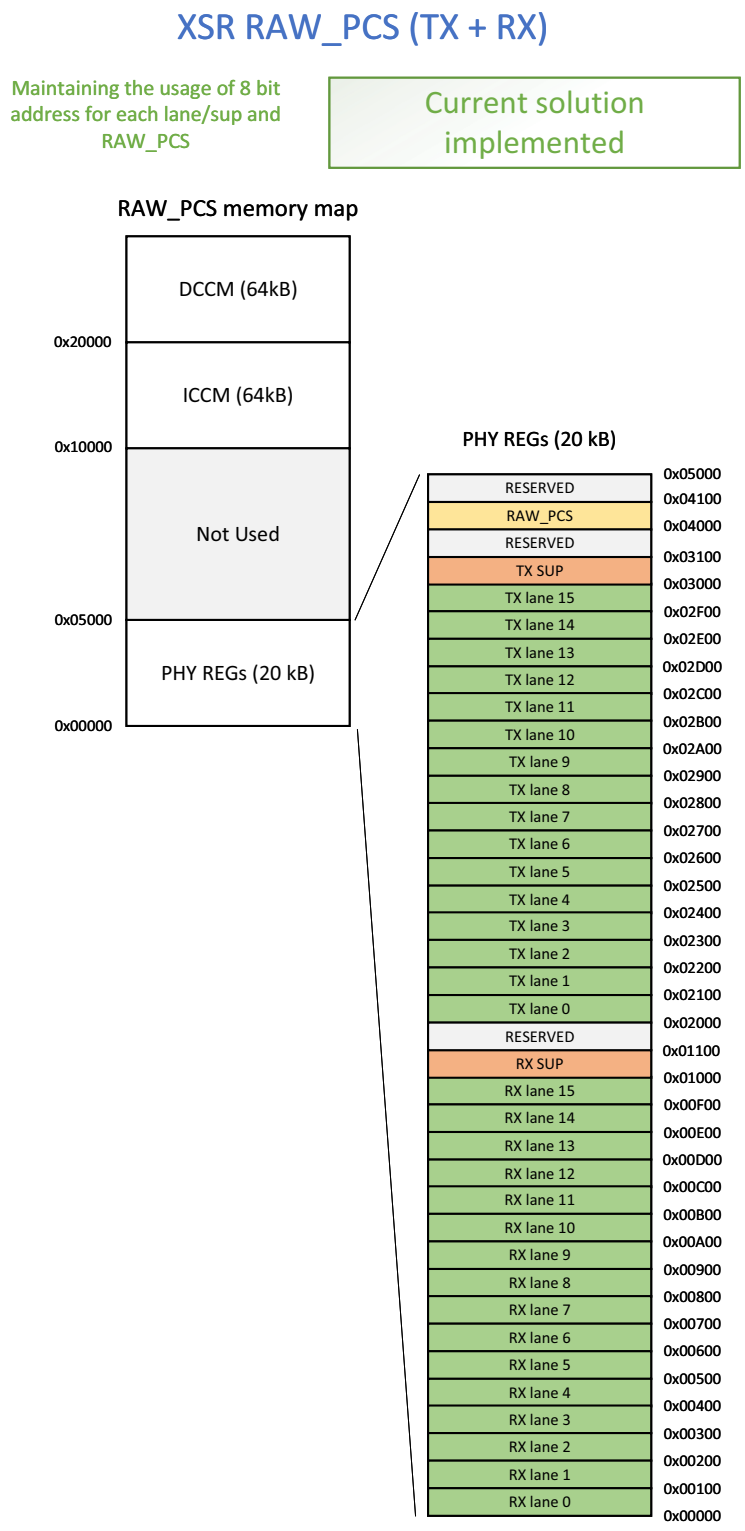
5.7.1 Address Map

Figure 5-12 shows the Raw PCS address map, including control registers and memories. This address map is applicable to the APB, AHB and PMD JTAG interfaces.



Note PHY registers cannot be accessed using the CCM AHB-Lite interface.

Figure 5-12 PHY Memory Address Map



5.7.1.1 RiW Override Value Registers

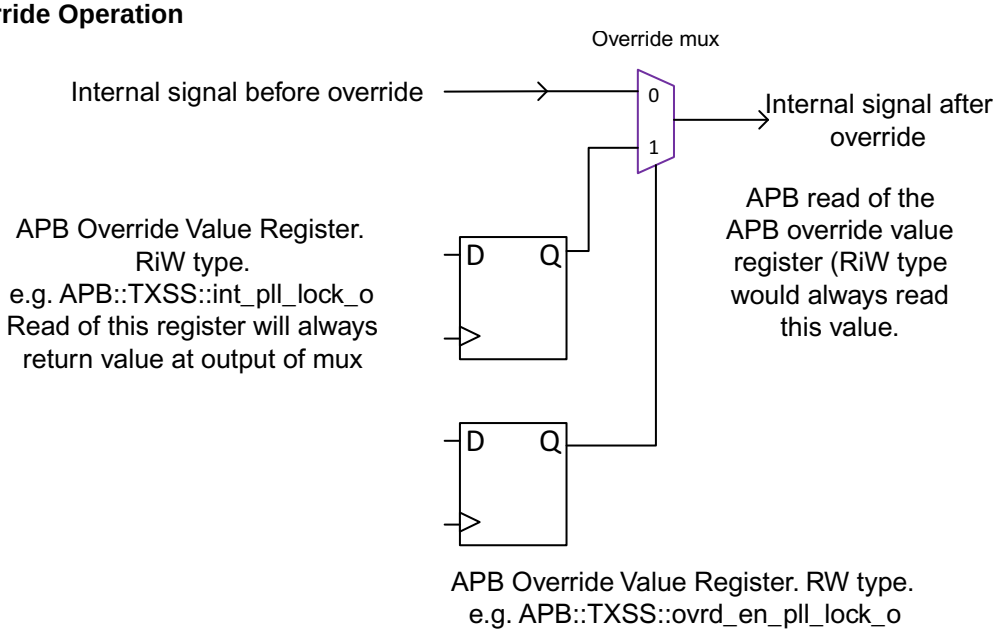
A register field of type RiW is a Read-internal/Write type register.

The RiW register type is used for enabling silicon debug and software workaround by overriding or observing the internal signal value.

> **Note** RiW register type applies to all registers that have the Volatile attribute set to true in the description.

Figure 5-13 shows the structure of the override mechanism using this RiW register.

Figure 5-13 Override Operation



The value returned on reading a RiW register depends on the status of override enable register. If the override is enabled, the read gives the value of the override value register. If the override is not enabled, the read gives the value of the internal signal.

5.7.2 APB Interface Signaling

The raw PCS pins under APB Interface class can be used to configure the PHY registers. See "APB Interface Signals" on page 41 for more information about these pins.

The APB interface is compatible with APB4—for details about the APB protocol, see the AMBA APB Protocol Specification v2.0—with the exception of not supporting the PPROT protection signal. For more information about these pins, see "APB Interface Signals" on page 41.

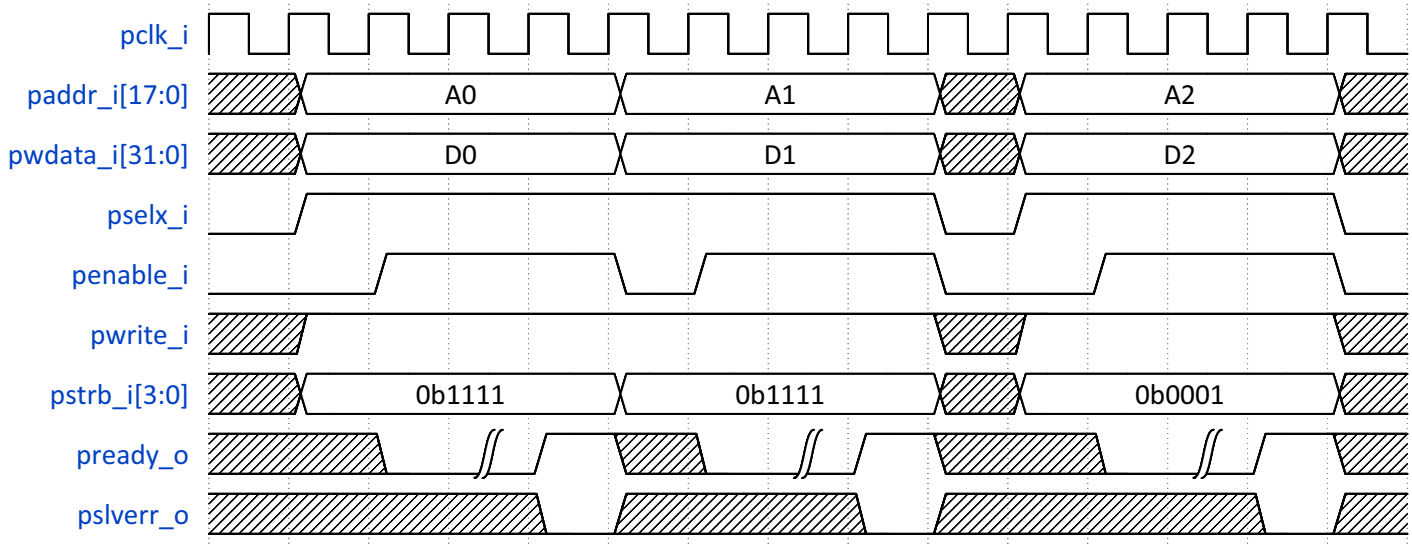
The following figures show timing diagram of APB Interface transactions.

- APB read and write takes several APB clock cycles to complete. See "APB Worst Case Access Latency" for worst case latency.
- `pstrb_i` can be used to do a byte level write transaction. For the read transaction, `pstrb_i` status is ignored.

- `pslverr_o` indicates invalid access, see "APB Slave Error Scenarios" .

Figure 5-14 shows the APB interface timing diagram for APB write transaction.

Figure 5-14 APB Write Transaction Timing Diagram



1. Transaction 1

   Writes all 32 bits of D0 into address A0. `pready_o` asserts after several clock cycles to indicate the completion of the transfer.

2. Transaction 2

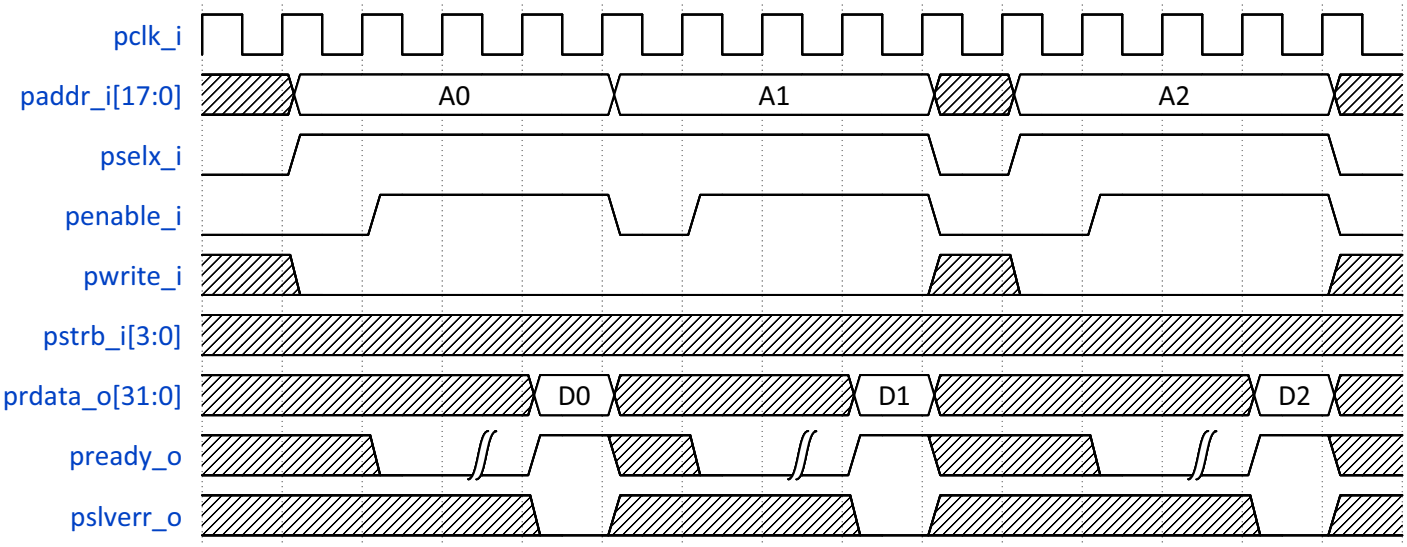
   The next write transfer into address A1 follows without de-asserting `pselx_i`. In this case, `penable_i` must be de-asserted for only one clock cycle.

3. Transaction 3

   Transfer into address A2 is similar to the first one with the exception of only the least significant byte of D2 is written into A2. `pready_o` is valid for only one APB clock cycle, once asserted. `pslverr_o` is valid when `pready_o` is asserted; it is valid for only one clock cycle.

Figure 5-15 shows timing diagram of APB Interface read transaction.

Figure 5-15 APB Read Transactions Timing Diagram



1. Transaction 1

   Reads address A0. `pready_o` asserts after several clock cycles to indicate the completion of the transfer.

2. Transaction 2

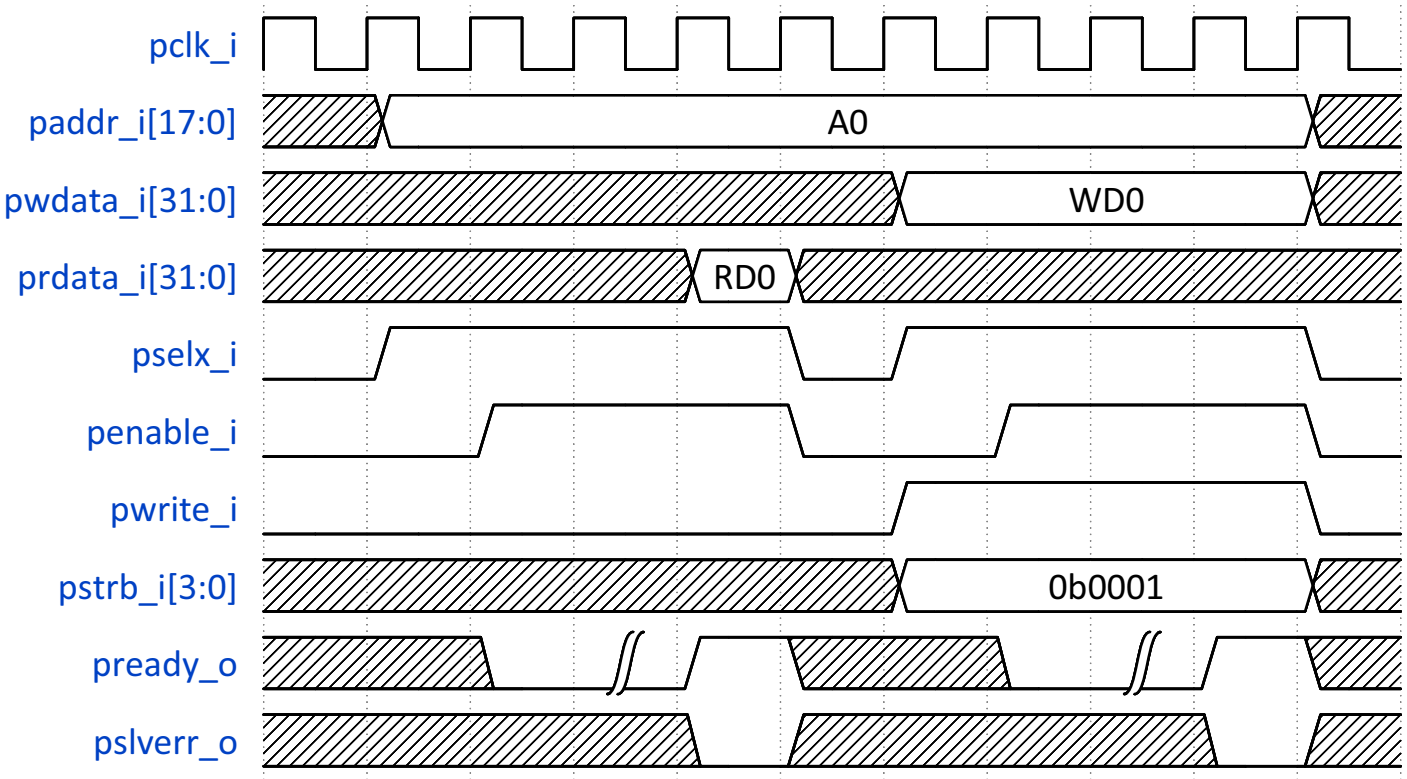
   Read transfer from address A1 follows without de-asserting `pselx_i`. `penable_i` must be de-asserted for only one clock cycle.

3. Transaction 3

   Read from A2 is similar to the first transfer. `pready_o` is only valid for one APB clock cycle. `prdata_o` and `pslverr_o` are valid only when `pready_o` is asserted; they are valid for only one clock cycle. The value of `pstrb_o` has no effect on the read transfers, all 32 bits are always read.

Figure 5-16 is an example of a read-modify-write transaction, where all 32 bits are read, and only the 8 least significant bits are modified and written. Both read and write operations are shown as three cycle transactions.

Figure 5-16 APB Read Write Transition Timing Diagram



5.7.2.1 APB Worst Case Access Latency

The APB access latency is a function of the `cpu_clk_i` and `pclk_i` input ports. The latency is defined as the number of clock cycles from the assertion of `penable_i` to the assertion of `pready_o`. It can be calculated as follows:

5.7.2.2 APB Slave Error Scenarios

The following conditions cause `pslverr_o` to assert indicating an access error:

- Accessing unused address space (for the unused addresses see Figure 5-12)
- Attempting to access ICCM while internal bootloader is running and not done
- Attempting to access ICCM/DCCM when `APB::RAW_PCS::RAW_PCS_CFG::ccm_fast_access_sel=0b1`

5.7.3 CCM AHB-Lite Interface

The Closely-Coupled Memories (CCM) AHB-Lite interface can be used to perform read/write operations on either the ICCM or the DCCM. It is compatible with AMBA3 AHB-Lite Protocol Specification v1.0. This

interface follows the overall raw PCS memory map (see "Address Map" on page 100). PHY registers cannot be accessed by the CCM AHB-Lite Interface, any request to access PHY registers or unused memory space will return an AHB-Lite error response. The CCM AHB-Lite Interface is synchronous to the CPU clock signal.

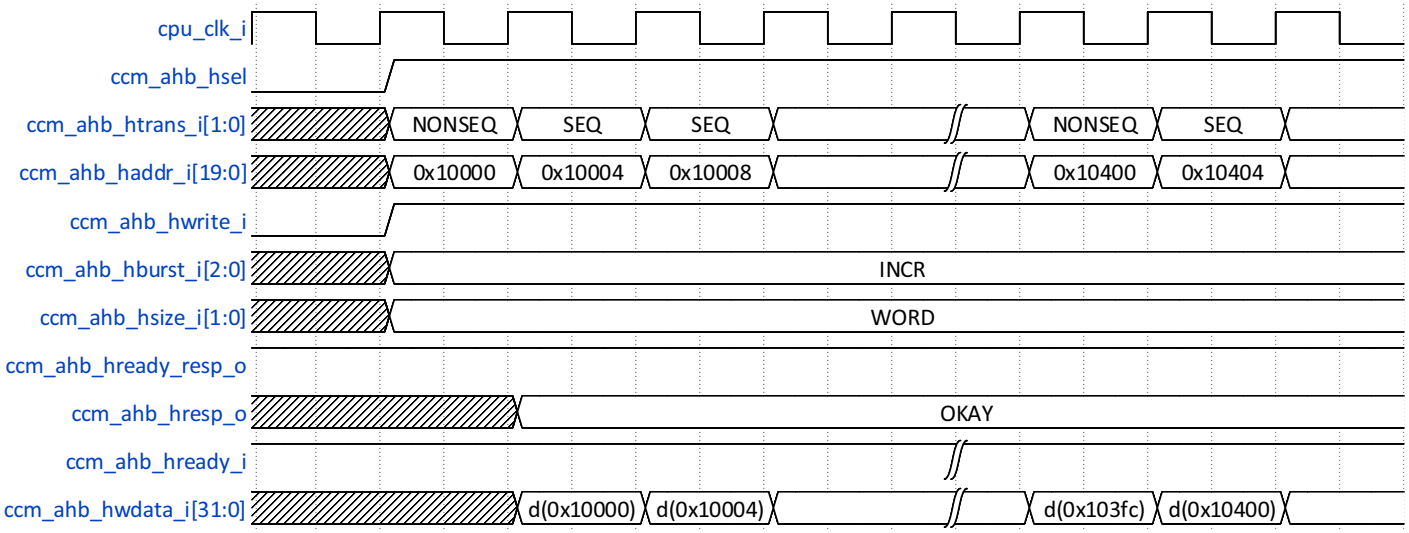
Requirements to use the CCM AHB-Lite Interface:

- The CPU clock must be running (`cpu_clk_i`)
- The PHY reset must be de-asserted (`rstn_i`)
- `APB::RAW_PCS::RAW_PCS_CFG::ccm_fast_access_sel=0b1`

The CCM AHB-Lite Interface can be used to Read the CCMs while the CPU is halted (`cpu_sys_halt_o = 1'b1`) or running (`cpu_sys_halt_o = 1'b0`). Make sure that while loading the firmware into the ICCM, the CPU is halted (`cpu_sys_halt_o = 1'b1`). The CCM AHB-Lite Interface supports AHB Lite burst transfers and, if used, it can sustain one data transfer every clock cycle. According to AHB-Lite protocol, burst size must not exceed 1 kB Address boundary. A single non-burst transfer can take 3 to 20 clock cycles depending on whether the CPU core is running (`cpu_sys_halt_o = 1'b0`) or not, and also on the value of `ccm_ahb_prio_i` which sets the priority of the access. The following are example timing diagrams, for more details, refer to AHB-Lite protocol specification

Figure 5-17 shows a CCM AHB-Lite Burst Write Transfer Timing Diagram. In this example firmware is loaded into ICCM using the CCM AHB-Lite burst transfer.

Figure 5-17 CCM AHB-Lite Burst Write Transfer Timing Diagram



Note

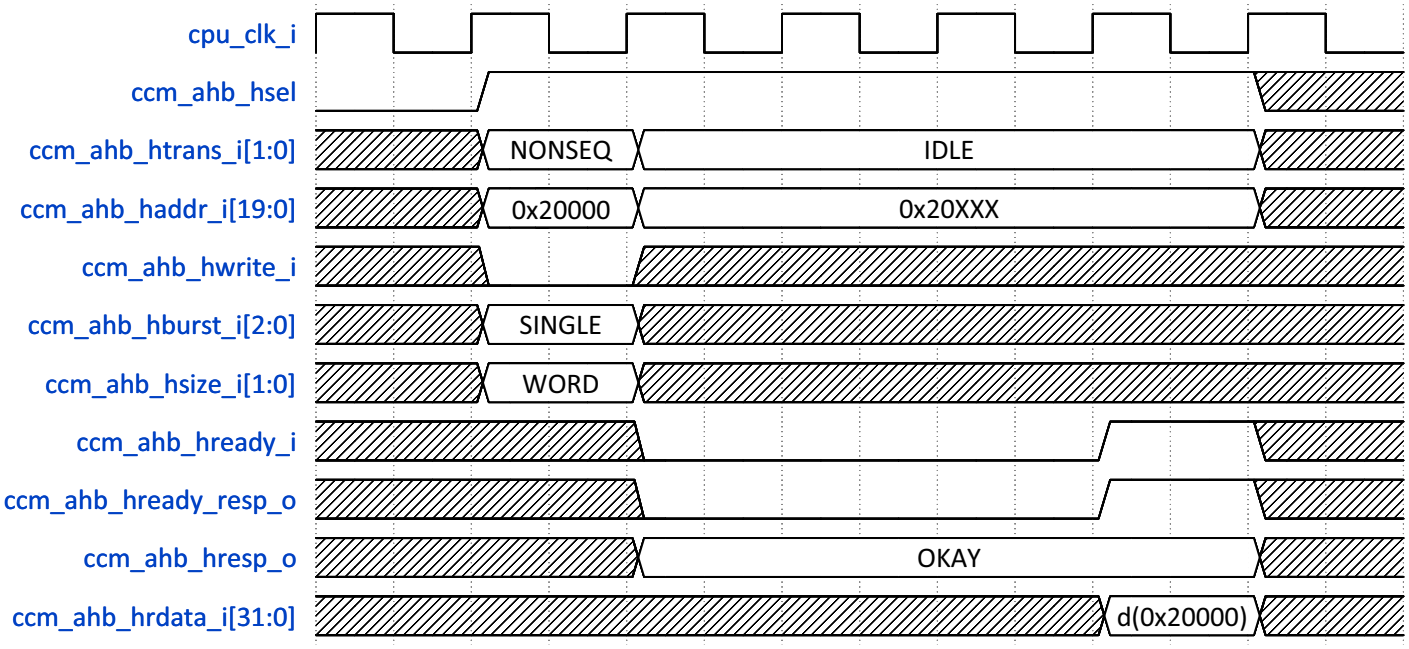
A single burst must not exceed 1 kB boundary by changing the ccm_ahb_htrans_i[1:0] to NONSEQ transfer type.

To load 64 kB of firmware, use 64 bursts of 1 kB of burst size.

By keeping ccm_ahb_prio_i asserted, you guarantee that this bus is given a high priority and avoid having extra cycles. This signal is not part of the AHB protocol.

Figure 5-18 shows a single read access through the CCM AHB-Lite interface.

Figure 5-18 CCM AHB-Lite Single Read Transfer Timing Diagram



DCCM is accessed; it is required to retain its base address in the address bus until the data transfer phase is complete.

As a requirement of the raw PCS implementation, ICCM and DCCM accesses must not be pipelined together. In general, pipelining is only supported for multiple accesses to the same memory (ICCM or DCCM). The base address must remain the same until the data phase is complete. This requirement is applicable to both burst and single transfers.

5.7.3.1 SRAM Interface

The microcontroller uses two closely coupled SRAM arrays called DCCM and ICCM for storage of program data and instructions. The microcontroller accesses the DCCM and ICCM simultaneously over separate interfaces. To support the firmware, the array sizes must be 16K x 32b (64 kB) for ICCM array and 16K x 32b (64 kB) for the DCCM array.

In order to optimize firmware spacing, the firmware is separated into two files:

- firmware file for the ICCM, *_iccm.fw, should be loaded into the ICCM;
- firmware file for the DCCM, *_dccm.fw, should be loaded into the DCCM.

To ensure timing closure and to minimize area required for SRAM, it is not recommended for the microcontroller SRAM to include ECC. The following is the recommended procedure during firmware operation when an extremely heightened level of SRAM fidelity is required:

1. Write firmware into SRAM.
2. Read firmware from SRAM and compare to ensure it is correctly programmed.
3. Execute Firmware.
4. Read Firmware from SRAM to compare again to ensure it is correctly programmed.

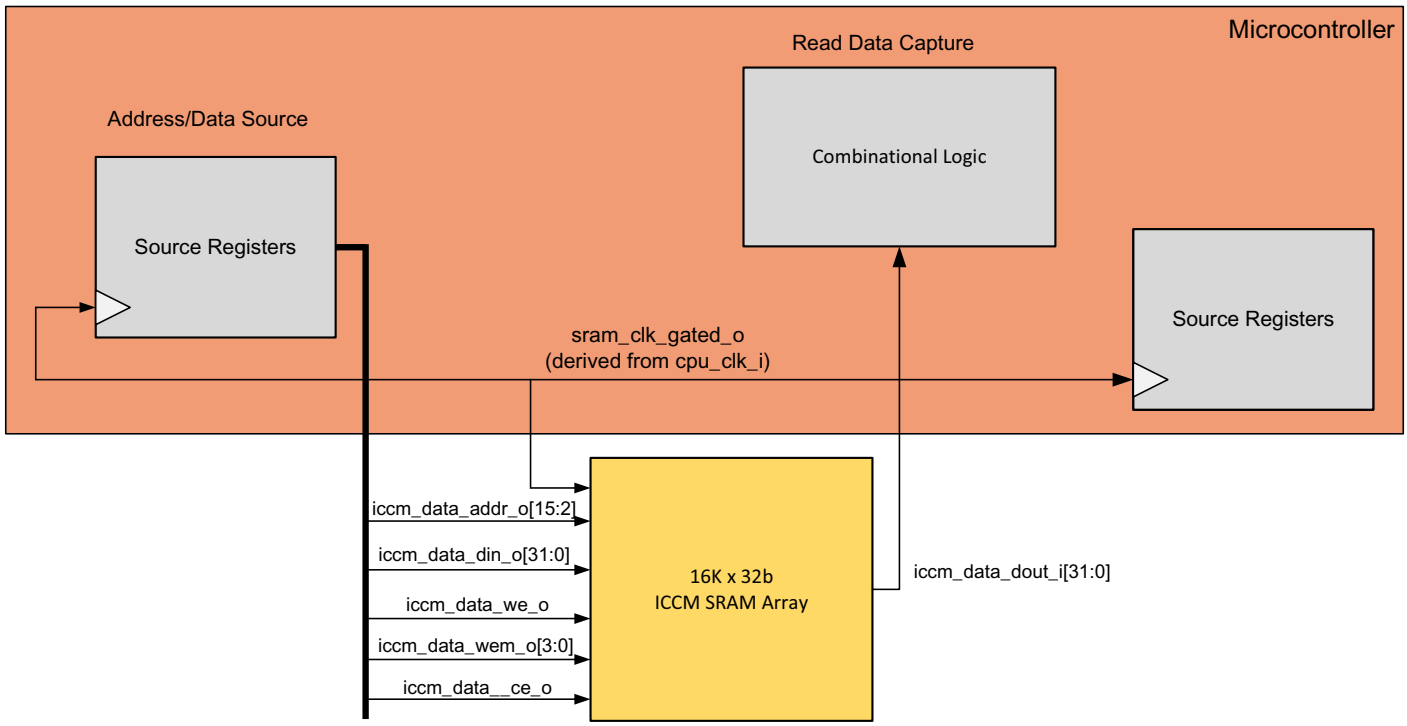
5. Proceed with operation.

For almost all systems, this is not expected to be required.

For the purposes of timing closure, the microcontroller SRAMs should be hardened together with the raw PCS, or abutted directly up against the hardened raw PCS. For best results, it is recommended hardening the SRAMs with the raw PCS. This allows for best placement and timing closure. When reviewing initial SRAM choices, a Clock-to-Q of less than one-third of the CPU clock cycle period is a recommended starting point. As the physical design development progresses, a better understanding of overall device performance and timing is obtained. Additional trade-offs may be made in the SRAM selection to best suit your system, while still meeting overall SRAM to raw PCS timing requirements.

The two memory arrays must be byte-writeable, fully synchronous designs. Each can be implemented as a single memory instance, or built from multiple memory instances. The memories must retain their output values when chip enable is not asserted. You must add latches if your memories change their outputs to the X or Z state when chip enable is not asserted. After address is applied, the raw PCS expects valid data in the next clock cycle.

Figure 5-19 Implementation Example of 64 kB ICCM



The microcontroller design is provided with a behavioral model of the ICCM and DCCM closely-coupled memories. Simulation on the behavioral model can be started immediately before the actual ICCM and DCCM arrays are constructed. Using the behavioral model establishes working tests in simulation to build confidence for simulations with the memory-based array design.

Figure 5-21 and Figure 5-22 show the write and read timing diagrams for the ICCM and DCCM SRAMs interfaces.

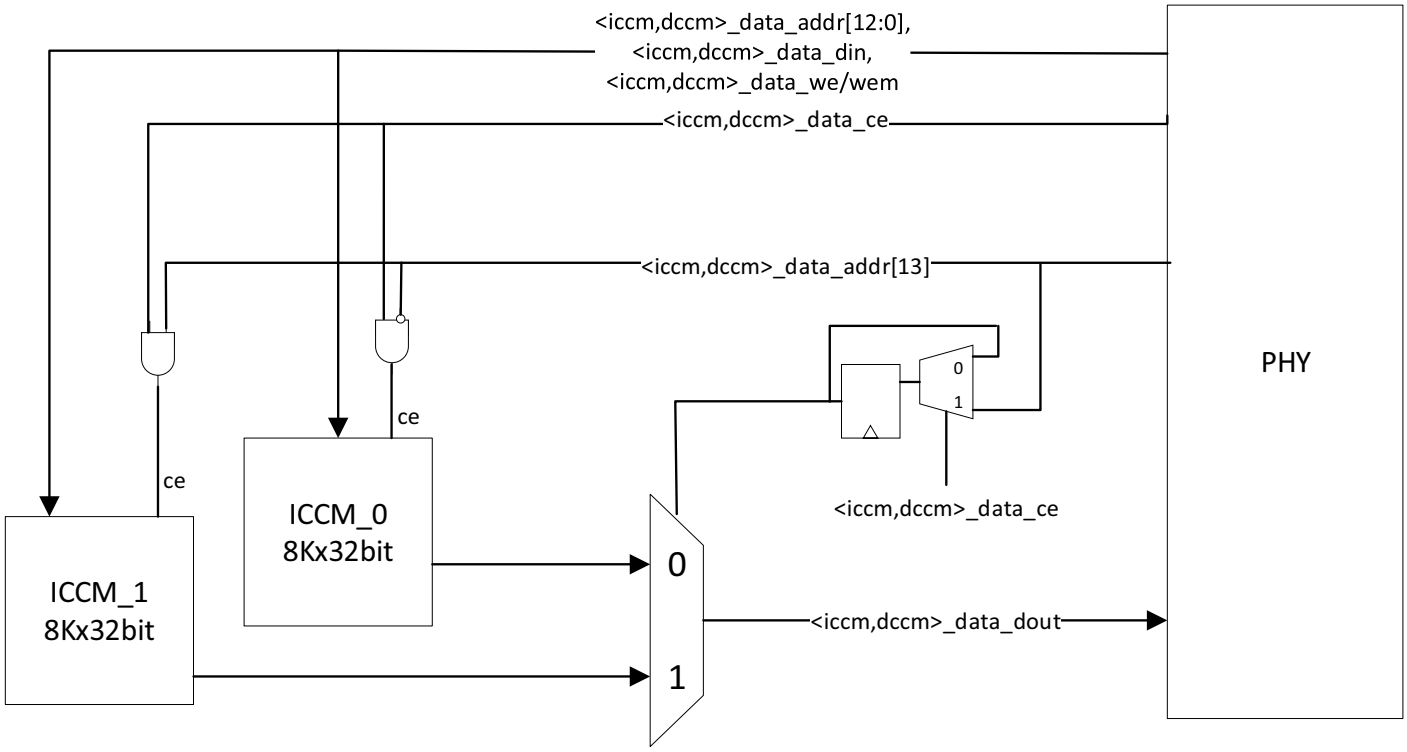
In the write timing diagram, writes can only happen when `<iccm,dccm>_data_ce_o` and `<iccm,dccm>_data_we_o` are high.

In the read timing diagram, the write enable signal `<iccm,dccm>_data_we_o` does not assert. For every cycle where `<iccm,dccm>_data_ce_o` asserts, and with no write enable present, these are true read cycles where the microcontroller expects read data to be available on the next clock cycle.

The microcontroller expects the ICCM and DCCM data output `<iccm,dccm>_data_dout` to be held to the previous value when `<iccm,dccm>_data_ce_o` is deasserted. This may require additional attention when integrating memories. Placing a physical RAM instance into a wrapper is straightforward if you can generate a single physical RAM instance of the required size. Sometimes the maximum available size of a physical RAM instance is smaller than the required memory size. In this case it is necessary to use a few instances together.

Sometimes the maximum available size of a physical RAM instance is smaller than the required memory size. In this case it is necessary to use a few instances together. Figure 5-20 shows one such example where two ICCMs of 32 KB each need to be integrated so as to result into a 64KB memory.

Figure 5-20 Integrating Two ICCM SRAM



Please note that Figure 5-20 is provided for illustration purposes only and should be used as a reference in case the memory is not 64KB or when integrating more than two SRAMs. A similar approach can be applied for DCCM as well. For clarity reasons some signals are not shown in Figure 5-20.

Figure 5-21 SRAM Write Timing Diagram (for ICCM and DCCM)

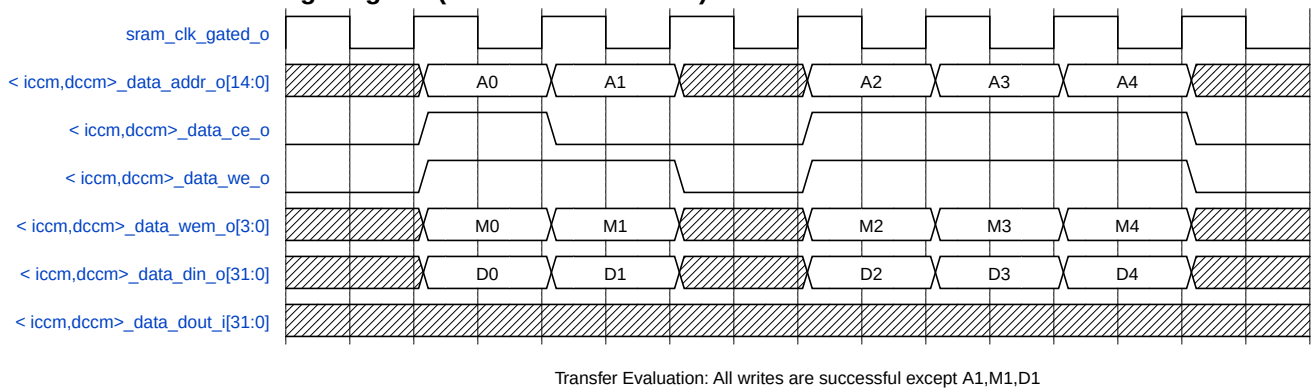
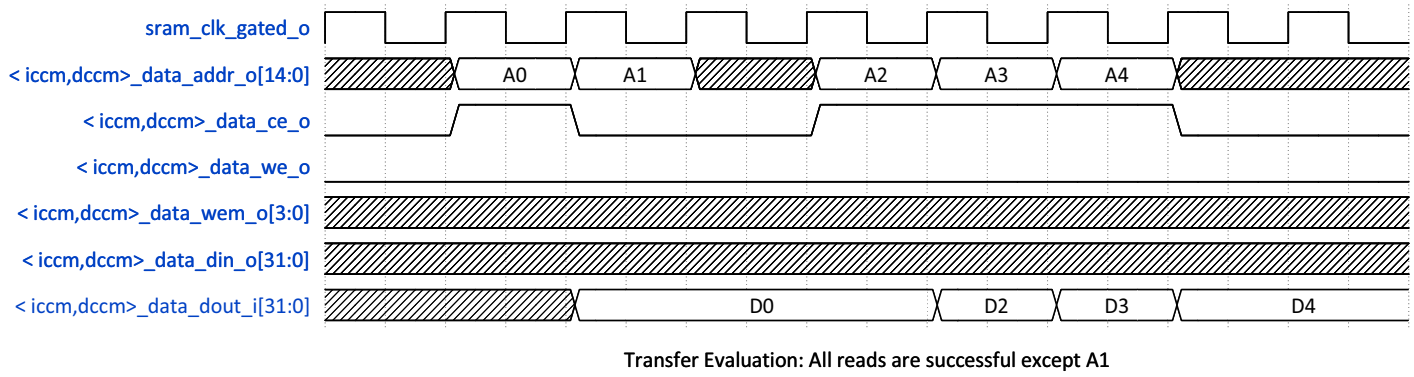


Figure 5-22 SRAM Read Timing Diagram (for ICCM and DCCM)



5.7.4 Raw PCS JTAG Interface

The RAWPCS JTAG interface provides access to an internal TAP controller which implements the APBSEL instruction used for writing and reading registers in the PHY. It can also be used to read/write SRAMs if `APB::RAW_PCS::RAW_PCS_CFG::ccm_fast_access_sel=0b0`, CPU clock is running and PHY reset is de-asserted. The JTAG interface is enabled when input `jtag_apb_sel_i` is asserted to 1. There is one JTAG interface per PHY; for configurations where there are multiple PHYs aggregated (instantiated), there is more than one JTAG interface. To control these with a single TAP controller, the JTAG interfaces can be daisy chained externally. For more information about the Raw PCS JTAG Interface signals see DFT JTAG Signals.

5.7.4.1 IR Codes

Table 5-9 describes the supported JTAG instructions. The LSB of the IR and DR registers is the first bit shifted into `jtag_tdi_i` input.

Table 5-9 Supported JTAG Instructions

Instruction	IR	DR	Operation
IDCODE	8'h01	32 bits	Shift out ID code. The 32-bit IDCODE can be found in the raw_pcs_id_code[31:0] register.

Instruction	IR	DR	Operation
APBSEL	8'h31	34 bits	APB PHY register access.
BYPASS	All others	1 bit	jtag_tdi to jtag_tdo bypass.

5.7.5 CPU JTAG Interface

The CPU JTAG Interface provides a debugger interface for the CPU withing the Raw PCS. This section describes how the CPU JTAG is used to control the processor and to access various resources:

- Processor-internal resources:
  - Core Registers
  - Auxiliary Registers
  - ICCM and DCCM Closely coupled memories
- Processor-visible external resources, peripherals (PHY Registers)

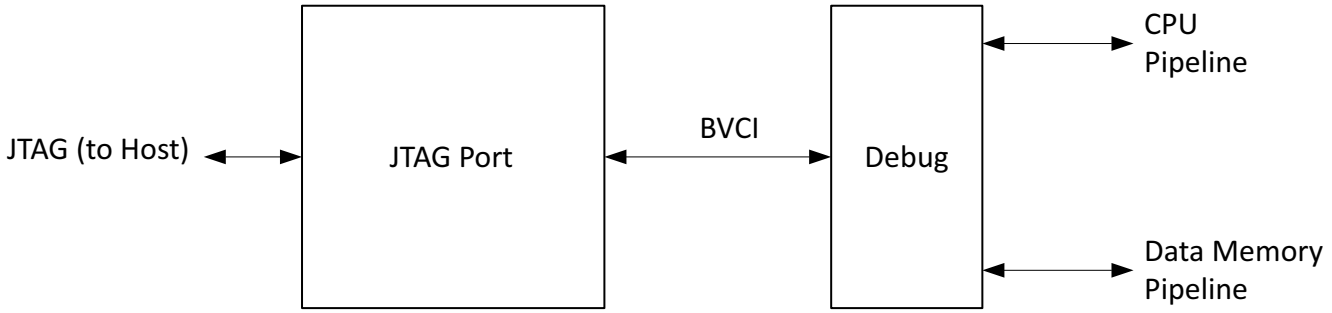
For more information about the CPU JTAG signals see "CPU Signals" on page 46. The JTAG module contains logic for communicating with the microcontroller processor and its memory system. The host interacts with this module through high-level commands that specify transaction parameters. The module supports the IEEE 1149.1 four-wire JTAG interface.

Note More information about CPU JTAG interface will be provided in a future release.

5.7.5.1 JTAG 1149.1 Port

Figure 5-23 shows how the JTAG communications port integrates within the microcontroller system. The port is linked to the host interface of the microcontroller processor and to system memory through the data-memory pipeline. The JTAG port is also known as TAP (Test Access Port).

Figure 5-23 JTAG 1149.1 Port with Microcontroller Processor



5.7.5.2 JTAG 1149.1 Programmer's Model

The JTAG module includes the internal registers as shown in Table 5-10. The host can perform a read or write transaction to a memory location or a microcontroller register using these internal registers. Six of the

seven registers are collectively referred to as data registers (IEEE STD 1149.1). The remaining register is the instruction register, which plays a central role in accessing all debug-interface registers.

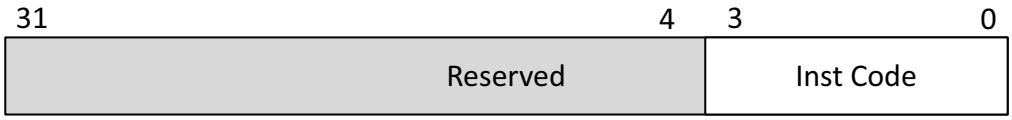
Table 5-10 JTAG 1149.1 Registers

Value	Code	JTAG Register	Type
N/A	N/A	INSTRUCTION REGISTER	Instruction
0x8	1000	JTAG STATUS REGISTER	Data
0x9	1001	TRANSACTION COMMAND REGISTER	Data
0xA	1010	ADDRESS REGISTER	Data
0xB	1011	DATA REGISTER	Data
0xC	1100	IDCODE REGISTER	Data
0xF	1111	BYPASS REGISTER	Data

5.7.5.2.1 JTAG 1149.1 Instruction Register

The instruction register is used to access all data registers. Each data register is addressed by a unique 4-bit instruction code, see Figure 5-24.

Figure 5-24 Instruction Code



The correct code must be written into the instruction register to access the required data register. The instruction shift register is loaded with 0x1 in capture-IR, so that external circuitry—by looking for a transition when in shift-IR—can detect a stuck-at fault in the JTAG chain.

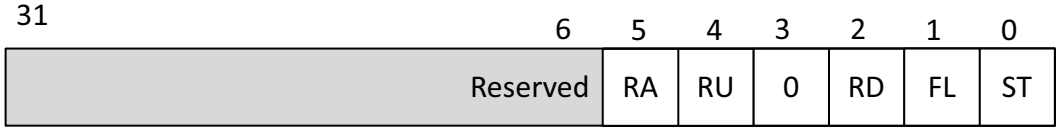
Because of this behavior, which is specified in IEEE 1149.1, the current contents of the instruction register cannot be read by the external circuitry. The register itself is initialized to point to the IDCODE register in test-logic-reset.

In addition to accessing the data registers, the instruction register is also used to select a test sequence to apply to the device.

5.7.5.2.2 JTAG 1149.1 Status Register (Instruction Code 0x8)

The JTAG status register is read-only and is used by the host device to obtain important information on the state of the JTAG module or the result of debug operations. See Figure 5-25 for the register bit mapping.

Figure 5-25 Instruction Code 0x8



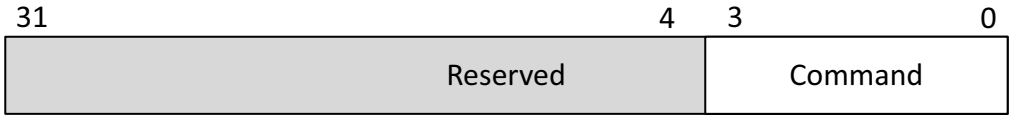
Each field in the JTAG status register reflects the following information:

- Bit 0 - The stalled (ST) flag indicates that the current transaction is still in progress.
- Bit 1 - When true, the failure (FL) flag indicates that a read (or write) has failed. For example, this flag is set if an access to an unimplemented core occurs. The failure flag is cleared automatically when a new transaction starts.
- Bit 2 - The ready (RD) flag indicates whether the JTAG module is available to accept another transaction command. This flag is set when a transaction has just completed or when the JTAG module is idle.
- Bit 3 - Always zero.
- Bit 4 - The run (RU) flag is set to one when the microcontroller processor is running. This bit is cleared to zero when the processor is halted. This flag is the inverse of the halt bit in the microcontroller STATUS32 register.
- Bit 5 - The Reset Applied (RA) flag is used by the debug host to determine that a target reset has occurred. The RA flag is set to 1 on a hard reset of the CPU and can be written only by the host.
- Bits[31:6] - Reserved.

5.7.5.2.3 JTAG 1149.1 Transaction-Command Register (Instruction Code 0x9)

The transaction-command register is used to specify the communication transaction to be performed.

Figure 5-26 Instruction Code 0x9



The JTAG module supports several different accesses or transactions, shown in Table 5-11 with their associated encodings.

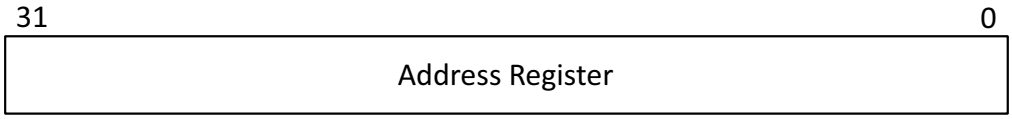
Table 5-11 JTAG Read and Write Transactions

Value	Code	Communication Transaction
0x0	0000	Write to a memory location.
0x1	0001	Write to a microcontroller core register.
0x2	0010	Write to a microcontroller auxiliary register.
0x3	0011	NOP. The register is initialized to this value.
0x4	0100	Read from a memory location.
0x5	0101	Read from a microcontroller core register.
0x6	0110	Read from a microcontroller auxiliary register.

5.7.5.2.4 JTAG 1149.1 Address Register (Instruction Code 0xA)

The address register is used to supply the address for read and write transactions to the microcontroller registers and system memory.

Figure 5-27 Instruction Code 0xA



Accesses to memory must always be given in bytes. Access to the microcontroller internal registers is specified by their register numbers. The value contained in this register is automatically incremented by four (a memory access) or one (a register access) when a read or write transaction completes. This feature is used to save cycle time when downloading or uploading a stream of data. For this reason, the register does not need to be rewritten with the next address value.

5.7.5.2.5 JTAG 1149.1 Data Register (Instruction Code 0xB)

The data register performs two functions. When data is written to this register, it is placed into a write buffer that drives two write data buses, one for the microcontroller host interface and another for the memory arbitrator interface. This bus is used to specify the data contents to be written when performing a write transaction.

Figure 5-28 Instruction Code 0xB

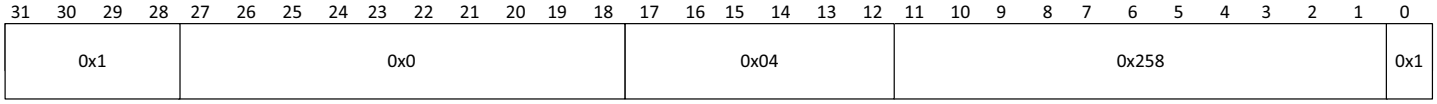


A read buffer is selected to read this register. The read buffer stores data retrieved from the target device during a read transaction. The appropriate read data bus (arbitrator or host interface) is selected according to which device the host is accessing.

5.7.5.2.6 JTAG 1149.1 IDCODE register (Instruction Code 0xC)

The IDCODE register is used by the JTAG emulator to identify the core. The IDCODE register is selected when a reset is applied to the JTAG module.

Figure 5-29 Instruction Code 0xC



Each field in the IDCODE register reflects the following information:

- Bits[31:28] – This field is set to 0x1.
- Bits[27:18] – This field is set to 0x0.
- Bits[17:12] – This field is set to 0x04.
- Bits[11:1] – This field is set to 0x258.
- Bit 0 – This field is fixed at 1, as specified in the IEEE 1149.1-2001 standard. This field is used with Bits[11:1] to allow automatic discovery when the chain is initialized.

  JEDEC never assigns the manufacturer's code 0x7F in group zero. The JTAG emulator can therefore shift the 32-bit IDCODE 0x000000FF into TDI at the beginning of the chain after reset. The standard

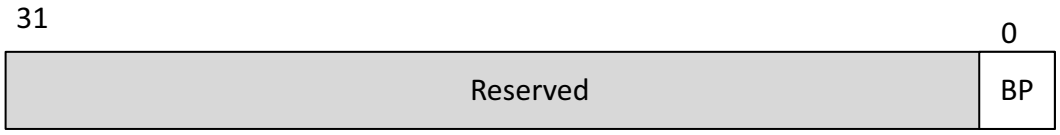
specifies that upon receiving a TCK when in the reset-test-logic state, the instruction register must be initialized to point to the IDCODE register if this register exists, and to the BYPASS register otherwise.

In capture-DR, the shift register is loaded with this 32-bit code if IDCODE is in the instruction register, and with a single bit of zero if BYPASS is. Thus the external circuitry can examine TDO in shift-DR and determine whether this TAP controller has only a one-bit bypass register (TDO = zero), or a 32-bit IDCODE register (TDO = 1). By shifting through and examining until 0x000000FF appears, the emulator can uniquely identify the number and kind of devices in the chain.

5.7.5.2.7 JTAG 1149.1 Bypass Register (Instruction Code 0xF)

The bypass register is required as part of the IEEE STD 1149.1 standard.

Figure 5-30 Instruction Code 0xF



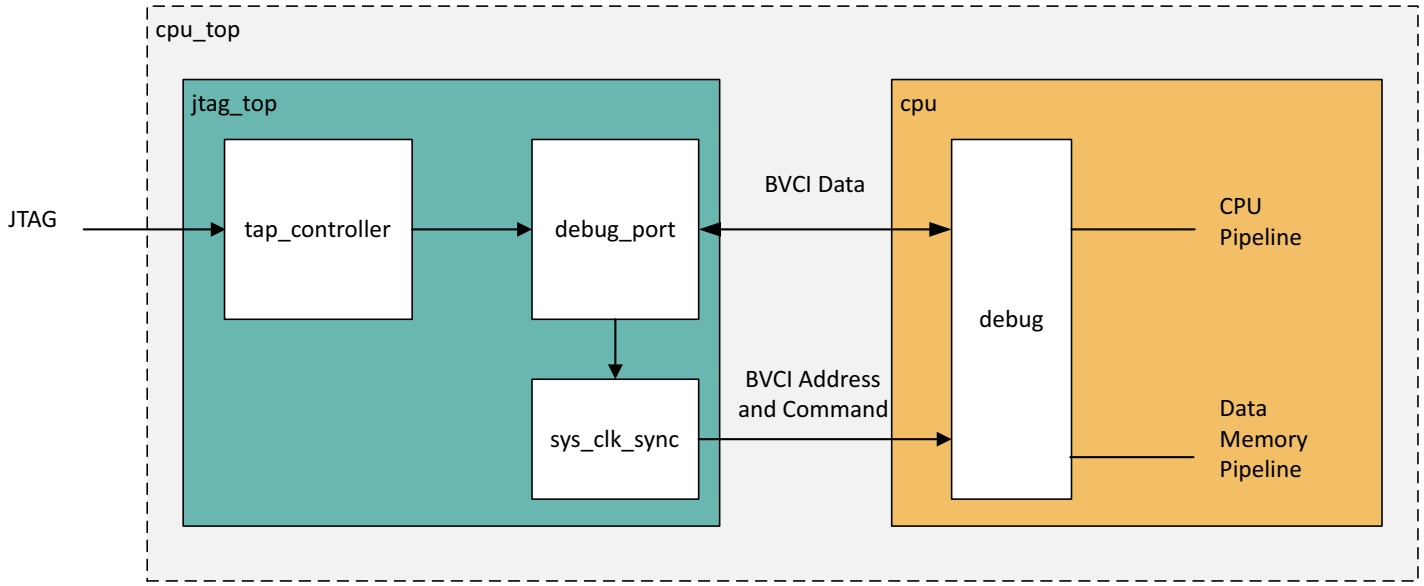
When this register is selected, the serial data input (TDI) is connected to the serial data output (TDO) through this register. The data on TDI is passed to TDO on the rising edge of the JTAG clock TCK when in shift-DR, and the register is initialized to 0 in capture-DR. In all other states, the data in this register is held.

5.7.5.3 JTAG 1149.1 TAP Controller

The Test Access Port (TAP) controller is central to the operation of the JTAG module as shown in Figure 5-31. All internal register accesses are performed serially using the TAP controller. An accompanying block, the debug port, performs the majority of internal tasks (accessing internal JTAG registers) and external tasks. The debug port and the TAP controller are clocked by TCK.

On the other side of the BVCI Debug Interface, the CPU debug module contains the address, data, command, and status registers, and handles the host interface and data memory pipeline transactions. This allows you to optionally replace the JTAG port with a custom interface to an external debugger or to another processor in a master-slave configuration.

Figure 5-31 Internal Structure of the JTAG 1149.1 Port



The TAP controller is an internal state machine that is controlled entirely by the host using the TMS and TCK interface signals. The controller is used to indirectly initiate communication transactions and access the internal JTAG registers.

5.8 TX and RX Power States

The PHY does not support power states. It only supports:

- Power Down (OFF)
- Mission Mode (ON)

5.9 Disabling a PHY

To disable a PHY, follow the method outlined in Table 5-12.

Table 5-12 Signal Connections If the PHY is Not Used

Signal	All Power Supplies (vp, vph, vpdig, gd) at Nominal Voltage and IDDQ* Mode Enabled [a]
PMA inputs from RAW PCS	Not required[b]
PMA outputs to RAW PCS	Not required[b]
rxX_p, rxX_m	Leave floating or tie low
txX_p, txX_m	Leave floating
resref	Leave floating
ref_pad_clk_p, ref_pad_clk_m	Tie low
ref_alt_clk_p, ref_alt_clk_m	Tie low

a. For enabling the IDDQ Mode, see “IDDQ Mode” on page 121

b. Assuming VDD_SOC is supplied and at nominal voltage

Note

- To prevent the PHY circuitry from inadvertently toggling, it is good practice to tie all the inputs to the ground.
- For REXT sharing, the request and ack handshake logic within each PHY must be powered and must allow the handshake signals from other PHYs to propagate through. For more information about resistor sharing, see Figure 8-10 “Reference Resistor Contention Bus - Multiple PHYs Sharing a Single Resistor” on page 146
- If the clock is being repeated, ensure that you do not power down the PMAs in the chain.
- For more information about IDDQ mode, see “IDDQ Mode” on page 121.

5.10 Firmware

The PHY can have all configurations available over IP Firmware. This reduces the level of complexity when configuring the PHY, including registers configurations, startup, change clock modes, change clock frequencies, and when performing all the needed calibrations to have the PHY in mission mode. Firmware is also responsible to handle the PIPE interface.

Before launching the Firmware make sure that the primary inputs are set to the desired configuration values. Reference clocks are set according to these configurations.

- PHY data rate: pin `rate[1:0]`
- Clock rates (PLL Frequencies): pin `sup_cntx_sel` see Appendix A, “PHY Configuration”
- `signal_type` (PAM4/NRZ)
- Clock modes: pin `clk_mode[2:0]`see Table 5-6 on page 93

Table 5-13 shows example PHY configurations using 100-MHz refclk.

Table 5-13 PHY Configuration Examples

Bitrate [Gbps]	Rate	PAM4	PLL CLK [GHz]=bitrate * rate / (PAM4+1)	Data Bits Valid=’LSB	dig_clk [GHz]=bitrate/ bus_width	sup_cntx_sel [6:0]	rate[2:0]
112	1/4	1	14	64	1.75	‘d12	3'b000
56	1/4	1	7	64	0.875	‘d7	3'b000
28	1/2	1	7	32	0.875	‘d7	3'b001
56	1/4	0	14	32	1.75	‘d12	3'b000
28	1/4	0	7	32	0.875	‘d7	3'b000
14	1/2	0	7	16	0.875	‘d7	3'b001
7	1	0	7	8	0.875	‘d7	3'b010
3.5	2x	0	7	4	0.875	‘d7	3'b011
1.75	4x	0	7	2	0.875	‘d7	3'b100

The full lookup tables are coded in the firmware, so select the `sup_cntx_sel` and the desired clock mode. For more information about configurations, see Appendix A, “PHY Configuration”.

6

Test and Diagnostics

This chapter details supported test procedures and diagnostic features. It includes the following sections:

- "Test Modes" on page 120
- "Diagnostics" on page 123

6.1 Test Modes

This section provides information about test modes.

The PHY supports the following test modes:

- Scan Mode
- IDDQ
- Loopback Modes

6.1.1 Scan Mode (Stuck-at, At-speed)

The PHY supports stuck-at and at-speed scan for its digital logic.

Set scan_mode =1'b1 to enter scan mode.

When entering scan mode, data in the analog interface is internally gated.

To achieve coverage of the metrics below, all other inputs must be controllable and all outputs must be observable.

Note

PMA .lib ETM modes:

- NORMAL—For Functional mode
- SCAN—For Stuck-at Shift and Capture mode

  In this mode, Shift and Capture happens at Scan Clock Frequency.
- SCAN_ASST—For At Speed Scan Testing

  In this mode, Shift occurs at Scan Clock Frequency, whereas Capture occurs at Functional mode frequency.

The following are scan features requirements in the PHY:

- The PHY provides independent scan clock inputs per internal clock domain.
- Primary clock inputs to the PHY must have scan clocks supplied to them externally in scan mode.
- All clock outputs from the PHY, except ref_repeat_clk_p/m, have a scan clock input muxed on them.
- Scan clocks must have 50% duty cycle.
- Synchronous inputs and outputs are at-speed scannable.
- All resets (primary input resets or internally generated resets) are over-ridden to primary scan reset.
- Input scan_set_rst in scan mode.
- Maximum scan shift frequency is 100 MHz.
- Maximum scan chain length is 350 flops.
- All scan chains are positive-edge triggered.
- All scan chains are terminated with lockup latches.
- All analog blocks are powered down in scan mode. To prevent propagation of logic X's in the scan chain, the outputs of all the analog blocks feeding the digital scannable logic are over-ridden to known states.
- Coverage of the soft-macro (RAW PCS) is place-and-route implementation dependent.
- Stuck-At-Fault (SAF) coverage of the PMA is greater than 97%.

- Transition-Delay-Fault (TDF) coverage of the PMA is greater than 85%.

The scan clocks in Table 8-9 on page 152 support a maximum shift frequency of 100 MHz. Stuck-at and At-speed is done according to Table 8-9 on page 152.

6.1.2 IDDQ Mode

In IDDQ test mode, all the analog blocks are powered down to allow leakage measurement.

Since the PHY has different PMA's, the IDDQ test can be done independently per PMA.

IDDQ current consumption measures are done at the end of the sequence, with the clock stopped.

6.1.2.1 RX IDDQ Test Sequence

Follow these steps to perform IDDQ measurements on the RX.

1. Set all supplies on.
2. Set all PHY clocks off, except apb.
3. Do apb reset toggle:
   a. presetn_i set to 0;
   b. presetn_i set to 1.
4. Do the following apb writes:
   a. rx_sup.reg_sup_ana_ovrdval_9_h.ana_refa_ana_clk_en_i.set(1'b0);
   b. rx_sup.reg_sup_ana_ovrdval_9_h.ana_ref_ro_ana_clk_en_i.set(1'b0);
   c. rx_sup.reg_sup_ana_ovrdval_9_h.ana_ref_ana_diffbuf_clk_en_i.set(1'b0).
5. Set apb clocks off.
6. Set the following reset signals to 0:
   a. raw_pcs_rstn_i;
   b. rx_rstn_i;
   c. presetn_i.
7. Wait 2us.
8. Carry out the measurement.

6.1.2.2 TX IDDQ Test Sequence

Follow these steps to perform IDDQ measurements on the TX.

1. Set all supplies on.
2. Set all PHY clocks off, except apb.
3. Do apb reset toggle:
   a. presetn_i set to 0;
   b. presetn_i set to 1.

4. Do the following apb writes:
    a. tx_sup.reg_sup_ana_ovrdval_9_h.ana_refa_ana_clk_en_i.set(1'b0);
    b. tx_sup.reg_sup_ana_ovrdval_9_h.ana_ref_ro_ana_clk_en_i.set(1'b0);
    c. tx_sup.reg_sup_ana_ovrdval_9_h.ana_ref_ana_diffbuf_clk_en_i.set(1'b0).
5. Set apb clocks off.
6. Set the following reset signals to 0:
    a. raw_pcs_rstn_i;
    b. tx_rstn_i;
    c. presetn_i.
7. Wait 2us.
8. Carry out the measurement.

6.2 Diagnostics

The PHY provides the following diagnostic features to enhance testbench characterization and ATE testing:

- Loopback functions
  - parallel loopback at the raw_pcs level
  - per-lane TX serial internal loopback
  - per-lane RX serial internal loopback
- BERT (independent per-lane)
  - 7th, 15th, and 31st order polynomial pattern generation and recognition
  - 10th pseudo random quaternary sequence pattern generation for PAM4
  - Generation of simple custom test patterns
  - Error insertion and detection
- TX non-destructive internal eye monitor (TX driver output)
- RX non-destructive internal eye monitor (RX AFE output)
- Analog Test Bus to monitor internal analog signals
- Digital monitor for internal digital clocks

6.2.1 Loopback Modes

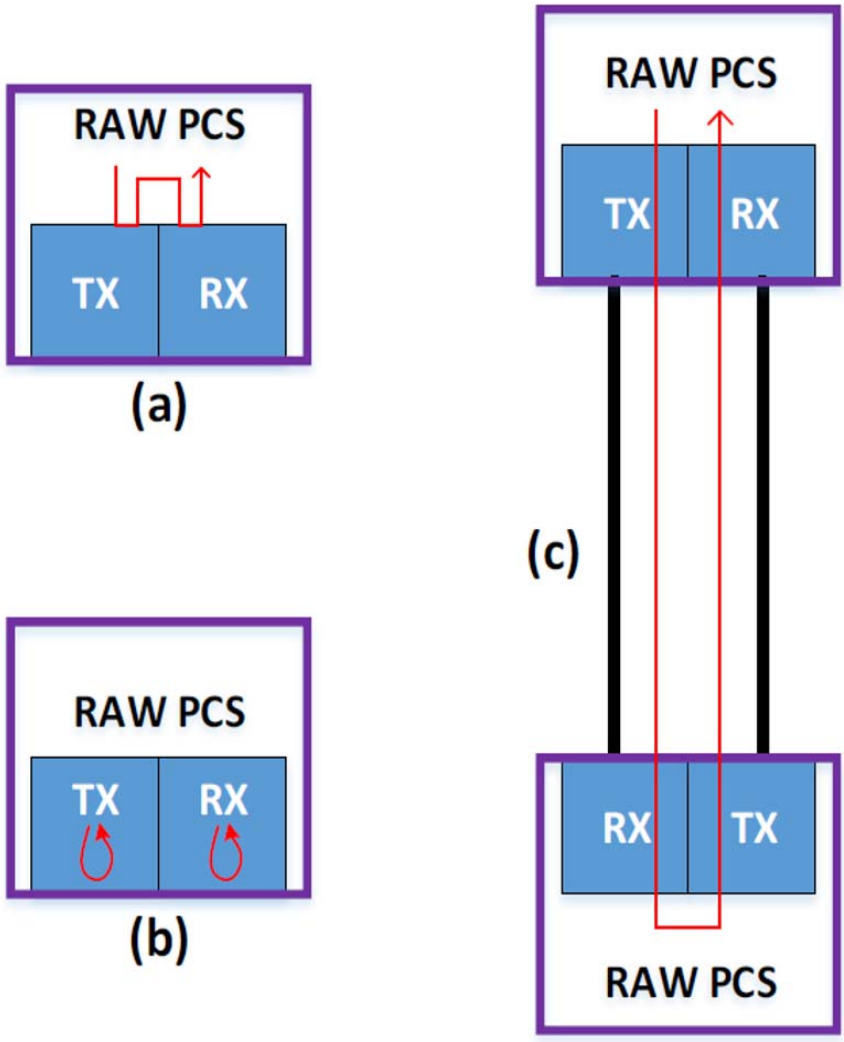
The Raw PCS supports loopback modes on the serial, and parallel sides.

The following are on-die and die-to-die loopback modes (described in the subsequent sections):

- Raw PCS level loopback
- Internal Per-lane loopback TX and RX individually
- Die-to-die loopback

Figure 6-1 illustrates the Loopback Modes.

Figure 6-1 Loopback Modes



6.2.1.1 Raw PCS Level Loopback

This loopback is also called near-end parallel loopback, or Shallow Loopback (marked (a) in Figure 6-1).

In the raw PCS level loopback mode the TX parallel input clock and data at the raw PCS input side are looped-back connected to RX parallel output clock and data.

To enable the shallow loopback mode, use the next register setting:

```
APB::RAW_PCS::tx_to_rx_lpbk_en:: =1'b1
```

6.2.1.2 Internal Per-Lane Loopback

This loopback is done per lane inner the lane RX or TX independently (marked (b) in Figure 6-1 on page 124).

This mode uses the built in PRBS Generator to generate data on the digital, sends it to ANA, which sends back to DIG to be matched on the Pattern matcher. Using this mode, the NRZ Datapath between analog and digital can be verified.

This mode requires an additional calibration to be done, by asserting the following registers the calibration is executed by the firmware.

Since the TX and RX are two different PMA's there is no need to calibrate both if not desired.

Note This loopback mode is only supported on NRZ mode. PAM4 is not supported.

6.2.1.3 Die-to-Die Loopback

The die-to-die loopback (marked (c) in Figure 6-1 on page 124) validates the data communication between two dies. This mode uses a near-end XSR and a far-end XSR. Loopback is programed so that the near-end TX activates the PRSBS Pattern Gen.

Data is sent through the TX to the opposite RX on the far-end, where it is sent back through the far-end die's TX, received by near-end die's RX, and then pattern matched.

Note For more information, refer to the Application Note for ATE testcases.

7

Verilog Model Description

This chapter includes the following sections:

- "Overview" on page 128
- "Verilog Function" on page 128
- "Serial Line Modeling" on page 129
- "Verilog Model Files and Simulators" on page 130

7.1 Overview

The product deliverables include a Verilog RTL for the raw PCS soft macro portion and Verilog behavioral models of the PMAs. Contrary to the Verilog RTL for the soft macro, which can be synthesized into a gate-level netlist by a synthesis tool, the Verilog behavioral models of the PMAs include a netlist of generic GTECH gates and a non-synthesizable behavioral model of the analog circuits.

In the set of deliverables, the raw PCS Verilog RTL and the PMAs Verilog behavioral models reside at:

- Raw PCS: ./Latest/macro/rtl
- PMA TX Verilog model: ./Latest/pma_tx/behavior
- PMA RX Verilog model: ./Latest/pma_rx/behavior

Together, they form the Verilog model of the PHY. The PHY model allows you to run functional verification of your logic along with the PHY.



Note This PHY model and testbench do not represent an example of exhaustive PHY testing.

7.2 Verilog Function

The Verilog model uses a four-state (0, 1, X, Z) simulator. Features that require analog voltage and current values transmitted over wires cannot be simulated with this model.

The following features are not accurately modeled by the PHY model:

- Receive and transmit I/O terminations.
- Termination resistor tuning and measurement.
- Analog test buses.
- Transmitter equalization settings (pre-cursor, swing, and post-cursor) on the transmit serial lines, TXP/N[<n>].
- Receiver equalizer settings.

7.3 Serial Line Modeling

The serial lines (TXP/N[<n>], RXP/N[<n>]) are fundamentally analog signals. Because each wire is limited to the simulator's four states, not all aspects of the analog signals can be reflected.

provides encodings of the analog transmitter model outputs.

Table 7-1 Transmitter Model Serial Encodings

TX Model Output Symbol	TXP[<n>]	TXN[<n>]
NRZ 1	1	0
NRZ 0	0	1
PAM4 +3	1	1
PAM4 +1	1	0
PAM4 -1	0	1
PAM4 -3	0	0

Table 7-2 provides the input decoding done by the analog receiver model.

Table 7-2 Receiver Model Serial Decodings

RXP[<n>]	RXN[<n>]	RX Model Input Symbol Decoding
1	0	NRZ 1
0	1	NRZ 0
1	1	PAM4 +3
1	0	PAM4 +1
0	1	PAM4 -1
0	0	PAM4 -3

7.4 Verilog Model Files and Simulators

The Verilog model is generated for simulations; it can be simulated using Synopsys VCS.

Table 7-3 describes each portion of the Verilog model and indicates the product deliverables directory where the portion is stored.

Table 7-3 PHY Model Database Structure

Directory	Description
./Latest/macro/include	Contains macros used in the raw_pcs, PMA and simulation files.
./Latest/macro/rtl	Contains Verilog RTL for the raw_pcs files. raw_pcs top-level wrapper includes soft macro logic as well as internal instantiation of the PMAs.
./Latest/pma_tx/behavior	Contains Verilog RTL for the behavioral models for the TX PMA. Top-level file for the TX xN PMA instantiating GTECH model of the digital and behavioral model of analog sub-blocks.
./Latest/pma_rx/behavior	Contains Verilog RTL for the behavioral models for the RX PMA. Top-level file for the RX xN PMA instantiating GTECH model of the digital and behavioral model of analog sub-blocks

Instructions for the PHY simulations are provided in the README file in the ./Latest/macro/sim directory. The simulation database structure for PHY simulations is provided in Table 7-4.

Table 7-4 Simulation Database Structure

Directory	Description
./Latest/macro/sim	VCS regression scripts for GTECH and gate-level simulations of the PHY, which includes the following: ■ run_single: Simulation run script for PMA model and raw_pcs, only one test. ■ run_all: runs regression, where several tests are launched to be simulated at the same time ■ README: Run instructions for the PHY testbench.
./Latest/macro/testbench	Sample test-cases for the behavioral and gate-level simulations.

8

Physical-Level Implementation

This chapter includes the following sections:

- "Floorplan Considerations" on page 132
- "Spacing Requirements" on page 134
- "Packaging/Pad Information" on page 135
- "Operating Conditions" on page 139
- "Power Supply Connectivity" on page 140
- "ESD" on page 141
- "Reference Resistor" on page 143
- "Raw PCS Synthesis" on page 147

8.1 Floorplan Considerations

When placing the assembled PMA keep in mind the next considerations.

8.1.1 PMA Rotation and Mirroring

The POLY gates within the PMA must be vertical according to foundry rules, see Figure 8-1.

The poly gate orientation is indicated by '_ns'/'_ew' suffix in PMA name.

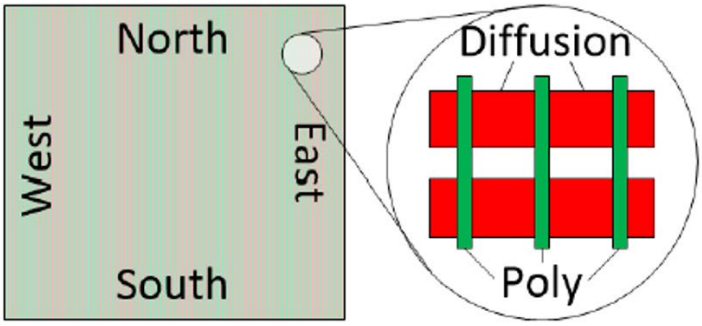
- '_ns' indicates that the poly gates are oriented vertically or 'north-south' in this PMA.
- '_ew' indicates that the poly gates are oriented horizontally or 'east-west' in this PMA.

Depending on the process, all poly gates may need to be oriented in the same direction on a single SoC.

The PMA may need customization in order to support rotation or mirroring. For more information about PMA rotation and mirroring, contact Synopsys support.

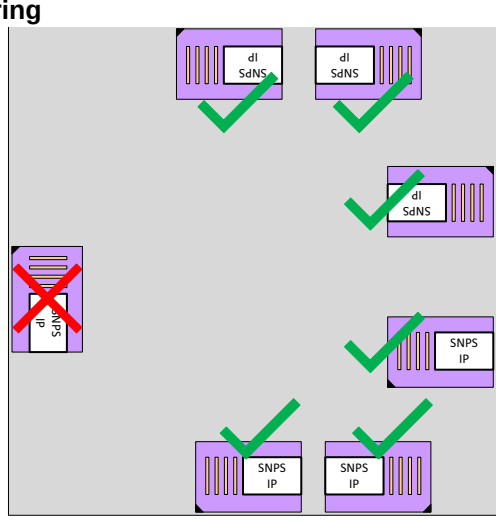
All diagrams and physical integration guidelines provided refer to the '_ns' poly orientation of the PMA and for process nodes which do NOT permit mixed orientation poly gates on an SoC.

Figure 8-1 Default Gate Orientation



The PMA hard macro is designed to be placed in either north/south or east/west side of the die, as shown in Figure 8-2.

Figure 8-2 PMA Rotation and Mirroring



The PMA may be rotated by 180°, but not 90° or 270° due to the vertical POLY requirement.

The PMA may be mirrored in either the X or Y direction as this does not change POLY orientation.

8.1.2 PMA Block Size

> **Note** Areas listed in Table 8-1 include only the active devices area of the PHY. Depending on the bump map used, some bumps may overhang outside the active area of the PHY. For accurate details, refer to the LEF view.

The size of the PMA is determined by the size of the following:

- PMA hard macro
- Flip-chip pads

Table 8-1 shows the estimated size of the hard macro building blocks for the TX and RX x8 PMAs. All dimensions are preliminary and subject to change.

Table 8-1 PMA Size

Configuration	Drawn x [um]	Drawn y [um]	Drawn Area [mm^2]	Shrink Factor	Actual Area [mm^2]
TX/RX PHY x8	N/A	N/A	1.289339566848	0.98	1.27

8.2 Spacing Requirements

8.2.1 IP Components with Themselves

Multiple instances of the PMA can be placed side-by-side on north and south.

When multiple instances of the PMA are placed adjacent to each other side-by-side on the SoC, the spacing requirement is the greater of either:

- A minimum of 5 µm from the PMA boundary on South, East and West directions
- Minimum spacing to meet bump-to-bump spacing and bump-pitch rules at SoC level
  - Sometimes due to design restrictions the bumps can overhang the side of the IP and/or do not meet ½ bump spacing rules.
  - For optimal placement options for multiple instances contact Synopsys Support.

8.2.2 IP Components and other Un-related Blocks or Logic

Synopsys recommends a 5 µm spacing for any core logic instance placement around the IP for all sides.

The north edge of the PHY can include signal routing and power meshes as required for connectivity to the PHY.

It is recommended routing and power-grid keep-out of 5 µm for the east, west and south edges of the PMA macro for all metal and RDL layers which are included in the PMA macro. These edges should only include density layer fill shapes as generated by the foundry fill utility.

There should be no SoC level routing over top of the PMA macro.

If there are bump pads which overhang the edge of the PMA macro the SoC level RDL routing should maintain the keep-out from the edge of these bumps

8.2.3 IP Component and Chip Edge

The PMA is designed for optimal placement on the south edge of the die.

To meet the UBM to chip-edge spacing rule the SoC level designer may need to place the PMA with an offset from chip edge. SoC level designer must verify if this placement follows the foundry design-rule and pad rule verification decks.

8.2.4 Other Placement Notes

8.2.4.1 PMA Fill

The PMA comes pre-filled. When you run density layer fill at the chip level, no fill shapes are to be added within the PMA.

8.2.4.2 Density Layer Fill

The PHY contains all necessary BEOL and FEOL fill shapes to meet all density rules unless otherwise specified in the package readme.

When the foundry fill utilities are run at SoC level, there should not be any additional fill shapes added over the PMA macro.

8.3 Packaging/Pad Information

This section provides information about the bump map, pad size, and associated diagrams.

8.3.1 Flip-Chip

PMA Bump pitch, bump type and sizing have been designed based on optimized bump placement for the IP. The bump pad RDL, passivation opening and UBM sizing are based on foundry required dimensions for the PMA bump pitch.

For PMA bump pitch and pin location information for package design refer to the RDL pin locations in the LEF delivery file.

For different bump pitch, pad type or pad sizing, contact Synopsys sales manager for customization option.

Note Ensure that the under-bump metalization (UBM) is the correct size. Foundry and packaging typically require that the UBM shape used for the bump is the same for all bumps within the same die. The size of the UBM shape is dictated by the minimum bump pitch in the die.

Attention The bump maps shown in this section are for reference purposes only. For dimensions, refer to the .lef in the release package.

Figure 8-3 shows the RX PMA bump map.

Figure 8-3 Example of RX Bump Map

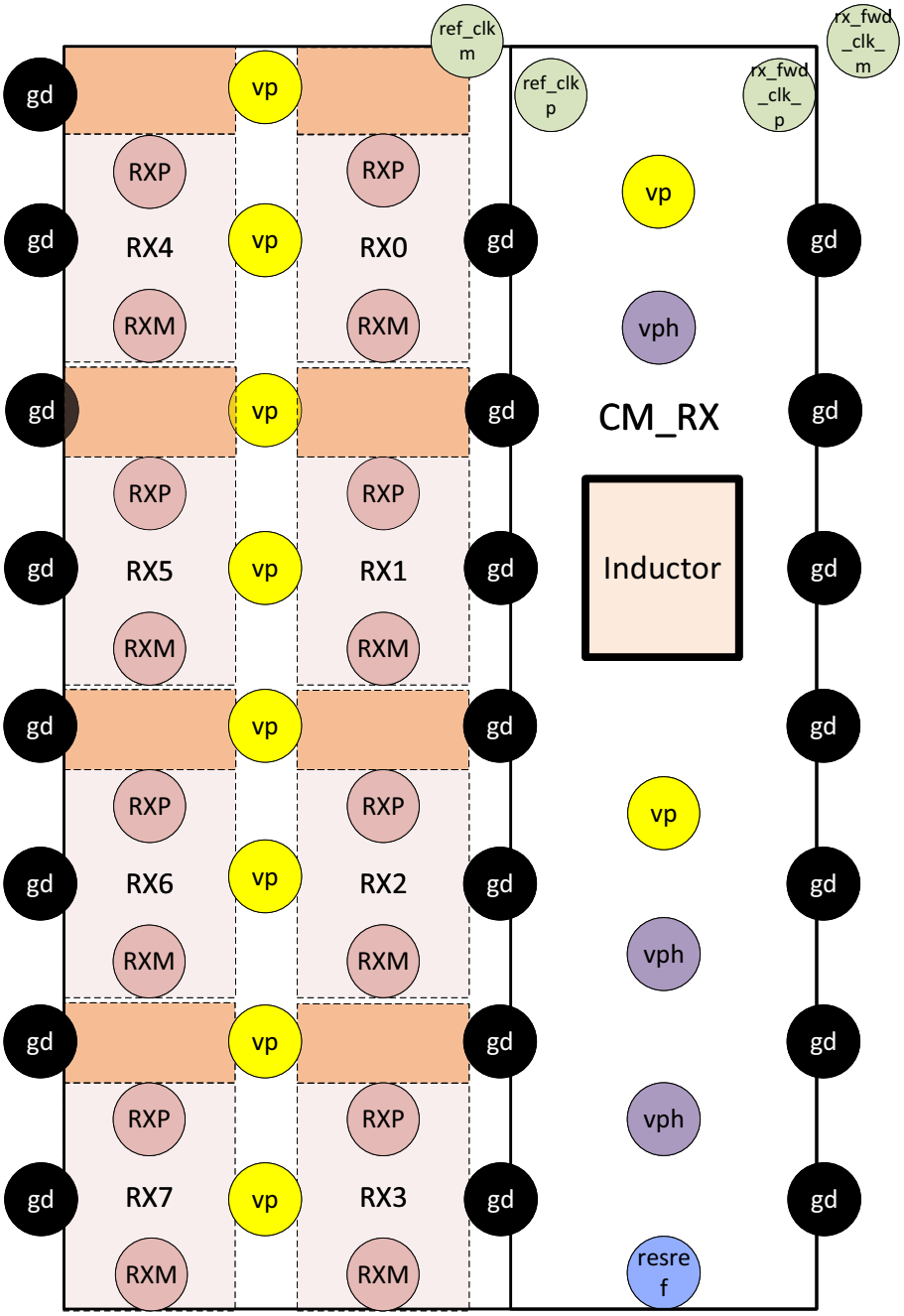
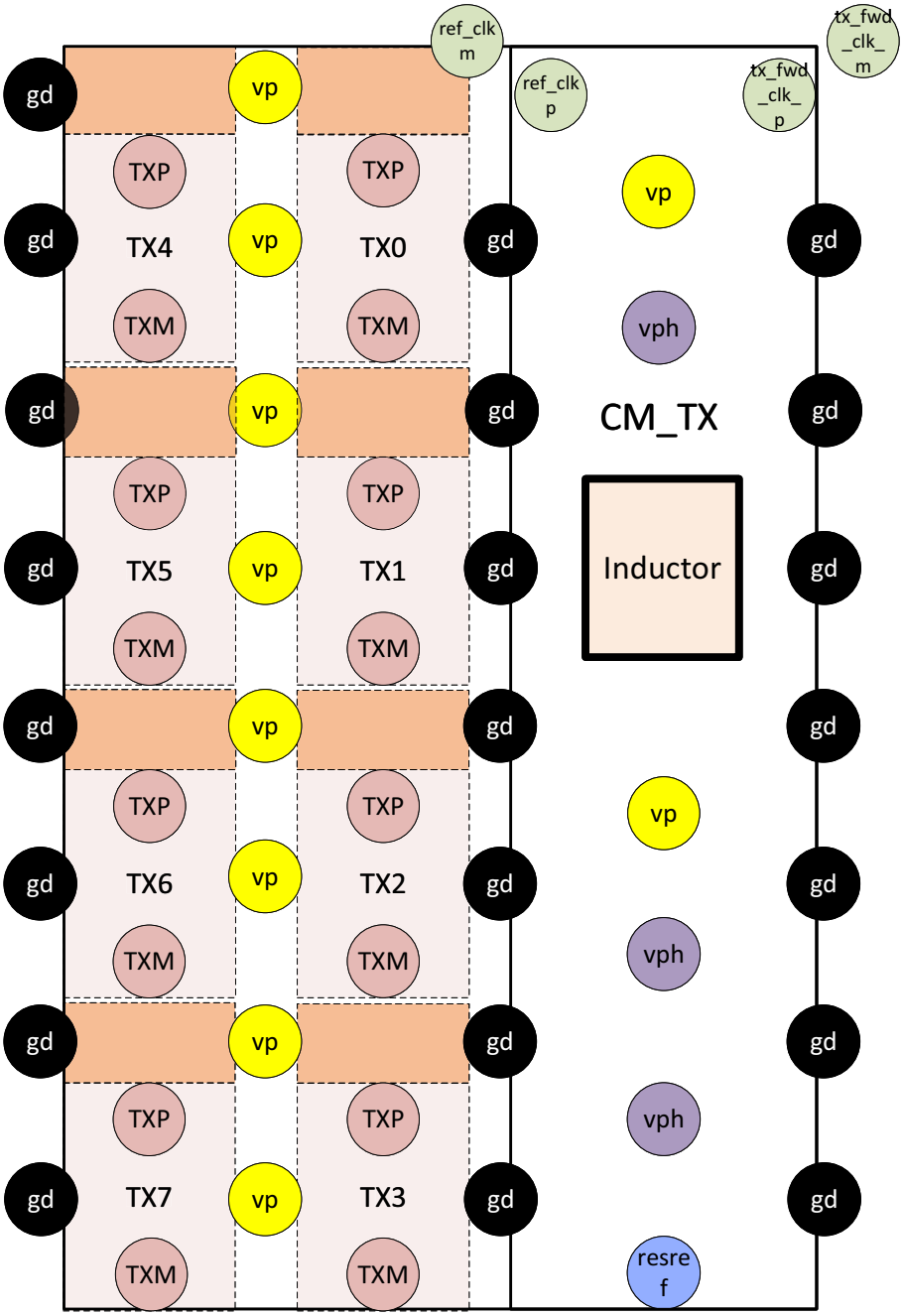


Figure 8-4 shows the TX PMA bump map.

Figure 8-4 Example of TX Bump Map



8.3.2 Bump Pad RDL and UBM Sizing

Foundry requirements state that pad RDL and UBM sizing is dictated by the minimum bump pitch found on the die.

The pads layout provided with the PMA have a bump pitch of 153<=P<163> µm. The RDL and UBM sizing used is suitable for a minimum bump pitch between NA µm.The bump RDL size is 85.7 µm and the UBM size is 81.6 µm.

Table 8-2 Bump Pitch Information

Bump Pitch (µm)	Bump Pitch Range (µm)	Bump Type	UBM (µm)	RDL (µm)	PM (µm)	AP (µm)	CBD/CB2_FC (µm)
153<=P<163>	NA	C4 (Controlled Collapse Chip connection)	81.6	85.7	NA	NA	NA

8.4 Operating Conditions

This section details the operating conditions of the PMA.

8.4.1 Temperature

Table 8-3 lists the PMA supported operating temperature range.

Table 8-3 PMA Operating Temperature Range

Operation	Temperature Range	Description
Performance	-20 to 105°C	Meets electrical specifications as specified in the databook.
Functional	-40 to 125°C	Analog and digital sub-blocks operational but will exhibit reduced margin to design targets; overall RX and TX may not meet specification at temperature extremes; PHY may negotiate to lower data rate or PHY may exhibit higher bit error rate (BER) performance.

Temperature drift after initial startup adaptation and calibration can affect the performance of PMA, particularly the receiver. The amount of temperature drift that the PMA can withstand before violating the protocol bit error rate limit depends on the receiver, the data rate, and external channel impairments. If there is a temperature drift, it may be necessary to re-adapt and re-calibrate the receiver. Sensitivity to this effect can be reduced by implementing continuous offset correction and adaptation (CCA).

This PHY implements continuous offset calibration and adaptation (CCA). This reduces but does not zero out the sensitivity of the receiver to changes in temperature. Changes in temperature may still result in reduced eye opening as seen by the receiver and will reduce margin for jitter tolerance.

8.4.2 Electromigration

The electromigration limit of the PMA is 88KPOH@105°C.

Note Operating at conditions that exceed the electromigration limits of PMA reduces the lifetime of the PMA. While the PMA can meet performance requirements at higher temperature, this is expected to be a short term condition, such as temporary failure in cooling and so on.

8.5 Power Supply Connectivity

This section provide details about power and ground isolation, as well as on-chip supply decoupling if indicated after running power integrity simulations.

8.5.1 Power and Ground Isolation

Power supply connectivity must be designed accounting for all noise sources, decap, parasitics, package, and so on. You must perform a power integrity analysis.

To prevent power supply noise from inducing jitter on the transmitted and received waveforms, ensure that the PHY power supplies are isolated from the power supplies used on the rest of the ASIC.

The PHY provides power pins for purpose of additional on-die decap, if needed, as determined by your power integrity analysis of the PHY plus the package plus the board. Due to jitter and isolation requirement these pins should not be used to connect the PHY to other on-die supplies.

The PHY ground, gd, is isolated on the die from the ASIC core ground gddig. Back-to-back diodes between the ASIC ground (connected to gddig pin) and the PHY ground, gd, are provided within the PHY macro as part of the ESD protection strategy (described in "ESD Protection Strategy" on page 141"), but no direct metal-metal connection is permitted.

Isolation of all power and ground on the die implies the only connection between the PHY hard macro and the rest of the ASIC are for signal wires. The signal integrity of this interface is maintained using a fully clock-forwarded architecture where both clock and data cross the power domains together, so that subtle differences in supply voltages have no impact on timing.

8.5.2 On-Chip Supply Decoupling

If power integrity simulations indicate that additional PHY power supply decoupling is needed, one option is to include that capacitance on-chip. This may also be advantageous for utilizing space in between PHY instantiations or along the die-edge.

Contact Synopsys for on-chip capacitance addition to ensure proper decoupling capacitance placement and connection to listed power supplies.



Note Power integrity of the macro supplies must be considered early in the design process and can impact both on-die implementation (if additional on-die decap is needed) as well as decap design and board and package implementation.

8.6 ESD

This section presents ESD considerations.

8.6.1 ESD Requirements

ESD performance is evaluated by the amount of voltage and current that the protection circuits can absorb during an ESD event while keeping voltages below damaging levels. The protection circuits are designed to meet the protection levels provided in Table 8-4.

Table 8-4 Minimum ESD Protection Level

Pins	Charged Device Model (CDM)	Human Body Model (HBM)
High-speed pins on the die-to-die link: txN_p/m_i, txN_p/m_o, tx_fwd_clk_m/p_o, rx_fwd_clk_m/p_o	1 A	100 V
All other pins	6 A	2 kV

8.6.2 ESD Protection Strategy

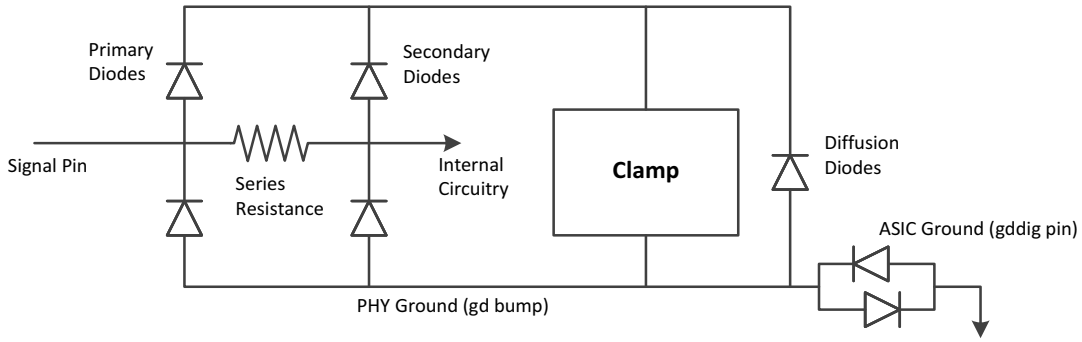
The ESD protection strategy of the PHY provides the following features:

- Large primary diodes between signal pad and PHY power and ground supplies
- A series resistor and secondary diodes between signal pad and PHY power and ground supplies
- An active clamp between the PHY power and ground supplies

  The clamp and primary diodes are placed close to each other within the PHY, minimizing the resistance between them and maximizing the effectiveness of the ESD circuits.
- Back-to-back diodes between PHY ground, gd, and interfacing (ASIC) ground, gddig
- Sufficient metal width in all ESD paths

Figure 8-5 shows the recommended ESD strategy to implement in

Figure 8-5 ESD Protection Strategy



During ESD events, the primary diodes direct most of the current to the PHY power/ground buses and through the clamp. This path has minimal resistance, keeping the signal pad voltage and voltage across the PHY power buses low. Some current flows through the secondary diodes and develops a further voltage

drop across the series resistor from the signal pad to the internal circuits. Back-to-back diodes enable current flow between ground domains-preventing large differences from developing between ground domains. The back-to-back diodes prevent damage between interfacing circuits due to power bus differences.

8.6.3 ESD Protection Strategy for Die-to-Die Pins

The ESD protection strategy of the PHY provides the following features:

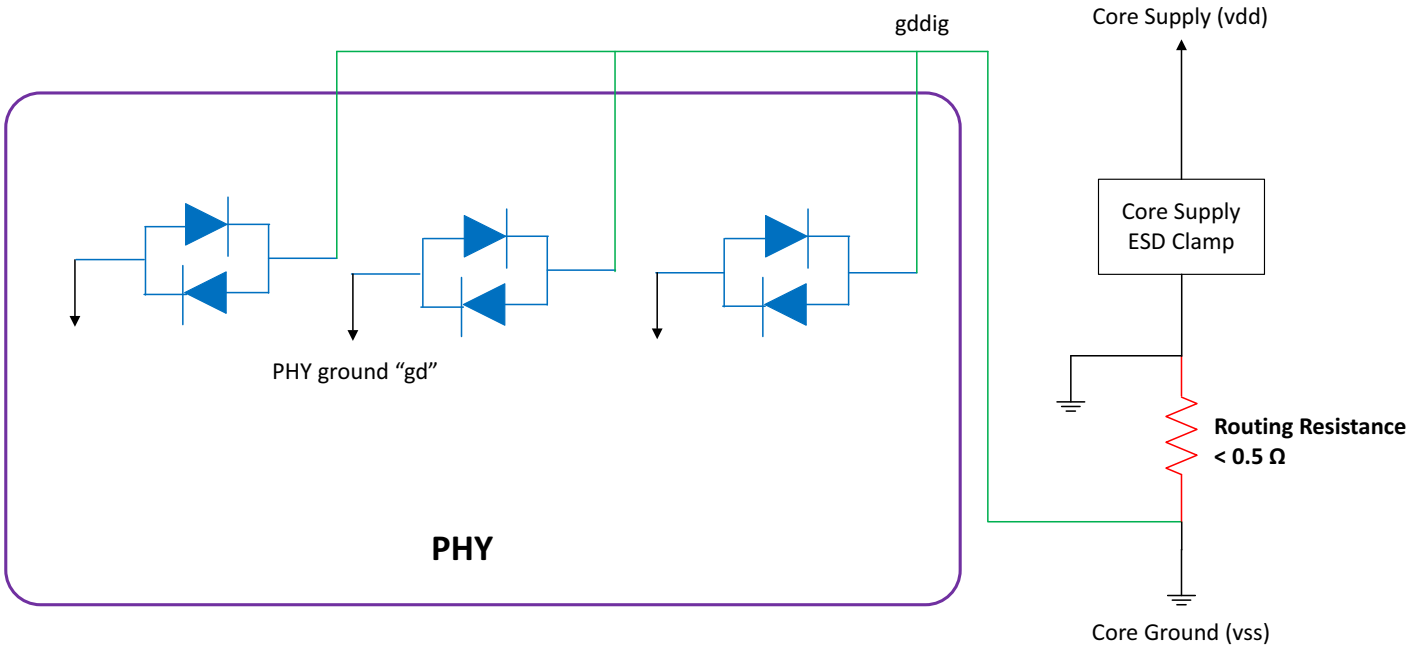
- Current path between signal pad and PHY power and ground supplies
- Back-to-back diodes between PHY ground, gd, and interfacing (ASIC) ground, gddig
- Sufficient metal width in all ESD paths

8.6.4 ESD Back-to-Back Diode Connection to Core VSS

Although gddig and gd are to be connected together within the package, for ESD protection of the PHY/ASIC interface a back to back diode connection between gddig and gd is provided within the PHY.

The gddig pin on the PHY must have a strong connection to the ASIC gddig supply grid. The gddig pin provided in the IP should be connected with a solid metal. There should be less than 0.5 ohm of resistance between the gddig pin on the PHY and the closest clamp within the ASIC (see Figure 8-6).

Figure 8-6 Back-to-Back Diodes in PHY



Note Add a vdd-to-vss clamp close to the back-to-back diodes with no more than 0.5 ohm routing resistance from the diodes to the clamp or with a value recommended by the foundry Design Rule Manual (DRM).

If non-ground supplies are connected together for ESD protection through back-to-back diodes, special care must be taken to check for, and guard against latch-up, power-up sequencing and leakage issues.

8.7 Reference Resistor

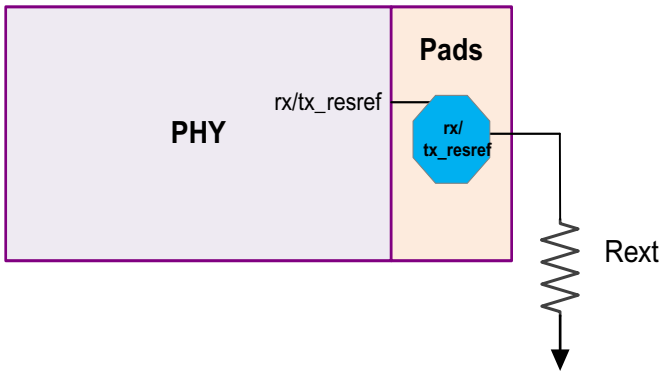
An external reference resistor is used for calibration of the RX and TX termination. The external resistor is automatically used for calibration when the PHY comes out of reset and when a manual tuning request is made. There is also the possibility to demand a manual tuning request. When a PHY is using the reference resistor, it forces current across the resistor. When a PHY is not using the reference resistor, the current is disabled.

Table 8-5 Reference Resistor Requirements

Item	Value Design	Guidelines
External Reference Resistor Resistance	200 ohm precision resistor (+/-1%)	None
External Reference Resistor Maximum Current	2 mA	None
External Reference Resistor parasitic capacitance C_{load}	< 20 pF	■ The maximum parasitic capacitance includes external loading of package trace and PCB as well as PHY loading. ■ Each PHY connected to the resref pin adds an additional 1.5 pF loading. For example, a single PMA supports up to 18.5 pF (20 pF1x1.5 pF) of external load. Sharing a single reference resistor between 4 PMAs can support up to 14 pF (20 pF -4x1.5 pF) of external load.
External Reference Resistor parasitic resistance through package and board R_{route}	1% of 200 ohm	Take care to minimize the resistance of the connection through the package to the external reference resistor as this affects the accuracy of the termination tuning. This specification is for total resistance through the package and board to the board's external resistor. The ground used on the board to terminate the external resistor should be as close as possible to the PMA ground voltage. Noise adjacent signals (for example, clocks) should be spaced away from resref or should be shielded with ground, if possible.

Figure 8-7 shows the reference resistor connection for a single PHY implementation.

Figure 8-7 Reference Resistor Connectivity - Single PHY



8.7.1 Reference Resistor Contention Bus

Each PHY requires access to a reference resistor. This access may be per PHY or it may be shared. A contention bus is used to control PHY access to the external resistor using res_req_in, res_req_out, res_ack_in, res_ack_out.

For a single-PHY implementation, connect the signals as shown in Figure 8-8.

This calibration is handled by firmware. Currently to reduce time it is only performed over TX. The TX calibration codes are applied to the RX. All of this procedure is transparent for the user.

Figure 8-8 Reference Resistor Contention Bus - Single PHY

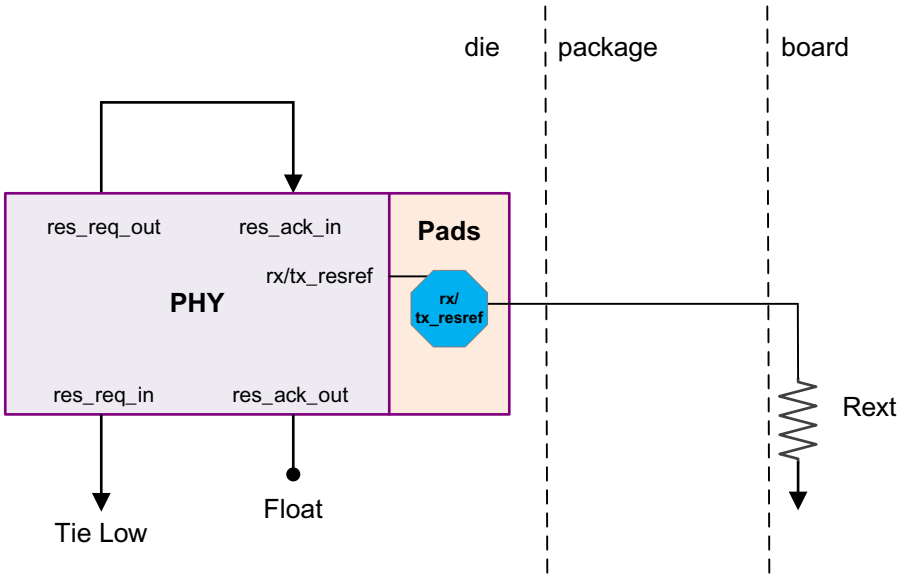
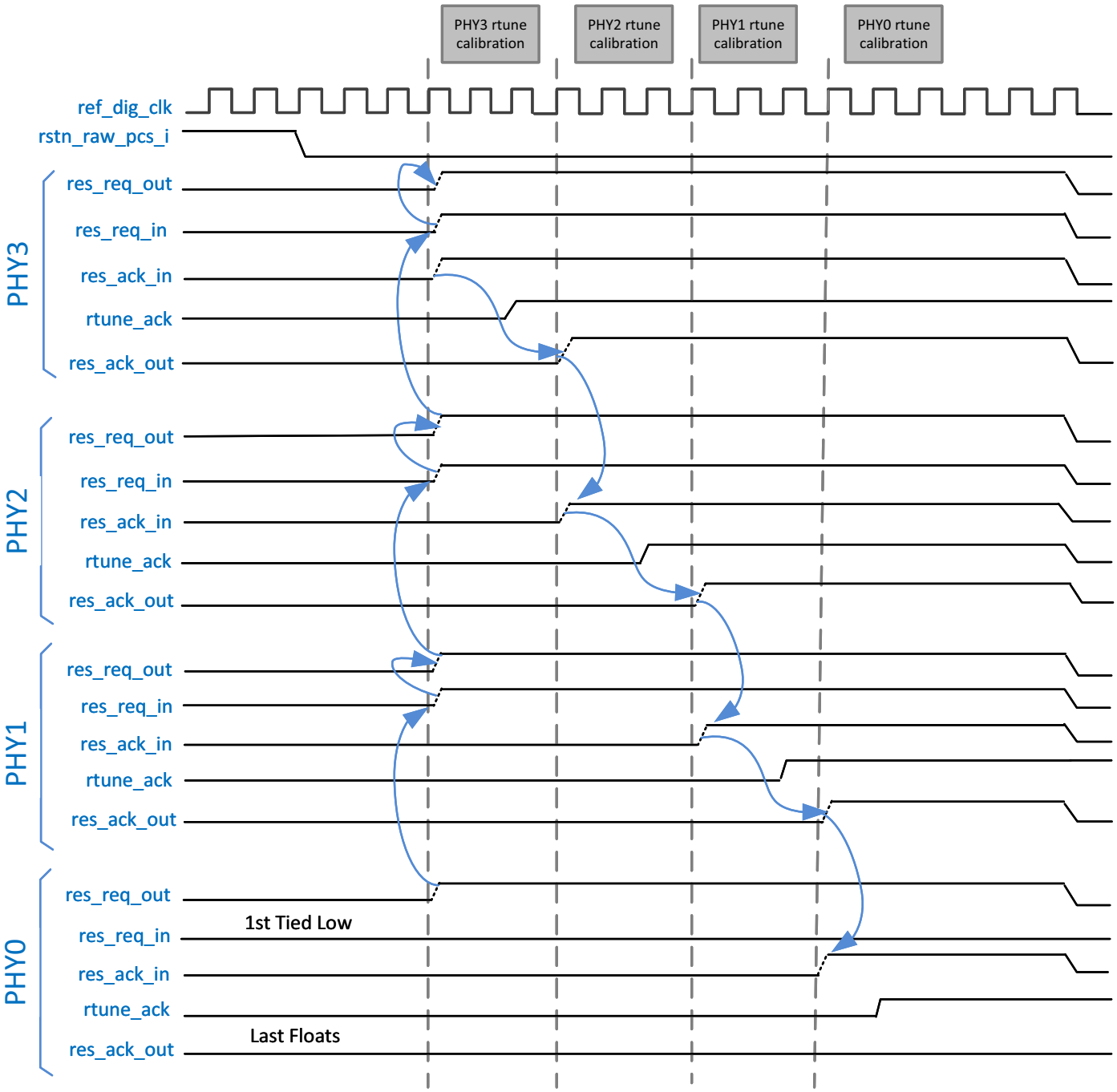


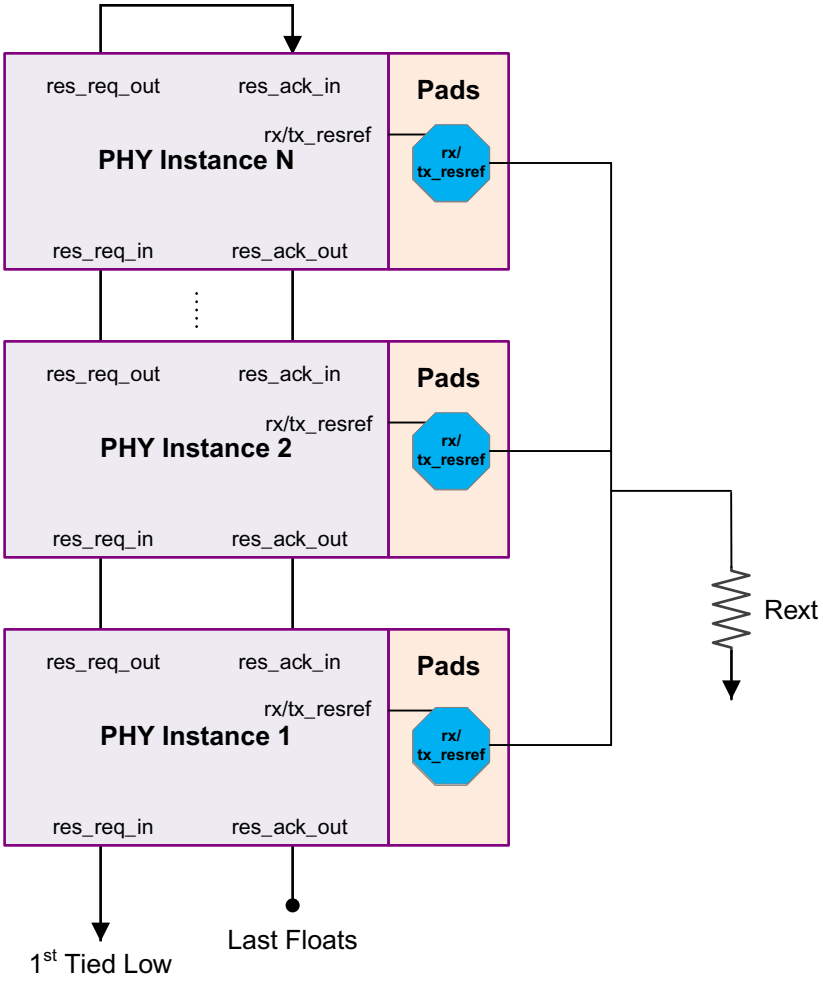
Figure 8-9 shows the reference resistor contention bus diagram.

Figure 8-9 Reference Resistor Contention Bus Diagrams



For sharing a reference resistor among several PHYs, connect the signals as shown in Figure 8-10.

Figure 8-10 Reference Resistor Contention Bus - Multiple PHYs Sharing a Single Resistor



Note Currently manual request for this procedure after startup and during mission mode is not supported.

8.8 Raw PCS Synthesis

The raw PCS is delivered as soft RTL, it will be hardened along with the rest of the RTL of the protocol-PCS controller and ASIC. Synthesizing the raw PCS requires a Verilog-2001 compatible synthesis tool.

8.8.1 Synthesis Database

The deliverables include the Design Compiler synthesis scripts in TCL format that can be used to synthesize the raw PCS soft macro logic using the timing model of the PMA. There are two PMAs and two timing models available.

Table 8-6 shows the synthesis database structure of the raw PCS soft macro.

Table 8-6 Synthesis Database Structure for Raw PCS Soft Macro

Directory	Description
./Latest/macro/synth	Synthesis directory for raw PCS ■ run_synth.sh: Shell script to run synthesis using Design Compiler TCL scripts. Formality will run automatically. To run synthesis without formality add -nofv flag You need to set the environment variable TARGET_DB_LIST to the installed standard cell db. This script must be run in the directory ./Latest/macro/synth
./Latest/macro/synth/scripts	Synthesis and Formality scripts: ■ Formality script for running logic equivalence check on the synthesized netlist of raw PCS ■ Set ideal networks on Analog signals ■ Design Complier synthesis script to synthesize dwc_d2d_sr112_phy_x-8_ns_raw_pcs_top_raw_pcs_top using RTL files and libs ■ DFT insertion script
./Latest/macro/constraints	Synthesis Constraints files

Table 8-7 shows the ATPG deliverables.

Table 8-7 ATPG Deliverables for the PHY

Directory	Description
./Latest/pma/atpg/<metal_stack>/ctl	CTL and SPF model
./Latest/pma/atpg/<metal_stack>/models	Verilog scan model of the PMA; only to be used by TetraMax (do not use for simulations)
./Latest/pma/atpg/<metal_stack>/rpt	TetraMax log file
./Latest/pma/atpg/<metal_stack>/tetramax	run_tmax.sh: Top-level shell script to run TetraMax TCL script (tmax_atpg.tcl) for generating vectors

Directory	Description
./Latest/pma/atpg/<metal_stack>/vectors	Synthesis Constraints files: ■ WGL vectors for tester (*_test.stuck.wgl and *_test.tdf.wgl) ■ Top-level Verilog ATPG serial testbench (*_test.serial.stuck.v and *_test.serial.tdf.v) ■ Top-level Verilog ATPG parallel testbench (*_test.para.tdf.v and *_test.para.stuck.v)

Note In case of not using all the scripts provided above, define the macro `SYNTH during synthesis flow.

You can refer to an example is shown in the analyze command in the file dwc_d2d_sr112_phy_x8_ns_raw_pcs_top_dc.tcl

8.8.2 Raw PCS Constraints

The synthesis constraints are provided in the form of TCL scripts used by Synopsys Design Compiler. These files (./Latest/macro/constraints/) describe the design intent and surrounding constraints for synthesis, clocking, timing, power, test, and environmental and operating conditions. The constraints include the input and output delays of interface pins, clock definitions, and any timing path exceptions in the design, such as false paths or multi-cycle paths.

In most cases, the raw PCS is synthesized in conjunction with other logic that resides on the opposite side of the ASIC interface boundary. In this situation, timing constraints can be automatically inferred from the adjoining logic and from the PMA timing model to which the raw PCS is connected. When synthesizing the raw PCS as a standalone block, signals on the ASIC interface must be constrained, so that sufficient time is budgeted for the adjoining logic to be connected later.

8.8.2.1 Hand-Placed Logic Constraints

There are primitive modules within the raw PCS that are used to instantiate a specific digital cell that must not be logically optimized during synthesis. These modules are typically used around asynchronous Clock Domain Crossing (CDC) points within the design. It is essential that these modules are processed correctly, not just synthesized as usual with the raw PCS logic. Failure to do so will likely create glitch-generating and re-convergent logic that can cause problems on the ASIC interface.

Table 8-8 lists the raw PCS primitive modules.

Table 8-8 Raw PCS Primitive Modules

Module	Function
*_param_rstgen	Asynchronous assert, synchronous de-assert reset synchronizer
*_xsr_sync	Single bit, single clock synchronizer
*_xsr_clk_and2_d2	Two-input clock AND gate

Module	Function
*_xsr_clk_mux2_d2	Two-input clock multiplexer Two-stage synchronizer, first stage is clocked at the negative edge
*_cdc_sync	Two-stage synchronizer
*_cdc_np_sync	Two-stage synchronizer ■ First stage is clocked at the negative edge of the clock ■ Second stage is clocked at the positive edge of the clock
*_cdc_reset_sync	Reset synchronizer
*_jtag_port	Clock Multiplexer during Scan only
*_sys_clk_sync	Clock Multiplexer used during Scan only

To prevent these modules from being optimized during synthesis, you can either replace the contents of these modules with an instantiation of the appropriate standard cell from a digital library, or synthesize each module stand-alone with a restricted library, so that only the correct cell is selected by the tool. In both cases, a set_size_only attribute must be applied to prevent re-optimization of the instance during regular raw PCS synthesis.

8.8.2.2 Clock Contraints

On the raw pcs we added already the needed scan muxes for the cpu_clk_i, cpu_jtag_tck_i and raw_pcs_jtag_tck_i and pclk_i.. Figure x shows the clocking structure for functional clocks for the interfaces.

The Raw PCS includes scan muxes for the `cpu_clk_i, cpu_jtag_tck_i, raw_pcs_jtag_tck_i` and `pclk_i`.

Figure 8-11 shows the clocking structure for the interfaces functional clocks.

Figure 8-11 Functional Clocks Clocking Structure

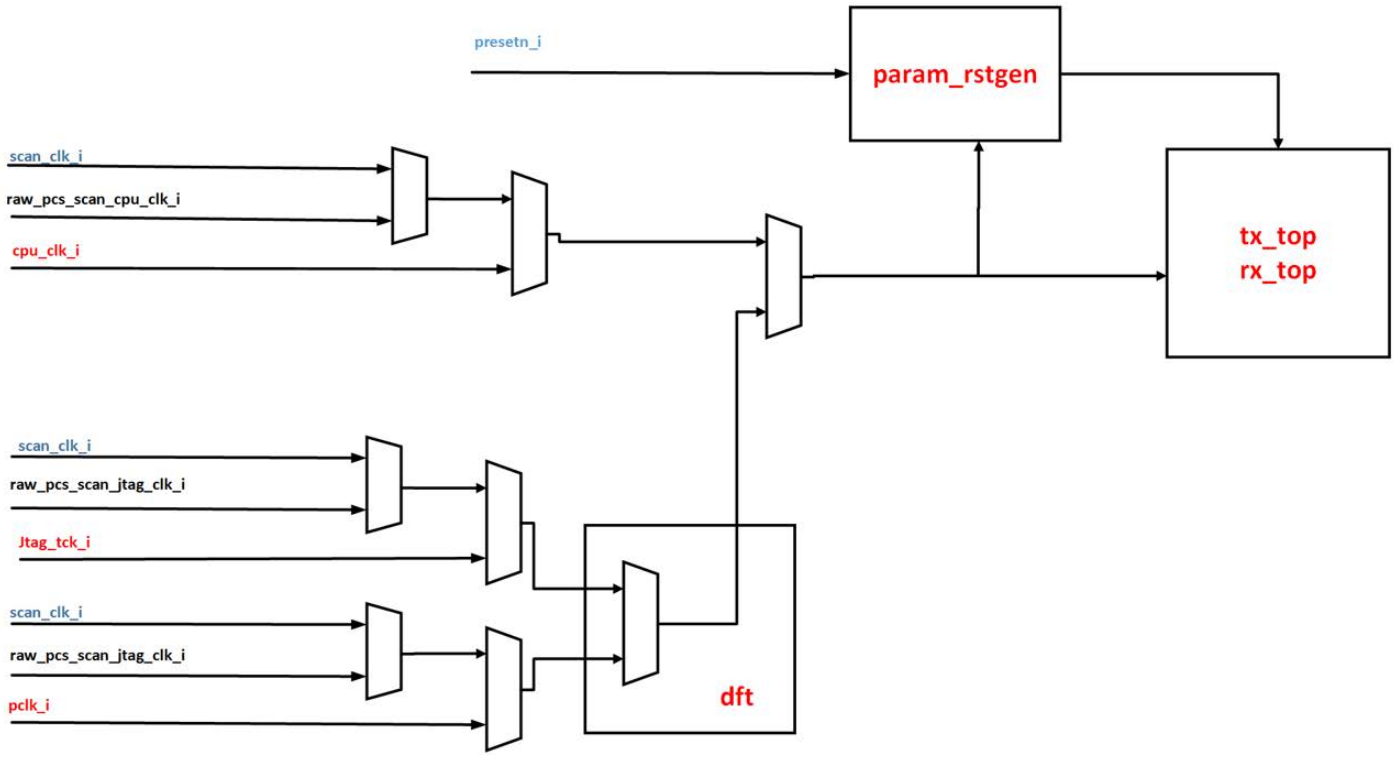


Figure 8-12 shows the functional clocks for RX datapath on Raw PCS clocking structure.

Figure 8-12 Functional Clocks RX Datapath

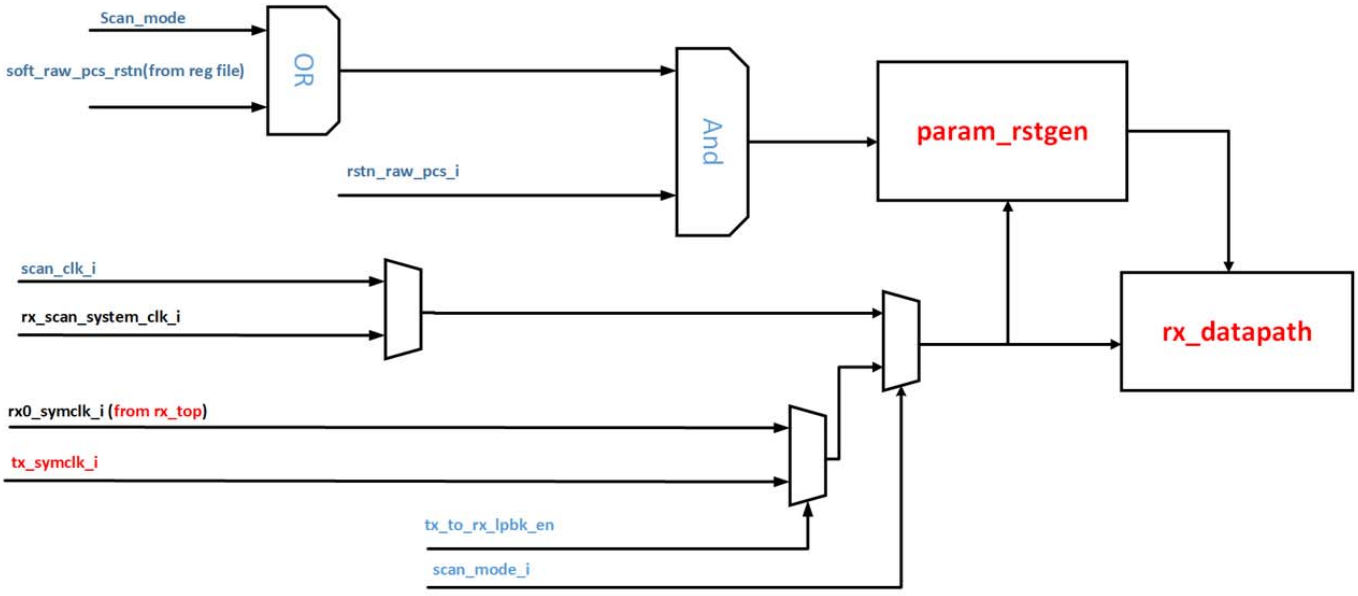
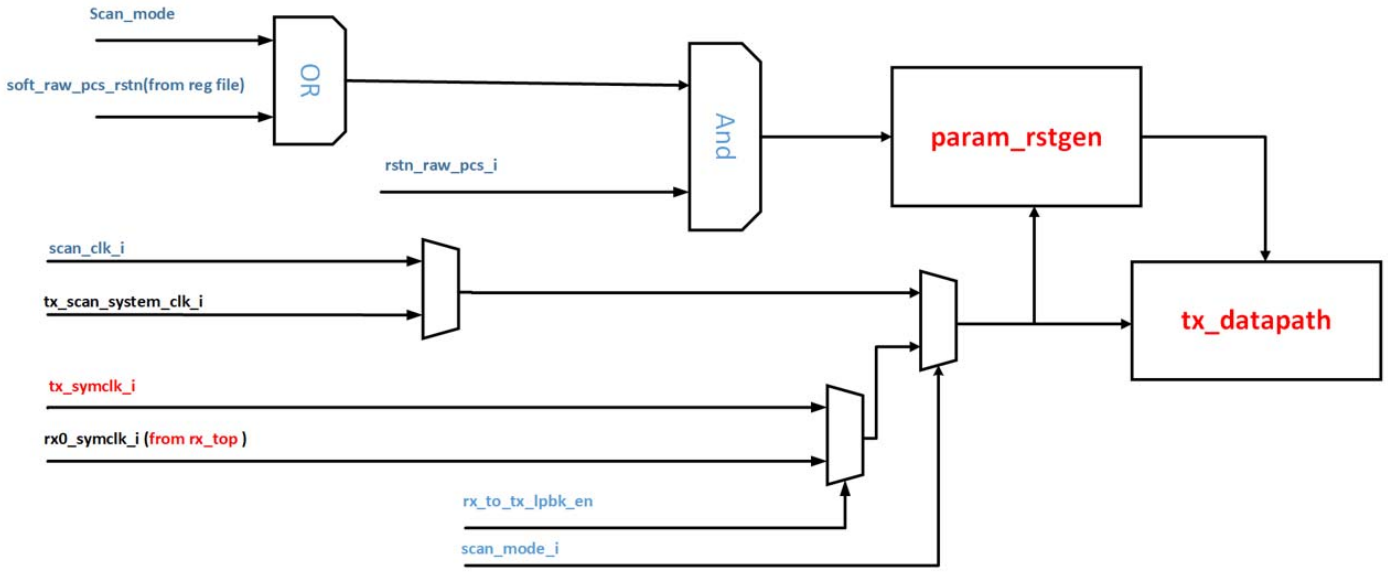


Figure 8-13 shows the functional clocks for TX datapath on Raw PCS clocking structure.

Figure 8-13 Functional Clocks TX Datapath



The clock definitions and associated false paths for the PHY are provided in the constraints folder.

The PHY supports two scan clock modes for stuck-at and at-speed (ASST) scan.

- scan_clk_sel=1

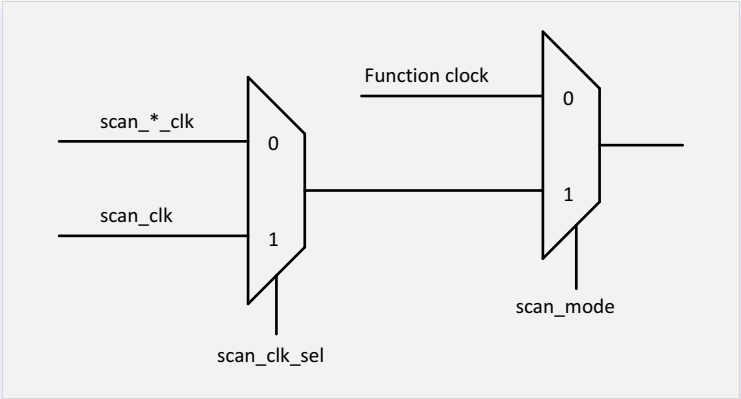
  The scan clock MUX selects the scan_clk input clock. This clock is used by entire PHY. Only stuck-at (shift and capture) is supported.

- scan_clk_sel=0

  The scan clock MUX selects the at-speed scan input clocks. Both stuck-at and at speed (shift and capture) are supported. The scan clock inputs are required to be one-hot.

The scan clock MUX is not glitchless. So scan_clk_sel should not change dynamically when the clocks are active.

Figure 8-14 Scan Clock MUX Structure



Scan clocks definitions at PHY (PMA plus Raw PCS) level are indicated in Table 8-9.

Table 8-9 PHY Scan Clock Constraints

Lock Name	Creation Port	ASST Max. Freq. [MHz]	Description
scan_sa_clk	scan_clk_i	100	Stuck at scan capture clock
scan_shift_clk	scan_clk_i	100	Stuck at & ASST scan shift clock
raw_pcs_scan_apb_clk_i	raw_pcs_scan_apb_clk_i	100	(APB, jtag and cpu_jtag) scan clock for tx,rx and raw_pcs
raw_pcs_scan_cpu_clk_i	raw_pcs_scan_cpu_clk_i	100	CPU scan clock
tx_scan_system_clk_i	tx_scan_system_clk_i	1750	ASST scan system clock on TX
tx_scan_system_div_clk_i	tx_scan_system_div_clk_i	875	ASST scan system divided clock on TX
tx_sup_scan_ref_clk_i	tx_sup_scan_ref_clk_i	875	ASST scan ref_clk on SUP TX
tx_sup_scan_mplla_word_clk_i	tx_sup_scan_mplla_word_clk_i	1750	ASST scan mplla_word_clk on SUP TX
rx_scan_system_clk_i	rx_scan_system_clk_i	1750	ASST scan system clock on RX
rx_scan_system_div_clk_i	rx_scan_system_div_clk_i	875	ASST scan system divided clock on RX
rx_sup_scan_ref_clk_i	rx_sup_scan_ref_clk_i	875	ASST scan ref_clk on SUP RX
rx_sup_scan_mplla_word_clk_i	rx_sup_scan_mplla_word_clk_i	1000	ASST scan mplla_word_clk on SUP RX

Table 8-10 lists the clock constraints.

Table 8-10 Clock Constraints

Clock	Frequency [MHz]
jtag_tck_i	100
pclk_i	100
cpu_clk_i	300
cpu_jtag_tck_i	100
pipe_tx_clk	1750
pipe_rx_clk	1750

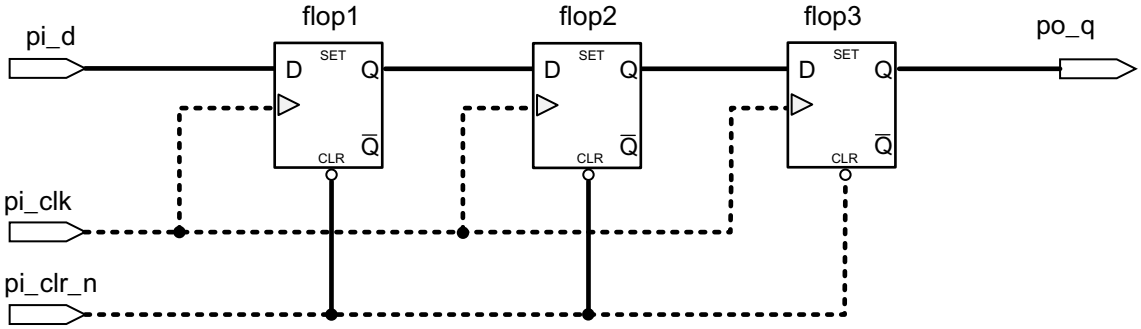
8.8.2.3 Syncronizers

Raw PCS instantiates the synchronizer modules used for clock domain crossing protection. The design includes two types of synchronizers, data and reset synchronizers.

Data synchronizers are used for single-bit signals that must be passed from one clock domain to another, an instance of one of the modules *_xsr_sync is used to synchronize the source signal into the receiving clock domain.

Figure 8-15 shows the structure of a single-bit 3-stage data synchronizer.

Figure 8-15 dwc_*_*_xsr_sync Module



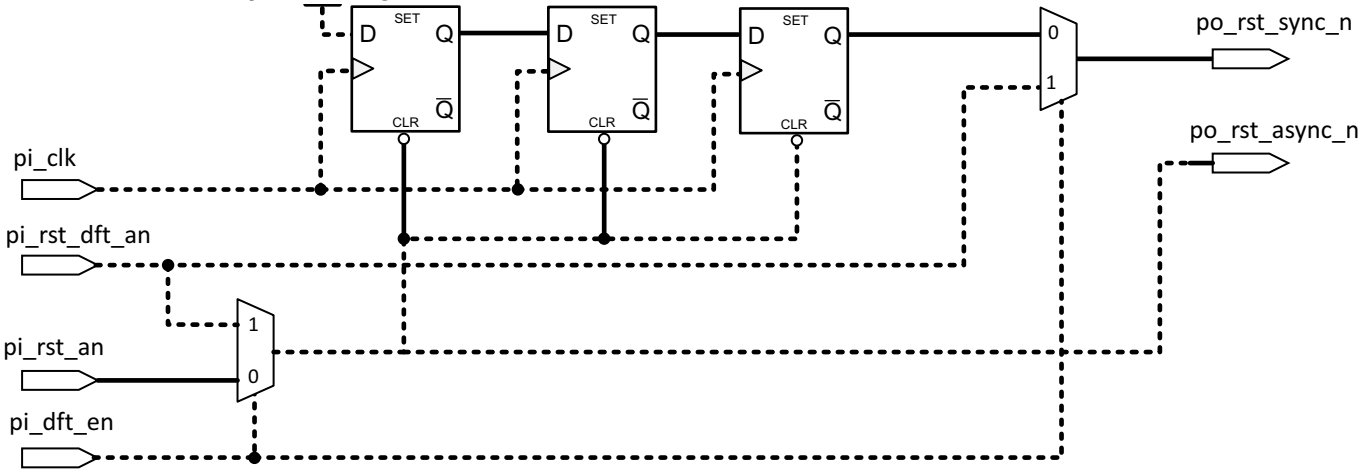
The input to the synchronizer first stage can be considered completely asynchronous. All paths to the flop1 register can be considered false_paths. The second and third registers in the dwc_*xsr_sync module are flop2 and flop3. To ensure stable synchronization, constrain the maximum delay to these cells, from flop1/Q to flop2/D and from flop2/Q to flop3/D to 20% of the sampling clock frequency. For ease of timing constraint definition, this delay can be implemented as 200 ps (20% of 1000 MHz) plus the intended clock tree depth of the respective sampling clock for the associated module instances. You can ignore the clock-tree delay if the paths are constrained explicitly from flop1/Q to flop2/D and from flop2/Q to flop3/D. Metastability behavior is not modeled by default in the synchronizers (for RTL simulations only). This behavior can be enabled by defining the macro VERIFY_EN.

Reset Synchronizers are used for synchronizing the asynchronous reset signal, an instance of the modules, dwc_*_param_rstgen is used. As shown in Figure 8-16, any time an external or internal reset condition is passed from one clock domain to another domain, the reset condition must pass through dwc_*xsr_*_param_rstgen module to do the following:

- Asynchronously capture the reset signal
- Stretch the reset pulse to be active for at least one full clock cycle in the receiving domain
- Synchronously de-assert the reset signal with respect to the receiving domain

Because assertion of a reset will cause all flip-flops in the synchronizer module to capture the pulse, the de-assertion edge of the asynchronous reset input has no effect on the final flip-flop that drives the reset into the receiving domain. All asynchronous paths through the d_s3 register of this module can be considered false.

Figure 8-16 dwc_*_*_param_rstgen



8.8.2.4 txX_clk Constraints

However, it is possible for the incoming ASIC txX_clk to drift relative to the internal txX_ana_dword_clk domain because of voltage and temperature variation. Because TX clock alignment is performed only on initial phase for FW. The variation on the txX_clk must be bound to ensure that this drift does not cause txX_clk to misalign relative to the txX_ana_dword_clk domain, resulting in the PHY's inability to capture incoming data. Because the ASIC txX_clk is derived from the output PMA MPLL clocks, the entire path from the PMA lib outputs, through the external logic and back to the PMA, the txX_clk input must be analyzed to determine the worst-case clock tree latency variation across all voltage/temperature (VT) corners.

This external variation must be less than 100ps when PMA internal TX FIFO is disabled, and less than (0.5 WORD clock + 100ps) ~400ps when FIFO is enabled.

By default the FIFO datapath is disabled because it causes latency increasing.

8.8.2.5 PMA .lib ETM Timing Modes

The deliverables include multi-mode .lib files for the PMA modules (TX and RX) that provide timing information for various modes of operation.

Table 8-11 outlines the ETM timing modes.

Table 8-11 PMA .lib ETM Timing Modes

Timing Mode	ETM Mode in .lib	Additional Constraints
Functional	NORMAL	N/A
Scan Shift	SCAN	False path all non scan_*_in/out ports
Stuck-At Scan capture	SCAN	False path all scan_*_in/out ports
At-Speed Scan Capture	SCAN_ASST	False path all scan_*_in/out ports

The PMA hard macro .lib file contains generated clock definitions for various functional clocks. If timing is performed using .lib ETM timing modes for SCAN_SHIFT/SCAN_CAPTURE modes without defining the functional clocks, it may be necessary to remove these generated clocks (using remove_clock constraints) to prevent clock propagation errors.

The PMA hard macro .lib file contains generated clock definitions for various functional clocks.

8.8.2.6 Unused Signals

Unused inputs to the raw PCS should be tied to ground potential.

Unused outputs from the raw PCS should be left floating.

9

Package-Level Implementation

This chapter includes the following sections:

- "Package Type" on page 158
- "Signal Integrity – RX and TX Differential Signal Guidelines" on page 158
- "Power Integrity – Supplies Requirements" on page 160

9.1 Package Type

The PMA is designed to support flip-chip packaging technology.

9.2 Signal Integrity – RX and TX Differential Signal Guidelines

You must design and simulate each of the RX and TX differential routes to ensure that signal integrity requirements are met.

Note For best signal integrity practices, refer to *DesignWare Cores SERDES PCB and Packaging Design Guide Application Note*. Also, see the "Package and PCB checklist" in Appendix A of *DesignWare Cores SERDES PCB and Packaging Design Guide Application Note*.

9.2.1 Package Guidelines

In a die-to-die application, the entirety of the channel is enclosed inside the package. As such, the package design is critical for signal integrity. This section details guidelines for the high-speed channels based on the Synopsys test chip package design.

Table 9-1 provides the guidelines for signals, package and board requirements.

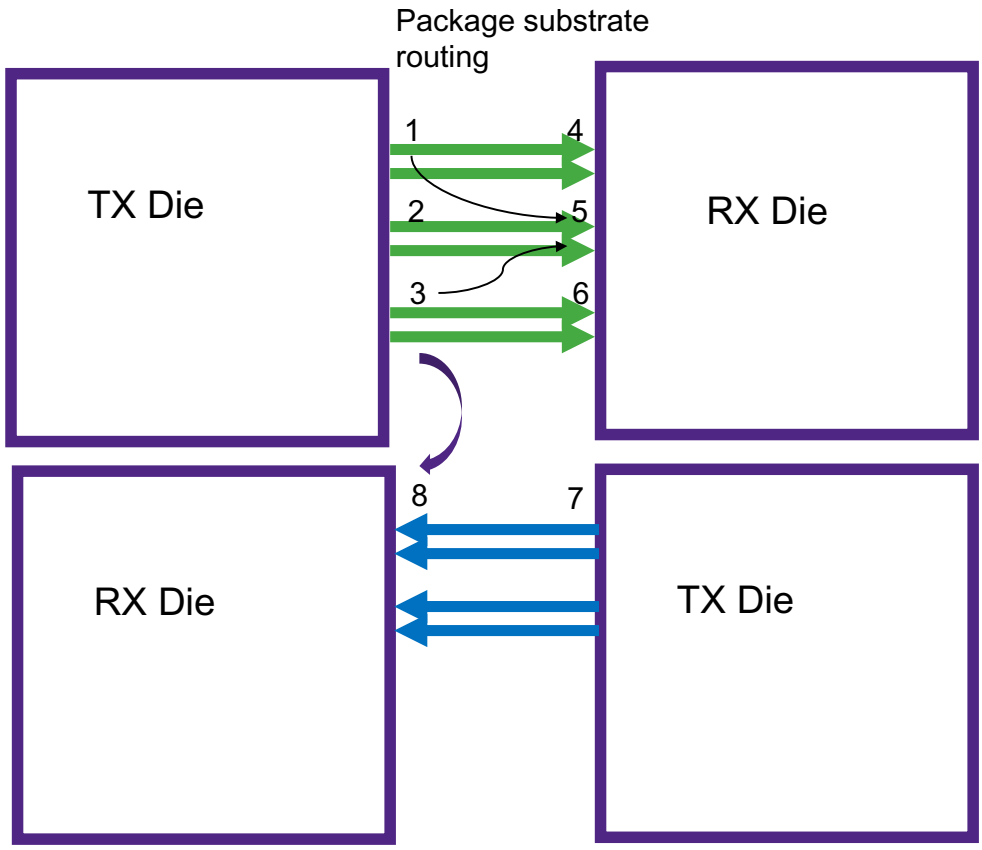
Table 9-1 Package Guidelines

Parameter	Package Guideline
TX/RX pair target nominal differential impedance	92.5 ohm
Maximum allowed skew among all pairs (including TX and RX pairs)	20 ps
Differential Insertion Loss	Less than 4 dB at 28 GHz for links less than or equal to 15 mm
Diff ILD (Insertion Loss Deviation)	0.5 dB from 0 Hz to 28 GHz
Differential Return Loss	Less than -15dB to 28 GHz
Common mode Return Loss	Less than -9 dB to 28 GHz
Aggregated crosstalk for (TX and forward clock) to RX FEXT	Less than -34 dB to 28 GHz
Aggregated Crosstalk TX to RX NEXT (IP macro to IP macro)	Less than -60dB at 28 GHz

Figure 9-1 illustrates the TX to RX FEXT and TX to RX NEXT routing.

In forwarded clock mode difbuff_clk_p and difbuff_clk_m should be used to transmit 1 GHz clock from TX and RX. In this case the requirements in Table 9-1 should apply for those routes, other than “Maximum allowed skew” requirements.

Figure 9-1 Package Substrate Routing



TX to RX FEXT refers to total far end crosstalk at victim RX from all aggressor pairs going from TX die to RX die, in this example, SDD15 + SDD35.

TX to RX NEXT refers to near end crosstalk at victim RX received from a strong TX signal travels in opposite direction, in this example, SDD38.

Aggregated crosstalk refers to crosstalk received from all aggressors from same category (near end or far end aggressors).

9.2.2 Signal Integrity Analysis

To enable signal integrity (SI) analysis of the RX and TX routes, s-parameter models as well as an AMI model is provided with the PHY deliverables.

These models will be provided in a future release.

9.2.2.1 S-Parameter Model

To assist in package design, an S-parameter model of the RX input and TX output are provided. These models should be simulated along with your extracted package model to ensure that the protocol specific requirements for return loss and other signal integrity requirements are met.

9.2.2.2 AMI Model

To assist link modeling, an IBIS Algorithmic Modeling Interface (IBIS-AMI) model of the PHY transmitter and receiver is provided along with a representative channel model.

The provided channel model is simply an example, you should measure the system board and connectors you are using to get a more accurate representation of the link performance.

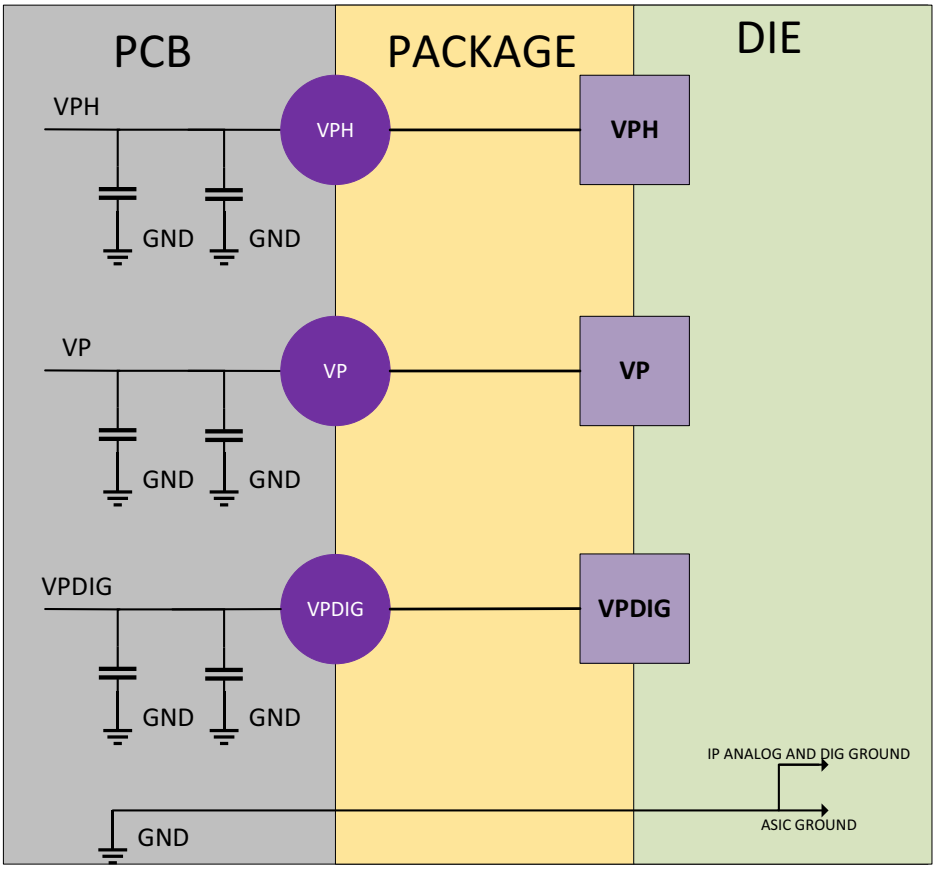
9.3 Power Integrity – Supplies Requirements

You must design and simulate the power distribution network (PDN) for each of the PHY supplies to ensure that power integrity requirements are met.

Note For more information about the requirements, see the requirements listed in the "Supply Guidelines" chapter of *DesignWare Cores SERDES PCB and Packaging Design Guide Application Note*. Also, see the "Package and PCB checklist" in Appendix A of *DesignWare Cores SERDES PCB and Packaging Design Guide Application Note*.

Vp/Vph for TX and RX PMAs are separate on the die, but are allowed to be merged on the package substrate. A diagram of the power supplies, which are common for TX and RX PMAs is shown in Figure 9-2.

Figure 9-2 Power Supply Diagram



9.3.1 Power Supplies Delivery to Bump Pads

You must perform a power integrity (PI) analysis to ensure that the DC and AC noise supply requirements in Table 9-2 are met on-die.

Table 9-2 Power Supply Requirements

Pin	Supply Voltage at PHY Bumps	DC Noise	AC Noise	Intermediate Range
Vph	1.2 V +10%/-3% 1.5 V ± 10% 1.8 V ± 10%	5% P2P of DC level	3% P2P of DC level for >100MHz	According to Figure 9-3
Vp	0.90 V ±5%	5% P2P of DC level	3% P2P of DC level for >1000MHz	According to Figure 9-4
Vpdig	0.8 V ±10%[a]	N/A	N/A	

a. SoC voltage is used for level shifting in the PHY

Note For all noise frequency ranges, the supply values should be in range of supply voltage at PHY bumps.

Current waveforms along with Cdie and Rdie are provided with the PHY deliverables. You should simulate the supply noise using this PI enablement along with a model of your package and board to ensure that ondie supply requirements are met.

The power supply requirements are on-die at the PHY pads.

Figure 9-3 shows the DC and AC limits for the VPH analog supply.

Figure 9-3 DC and AC Limits for the VPH Analog Supply

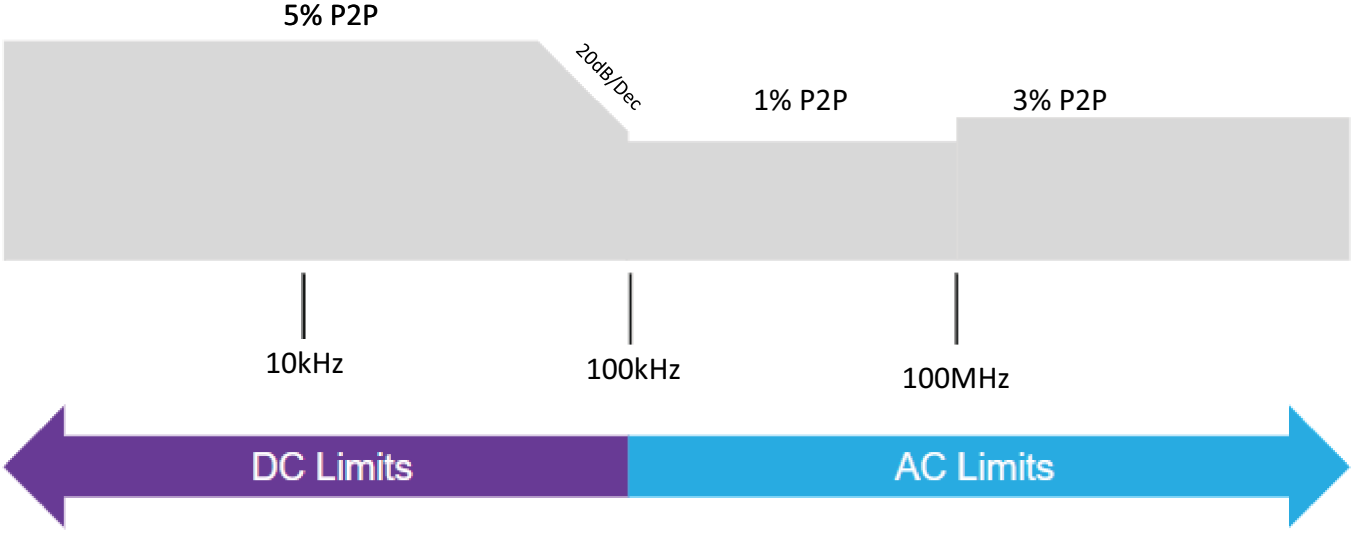


Table 9-3 lists the DC and AC limits for the VPH analog supply.

Table 9-3 DC and AC Limits for VPH Analog Supply

	DC Limits	AC Noise Limits
Voltage Range	5% of VPH	■ Noise level greater than 100 MHz: Maximum 3% (peak-to-peak) of the DC level ■ Noise level in the 100 kHz to 100 MHz range: Maximum 1% (peak-to-peak) of the DC level
Frequency Range	0 to 100 kHz	

The following are DC limit recommendations:

- The overshoot or undershoot of the low-frequency supply caused due to the board filter network should be lower than 100 kHz.
- The overall DC budget should account for the low frequency overshoot/undershoot in addition to the board plus package IR drop.
- There is a transition zone between 10 kHz and 100 kHz frequencies in which the supply noise can increase as the frequency decreases at a rate of 20 dB/Dec up to 5% of VPH.

The following are AC limit recommendations:

- Use an LDO as a switching supply.
- When using the switching supply for the VPH, its tone and harmonics are within the more sensitive region. Make sure that you meet the tighter noise limits.
- Do not share directly with other noisy circuitry.

Figure 9-4 shows the DC and AC limits for the VP supply.

Figure 9-4 DC and AC Limits for the VP Analog Supply

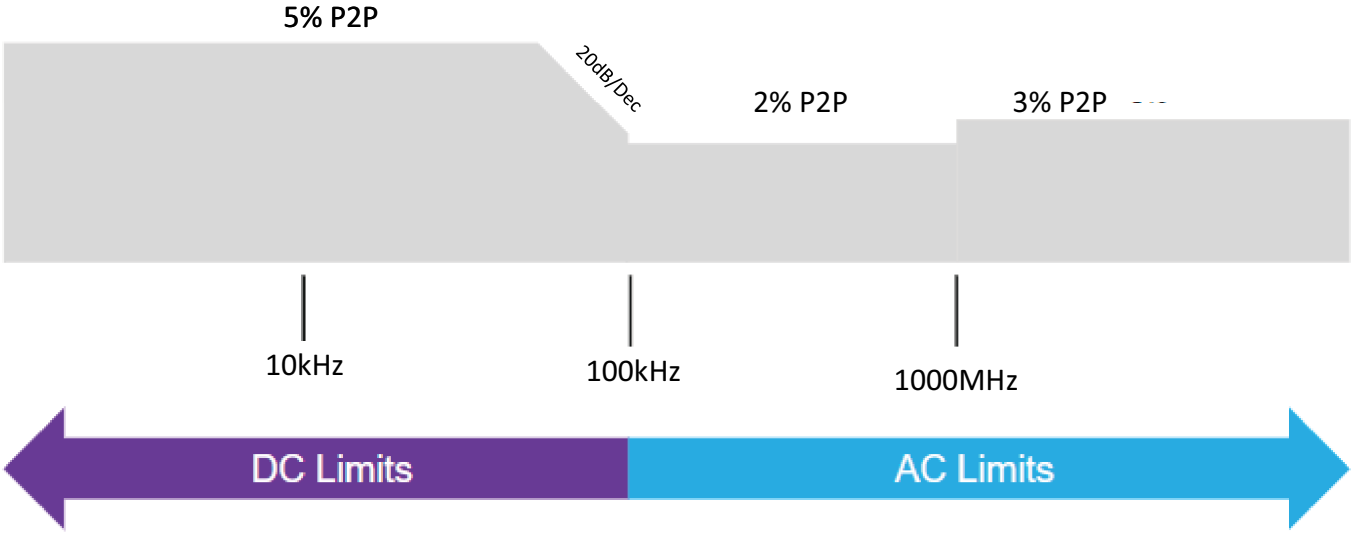


Table 9-4 lists the DC and AC limits for the VP analog supply.

Table 9-4 DC and AC Limits for VP Analog Supply

	DC Limits	AC Noise Limits
Voltage Range	5% of VP	■ Noise level greater than 1000 MHz: Maximum 3% (peak-to-peak) of the DC level ■ Noise level in the 100 kHz to 1000 MHz range: Maximum 2% (peak-to-peak) of the DC level
Frequency Range	0 to 1000 MHz	

The following are DC limit recommendations:

- The overshoot or undershoot of the low-frequency supply caused due to the board filter network should be lower than 100 kHz.
- The overall DC budget should account for the low frequency overshoot/undershoot in addition to the board plus package IR drop.
- There is a transition zone between 10 kHz and 100 kHz frequencies in which the supply noise can increase as the frequency decreases at a rate of 20 dB/Dec up to 5% of VP.

To met the AC limit recommendations, you can use a switching supply until the overall noise limits (including, the self noise) are met.

9.3.2 Package Guidelines – Avoidance of Substrate Routing Above Inductors

There are on-die inductors implemented in the PHY: Tcoil inductors in transmit and receive paths, and in MPLL. In the silicon interposer or package layout, Synopsys requires that you avoid any routing of signals, for example, TX or Rx differential lines or reference clock lines, or any other toggling signals immediately beneath the MPLL on-chip inductors due to noise coupling concerns. Synopsys recommends to have a ground metal reference copper or an IO Supply reference copper populated under MPLL inductor at 50 um or more vertical spacing on package substrate or interposer. This helps to shield inductor from noise coupling and at the same time ensure on chip inductance not affected.

Figure 9-5 shows the inductor location within the common module (CM). For the coordinates of the keep-out areas illustrated in Figure 8-3 on page 136 and Figure 8-4 on page 137, contact Synopsys.

Figure 9-5 On-Die Inductor Location

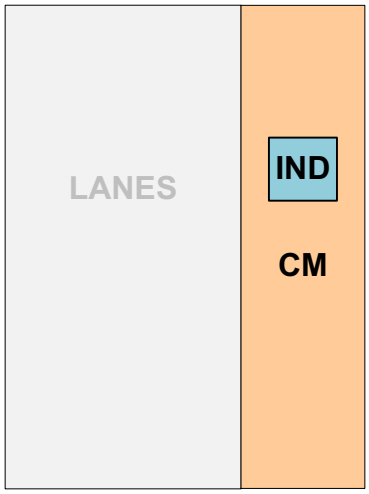


Table 9-5 lists the inductor coordinates.

Table 9-5 Inductor Coordinates

Position	Coordinate
Lower Left	(445.1745, 637.048)
Upper Right	(575.1745, 766.855)

10

Reference Clock Implementation

The PHY is designed to handle a wide range of input clock frequencies to support a multiple applications. This chapter includes the following sections:

- "Clocking Architecture" on page 166
- "Reference Clock Requirements" on page 167
- "Reference Clock Implementation" on page 169

10.1 Clocking Architecture

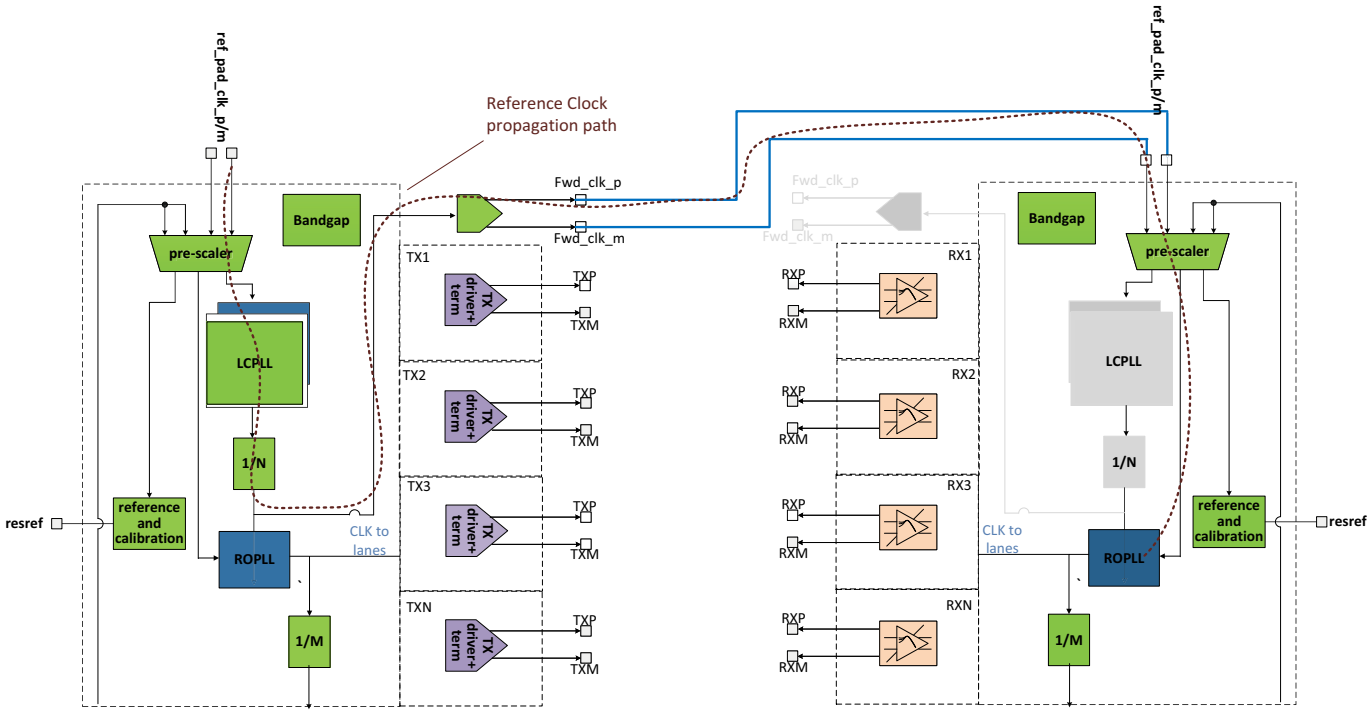
High-speed clock is generated by the common clock module in the support block (SUP). The support clock consists of two PLL's MPLL and ring-oscillator phase-locked loop.

MPLL generates clock from 4.6875–5.5296 GHz. ROPLL uses divided clock (divide by 5) and can support frequencies from 7-14 GHz.

The high-frequency reference clock allows a high ROPLL bandwidth to optimize the jitter compared to power performance of the ROPLL.

The MPLL generates the high-frequency clocks used by the ROPLL which are subsequently used by lanes (TX and RX). MPLL clock used in TX can be send to RX in clock forwarded configuration (see Figure 10-1).

Figure 10-1 RX and TX Configuration for Clock Forwarded Mode



You need to configure the MPLL, so that it runs at the correct frequency for your application and protocols.

To configure the MPLL use Appendix A, "PHY Configuration" to determine the correct setting of the mpll_multiplier[11:0]. The divider has a resolution of 0.25 input clock cycles and thus mpll_multiplier[0] has a binary weighting of 0.25, mpll_multiplier[1] has a weighting of 0.5, mpll_multiplier[2] has a weighting of 1, and so on.

Use the following formulas to better understand how to configure MPLL and ROPLL frequencies.

MPLL Configuration

$$F_{ropll_ana_refclk} = \frac{F_{reflck}}{N * 2^{ref_ana_mplla_div[2:0]}} \times 2^{(1+\text{mplla_ana_word_clk_div4_en})} \times \left(\frac{mplla_ana\ multiplier[11:0]}{4}\right)$$

N = 5*mplla_ana_word_clk_div5_en+4*mplla_ana_word_clk_div4_en

For RX in clock forwarding mode the same F_{ropll_refclk} is used to define FVCO Frequency of RX Ropll.

ROPLL Configuration

$$F_{VCO} = \frac{F_{ropll_ana_refclk}}{ropll_\text{ana}_refdiv[3:0]} \times ropll_ana_fbdiv[5:0] \times (1 + ropll_ana_fbdiv[6])$$

The ROPLL VCO frequency is configured as follows, where ropll_refclk is the reference clock sent to the ROPLL from the MPLL.

10.2 Reference Clock Requirements

The PHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

10.2.1 Generic Reference Clock Requirement for Protocols

Differential reference clock requirements are provided in Table 10-1.

Table 10-1 Reference Clock Input Requirements

Symbol	Parameter	Min	Max	Units	Conditions
FREF_CLK	Reference clock frequency	78.125	200	MHz	Not continuous; per protocol as noted in the databook.
FREF_OFFSET	Reference clock frequency offset	-100	100	ppm	
RMSJREF_CLK 28G-VSR	Reference clock random jitter		1.0	ps_{rms}	Integrated RJ from 12 kHz to 20 MHz.
			0.8		Integrated Rj from 2 MHz to 20 MHz.
DJREF_CLK 28G-VSR	Reference clock deterministic jitter		1.7 3.4	ps, pp	0.75-10 MHz (offset)[a] 0.2-50 MHz (offset)[a]
RMSJREF_CLK XSR-112G	Reference clock random jitter		0.15	ps_{rms}	Integrated RJ from 12 kHz to 20 MHz.
			0.1		Integrated Rj from 2 MHz to 20 MHz.
DJREF_CLK XSR-112G	Reference clock deterministic jitter		0.2 0.45	ps, pp	0.5-10 MHz (offset)[a] 0.1-50 MHz (offset)[a]
DCREF_CLK	Duty cycle	40	60	%	
VCMREF_CLK	Common mode input level	0	VP	V	Differential inputs.
VDREF_CLK	Differential input swing for ref_pad_clk_{p,m}	300		mVpp	Differential inputs must meet VDREF_CLK, VDSEREF_CLK_DC and VDSEREF_CLK_AC.

Symbol	Parameter	Min	Max	Units	Conditions
VDREF_CLK_ALT	Differential input swing for ref_alt_clk_{p,m}	200		mVpp	Differential inputs must meet VDREF_CLK_ALT, VDSEREF_CLK_DC and VDSEREF_CLK_AC.
VDSEREF_CLK	Input Voltage Level	-0.15	0.85	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK or both VDREF_CLK_ALT and VDSEREF_CLK.
SKEWDIF_CLK	Differential clock input skew		100	ps	
SWREF_CLK	Differential clock input slew rate	0.6		V/ns	20–80%

a. DJ noise limits are for noise within an offset from the divided reference clock and harmonics of the divided reference clock.

10.2.2 Repeat Clock Specification

The PHY is able to repeat the input reference clocks through up to four PHY macros. A reference clock repeater is provided to output the reference clock through ref_repeat_clk_p and ref_repeat_clk_m. The specifications for the repeater are provided in Table 10-2.

Note A PHY that repeats the input reference clock should NOT be power gated (that is, pg_mode = 1'b0). Power gating the PHY disables the repeating function.

Table 10-2 Repeat Clock Output Specification

Symbol	Parameter	Min	Max	Units	Conditions
FREF_CLK	Reference clock frequency	78.125	200	MHz	
DJREF_CLK	Increase in reference clock deterministic jitter for each repeat		20	fs	Output referred after filtering by Tx PLL and Rx CDR transfer functions. Due to self noise from the PMA. Any additional external noise is additive.
DCREF_CLK	Duty cycle	40	60	%	No degradation is allowed.
VCMREF_CLK	Common mode output	VP-0.2	VP-0.1	V	Differential CML output.
VDREF_CLK	Differential output swing	400	800	mVpp	Recommended constant swing design, which allows for 400 mVpp swing all the time.

Symbol	Parameter	Min	Max	Units	Conditions
SKEWDIF_CLK	Differential clock output skew		100	ps	No extra skew is allowed; instead, the PHY corrects the incoming clock skew (if needed).
CL_CLK	Single-ended output loading		4	pF	Specifies the maximum loading that the PHY can drive.
SWREPT_CLK	Differential clock output slew rate at far end	0.6		V/ns	Measured over 20% to 80% of signal transition.
LVL_OFF_CLK	Differential signal level when disabled		GD		Disabled clock repeater is pulled to 0 V.
REPT_CLK	Maximum repeat times allowed		3		Repeat through three PMAs to the final fourth PMA.

10.3 Reference Clock Implementation

There are two types of reference clock implementations:

- External off-chip reference clocks using ref_pad_clk_p/ref_pad_clk_m
- On-chip reference clocks using ref_alt_clk_p/m

Both reference clock implementations support either a differential clock or a CMOS single-ended clock. The reference clock input is an unterminated AC coupled input.

If the reference clock is a single-ended CMOS clock, one of the clock inputs (such as ref_pad_clk_p) must be a full-swing CMOS clock, and the other input (such as ref_pad_clk_m) should be tied to either high or low.

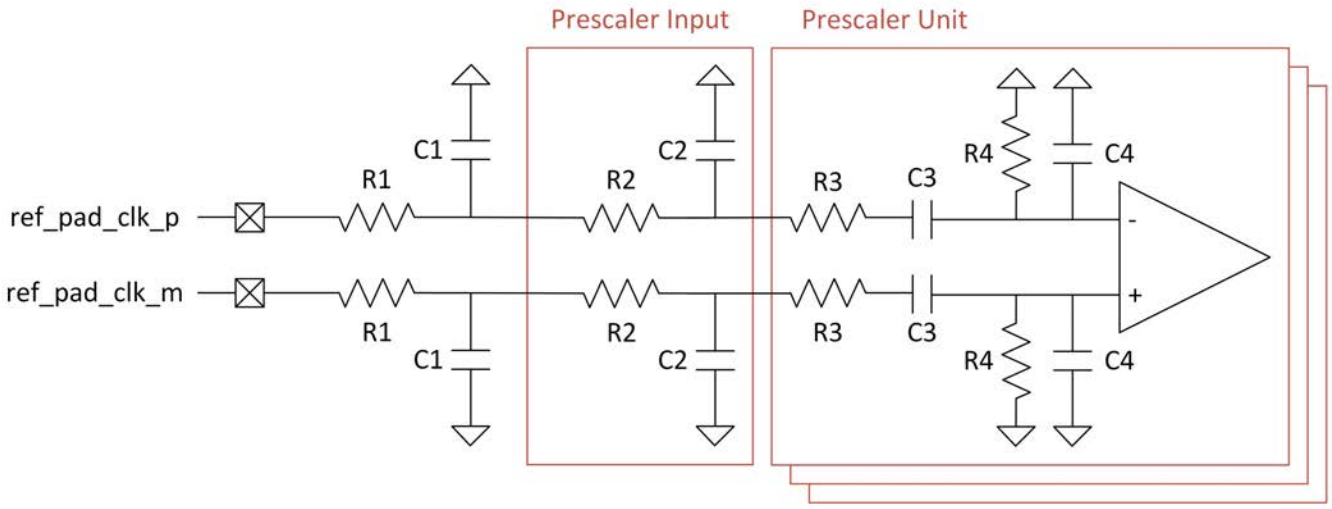
If the reference clock is differential, it may be CML or LVDS, or any differential specification that meets the requirements outlined in Table 10-1.

Any reference clock inputs that are not used should be tied to ground.

10.3.1 External Off-Chip Clock Sources

When routing reference clocks in from off-chip, a signal integrity (SI) analysis should be performed using a model of the clock input, package, board traces, and clock source. This analysis should confirm that the reference clock input specification is met across the external on-chip reference clock input (ref_pad_clk_p/ref_pad_clk_m), as outlined in "Reference Clock Requirements" on page 167. Figure 10-2 shows a model of the reference clock input, with parameters provided in Table 10-3.

Figure 10-2 Model of Reference Clock Pad Input Load



Note Loading of the prescaler input depends on the application-specific number of the prescaler units.

Table 10-3 Reference Clock External Pad Input Parasitics

Location	Parasitic Device	Value	Unit
Within PHY: bump-to-prescaler input	R1	<= 20	ohm
Within PHY: bump-to-prescaler input	C1	<= 0.2	pF
Within prescaler: second ESD Prot resistor[a]	R2	250	ohm
Within prescaler: second ESD Prot, clock routes, and input cap of multiplexer	C2	180	fF
Within prescaler unit: 3-1 multiplexer turn-on resistor	R3	150	ohm
Within prescaler unit: AC cap	C3	880	fF
Within prescaler unit: DC Bias resistor	R4	460	kohm
Within prescaler unit: input cap of prescaler input RX	C4	<= 120	fF

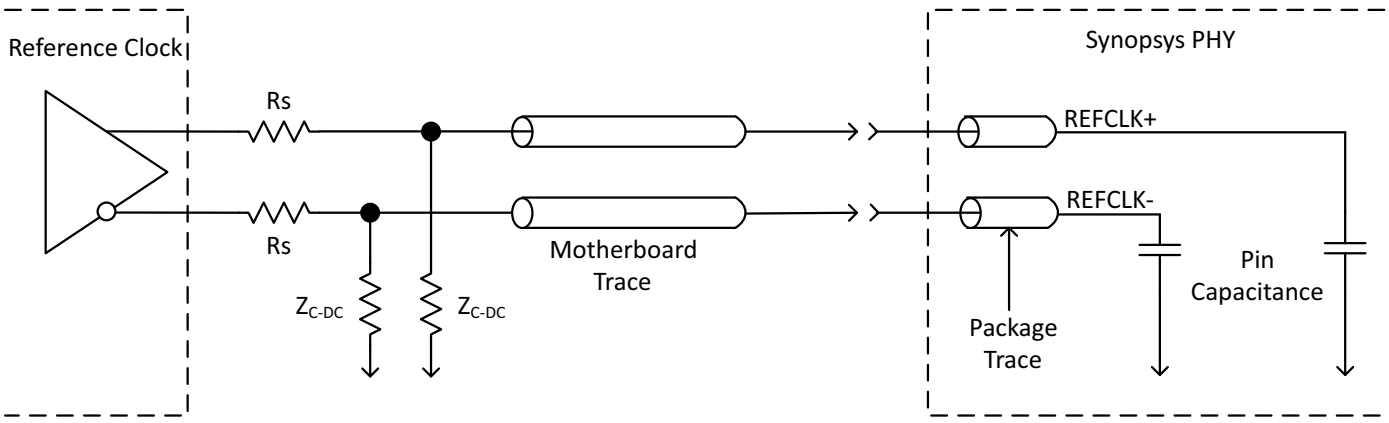
a. Required for 6A peak discharge current ESD rating.

To improve Common-Mode Rejection Ratio (CMRR), differential traces should be as well matched as possible.

To enable connecting reference clocks to multiple PHYs at the same time, the reference clock inputs are not terminated. If termination is used, place it on your board as close to the chip as possible. For PCIe, see the PCIe CEM specification for details on how to implement the on-board termination.

If you use off-chip reference clock (that is, ref_pad_clk_p or ref_pad_clk_m), you must terminate the off-chip clock source to suppress reflection. Also, make sure that the clock swing at the Synopsys PHY input is within reference clock specification. An example of this implementation is shown in Figure 10-3.

Figure 10-3 Off-Chip Clock Source Termination



10.3.2 On-Chip Clock Sources

Many applications require an on-chip reference clock. The challenge is to get the reference clock to the PHY without adding supply-noise-induced jitter to the clock.

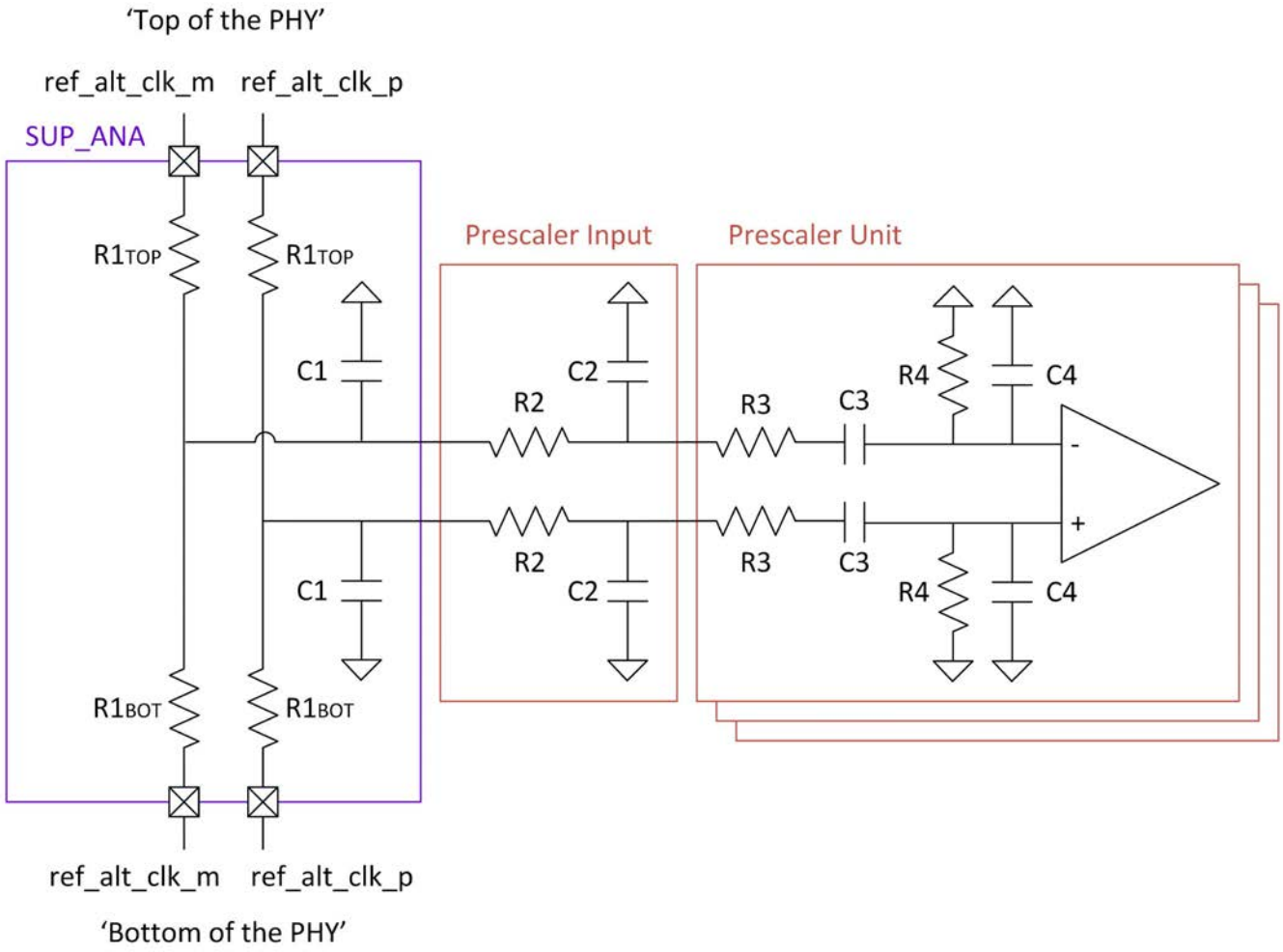
The following general guidelines should be followed when routing an on-chip reference clock:

- Route manually (do not use PNR tool router)
- Design differential routing to optimize RC
- Consider higher-level metal to get best RC factor (lower-level metals tend to be too resistive)
- Use shielding on both sides of diff routes
- Avoid noisy signals directly above and below the routes
- If you do route signals near the routes, do not have them parallel—use perpendicular routes only, so that noise is common mode
- Simulate

Figure 10-4 shows a model of the reference clock input with parameters provided in Table 10-4.

When routing reference clocks on-chip, you should consider the parasitic loading of the clock distribution lines and the PHY clock input. This analysis should confirm that the reference clock input specification is met across the on-chip reference clock input (ref_alt_clk_p0/m and ref_alt_clk_p1/m) as outlined in "Reference Clock Requirements" on page 167.

Figure 10-4 Model of Reference Clock ASIC Side ALT Input Loading



Note Loading of the prescaler input depends on the application-specific number of the prescaler units.

Table 10-4 Reference Clock ASIC-Side ALT Input Parasitics

Location	Parasitic Device	Value	Units
Clock routes before prescaler from "Top of the PFY"	R1TOP	25	ohm
Clock routes before prescaler from "Bottom of the PFY"	R1BOT	90	ohm
Clock routes before prescaler	C1	200	fF
Within prescaler: clock routes	R2	280	ohm
Within prescaler: clock routes and input cap multiplexer	C2	80	fF

Location	Parasitic Device	Value	Units
Within prescaler: 3-1 multiplexer turn-on resistor	R3	150	ohm
Within prescaler: AC cap	C3	880	fF
Within prescaler: DC bias resistor	R4	460	kohm
Within prescaler: input cap of prescaler input RX	C4	<= 120	fF

Note To improve common-mode rejection ratio (CMRR), differential traces should be as well matched as possible.

10.3.3 Reference Clock Repeater

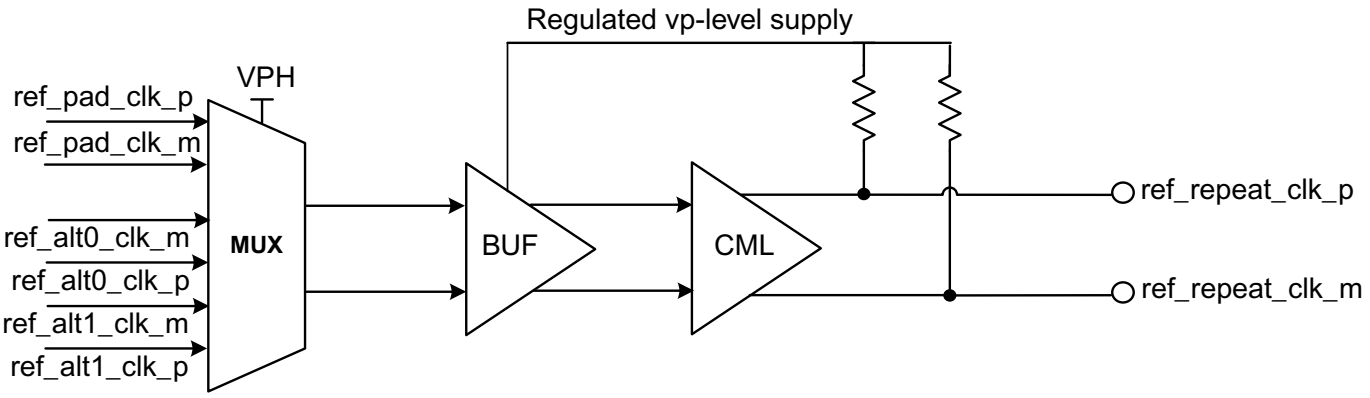
There is a reference clock repeater output (ref_repeat_clk_m/ref_repeat_clk_p) that allows the PHY to buffer the reference clock and send it on to another PHY. The driver is a VP supply referenced CML output clock driver, as shown in Figure 10-5. The repeater is designed to meet the specifications given in "Repeat Clock Specification" on page 168.

The following general guidelines should be followed when routing the repeater output (ref_repeat_clk_p/ref_repeat_clk_m) from one PHY to another PHY.

- Route manually (do not use PNR tool router) with following recommended configuration:
  - Signal routing width = 1 µm, highest metal
  - Spacing (ref_repeat_clk_p - ref_repeat_clk_m) = 2 µm
  - Spacing (signal-to-shield) = 2.5 µm
  - Ground Shield Width = minimum per DRC
  - GDshield - (2.5 µm space) - ref_repeat_clk_p - (2 µm space) - ref_repeat_clk_m - (2.5 µm space) - Gdshield
  - Density layer fill for above and below routes (that is, +1/-1 of routing layer)
- Design differential routing to optimize RC.
- Consider the highest-level metal to get the best RC factor (lower-level metals tend to be too resistive).
- Use shielding on both sides of differential routes.
- Avoid noisy signals directly above and below the routes.
- If you do route signals near the routes, do not have them parallel. Use perpendicular routes only, so that noise is common mode and has a minimal impact on the differential signal.
- Simulate.

Ref_clk repeat buffer and single-ended buffers are not in an always-on power domain. If powergating is enabled, these outputs are powered down. Do not use these signals for logic or PHYs intended to remain operational during power-gating.

Figure 10-5 Reference Clock Repeater



To improve common-mode rejection ratio (CMRR), differential traces should be as well matched as possible.

10.3.4 Reference Clock Distribution

To minimize the amount of jitter on the reference clock, Synopsys recommends STAR configuration in the clock distribution.

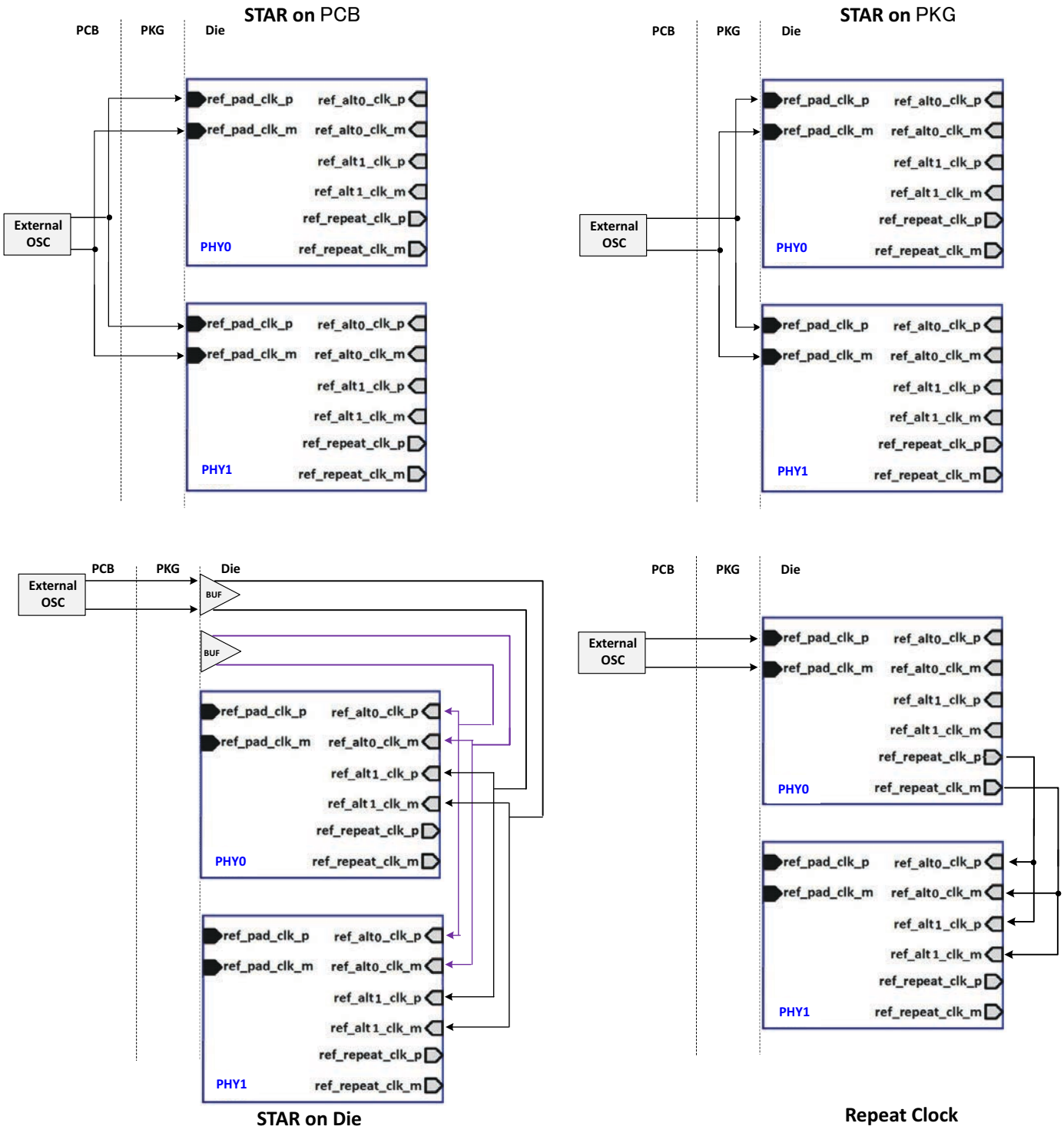
Table 10-5 provides impact of each of the reference clock configurations.

Table 10-5 Clock Distribution Impact

No	Item	Clock Distribution Example				Note
		STAR on PCB	STAR on Package	STAR on Die	Repeat Clock	
1	REFCLK input	PAD	PAD	Core	PAD/Core	
2	Jitter	-	-	Increase	Increase	Jitter increases if you use active devices.
3	ATE Testbench	As is	As is	As is	Customize	Two PHYs setting becomes different if you use repeat clock. Then, you need to customize your ATE testbench.
4	REFCLK SI analysis	ref_pad_clk only	ref_pad_clk only	ref_alt_clk only	ref_pad_clk ref_alt_clk	
5	Additional buffer	-	-	Necessary	-	Additional REFCLK buffer is necessary for STAR on Die.
6	Additional ESD consideration	-	-	Necessary	(it depends)	If VP supply is not driving ref_alt_clk, additional ESD consideration (CDM) is necessary.

Figure 10-6 provides examples of the clock distribution on board, in the package and on the die.

Figure 10-6 Reference Clock Distribution



Connect *repeat* clock to either *alt0* or *alt1* (not both). Unused *alt* signal connect to gnd.

Set ref_repeat_clk_sel[1:0] according to your selection.

11

Register Descriptions

This chapter details all possible registers in the IP. They are arranged hierarchically into maps and blocks (banks).

Exists Expression

The Exist expressions (if present) indicate the combination of configuration parameters required for a register, field, or block to exist in the memory map. The expression is only valid in the local context and does not indicate the conditions for existence of the parent. For example, the Exists expression for a bit field in a register assumes that the register exists and does not include the conditions for existence of the register.

Offset

The term *Offset* is synonymous with *Address.*

Memory Access Attributes

The Memory Access attribute is defined as `<ReadBehavior>/<WriteBehavior>` which are defined in the following table.

Table 11-1 Possible Read and Write Behaviors

Read (or Write) Behavior	Description
RC	A read clears this register field.
RS	A read sets this register field.
RM	A read modifies the contents of this register field.
Wo	You can only write once to this register field.
W1C	A write of 1 clears this register field.
W1S	A write of 1 sets this register field.
W1T	A write of 1 toggles this register field.
W0C	A write of 0 clears this register field.
W0S	A write of 0 sets this register field.

Read (or Write) Behavior	Description
W0T	A write of 0 toggles this register field.
WC	Any write clears this register field.
WS	Any write sets this register field.
WM	Any write toggles this register field.
no Read Behavior attribute	You cannot read this register. It is Write-Only.
no Write Behavior attribute	You cannot write to this register. It is Read-Only.

Table 11-2 Memory Access Examples

Memory Access	Description
R	Read-only register field.
W	Write-only register field.
R/W	Read/write register field.
R/W1C	You can read this register field. Writing 1 clears it.
RC/W1C	Reading this register field clears it. Writing 1 clears it.
R/Wo	You can read this register field. You can only write to it once.

Special Optional Attributes

Some register fields might use the following optional attributes.

Table 11-3 Optional Attributes

Attribute	Description
Volatile	As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the IP updates the register field contents.

Attribute	Description
Testable	As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.
Reset Mask	As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.
* Varies	Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.

Register definitions for each component memory map.

Table 11-4 Registers for the regmap_XSR_X8 Memory Map

Register	Offset	Description
Receiver Block Registers RX_RX_j (for j = 0; j <= 7) Exists: Always		
"ANA_OVRDVAL_0" on page 194	0x0 + j*0x100 + 0x000	(RXS) Internal Override Value Register 0
"ANA_OVRDVAL_1" on page 196	0x0 + j*0x100 + 0x004	(RXS) Internal Override Value Register 1
"ANA_OVRDVAL_2" on page 198	0x0 + j*0x100 + 0x008	(RXS) Internal Override Value Register 2
"ANA_OVRDVAL_3" on page 200	0x0 + j*0x100 + 0x00C	(RXS) Internal Override Value Register 3
"ANA_OVRDVAL_7" on page 203	0x0 + j*0x100 + 0x010	(RXS) Internal Override Value Register 7

Register	Offset	Description
"ANA_OVRDVAL_8" on page 207	0x0 + j*0x100 + 0x014	(RXS) Internal Override Value Register 8
"ANA_OVRDVAL_10" on page 208	0x0 + j*0x100 + 0x018	(RXS) Internal Override Value Register 10
"ANA_OVRDVAL_11" on page 210	0x0 + j*0x100 + 0x01C	(RXS) Internal Override Value Register 11
"ANA_OVRDVAL_13" on page 212	0x0 + j*0x100 + 0x020	(RXS) Internal Override Value Register 13
"ANA_OVRDVAL_14" on page 214	0x0 + j*0x100 + 0x024	(RXS) Internal Override Value Register 14
"ANA_OVRDVAL_16" on page 216	0x0 + j*0x100 + 0x028	(RXS) Internal Override Value Register 16
"ANA_OVRDVAL_17" on page 217	0x0 + j*0x100 + 0x02C	(RXS) Internal Override Value Register 17
"ANA_OVRDEN_0" on page 220	0x0 + j*0x100 + 0x030	(RXS) Internal Override Enable Register 0
"ANA_OVRDEN_1" on page 223	0x0 + j*0x100 + 0x034	(RXS) Internal Override Enable Register 1
"ANA_OVRDEN_2" on page 226	0x0 + j*0x100 + 0x038	(RXS) Internal Override Enable Register 2
"ANA_OVRDEN_3" on page 229	0x0 + j*0x100 + 0x03C	(RXS) Internal Override Enable Register 3
"PIN_OVRDEN_0" on page 232	0x0 + j*0x100 + 0x040	(RXS) Pin Override Enable Register 0
"PIN_OVRDVAL_0" on page 233	0x0 + j*0x100 + 0x044	(RXS) Pin Override Value Register 0

Register	Offset	Description
"GENERIC_CONF_0" on page 235	0x0 + j*0x100 + 0x048	Generic Settings
"PATGEN_0" on page 238	0x0 + j*0x100 + 0x04C	Pattern Generator Configuration Settings Register 0
"PATGEN_1" on page 240	0x0 + j*0x100 + 0x050	Pattern Generator Configuration Settings Register 1
"PATGEN_2" on page 241	0x0 + j*0x100 + 0x054	Pattern Generator Configuration Settings Register 2
"PATGEN_3" on page 242	0x0 + j*0x100 + 0x058	Pattern Generator Configuration Settings Register 3
"PATGEN_4" on page 243	0x0 + j*0x100 + 0x05C	Pattern Generator Configuration Settings Register 4
"PATGEN_5" on page 244	0x0 + j*0x100 + 0x060	Pattern Generator Configuration Settings Register 5
"PATGEN_6" on page 245	0x0 + j*0x100 + 0x064	Pattern Generator Configuration Settings Register 6
"PATGEN_7" on page 246	0x0 + j*0x100 + 0x068	Pattern Generator Configuration Settings Register 7
"PATGEN_8" on page 247	0x0 + j*0x100 + 0x06C	Pattern Generator Configuration Settings Register 8
"PATGEN_9" on page 249	0x0 + j*0x100 + 0x070	Pattern Generator Configuration Settings Register 9
"GENERIC_CONF_1" on page 250	0x0 + j*0x100 + 0x074	Generic Settings 1
"GENERIC_CONF_2" on page 252	0x0 + j*0x100 + 0x078	Generic Settings 2

Register	Offset	Description
“GENERIC_CONF_3” on page 254	0x0 + j*0x100 + 0x07C	Generic Settings 3
“GENERIC_CONF_4” on page 256	0x0 + j*0x100 + 0x080	Generic Settings 4
“GENERIC_CONF_5” on page 259	0x0 + j*0x100 + 0x084	Generic Settings 5
“GENERIC_CONF_6” on page 261	0x0 + j*0x100 + 0x088	Generic Settings 6
“GENERIC_CONF_18” on page 262	0x0 + j*0x100 + 0x08C	Generic Settings 18
“GENERIC_CONF_19” on page 263	0x0 + j*0x100 + 0x090	Generic Settings 19
“GENERIC_CONF_7” on page 265	0x0 + j*0x100 + 0x094	Generic Settings 7
“GENERIC_CONF_17” on page 266	0x0 + j*0x100 + 0x098	Generic Settings 17
“GENERIC_CONF_8” on page 267	0x0 + j*0x100 + 0x09C	Generic Settings 8
“GENERIC_CONF_9” on page 269	0x0 + j*0x100 + 0x0A0	Generic Settings 9
“GENERIC_CONF_10” on page 271	0x0 + j*0x100 + 0x0A4	Generic Settings 10
“GENERIC_CONF_11” on page 273	0x0 + j*0x100 + 0x0A8	Generic Settings 11
“GENERIC_CONF_12” on page 276	0x0 + j*0x100 + 0x0AC	Generic Settings 12

Register	Offset	Description
“GENERIC_CONF_13” on page 277	0x0 + j*0x100 + 0x0B0	Generic Settings 13
“GENERIC_CONF_14” on page 279	0x0 + j*0x100 + 0x0B4	Generic Settings 14
“GENERIC_CONF_15” on page 280	0x0 + j*0x100 + 0x0B8	Generic Settings 15
“GENERIC_CONF_16” on page 281	0x0 + j*0x100 + 0x0BC	Generic Settings 16
“GENERIC_CONF_20” on page 283	0x0 + j*0x100 + 0x0C0	Generic Settings 20
“GENERIC_CONF_21” on page 285	0x0 + j*0x100 + 0x0C4	Generic Settings 21
“GENERIC_CONF_22” on page 286	0x0 + j*0x100 + 0x0C8	Generic Settings 22
“GENERIC_CONF_23” on page 288	0x0 + j*0x100 + 0x0CC	Generic Settings 23
“GENERIC_CONF_24” on page 289	0x0 + j*0x100 + 0x0D0	Generic Settings 24
“STATUS_0” on page 290	0x0 + j*0x100 + 0x0D4	(RXS) Digital Status Register 0
“STATUS_1” on page 291	0x0 + j*0x100 + 0x0D8	(RXS) Digital Status Register 1
“STATUS_7” on page 292	0x0 + j*0x100 + 0x0DC	(RXS) Digital Status Register 2
“STATUS_8” on page 293	0x0 + j*0x100 + 0x0E0	(RXS) Digital Status Register 3

Register	Offset	Description
"STATUS_9" on page 294	0x0 + j*0x100 + 0x0E4	(RXS) Digital Status Register 4
"STATUS_10" on page 295	0x0 + j*0x100 + 0x0E8	(RXS) Digital Status Register 5
"STATUS_11" on page 296	0x0 + j*0x100 + 0x0EC	(RXS) Digital Status Register 6
"STATUS_12" on page 297	0x0 + j*0x100 + 0x0F0	(RXS) Digital Status Register 7
"STATUS_13" on page 298	0x0 + j*0x100 + 0x0F4	(RXS) Digital Status Register 8
"STATUS_14" on page 299	0x0 + j*0x100 + 0x0F8	(RXS) Digital Status Register 9
"STATUS_15" on page 300	0x0 + j*0x100 + 0x0FC	(RXS) Digital Status Register 10
Supplemental Receiver Block Registers SUP_RX_SUP Exists: Always		
"SUP_CFG_0" on page 302	0x1000	(SUP) Configuration Register
"SUP_OVRDEN" on page 305	0x1004	(SUP) Internal Logic Override Enable Register
"SUP_OVRDVAL" on page 306	0x1008	(SUP) Internal Logic Override Value Register
"SUP_STAT" on page 307	0x100c	(SUP) Status Register
"ANA_OVRDVAL_0" on page 308	0x1010	OSC Configuration Register 0
"ANA_OVRDVAL_1" on page 310	0x1014	ANA Configuration Register 1
"ANA_OVRDVAL_2" on page 313	0x1018	CRU Access Register 2
"ANA_OVRDVAL_3" on page 315	0x101c	CRU Access Register 3
"ANA_OVRDVAL_4" on page 316	0x1020	DIFFBUF Configuration Register 4
"ANA_OVRDVAL_5" on page 318	0x1024	MPLL Analog Configuration Register 5
"ANA_OVRDVAL_6" on page 320	0x1028	MPLL Startup Configuration Register 6
"ANA_OVRDVAL_7" on page 322	0x102c	MPLL Dividers Configuration Register 7

Register	Offset	Description
“ANA_OVRDVAL_8” on page 324	0x1030	MPLL PMIX Configuration Register 8
“ANA_OVRDVAL_9” on page 326	0x1034	REF Configuration Register 9
“ANA_OVRDVAL_10” on page 329	0x1038	ROPLL Startup Configuration Register 10
“ANA_OVRDVAL_11” on page 332	0x103c	ROPLL Dividers Configuration Register 11
“ANA_OVRDVAL_12” on page 335	0x1040	RTRX Configuration Register 12
“ANA_OVRDEN_0” on page 337	0x1044	(SUP) Internal Override Enable Register 0
“ANA_OVRDEN_1” on page 340	0x1048	(SUP) Internal Override Enable Register 1
“ANA_OVRDEN_2” on page 343	0x104c	(SUP) Internal Override Enable Register 2
“ANA_OVRDEN_3” on page 346	0x1050	(SUP) Internal Override Enable Register 3
“ANA_OVRDVAL_13” on page 348	0x1054	SUP ANA Output Register 13
“ANA_OVRDVAL_14” on page 350	0x1058	SUP ANA Output Register 14
“SUP_CFG_1” on page 351	0x105c	(SUP) Configuration Register 1
“SUP_CFG_2” on page 353	0x1060	(SUP) Configuration Register 2
“SUP_CFG_3” on page 354	0x1064	(SUP) Configuration Register 3
“SUP_CFG_4” on page 355	0x1068	(SUP) Configuration Register 4
“SUP_CFG_5” on page 357	0x106c	(SUP) Configuration Register 5
“SUP_CFG_6” on page 359	0x1070	(SUP) Configuration Register 6
“SUP_CFG_7” on page 360	0x1074	(SUP) Configuration Register 7
“SUP_CFG_8” on page 361	0x1078	(SUP) Configuration Register 8
“SUP_CFG_9” on page 363	0x107c	(SUP) Configuration Register 9
“SUP_CFG_10” on page 364	0x1080	(SUP) Configuration Register 10
“SUP_CFG_11” on page 365	0x1084	(SUP) Configuration Register 11
“MPLL_STAT_0” on page 366	0x1088	MPLL Configuration Register 0
“MPLL_STAT_1” on page 367	0x108c	MPLL Configuration Register 1
“MPLL_STAT_2” on page 369	0x1090	MPLL Configuration Register 2
“MPLL_STAT_3” on page 370	0x1094	MPLL Configuration Register 3
“SUP_CFG_12” on page 371	0x1098	(SUP) Configuration Register 12
“SUP_CFG_13” on page 372	0x109c	(SUP) Configuration Register 13
“SUP_CFG_14” on page 374	0x10a0	(SUP) Configuration Register 14

Register	Offset	Description
"rt_stat_0" on page 375	0x10a4	rt_stat Configuration register 0
"rt_stat_1" on page 376	0x10a8	rt_stat Configuration register 1
"SUP_CFG_15" on page 377	0x10ac	(SUP) Configuration Register 15
"SUP_CFG_16" on page 378	0x10b0	(SUP) Configuration Register 16
"SUP_CFG_17" on page 379	0x10b4	(SUP) Configuration Register 17
"SUP_CFG_18" on page 380	0x10b8	(SUP) Configuration Register 18
"SUP_CFG_19" on page 381	0x10bc	(SUP) Configuration Register 19
"SUP_CFG_20" on page 382	0x10c0	(SUP) Configuration Register 20
"SUP_CFG_21" on page 383	0x10c4	(SUP) Configuration Register 21
Transmitter Block Registers TX_TX_j (for j = 0; j <= 7) Exists: Always		
"TXS_CFG_0" on page 385	0x2000 + j*0x100 + 0x00	(TXS) Configuration Register 0
"TXS_CFG_1" on page 387	0x2000 + j*0x100 + 0x04	(TXS) Configuration Register 1
"TXS_CFG_2" on page 388	0x2000 + j*0x100 + 0x08	(TXS) Configuration Register 2
"TXS_CFG_3" on page 390	0x2000 + j*0x100 + 0x0C	(TXS) Configuration Register 3
"TXS_CFG_4" on page 391	0x2000 + j*0x100 + 0x10	(TXS) Configuration Register 4
"TXS_CFG_5" on page 393	0x2000 + j*0x100 + 0x14	(TXS) Configuration Register 5
"PIN_OVRDEN_0" on page 395	0x2000 + j*0x100 + 0x18	(TXS) Port Override Enable Register 0
"PIN_OVRDVAL_0" on page 397	0x2000 + j*0x100 + 0x1C	(TXS) Port Override Value Register 0

Register	Offset	Description
“PIN_OVRDVAL_1” on page 399	0x2000 + j*0x100 + 0x20	(TXS) Port Override Value Register 1
“PIN_OVRDVAL_2” on page 400	0x2000 + j*0x100 + 0x24	(TXS) Port Override Value Register 2
“TST_CFG_0” on page 401	0x2000 + j*0x100 + 0x28	(TXS) Test Mode Configuration Register 0
“TST_CFG_1” on page 403	0x2000 + j*0x100 + 0x2C	(TXS) Test Mode Configuration Register 1
“TST_CFG_2” on page 404	0x2000 + j*0x100 + 0x30	(TXS) Test Mode Configuration Register 2
“TST_CFG_3” on page 405	0x2000 + j*0x100 + 0x34	(TXS) Test Mode Configuration Register 3
“TST_CFG_4” on page 406	0x2000 + j*0x100 + 0x38	(TXS) Test Mode Configuration Register 4
“TST_CFG_5” on page 407	0x2000 + j*0x100 + 0x3C	(TXS) Test Mode Configuration Register 5
“TST_CFG_6” on page 408	0x2000 + j*0x100 + 0x40	(TXS) Test Mode Configuration Register 6
“TST_CFG_7” on page 409	0x2000 + j*0x100 + 0x44	(TXS) Test Mode Configuration Register 7
“TST_CFG_8” on page 410	0x2000 + j*0x100 + 0x48	(TXS) Test Mode Configuration Register 8
“TST_CFG_9” on page 412	0x2000 + j*0x100 + 0x4C	(TXS) Test Mode Configuration Register 9
“TST_CFG_10” on page 413	0x2000 + j*0x100 + 0x50	(TXS) Test Mode Configuration Register 10

Register	Offset	Description
"TST_CFG_11" on page 414	0x2000 + j*0x100 + 0x54	(TXS) Test Mode Configuration Register 11
"TST_CFG_12" on page 415	0x2000 + j*0x100 + 0x58	(TXS) Test Mode Configuration Register 12
"TST_CFG_13" on page 417	0x2000 + j*0x100 + 0x5C	(TXS) Test Mode Configuration Register 13
"TST_CFG_14" on page 418	0x2000 + j*0x100 + 0x60	(TXS) Test Mode Configuration Register 14
"TST_CFG_15" on page 419	0x2000 + j*0x100 + 0x64	(TXS) Test Mode Configuration Register 15
"ANA_OVRDEN_0" on page 421	0x2000 + j*0x100 + 0x68	(TXS) Internal Override Enable Register 0
"ANA_OVRDEN_1" on page 424	0x2000 + j*0x100 + 0x6C	(TXS) Internal Override Enable Register 1
"ANA_OVRDEN_2" on page 427	0x2000 + j*0x100 + 0x70	(TXS) Internal Override Enable Register 2
"ANA_OVRDEN_3" on page 430	0x2000 + j*0x100 + 0x74	(TXS) Internal Override Enable Register 3
"ANA_OVRDVAL_0" on page 431	0x2000 + j*0x100 + 0x78	(TXS) Internal Override Value Register 0
"ANA_OVRDVAL_1" on page 433	0x2000 + j*0x100 + 0x7C	(TXS) Internal Override Value Register 1
"ANA_OVRDVAL_2" on page 435	0x2000 + j*0x100 + 0x80	(TXS) Internal Override Value Register 2
"ANA_OVRDVAL_3" on page 437	0x2000 + j*0x100 + 0x84	(TXS) Internal Override Value Register 3

Register	Offset	Description
"ANA_OVRDVAL_4" on page 439	0x2000 + j*0x100 + 0x88	(TXS) Internal Override Value Register 4
"ANA_OVRDVAL_5" on page 443	0x2000 + j*0x100 + 0x8C	(TXS) Internal Override Value Register 5
"ANA_OVRDVAL_6" on page 447	0x2000 + j*0x100 + 0x90	(TXS) Internal Override Value Register 6
"ANA_OVRDVAL_7" on page 449	0x2000 + j*0x100 + 0x94	(TXS) Internal Override Value Register 7
"ANA_OVRDVAL_8" on page 450	0x2000 + j*0x100 + 0x98	(TXS) Internal Override Value Register 8
"ANA_OVRDVAL_9" on page 452	0x2000 + j*0x100 + 0x9C	(TXS) Internal Override Value Register 9
"ANA_OVRDVAL_10" on page 453	0x2000 + j*0x100 + 0xA0	(TXS) Internal Override Value Register 10
"TXS_STAT_0" on page 455	0x2000 + j*0x100 + 0xA4	(TXS) Status Register 0
"TXS_STAT_1" on page 456	0x2000 + j*0x100 + 0xA8	(TXS) Status Register 1
"TXS_STAT_2" on page 457	0x2000 + j*0x100 + 0xAC	(TXS) Status Register 2
"TXS_STAT_3" on page 458	0x2000 + j*0x100 + 0xB0	(TXS) Status Register 3
"TXS_STAT_4" on page 459	0x2000 + j*0x100 + 0xB4	(TXS) Status Register 4
"TXS_STAT_5" on page 460	0x2000 + j*0x100 + 0xB8	(TXS) Status Register 5

Register	Offset	Description
"TXS_STAT_6" on page 461	0x2000 + j*0x100 + 0xBC	(TXS) Status Register 6
"TXS_STAT_7" on page 462	0x2000 + j*0x100 + 0xC0	(TXS) Status Register 7
"TXS_STAT_8" on page 463	0x2000 + j*0x100 + 0xC4	(TXS) Status Register 8
"TXS_STAT_9" on page 464	0x2000 + j*0x100 + 0xC8	(TXS) Status Register 9
"TXS_STAT_10" on page 465	0x2000 + j*0x100 + 0xCC	(TXS) Status Register 10
Supplemental Transmitter Block Registers SUP_TX_SUP Exists: Always		
"SUP_CFG_0" on page 467	0x3000	(SUP) Configuration Register
"SUP_OVRDEN" on page 470	0x3004	(SUP) Internal Logic Override Enable Register
"SUP_OVRDVAL" on page 471	0x3008	(SUP) Internal Logic Override Value Register
"SUP_STAT" on page 472	0x300c	(SUP) Status Register
"ANA_OVRDVAL_0" on page 473	0x3010	OSC Configuration Register 0
"ANA_OVRDVAL_1" on page 475	0x3014	ANA Configuration Register 1
"ANA_OVRDVAL_2" on page 478	0x3018	CRU Access Register 2
"ANA_OVRDVAL_3" on page 480	0x301c	CRU Access Register 3
"ANA_OVRDVAL_4" on page 481	0x3020	DIFFBUF Configuration Register 4
"ANA_OVRDVAL_5" on page 483	0x3024	MPLL Analog Configuration Register 5
"ANA_OVRDVAL_6" on page 485	0x3028	MPLL Startup Configuration Register 6
"ANA_OVRDVAL_7" on page 487	0x302c	MPLL Dividers Configuration Register 7
"ANA_OVRDVAL_8" on page 489	0x3030	MPLL PMIX Configuration Register 8
"ANA_OVRDVAL_9" on page 491	0x3034	REF Configuration Register 9
"ANA_OVRDVAL_10" on page 494	0x3038	ROPLL Startup Configuration Register 10
"ANA_OVRDVAL_11" on page 497	0x303c	ROPLL Dividers Configuration Register 11

Register	Offset	Description
“ANA_OVRDVAL_12” on page 500	0x3040	RTRX Configuration Register 12
“ANA_OVRDEN_0” on page 502	0x3044	(SUP) Internal Override Enable Register 0
“ANA_OVRDEN_1” on page 505	0x3048	(SUP) Internal Override Enable Register 1
“ANA_OVRDEN_2” on page 508	0x304c	(SUP) Internal Override Enable Register 2
“ANA_OVRDEN_3” on page 511	0x3050	(SUP) Internal Override Enable Register 3
“ANA_OVRDVAL_13” on page 513	0x3054	SUP ANA Output Register 13
“ANA_OVRDVAL_14” on page 515	0x3058	SUP ANA Output Register 14
“SUP_CFG_1” on page 516	0x305c	(SUP) Configuration Register 1
“SUP_CFG_2” on page 518	0x3060	(SUP) Configuration Register 2
“SUP_CFG_3” on page 519	0x3064	(SUP) Configuration Register 3
“SUP_CFG_4” on page 520	0x3068	(SUP) Configuration Register 4
“SUP_CFG_5” on page 522	0x306c	(SUP) Configuration Register 5
“SUP_CFG_6” on page 524	0x3070	(SUP) Configuration Register 6
“SUP_CFG_7” on page 525	0x3074	(SUP) Configuration Register 7
“SUP_CFG_8” on page 526	0x3078	(SUP) Configuration Register 8
“SUP_CFG_9” on page 528	0x307c	(SUP) Configuration Register 9
“SUP_CFG_10” on page 529	0x3080	(SUP) Configuration Register 10
“SUP_CFG_11” on page 530	0x3084	(SUP) Configuration Register 11
“MPLL_STAT_0” on page 531	0x3088	MPLL Configuration Register 0
“MPLL_STAT_1” on page 532	0x308c	MPLL Configuration Register 1
“MPLL_STAT_2” on page 534	0x3090	MPLL Configuration Register 2
“MPLL_STAT_3” on page 535	0x3094	MPLL Configuration Register 3
“SUP_CFG_12” on page 536	0x3098	(SUP) Configuration Register 12
“SUP_CFG_13” on page 537	0x309c	(SUP) Configuration Register 13
“SUP_CFG_14” on page 539	0x30a0	(SUP) Configuration Register 14
“rt_stat_0” on page 540	0x30a4	rt_stat Configuration register 0
“rt_stat_1” on page 541	0x30a8	rt_stat Configuration register 1
“SUP_CFG_15” on page 542	0x30ac	(SUP) Configuration Register 15
“SUP_CFG_16” on page 543	0x30b0	(SUP) Configuration Register 16

Register	Offset	Description
“SUP_CFG_17” on page 544	0x30b4	(SUP) Configuration Register 17
“SUP_CFG_18” on page 545	0x30b8	(SUP) Configuration Register 18
“SUP_CFG_19” on page 546	0x30bc	(SUP) Configuration Register 19
“SUP_CFG_20” on page 547	0x30c0	(SUP) Configuration Register 20
“SUP_CFG_21” on page 548	0x30c4	(SUP) Configuration Register 21
Raw PCS Block Registers RAW_PCS_RAW_PCS Exists: Always		
“CPU_CFG” on page 550	0x4000	RAW PCS Configuration Register 0
“CPU_STAT” on page 552	0x4004	CPU Status Register
“raw_pcs_CFG” on page 553	0x4008	RAW PCS Configuration Register 1
“SUP_CFG” on page 554	0x400c	SUP Block Configuration Register
“FW_CFG_0” on page 555	0x4010	Firmware Configuration Register 0
“FW_CFG_1” on page 557	0x4014	Firmware Configuration Register 1
“LANES_STATUS” on page 558	0x4018	Lane Status Register
“JTAG_ID_CODE” on page 560	0x401c	JTAG ID Code Configuration Register
“PIPE_CONFIG_0” on page 561	0x4020	PIPE Interface Configuration Register 0
“PIPE_CONFIG_1” on page 562	0x4024	PIPE Interface Configuration Register 1
“PIPE_CONFIG_2” on page 563	0x4028	PIPE Interface Configuration Register 2
“EQ_CONFIG” on page 564	0x402c	Equalization Configuration Register
“SPARE_FW_0” on page 565	0x4030	Spare Register 0
“SPARE_FW_1” on page 566	0x4034	Spare Register 1

11.1 RX_RX_j Registers

11.1.1 ANA_OVRDVAL_0

- **Name:** (RXS) Internal Override Value Register 0
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 0.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x000

31	30	29	28	27:21	20:14	13:7	6:0
ana_rx_ana_pmix_data270_en_i	ana_rx_ana_pmix_data180_en_i	ana_rx_ana_pmix_data90_en_i	ana_rx_ana_pmix_data0_en_i	ana_rx_ana_pmix_data270_code_i	ana_rx_ana_pmix_data180_code_i	ana_rx_ana_pmix_data90_code_i	ana_rx_ana_pmix_data0_code_i

Table 11-5 Fields for Register: ANA_OVRDVAL_0

Bits	Name	Memory Access	Description
31	ana_rx_ana_pmix_data270_en_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_data270_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
30	ana_rx_ana_pmix_data180_en_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_data180_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
29	ana_rx_ana_pmix_data90_en_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_data90_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
28	ana_rx_ana_pmix_data0_en_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_data0_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
27:21	ana_rx_ana_pmix_data270_code_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_data270_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x70 **Volatile:** true
20:14	ana_rx_ana_pmix_data180_code_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_data180_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x50 **Volatile:** true
13:7	ana_rx_ana_pmix_data90_code_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_data90_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x30 **Volatile:** true
6:0	ana_rx_ana_pmix_data0_code_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_data0_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x10 **Volatile:** true

11.1.2 ANA_OVRDVAL_1

- **Name:** (RXS) Internal Override Value Register 1
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 1.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x004

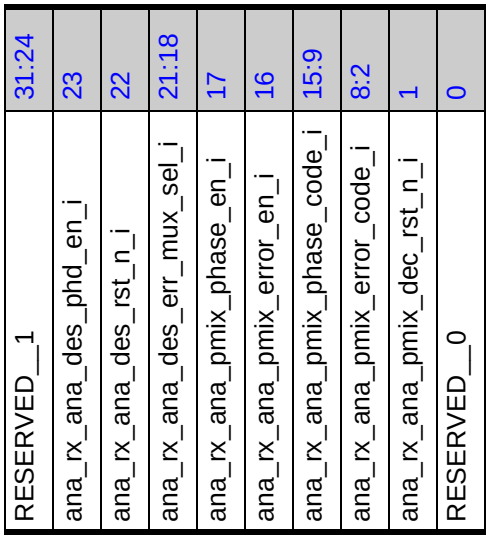


Table 11-6 Fields for Register: ANA_OVRDVAL_1

Bits	Name	Memory Access	Description
31:24	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
23	ana_rx_ana_des_phd_en_i	R/W	Override value set on (RXS) analog port rx_ana_des_phd_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
22	ana_rx_ana_des_rst_n_i	R/W	Override value set on (RXS) analog port rx_ana_des_rst_n. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
21:18	ana_rx_ana_des_err_mux_sel_i	R/W	Override value set on (RXS) analog port rx_ana_des_err_mux_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
17	ana_rx_ana_pmix_phase_en_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_phase_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_rx_ana_pmix_error_en_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_error_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:9	ana_rx_ana_pmix_phase_code_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_phase_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8:2	ana_rx_ana_pmix_error_code_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_error_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1	ana_rx_ana_pmix_dec_rst_n_i	R/W	Override value set on (RXS) analog port rx_ana_pmix_dec_rst_n. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.1.3 ANA_OVRDVAL_2

- **Name:** (RXS) Internal Override Value Register 2
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 2.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x008

31	30:19	18	17	16	15:14	13:12	11:6	5:0
RESERVED__0	ana_rx_ana_slc_en_i	ana_rx_ana_slc_phs_en_i	ana_rx_ana_slc_err_en_i	ana_rx_ana_cal_res_div_en_i	ana_rx_ana_cal_dac_vcm_i	ana_rx_ana_cal_dac_range_i	ana_rx_ana_cal_muxb_sel_i	ana_rx_ana_cal_muxa_sel_i

Table 11-7 Fields for Register: ANA_OVRDVAL_2

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:19	ana_rx_ana_slc_en_i	R/W	Override value set on (RXS) analog port rx_ana_slc_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
18	ana_rx_ana_slc_phs_en_i	R/W	Override value set on (RXS) analog port rx_ana_slc_phs_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
17	ana_rx_ana_slc_err_en_i	R/W	Override value set on (RXS) analog port rx_ana_slc_err_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_rx_ana_cal_res_div_en_i	R/W	Override value set on (RXS) analog port rx_ana_cal_res_div_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:14	ana_rx_ana_cal_dac_vcm_i	R/W	Override value set on (RXS) analog port rx_ana_cal_dac_vcm. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
13:12	ana_rx_ana_cal_dac_range_i	R/W	Override value set on (RXS) analog port rx_ana_cal_dac_range. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
11:6	ana_rx_ana_cal_muxb_sel_i	R/W	Override value set on (RXS) analog port rx_ana_cal_muxb_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5:0	ana_rx_ana_cal_muxa_sel_i	R/W	Override value set on (RXS) analog port rx_ana_cal_muxa_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.1.4 ANA_OVRDVAL_3

- **Name:** (RXS) Internal Override Value Register 3
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 3.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x00C

31:27	26	25	24:23	22	21:20	19	18	17:12	11	10:2	1	0
RESERVED__0	ana_rx_ana_dll_phd_early_late_o	ana_rx_ana_cal_result_o	ana_rx_ana_cal_dac_extra_range_i	ana_rx_ana_cal_dac_dec_rst_n_i	ana_rx_ana_cal_mode_i	ana_rx_ana_cal_lpfbyp_en_i	ana_rx_ana_cal_dac_en_i	ana_rx_ana_cal_dac_addr_i	ana_rx_ana_cal_dac_dec_clk_i	ana_rx_ana_cal_dac_code_i	ana_rx_ana_cal_comp_en_i	ana_rx_ana_cal_comp_clk_i

Table 11-8 Fields for Register: ANA_OVRDVAL_3

Bits	Name	Memory Access	Description
31:27	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
26	ana_rx_ana_dll_phd_early_late_o	R/W	Override value set on (RXS) analog port rx_ana_dll_phd_early_late. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_rx_ana_cal_result_o	R/W	Override value set on (RXS) analog port rx_ana_cal_result. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
24:23	ana_rx_ana_cal_dac_extra_range_i	R/W	Override value set on (RXS) analog portrx_ana_cal_dac_extra_range. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
22	ana_rx_ana_cal_dac_dec_rst_n_i	R/W	Override value set on (RXS) analog port rx_ana_cal_dac_dec_rst_n. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
21:20	ana_rx_ana_cal_mode_i	R/W	Override value set on (RXS) analog port rx_ana_cal_mode. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
19	ana_rx_ana_cal_lpfbyp_en_i	R/W	Override value set on (RXS) analog port rx_ana_cal_lpfbyp_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
18	ana_rx_ana_cal_dac_en_i	R/W	Override value set on (RXS) analog port rx_ana_cal_dac_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
17:12	ana_rx_ana_cal_dac_addr_i	R/W	Override value set on (RXS) analog port rx_ana_cal_dac_addr. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11	ana_rx_ana_cal_dac_dec_clk_i	R/W	Override value set on (RXS) analog port rx_ana_cal_dac_dec_clk. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:2	ana_rx_ana_cal_dac_code_i	R/W	Override value set on (RXS) analog port rx_ana_cal_dac_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x100 **Volatile:** true

Bits	Name	Memory Access	Description
1	ana_rx_ana_cal_comp_en_i	R/W	Override value set on (RXS) analog port rx_ana_cal_comp_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_rx_ana_cal_comp_clk_i	R/W	Override value set on (RXS) analog port rx_ana_cal_comp_clk. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.1.5 ANA_OVRDVAL_7

- **Name:** (RXS) Internal Override Value Register 7
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 7.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x010

31	30	29	28	27	26	25	24	23	22:19	18	17	16	15	14:13	12	11:9	8	7:6	5	4:3	2	1:0
ana_rx_ana_afe_vcm_opamp_startup_en_i	ana_rx_ana_afe_vcm_opamp_en_i	ana_rx_ana_afe_rst_n_i	ana_rx_ana_afe_lb_mux_vcm_sel_i	ana_rx_ana_afe_lb_mux_gd_sel_i	ana_rx_ana_afe_lb_mux_data_sel_i	ana_rx_ana_afe_ctle_offset_en_i	ana_rx_ana_afe_ctle_main_en_i	ana_rx_ana_afe_cp_en_i	ana_rx_ana_afe_att_rdc_i	ana_rx_ana_afe_att_offset_range_i	ana_rx_ana_afe_att_offset_en_i	ana_rx_ana_afe_ac_cap_lf_path_en_i	ana_rx_ana_ibias_ref_en_i	ana_rx_ana_ibias_ref_cont_i	ana_rx_ana_ibias_lb_en_i	ana_rx_ana_ibias_lb_cont_i	ana_rx_ana_ibias_ctle_en_i	ana_rx_ana_ibias_ctle_cont_i	ana_rx_ana_ibias_cal_en_i	ana_rx_ana_ibias_cal_cont_i	ana_rx_ana_ibias_att_en_i	ana_rx_ana_ibias_att_cont_i

Table 11-9 Fields for Register: ANA_OVRDVAL_7

Bits	Name	Memory Access	Description
31	ana_rx_ana_afe_vcm_opamp_startup_en_i	R/W	Override value set on (RXS) analog port rx_ana_afe_vcm_opamp_startup_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
30	ana_rx_ana_afe_vcm_opamp_en_i	R/W	Override value set on (RXS) analog port rx_ana_afe_vcm_opamp_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
29	ana_rx_ana_afe_rst_n_i	R/W	Override value set on (RXS) analog port rx_ana_afe_rst_n. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
28	ana_rx_ana_afe_lb_mux_vcm_sel_i	R/W	Override value set on (RXS) analog port rx_ana_afe_lb_mux_vcm_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
27	ana_rx_ana_afe_lb_mux_gd_sel_i	R/W	Override value set on (RXS) analog port rx_ana_afe_lb_mux_gd_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
26	ana_rx_ana_afe_lb_mux_data_sel_i	R/W	Override value set on (RXS) analog port rx_ana_afe_lb_mux_data_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_rx_ana_afe_ctle_offset_en_i	R/W	Override value set on (RXS) analog port rx_ana_afe_ctle_offset_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	ana_rx_ana_afe_ctle_main_en_i	R/W	Override value set on (RXS) analog port rx_ana_afe_ctle_main_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
23	ana_rx_ana_afe_cp_en_i	R/W	Override value set on (RXS) analog port rx_ana_afe_cp_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
22:19	ana_rx_ana_afe_att_rdc_i	R/W	Override value set on (RXS) analog port rx_ana_afe_att_rdc. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
18	ana_rx_ana_afe_att_offset_range_i	R/W	Override value set on (RXS) analog port rx_ana_afe_att_offset_range. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
17	ana_rx_ana_afe_att_offset_en_i	R/W	Override value set on (RXS) analog port rx_ana_afe_att_offset_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_rx_ana_afe_ac_cap_lf_path_en_i	R/W	Override value set on (RXS) analog port rx_ana_afe_ac_cap_lf_path_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15	ana_rx_ana_ibias_ref_en_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_ref_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14:13	ana_rx_ana_ibias_ref_cont_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_ref_cont. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
12	ana_rx_ana_ibias_lb_en_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_lb_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11:9	ana_rx_ana_ibias_lb_cont_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_lb_cont. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x6 **Volatile:** true
8	ana_rx_ana_ibias_ctle_en_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_ctle_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
7:6	ana_rx_ana_ibias_ctle_cont_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_ctle_cont. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
5	ana_rx_ana_ibias_cal_en_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_cal_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4:3	ana_rx_ana_ibias_cal_cont_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_cal_cont. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
2	ana_rx_ana_ibias_att_en_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_att_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1:0	ana_rx_ana_ibias_att_cont_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_att_cont. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true

11.1.6 ANA_OVRDVAL_8

- **Name:** (RXS) Internal Override Value Register 8
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 8.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x014

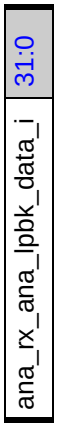


Table 11-10 Fields for Register: ANA_OVRDVAL_8

Bits	Name	Memory Access	Description
31:0	ana_rx_ana_lpbk_data_i	R/W	Override value set on (RXS) analog port rx_ana_lpbk_data. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.1.7 ANA_OVRDVAL_10

- **Name:** (RXS) Internal Override Value Register 10
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 10.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x018

31	30	29:28	27:26	25:19	18:16	15:0
ana_rx_ana_afe_term_ctrl_fix_vp_i	ana_rx_ana_afe_vcm_opamp_pdnoutlow_i	ana_rx_ana_afe_vcm_sel_i	ana_rx_ana_atb_meas_dll_pd_mux_i	ana_rx_ana_atb_meas_ibias_ref_i	ana_rx_ana_atb_meas_cal_mux_i	ana_rx_ana_atb_meas_i

Table 11-11 Fields for Register: ANA_OVRDVAL_10

Bits	Name	Memory Access	Description
31	ana_rx_ana_afe_term_ctrl_fix_vp_i	R/W	Override value set on (RXS) analog port rx_ana_term_ctrl_fix_vp. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
30	ana_rx_ana_afe_vcm_opamp_pdnoutlow_i	R/W	Override value set on (RXS) analog port rx_ana_afe_vcm_opamp_pdnoutlow. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
29:28	ana_rx_ana_afe_vcm_sel_i	R/W	Override value set on (RXS) analog port rx_ana_afe_vcm_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
27:26	ana_rx_ana_atb_meas_dll_pd_mux_i	R/W	Override value set on (RXS) analog port rx_ana_atb_meas_dll_pd_mux. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25:19	ana_rx_ana_atb_meas_ibias_ref_i	R/W	Override value set on (RXS) analog port rx_ana_atb_meas_ibias_ref. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
18:16	ana_rx_ana_atb_meas_cal_mux_i	R/W	Override value set on (RXS) analog port rx_ana_atb_meas_cal_mux. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:0	ana_rx_ana_atb_meas_i	R/W	Override value set on (RXS) analog port rx_ana_atb_meas. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.1.8 ANA_OVRDVAL_11

- **Name:** (RXS) Internal Override Value Register 11
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 11.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x01C

31	30	29	28	27:24	23	22:21	20:16	15	14:0
ana_rx_ana_dll_delay_clk_i	ana_rx_ana_dll_comp_dac_clk_i	ana_rx_ana_dll_vreg_dac_en_i	ana_rx_ana_dll_vreg_en_i	ana_rx_ana_afe_ctle_rtrim_i	ana_rx_ana_afe_ctle_lb_en_i	ana_rx_ana_afe_ctle_offset_range_i	ana_rx_ana_afe_ctle_boost_i	ana_rx_ana_afe_att_lvl_i	ana_rx_ana_afe_term_ctrl_i

Table 11-12 Fields for Register: ANA_OVRDVAL_11

Bits	Name	Memory Access	Description
31	ana_rx_ana_dll_delay_clk_i	R/W	Override value set on (RXS) analog port rx_ana_dll_delay_clk. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
30	ana_rx_ana_dll_comp_dac_clk_i	R/W	Override value set on (RXS) analog port rx_ana_dll_comp_dac_clk. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
29	ana_rx_ana_dll_vreg_dac_en_i	R/W	Override value set on (RXS) analog portrx_ana_dll_vreg_dac_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
28	ana_rx_ana_dll_vreg_en_i	R/W	Override value set on (RXS) analog port rx_ana_dll_vreg_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
27:24	ana_rx_ana_afe_ctle_rtrim_i	R/W	Override value set on (RXS) analog port rx_ana_afe_ctle_rtrim. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x8 **Volatile:** true
23	ana_rx_ana_afe_ctle_lb_en_i	R/W	Override value set on (RXS) analog port rx_ana_afe_ctle_lb_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
22:21	ana_rx_ana_afe_ctle_offset_range_i	R/W	Override value set on (RXS) analog port rx_ana_afe_ctle_offset_range. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
20:16	ana_rx_ana_afe_ctle_boost_i	R/W	Override value set on (RXS) analog port rx_ana_afe_ctle_boost. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15	ana_rx_ana_afe_att_lvl_i	R/W	Override value set on (RXS) analog port rx_ana_afe_att_lvl. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
14:0	ana_rx_ana_afe_term_ctrl_i	R/W	Override value set on (RXS) analog port rx_ana_afe_term_ctrl. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.1.9 ANA_OVRDVAL_13

- **Name:** (RXS) Internal Override Value Register 13
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 13.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x020

31	30:28	27	26:25	24:16	15:14	13	12:10	9:0
ana_rx_ana_dll_delay_rst_n_i	ana_rx_ana_dll_delay_leg_en_i	ana_rx_ana_dll_dac_dec_rst_n_i	ana_rx_ana_dll_comp_dac_range_i	ana_rx_ana_dll_comp_dac_code_i	ana_rx_ana_dll_comp_dac_vcm_i	ana_rx_ana_dll_comp_dac_en_i	ana_rx_ana_dll_delay_fine_en_i	ana_rx_ana_dll_addr_i

Table 11-13 Fields for Register: ANA_OVRDVAL_13

Bits	Name	Memory Access	Description
31	ana_rx_ana_dll_delay_rst_n_i	R/W	Override value set on (RXS) analog port rx_ana_delay_rst_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
30:28	ana_rx_ana_dll_delay_leg_en_i	R/W	Override value set on (RXS) analog port rx_ana_delay_leg_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
27	ana_rx_ana_dll_dac_dec_rst_n_i	R/W	Override value set on (RXS) analog port rx_ana_dll_dac_dec_rst_n. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
26:25	ana_rx_ana_dll_comp_dac_range_i	R/W	Override value set on (RXS) analog port rx_ana_dll_comp_dac_range. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
24:16	ana_rx_ana_dll_comp_dac_code_i	R/W	Override value set on (RXS) analog port rx_ana_dll_comp_dac_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x100 **Volatile:** true
15:14	ana_rx_ana_dll_comp_dac_vcm_i	R/W	Override value set on (RXS) analog port rx_ana_dll_comp_dac_vcm. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
13	ana_rx_ana_dll_comp_dac_en_i	R/W	Override value set on (RXS) analog port rx_ana_dll_comp_dac_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
12:10	ana_rx_ana_dll_delay_fine_en_i	R/W	Override value set on (RXS) analog port rx_ana_dll_delay_fine_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9:0	ana_rx_ana_dll_addr_i	R/W	Override value set on (RXS) analog port rx_ana_dll_addr. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x3ff **Volatile:** true

11.1.10 ANA_OVRDVAL_14

- **Name:** (RXS) Internal Override Value Register 14
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 14.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x024

31	30	29:27	26:25	24:16	15	14:10	9:5	4	3:0
RESERVED__0	ana_rx_ana_dll_vreg_byp_i	ana_rx_ana_dll_vreg_dac_coarse_i	ana_rx_ana_dll_vreg_dac_range_i	ana_rx_ana_dll_vreg_dac_code_i	ana_rx_ana_dll_phd_en_i	ana_rx_ana_dll_phd_dig_sel_i	ana_rx_ana_dll_phd_ana_sel_i	ana_rx_ana_dll_ac_en_i	ana_rx_ana_dll_dcc_en_i

Table 11-14 Fields for Register: ANA_OVRDVAL_14

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	ana_rx_ana_dll_vreg_byp_i	R/W	Override value set on (RXS) analog port rx_ana_dll_vreg_byp. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
29:27	ana_rx_ana_dll_vreg_dac_coarse_i	R/W	Override value set on (RXS) analog port rx_ana_dll_vreg_dac_coarse. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
26:25	ana_rx_ana_dll_vreg_dac_range_i	R/W	Override value set on (RXS) analog port rx_ana_dll_vreg_dac_range. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
24:16	ana_rx_ana_dll_vreg_dac_code_i	R/W	Override value set on (RXS) analog port rx_ana_dll_vreg_dac_code. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15	ana_rx_ana_dll_phd_en_i	R/W	Override value set on (RXS) analog port rx_ana_dll_phd_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14:10	ana_rx_ana_dll_phd_dig_sel_i	R/W	Override value set on (RXS) analog port rx_ana_dll_phd_dig_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9:5	ana_rx_ana_dll_phd_ana_sel_i	R/W	Override value set on (RXS) analog port rx_ana_dll_phd_ana_sel. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4	ana_rx_ana_dll_ac_en_i	R/W	Override value set on (RXS) analog port rx_ana_dll_ac_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3:0	ana_rx_ana_dll_dcc_en_i	R/W	Override value set on (RXS) analog port rx_ana_dll_dcc_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.1.11 ANA_OVRDVAL_16

- **Name:** (RXS) Internal Override Value Register 16
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 16.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x028



Table 11-15 Fields for Register: ANA_OVRDVAL_16

Bits	Name	Memory Access	Description
31:0	ana_rx_ana_spare_in_i	R/W	Override value set on (RXS) analog port rx_ana_spare_in. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.1.12 ANA_OVRDVAL_17

- **Name:** (RXS) Internal Override Value Register 17
- **Description:** Register bits act as override value at the mux which can override values on (RXS) analog-digital interface ports. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 17.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x02C

31:15	14	13	12	11	10:9	8	7	6	5	4	3:1	0
RESERVED__2	ana_rx_ana_cal_bias_en_i	ana_rx_ana_lpbk_xor_en_i	ana_rx_ana_lpbk_drv_en_i	ana_rx_ana_ibias_xor_en_i	ana_rx_ana_ibias_xor_cont_i	ana_rx_ana_lpbk_ser_rst_n_i	ana_rx_ana_lpbk_ser_data_en_i	RESERVED__1	ana_rx_ana_atb_s_en_i	ana_rx_ana_afe_dig_clk_i	RESERVED__0	ana_ltoh_pwrdn_h_i

Table 11-16 Fields for Register: ANA_OVRDVAL_17

Bits	Name	Memory Access	Description
31:15	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
14	ana_rx_ana_cal_bias_en_i	R/W	Override value set on (RXS) analog port rx_ana_cal_bias_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
13	ana_rx_ana_lpbk_xor_en_i	R/W	Override value set on (RXS) analog port rx_ana_lpbk_xor_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
12	ana_rx_ana_lpbk_drv_en_i	R/W	Override value set on (RXS) analog port rx_ana_lpbk_drv_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11	ana_rx_ana_ibias_xor_en_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_xor_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:9	ana_rx_ana_ibias_xor_cont_i	R/W	Override value set on (RXS) analog port rx_ana_ibias_xor_cont. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
8	ana_rx_ana_lpbk_ser_rst_n_i	R/W	Override value set on (RXS) analog port rx_ana_lpbk_ser_rst_n. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7	ana_rx_ana_lpbk_ser_data_en_i	R/W	Override value set on (RXS) analog port rx_ana_lpbk_ser_data_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
6	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
5	ana_rx_ana_atb_s_en_i	R/W	Override value set on (RXS) analog port rx_ana_atb_s_en. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4	ana_rx_ana_afe_dig_clk_i	R/W	Override value set on (RXS) analog port rx_ana_afe_dig_clk. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
3:1	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
0	ana_ltoh_pwrdn_h_i	R/W	Override value set on (RXS) analog port ltoh_pwrdn_h. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true

11.1.13 ANA_OVRDEN_0

- **Name:** (RXS) Internal Override Enable Register 0
- **Description:** Register bits act as select signal of the mux which overrides value on signals and internal analog-digital interface in (RXS). Register 0.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x030

31:30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED__1	RESERVED__0	ovrd_en_ana_rx_ana_cal_res_div_en_i	ovrd_en_ana_rx_ana_cal_dac_vcm_i	ovrd_en_ana_rx_ana_cal_dac_extra_range_i	ovrd_en_ana_rx_ana_cal_dac_dec_rst_n_i	ovrd_en_ana_rx_ana_cal_result_o	ovrd_en_ana_rx_ana_cal_mode_i	ovrd_en_ana_rx_ana_cal_lpfbyp_en_i	ovrd_en_ana_rx_ana_cal_dac_en_i	ovrd_en_ana_rx_ana_cal_dac_addr_i	ovrd_en_ana_rx_ana_cal_dac_dec_clk_i	ovrd_en_ana_rx_ana_cal_dac_code_i	ovrd_en_ana_rx_ana_cal_comp_en_i	ovrd_en_ana_rx_ana_cal_comp_clk_i	ovrd_en_ana_rx_ana_cal_dac_range_i	ovrd_en_ana_rx_ana_cal_muxb_sel_i	ovrd_en_ana_rx_ana_cal_muxa_sel_i	ovrd_en_ana_rx_ana_pmix_phase_en_i	ovrd_en_ana_rx_ana_pmix_error_en_i	ovrd_en_ana_rx_ana_pmix_phase_code_i	ovrd_en_ana_rx_ana_pmix_error_code_i	ovrd_en_ana_rx_ana_pmix_dec_rst_n_i	ovrd_en_ana_rx_ana_pmix_data270_en_i	ovrd_en_ana_rx_ana_pmix_data180_en_i	ovrd_en_ana_rx_ana_pmix_data90_en_i	ovrd_en_ana_rx_ana_pmix_data0_en_i	ovrd_en_ana_rx_ana_pmix_data270_code_i	ovrd_en_ana_rx_ana_pmix_data180_code_i	ovrd_en_ana_rx_ana_pmix_data90_code_i	ovrd_en_ana_rx_ana_pmix_data0_code_i

Table 11-17 Fields for Register: ANA_OVRDEN_0

Bits	Name	Memory Access	Description
31:30	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	ovrd_en_ana_rx_ana_cal_res_div_en_i	R/W	Override enable for (RXS) port rx_ana_cal_res_div_en_i. **Value After Reset:** 0x1
27	ovrd_en_ana_rx_ana_cal_dac_vcm_i	R/W	Override enable for (RXS) port rx_ana_cal_dac_vcm_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
26	ovrd_en_ana_rx_ana_cal_dac_extra_range_i	R/W	Override enable for (RXS) port rx_ana_cal_dac_extra_range_i. **Value After Reset:** 0x1
25	ovrd_en_ana_rx_ana_cal_dac_dec_rst_n_i	R/W	Override enable for (RXS) port rx_ana_cal_dac_dec_rst_n_i. **Value After Reset:** 0x1
24	ovrd_en_ana_rx_ana_cal_result_o	R/W	Override enable for (RXS) port rx_ana_cal_result_o. **Value After Reset:** 0x0
23	ovrd_en_ana_rx_ana_cal_mode_i	R/W	Override enable for (RXS) port rx_ana_cal_mode_i. **Value After Reset:** 0x1
22	ovrd_en_ana_rx_ana_cal_lpfbyp_en_i	R/W	Override enable for (RXS) port rx_ana_cal_lpfbyp_en_i. **Value After Reset:** 0x1
21	ovrd_en_ana_rx_ana_cal_dac_en_i	R/W	Override enable for (RXS) port rx_ana_cal_dac_en_i. **Value After Reset:** 0x1
20	ovrd_en_ana_rx_ana_cal_dac_addr_i	R/W	Override enable for (RXS) port rx_ana_cal_dac_addr_i. **Value After Reset:** 0x1
19	ovrd_en_ana_rx_ana_cal_dac_dec_clk_i	R/W	Override enable for (RXS) port rx_ana_cal_dac_dec_clk_i. **Value After Reset:** 0x0
18	ovrd_en_ana_rx_ana_cal_dac_code_i	R/W	Override enable for (RXS) port rx_ana_cal_dac_code_i. **Value After Reset:** 0x1
17	ovrd_en_ana_rx_ana_cal_comp_en_i	R/W	Override enable for (RXS) port rx_ana_cal_comp_en_i. **Value After Reset:** 0x1
16	ovrd_en_ana_rx_ana_cal_comp_clk_i	R/W	Override enable for (RXS) port rx_ana_cal_comp_clk_i. **Value After Reset:** 0x0
15	ovrd_en_ana_rx_ana_cal_dac_range_i	R/W	Override enable for (RXS) port rx_ana_cal_dac_range_i. **Value After Reset:** 0x1
14	ovrd_en_ana_rx_ana_cal_muxb_sel_i	R/W	Override enable for (RXS) port rx_ana_cal_muxb_sel_i. **Value After Reset:** 0x1
13	ovrd_en_ana_rx_ana_cal_muxa_sel_i	R/W	Override enable for (RXS) port rx_ana_cal_muxa_sel_i. **Value After Reset:** 0x1
12	ovrd_en_ana_rx_ana_pmix_phase_en_i	R/W	Override enable for (RXS) port rx_ana_pmix_phase_en_i. **Value After Reset:** 0x1
11	ovrd_en_ana_rx_ana_pmix_error_en_i	R/W	Override enable for (RXS) port rx_ana_pmix_error_en_i. **Value After Reset:** 0x1
10	ovrd_en_ana_rx_ana_pmix_phase_code_i	R/W	Override enable for (RXS) port rx_ana_pmix_phase_code_i. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
9	ovrd_en_ana_rx_ana_pmix_error_code_i	R/W	Override enable for (RXS) port rx_ana_pmix_error_code_i. **Value After Reset:** 0x1
8	ovrd_en_ana_rx_ana_pmix_dec_rst_n_i	R/W	Override enable for (RXS) port rx_ana_pmix_dec_rst_n_i. **Value After Reset:** 0x1
7	ovrd_en_ana_rx_ana_pmix_data270_en_i	R/W	Override enable for (RXS) port rx_ana_pmix_data270_en_i. **Value After Reset:** 0x1
6	ovrd_en_ana_rx_ana_pmix_data180_en_i	R/W	Override enable for (RXS) port rx_ana_pmix_data180_en_i. **Value After Reset:** 0x1
5	ovrd_en_ana_rx_ana_pmix_data90_en_i	R/W	Override enable for (RXS) port rx_ana_pmix_data90_en_i. **Value After Reset:** 0x1
4	ovrd_en_ana_rx_ana_pmix_data0_en_i	R/W	Override enable for (RXS) port rx_ana_pmix_data0_en_i. **Value After Reset:** 0x1
3	ovrd_en_ana_rx_ana_pmix_data270_code_i	R/W	Override enable for (RXS) port rx_ana_pmix_data270_code_i. **Value After Reset:** 0x0
2	ovrd_en_ana_rx_ana_pmix_data180_code_i	R/W	Override enable for (RXS) port rx_ana_pmix_data180_code_i. **Value After Reset:** 0x0
1	ovrd_en_ana_rx_ana_pmix_data90_code_i	R/W	Override enable for (RXS) port rx_ana_pmix_data90_code_i. **Value After Reset:** 0x0
0	ovrd_en_ana_rx_ana_pmix_data0_code_i	R/W	Override enable for (RXS) port rx_ana_pmix_data0_code_i. **Value After Reset:** 0x0

11.1.14 ANA_OVRDEN_1

- **Name:** (RXS) Internal Override Enable Register 1
- **Description:** Register bits act as select signal of the mux which overrides value on signals and internal analog-digital interface in (RXS). Register 1.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x034

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED__0	ovrd_en_ana_rx_ana_cal_bias_en_i	ovrd_en_ana_rx_ana_afe_vcm_opamp_startup_en_i	ovrd_en_ana_rx_ana_afe_vcm_opamp_en_i	ovrd_en_ana_rx_ana_afe_rst_n_i	ovrd_en_ana_rx_ana_afe_lb_mux_vcm_sel_i	ovrd_en_ana_rx_ana_afe_lb_mux_gd_sel_i	ovrd_en_ana_rx_ana_afe_lb_mux_data_sel_i	ovrd_en_ana_rx_ana_afe_ctle_offset_en_i	ovrd_en_ana_rx_ana_lpbk_data_i	ovrd_en_ana_rx_ana_in_ac_en_i	ovrd_en_ana_rx_ana_afe_ctle_main_en_i	ovrd_en_ana_rx_ana_afe_cp_en_i	ovrd_en_ana_rx_ana_afe_att_rdc_i	ovrd_en_ana_rx_ana_afe_att_offset_range_i	ovrd_en_ana_rx_ana_afe_att_offset_en_i	ovrd_en_ana_rx_ana_afe_ac_cap_lf_path_en_i	ovrd_en_ana_rx_ana_ibias_ref_en_i	ovrd_en_ana_rx_ana_ibias_ref_cont_i	ovrd_en_ana_rx_ana_ibias_lb_en_i	ovrd_en_ana_rx_ana_ibias_lb_cont_i	ovrd_en_ana_rx_ana_ibias_ctle_en_i	ovrd_en_ana_rx_ana_ibias_ctle_cont_i	ovrd_en_ana_rx_ana_ibias_cal_en_i	ovrd_en_ana_rx_ana_ibias_cal_cont_i	ovrd_en_ana_rx_ana_ibias_att_en_i	ovrd_en_ana_rx_ana_ibias_att_cont_i	ovrd_en_ana_rx_ana_afe_term_ctrl_fix_vp_i	ovrd_en_ana_rx_ana_afe_vcm_opamp_pdnoutlow_i	ovrd_en_ana_rx_ana_des_phd_en_i	ovrd_en_ana_rx_ana_des_err_mux_sel_i	ovrd_en_ana_rx_ana_des_rst_n_i

Table 11-18 Fields for Register: ANA_OVRDEN_1

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	ovrd_en_ana_rx_ana_cal_bias_en_i	R/W	Override enable for (RXS) port rx_ana_cal_bias_en_i. **Value After Reset:** 0x1
29	ovrd_en_ana_rx_ana_afe_vcm_opamp_startup_en_i	R/W	Override enable for (RXS) port rx_ana_afe_vcm_opamp_startup_en_i. **Value After Reset:** 0x1
28	ovrd_en_ana_rx_ana_afe_vcm_opamp_en_i	R/W	Override enable for (RXS) port rx_ana_afe_vcm_opamp_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
27	ovrd_en_ana_rx_ana_afe_rst_n_i	R/W	Override enable for (RXS) port rx_ana_afe_rst_n_i. **Value After Reset:** 0x1
26	ovrd_en_ana_rx_ana_afe_lb_mux_vcm_sel_i	R/W	Override enable for (RXS) port rx_ana_afe_lb_mux_vcm_sel_i. **Value After Reset:** 0x1
25	ovrd_en_ana_rx_ana_afe_lb_mux_gd_sel_i	R/W	Override enable for (RXS) port rx_ana_afe_lb_mux_gd_sel_i. **Value After Reset:** 0x1
24	ovrd_en_ana_rx_ana_afe_lb_mux_data_sel_i	R/W	Override enable for (RXS) port rx_ana_afe_lb_mux_data_sel_i. **Value After Reset:** 0x1
23	ovrd_en_ana_rx_ana_afe_ctle_offset_en_i	R/W	Override enable for (RXS) port rx_ana_afe_ctle_offset_en_i. **Value After Reset:** 0x1
22	ovrd_en_ana_rx_ana_lpbk_data_i	R/W	Override enable for (RXS) port rx_ana_lpbk_data_i. **Value After Reset:** 0x0
21	ovrd_en_ana_rx_ana_in_ac_en_i	R/W	Override enable for (RXS) port rx_ana_in_ac_en_i. **Value After Reset:** 0x1
20	ovrd_en_ana_rx_ana_afe_ctle_main_en_i	R/W	Override enable for (RXS) port rx_ana_afe_ctle_main_en_i. **Value After Reset:** 0x1
19	ovrd_en_ana_rx_ana_afe_cp_en_i	R/W	Override enable for (RXS) port rx_ana_afe_cp_en_i. **Value After Reset:** 0x1
18	ovrd_en_ana_rx_ana_afe_att_rdc_i	R/W	Override enable for (RXS) port rx_ana_afe_att_rdc_i. **Value After Reset:** 0x1
17	ovrd_en_ana_rx_ana_afe_att_offset_range_i	R/W	Override enable for (RXS) port rx_ana_afe_att_offset_range_i. **Value After Reset:** 0x1
16	ovrd_en_ana_rx_ana_afe_att_offset_en_i	R/W	Override enable for (RXS) port rx_ana_afe_att_offset_en_i. **Value After Reset:** 0x1
15	ovrd_en_ana_rx_ana_afe_ac_cap_lf_path_en_i	R/W	Override enable for (RXS) port rx_ana_afe_ac_cap_lf_path_en_i. **Value After Reset:** 0x1
14	ovrd_en_ana_rx_ana_ibias_ref_en_i	R/W	Override enable for (RXS) port rx_ana_ibias_ref_en_i. **Value After Reset:** 0x1
13	ovrd_en_ana_rx_ana_ibias_ref_cont_i	R/W	Override enable for (RXS) port rx_ana_ibias_ref_cont_i. **Value After Reset:** 0x1
12	ovrd_en_ana_rx_ana_ibias_lb_en_i	R/W	Override enable for (RXS) port rx_ana_ibias_lb_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
11	ovrd_en_ana_rx_ana_ibias_lb_cont_i	R/W	Override enable for (RXS) port rx_ana_ibias_lb_cont_i. **Value After Reset:** 0x1
10	ovrd_en_ana_rx_ana_ibias_ctle_en_i	R/W	Override enable for (RXS) port rx_ana_ibias_ctle_en_i. **Value After Reset:** 0x1
9	ovrd_en_ana_rx_ana_ibias_ctle_cont_i	R/W	Override enable for (RXS) port rx_ana_ibias_ctle_cont_i. **Value After Reset:** 0x1
8	ovrd_en_ana_rx_ana_ibias_cal_en_i	R/W	Override enable for (RXS) port rx_ana_ibias_cal_en_i. **Value After Reset:** 0x1
7	ovrd_en_ana_rx_ana_ibias_cal_cont_i	R/W	Override enable for (RXS) port rx_ana_ibias_cal_cont_i. **Value After Reset:** 0x1
6	ovrd_en_ana_rx_ana_ibias_att_en_i	R/W	Override enable for (RXS) port rx_ana_ibias_att_en_i. **Value After Reset:** 0x1
5	ovrd_en_ana_rx_ana_ibias_att_cont_i	R/W	Override enable for (RXS) port rx_ana_ibias_att_cont_i. **Value After Reset:** 0x1
4	ovrd_en_ana_rx_ana_afe_term_ctrl_fix_vp_i	R/W	Override enable for (RXS) port rx_ana_afe_term_ctrl_fix_vp_i. **Value After Reset:** 0x1
3	ovrd_en_ana_rx_ana_afe_vcm_opamp_pdnoutlow_i	R/W	Override enable for (RXS) port rx_ana_afe_vcm_opamp_pdnoutlow_i. **Value After Reset:** 0x1
2	ovrd_en_ana_rx_ana_des_phd_en_i	R/W	Override enable for (RXS) port rx_ana_des_phd_en_i. **Value After Reset:** 0x1
1	ovrd_en_ana_rx_ana_des_err_mux_sel_i	R/W	Override enable for (RXS) port rx_ana_des_err_mux_sel_i. **Value After Reset:** 0x1
0	ovrd_en_ana_rx_ana_des_rst_n_i	R/W	Override enable for (RXS) port rx_ana_des_rst_n_i. **Value After Reset:** 0x1

11.1.15 ANA_OVRDEN_2

- **Name:** (RXS) Internal Override Enable Register 2
- **Description:** Register bits act as select signal of the mux which overrides value on signals and internal analog-digital interface in (RXS). Register 2.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x038

31:23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED__4	ovrd_en_ana_rx_ana_dll_phd_early_late_o	ovrd_en_ana_rx_ana_dll_vreg_dac_range_i	ovrd_en_ana_rx_ana_dll_vreg_dac_code_i	ovrd_en_ana_rx_ana_lpbk_xor_en_i	ovrd_en_ana_rx_ana_lpbk_drv_en_i	ovrd_en_ana_rx_ana_ibias_xor_en_i	ovrd_en_ana_rx_ana_ibias_xor_cont_i	ovrd_en_ana_rx_ana_afe_term_ctrl_i	ovrd_en_ana_rx_ana_lpbk_ser_data_en_i	ovrd_en_ana_rx_ana_afe_ctle_rtrim_i	RESERVED__3	ovrd_en_ana_rx_ana_afe_ctle_lb_en_i	ovrd_en_ana_rx_ana_afe_ctle_offset_range_i	ovrd_en_ana_rx_ana_afe_ctle_boost_i	ovrd_en_ana_rx_ana_afe_vcm_sel_i	RESERVED__2	ovrd_en_ana_rx_ana_afe_att_lvl_i	RESERVED__1	ovrd_en_ana_rx_ana_atb_meas_dll_pd_mux_i	RESERVED__0	ovrd_en_ana_rx_ana_atb_meas_ibias_ref_i	ovrd_en_ana_rx_ana_atb_meas_cal_mux_i	ovrd_en_ana_rx_ana_atb_meas_i

Table 11-19 Fields for Register: ANA_OVRDEN_2

Bits	Name	Memory Access	Description
31:23	RESERVED__4	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
22	ovrd_en_ana_rx_ana_dll_phd_early_late_o	R/W	Override enable for (RXS) port rx_ana_dll_phd_early_late_o. **Value After Reset:** 0x1
21	ovrd_en_ana_rx_ana_dll_vreg_dac_range_i	R/W	Override enable for (RXS) port rx_ana_dll_vreg_dac_range_i. **Value After Reset:** 0x1
20	ovrd_en_ana_rx_ana_dll_vreg_dac_code_i	R/W	Override enable for (RXS) port rx_ana_dll_vreg_dac_code_i. **Value After Reset:** 0x0
19	ovrd_en_ana_rx_ana_lpbk_xor_en_i	R/W	Override enable for (RXS) port rx_ana_lpbk_xor_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
18	ovrd_en_ana_rx_ana_lpbk_drv_en_i	R/W	Override enable for (RXS) port rx_ana_lpbk_drv_en_i. **Value After Reset:** 0x1
17	ovrd_en_ana_rx_ana_ibias_xor_en_i	R/W	Override enable for (RXS) port rx_ana_ibias_xor_en_i. **Value After Reset:** 0x1
16	ovrd_en_ana_rx_ana_ibias_xor_cont_i	R/W	Override enable for (RXS) port rx_ana_ibias_xor_cont_i. **Value After Reset:** 0x1
15	ovrd_en_ana_rx_ana_afe_term_ctrl_i	R/W	Override enable for (RXS) port rx_ana_afe_term_ctrl_i. **Value After Reset:** 0x1
14	ovrd_en_ana_rx_ana_lpbk_ser_data_en_i	R/W	Override enable for (RXS) port rx_ana_lpbk_ser_data_en_i. **Value After Reset:** 0x1
13	ovrd_en_ana_rx_ana_afe_ctle_rtrim_i	R/W	Override enable for (RXS) port rx_ana_afe_ctle_rtrim_i. **Value After Reset:** 0x1
12	RESERVED__3	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
11	ovrd_en_ana_rx_ana_afe_ctle_lb_en_i	R/W	Override enable for (RXS) port rx_ana_afe_ctle_lb_en_i. **Value After Reset:** 0x1
10	ovrd_en_ana_rx_ana_afe_ctle_offset_range_i	R/W	Override enable for (RXS) port rx_ana_afe_ctle_offset_range_i. **Value After Reset:** 0x1
9	ovrd_en_ana_rx_ana_afe_ctle_boost_i	R/W	Override enable for (RXS) port rx_ana_afe_ctle_boost_i. **Value After Reset:** 0x1
8	ovrd_en_ana_rx_ana_afe_vcm_sel_i	R/W	Override enable for (RXS) port rx_ana_afe_vcm_sel_i. **Value After Reset:** 0x1
7	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
6	ovrd_en_ana_rx_ana_afe_att_lvl_i	R/W	Override enable for (RXS) port rx_ana_afe_att_lvl_i. **Value After Reset:** 0x1
5	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	ovrd_en_ana_rx_ana_atb_meas_dll_pd_mux_i	R/W	Override enable for (RXS) port rx_ana_atb_meas_dll_pd_mux_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
3	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
2	ovrd_en_ana_rx_ana_atb_meas_ibias_ref_i	R/W	Override enable for (RXS) port rx_ana_atb_meas_ibias_ref_i. **Value After Reset:** 0x1
1	ovrd_en_ana_rx_ana_atb_meas_cal_mux_i	R/W	Override enable for (RXS) port rx_ana_atb_meas_cal_mux_i. **Value After Reset:** 0x1
0	ovrd_en_ana_rx_ana_atb_meas_i	R/W	Override enable for (RXS) port rx_ana_atb_meas_i. **Value After Reset:** 0x1

11.1.16 ANA_OVRDEN_3

- **Name:** (RXS) Internal Override Enable Register 3
- **Description:** Register bits act as select signal of the mux which overrides value on signals and internal analog-digital interface in (RXS). Register 3.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x03C

31:28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED__0	ovrd_en_ana_rx_ana_dll_delay_clk_i	ovrd_en_ana_rx_ana_dll_comp_dac_clk_i	ovrd_en_ana_rx_ana_dll_delay_rst_n_i	ovrd_en_ana_rx_ana_dll_delay_leg_en_i	ovrd_en_ana_rx_ana_slc_en_i	ovrd_en_ana_rx_ana_slc_phs_en_i	ovrd_en_ana_rx_ana_slc_err_en_i	ovrd_en_ana_rx_ana_lpbk_ser_rst_n_i	ovrd_en_ana_rx_ana_dll_vreg_dac_en_i	ovrd_en_ana_rx_ana_spare_in_i	ovrd_en_ana_rx_ana_dll_vreg_byp_i	ovrd_en_ana_rx_ana_dll_vreg_dac_coarse_i	ovrd_en_ana_rx_ana_dll_vreg_en_i	ovrd_en_ana_rx_ana_dll_phd_en_i	ovrd_en_ana_rx_ana_dll_phd_dig_sel_i	ovrd_en_ana_rx_ana_dll_phd_ana_sel_i	ovrd_en_ana_rx_ana_dll_ac_en_i	ovrd_en_ana_rx_ana_dll_dcc_en_i	ovrd_en_ana_rx_ana_dll_dac_dec_rst_n_i	ovrd_en_ana_rx_ana_dll_comp_dac_range_i	ovrd_en_ana_rx_ana_dll_comp_dac_code_i	ovrd_en_ana_rx_ana_dll_delay_fine_en_i	ovrd_en_ana_rx_ana_dll_comp_dac_en_i	ovrd_en_ana_rx_ana_dll_comp_dac_vcm_i	ovrd_en_ana_rx_ana_dll_addr_i	ovrd_en_ana_rx_ana_atb_s_en_i	ovrd_en_ana_rx_ana_afe_dig_clk_i	ovrd_en_ana_ltoh_pwrdn_h_i

Table 11-20 Fields for Register: ANA_OVRDEN_3

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27	ovrd_en_ana_rx_ana_dll_delay_clk_i	R/W	Override enable for (RXS) port rx_ana_dll_delay_clk_i. **Value After Reset:** 0x1
26	ovrd_en_ana_rx_ana_dll_comp_dac_clk_i	R/W	Override enable for (RXS) port rx_ana_dll_comp_dac_clk_i. **Value After Reset:** 0x1
25	ovrd_en_ana_rx_ana_dll_delay_rst_n_i	R/W	Override enable for (RXS) port rx_ana_dll_delay_rst_n_i. **Value After Reset:** 0x1
24	ovrd_en_ana_rx_ana_dll_delay_leg_en_i	R/W	Override enable for (RXS) port rx_ana_dll_delay_leg_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
23	ovrd_en_ana_rx_ana_slc_en_i	R/W	Override enable for (RXS) port rx_ana_slc_en_i. **Value After Reset:** 0x1
22	ovrd_en_ana_rx_ana_slc_phs_en_i	R/W	Override enable for (RXS) port rx_ana_slc_phs_en_i. **Value After Reset:** 0x1
21	ovrd_en_ana_rx_ana_slc_err_en_i	R/W	Override enable for (RXS) port rx_ana_slc_err_en_i. **Value After Reset:** 0x1
20	ovrd_en_ana_rx_ana_lpbk_ser_rst_n_i	R/W	Override enable for (RXS) port rx_ana_lpbk_ser_rst_n_i. **Value After Reset:** 0x1
19	ovrd_en_ana_rx_ana_dll_vreg_dac_en_i	R/W	Override enable for (RXS) port rx_ana_dll_vreg_dac_en_i. **Value After Reset:** 0x1
18	ovrd_en_ana_rx_ana_spare_in_i	R/W	Override enable for (RXS) port rx_ana_spare_in_i. **Value After Reset:** 0x1
17	ovrd_en_ana_rx_ana_dll_vreg_byp_i	R/W	Override enable for (RXS) port rx_ana_dll_vreg_byp_i. **Value After Reset:** 0x1
16	ovrd_en_ana_rx_ana_dll_vreg_dac_coarse_i	R/W	Override enable for (RXS) port rx_ana_dll_vreg_dac_coarse_i. **Value After Reset:** 0x1
15	ovrd_en_ana_rx_ana_dll_vreg_en_i	R/W	Override enable for (RXS) port rx_ana_dll_vreg_en_i. **Value After Reset:** 0x1
14	ovrd_en_ana_rx_ana_dll_phd_en_i	R/W	Override enable for (RXS) port rx_ana_dll_phd_en_i. **Value After Reset:** 0x1
13	ovrd_en_ana_rx_ana_dll_phd_dig_sel_i	R/W	Override enable for (RXS) port rx_ana_dll_phd_dig_sel_i. **Value After Reset:** 0x1
12	ovrd_en_ana_rx_ana_dll_phd_ana_sel_i	R/W	Override enable for (RXS) port rx_ana_dll_phd_ana_sel_i. **Value After Reset:** 0x1
11	ovrd_en_ana_rx_ana_dll_ac_en_i	R/W	Override enable for (RXS) port rx_ana_dll_ac_en_i. **Value After Reset:** 0x1
10	ovrd_en_ana_rx_ana_dll_dcc_en_i	R/W	Override enable for (RXS) port rx_ana_dll_dcc_en_i. **Value After Reset:** 0x1
9	ovrd_en_ana_rx_ana_dll_dac_dec_rst_n_i	R/W	Override enable for (RXS) port rx_ana_dll_dac_dec_rst_n_i. **Value After Reset:** 0x1
8	ovrd_en_ana_rx_ana_dll_comp_dac_range_i	R/W	Override enable for (RXS) port rx_ana_dll_comp_dac_range_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
7	ovrd_en_ana_rx_ana_dll_comp_dac_code_i	R/W	Override enable for (RXS) port rx_ana_dll_comp_dac_code_i. **Value After Reset:** 0x1
6	ovrd_en_ana_rx_ana_dll_delay_fine_en_i	R/W	Override enable for (RXS) port rx_ana_dll_delay_fine_en_i. **Value After Reset:** 0x1
5	ovrd_en_ana_rx_ana_dll_comp_dac_en_i	R/W	Override enable for (RXS) port rx_ana_dll_comp_dac_en_i. **Value After Reset:** 0x1
4	ovrd_en_ana_rx_ana_dll_comp_dac_vcm_i	R/W	Override enable for (RXS) port rx_ana_dll_comp_dac_vcm_i. **Value After Reset:** 0x1
3	ovrd_en_ana_rx_ana_dll_addr_i	R/W	Override enable for (RXS) port rx_ana_dll_addr_i. **Value After Reset:** 0x1
2	ovrd_en_ana_rx_ana_atb_s_en_i	R/W	Override enable for (RXS) port rx_ana_atb_s_en_i. **Value After Reset:** 0x1
1	ovrd_en_ana_rx_ana_afe_dig_clk_i	R/W	Override enable for (RXS) port rx_ana_afe_dig_clk_i. **Value After Reset:** 0x0
0	ovrd_en_ana_ltoh_pwrdn_h_i	R/W	Override enable for (RXS) port ltoh_pwrdn_h_i. **Value After Reset:** 0x0

11.1.17 PIN_OVRDEN_0

- **Name:** (RXS) Pin Override Enable Register 0
- **Description:** Register bits act as select signal of the mux which overrides value on signals at internal pins of (RXS). Register 4.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x040

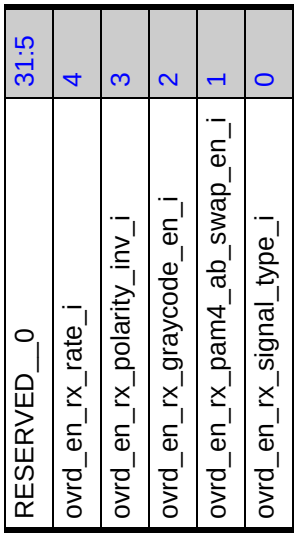


Table 11-21 Fields for Register: PIN_OVRDEN_0

Bits	Name	Memory Access	Description
31:5	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	ovrd_en_rx_rate_i	R/W	Override enable for (RXS) port rate_i. **Value After Reset:** 0x0
3	ovrd_en_rx_polarity_inv_i	R/W	Override enable for (RXS) port polarity_inv_i. **Value After Reset:** 0x0
2	ovrd_en_rx_graycode_en_i	R/W	Override enable for (RXS) port graycode_en_i. **Value After Reset:** 0x0
1	ovrd_en_rx_pam4_ab_swap_en_i	R/W	Override enable for (RXS) port pam4_ab_swap_en_i. **Value After Reset:** 0x0
0	ovrd_en_rx_signal_type_i	R/W	Override enable for (RXS) port signal_type_i. **Value After Reset:** 0x0

11.1.18 PIN_OVRDVAL_0

- **Name:** (RXS) Pin Override Value Register 0
- **Description:** Register bits that act as override value at the mux which can override values on (RXS). Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal. Register 0.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x044

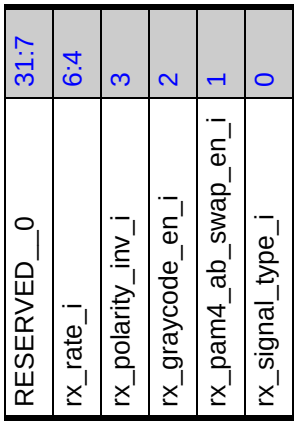


Table 11-22 Fields for Register: PIN_OVRDVAL_0

Bits	Name	Memory Access	Description
31:7	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
6:4	rx_rate_i	R/W	Setup data rate for the RX DES PMA. For further information consult the table of rate[1:0] pin. **Values:** ■ 0x0 (_1/4): Quarter rate ■ 0x1 (_1/2): Half rate ■ 0x2 (_FR): Full rate ■ 0x3 (OS2X): 2x oversampling ■ 0x4 (OS4X): 4x oversampling **Value After Reset:** 0x0 **Volatile:** true
3	rx_polarity_inv_i	R/W	Polarity inversion for the PAM4/NRZ Symbols : **Values:** ■ 0x0 (NOINV): no inversion ■ 0x1 (INVERT): invert all PAM4/NRZ symbols **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
2	rx_graycode_en_i	R/W	Enables/Disables hard-macro RX graycode. Used for pattern-matcher. **Value After Reset:** 0x0 **Volatile:** true
1	rx_pam4_ab_swap_en_i	R/W	When in PAM4 mode, enables swapping of most significant bit and least significant bit symbol: ■ sym 3 -> 3 ■ sym 1 -> -1 ■ sym -1 -> 1 ■ sym -3 -> -3 **Value After Reset:** 0x0 **Volatile:** true
0	rx_signal_type_i	R/W	Selects the Signal Type. **Values:** ■ 0x1 (PAM4): PAM4 ■ 0x0 (NRZ): NRZ **Value After Reset:** 0x0 **Volatile:** true

11.1.19 GENERIC_CONF_0

- **Name:** Generic Settings
- **Description:** Generic Configuration Settings. Register 0.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x048

31:30	29	28	27:23	22:12	11	10:8	7:2	1	0
RESERVED__0	rx_data_mode	dll_early_late_polarity_inv	cal_comp_clkdiv	cal_stat_ld_val	cal_stat_start	cal_stat_clk_mux	cal_stat_data_mux	cal_result_polarity_inv	data_sample

Table 11-23 Fields for Register: GENERIC_CONF_0

Bits	Name	Memory Access	Description
31:30	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	rx_data_mode	R/W	rx_data_mode is unused. **Value After Reset:** 0x0
28	dll_early_late_polarity_inv	R/W	Polarity invert option for dll_early_late from RX ANA. **Value After Reset:** 0x0
27:23	cal_comp_clkdiv	R/W	Divide by 'N' on ref_dig_clk to generate cal_comp_clk. **Value After Reset:** 0x9
22:12	cal_stat_ld_val	R/W	Sample Counter Load Value. **Value After Reset:** 0xf
11	cal_stat_start	R/W	Set 1b1 to start the Sample Counter. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
10:8	cal_stat_clk_mux	R/W	Selects the Clock for Cal Stat Calibration. Available options: **Values:** ■ 0x0 (PAM4): clk_32ui_pam4 ■ 0x1 (NRZ): clk_32ui_nrz ■ 0x2 (ERR): clk_32ui_err ■ 0x3 (COMP): cal_comp_clk ■ 0x4 (DLL): dll_clk **Value After Reset:** 0x0
7:2	cal_stat_data_mux	R/W	Selects the Data for Cal Stats Calibration. Available options: **Values:** ■ 0x0 (DATA0_HI): data0_pam4_32ui_hi[7:0] ■ 0x1 (DATA0_MD): data0_pam4_32ui_md[7:0] ■ 0x2 (DATA0_LO): data0_pam4_32ui_lo[7:0] ■ 0x3 (DATA90_HI): data90_pam4_32ui_hi[7:0] ■ 0x4 (DATA90_MD0): data90_pam4_32ui_md[7:0] ■ 0x5 (DATA90_LO): data90_pam4_32ui_lo[7:0] ■ 0x6 (DATA180_HI): data180_pam4_32ui_hi[7:0] ■ 0x7 (DATA180_MD): data180_pam4_32ui_md[7:0] ■ 0x8 (DATA180_LO): data180_pam4_32ui_lo[7:0] ■ 0x9 (DATA270_HI): data270_pam4_32ui_hi[7:0] ■ 0xa (DATA270_MD): data270_pam4_32ui_md[7:0] ■ 0xb (DATA270_LO): data270_pam4_32ui_lo[7:0] ■ 0xc (DATA0_NRZ): data0_nrz_32ui[7:0] ■ 0xd (DATA90_NRZ): data90_nrz_32ui[7:0] ■ 0xe (DATA180_NRZ): data18_nrz_32ui[7:0] ■ 0xf (DATA270_NRZ): data270_nrz_32ui[7:0] ■ 0x10 (ERR): err_32ui_hi[7:0] ■ 0x11 (PHS): phase_pam4[7:0] ■ 0x12 (RSV): RESERVED ■ 0x13 (RSLT): cal_result ■ 0x14 (DEL): dll_early_late ■ 0x15 (DATA0_HI_ERR): data0_hi_err_c_pol ■ 0x16 (DATA0_MD_ERR): data0_md_err_c_pol ■ 0x17 (DATA0_LO_ERR): data0_lo_err_c_pol ■ 0x18 (DATA90_HI_ERR): data90_hi_err_c_pol ■ 0x19 (DATA90_MD_ERR): data90_md_err_c_pol ■ 0x1a (DATA90_LO_ERR): data90_lo_err_c_pol ■ 0x1b (DATA180_HI_ERR): data180_hi_err_c_pol ■ 0x1c (DATA180_MD_ERR): data180_md_err_c_pol

Bits	Name	Memory Access	Description
7:2...(cont.)	cal_stat_data_mux.	R/W	■ 0x1d (DATA180_LO_ERR): data180_lo_err_c_pol ■ 0x1e (DATA270_HI_ERR): data270_hi_err_c_pol ■ 0x1f (DATA270_MD_ERR): data270_md_err_c_pol ■ 0x20 (DATA270_LO_ERR): data270_lo_err_c_pol ■ 0x21 (_180_0): rx_des_clk_ph_180by0 ■ 0x22 (_270_0): rx_des_clk_ph_270by0 ■ 0x23 (_90_0): rx_des_clk_ph_90by270 ■ 0x24 (ERRBY0): rx_des_clk_ph_errby0 ■ 0x25 (ERRBY90): rx_desd_clk_ph_phby90 **Value After Reset:** 0x0
1	cal_result_polarity_inv	R/W	Polarity invert option for cal_result from RX ANA. **Value After Reset:** 0x0
0	data_sample	R/W	data_sample is unused. **Value After Reset:** 0x1

11.1.20 PATGEN_0

- **Name:** Pattern Generator Configuration Settings Register 0
- **Description:** Configures Pattern Generator Settings. Register 0.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x04C

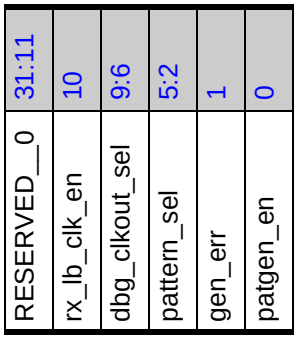


Table 11-24 Fields for Register: PATGEN_0

Bits	Name	Memory Access	Description
31:11	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
10	rx_lb_clk_en	R/W	Loopback Clock Enable Signal. **Value After Reset:** 0x1
9:6	dbg_clkout_sel	R/W	Select signal to bring-out different internal clocks for observation on (PIN::(RXS)::rx_dbg_clk_o). Available options: **Values:** ■ 0x0 (_0): ZERO ■ 0x1 (SYMB_8): RX_SYMCLK Clock/8 ■ 0x2 (SYMB_16): RX_SYMCLK Clock/16 ■ 0x3 (RSVD): (0x4, 0x5 and 0x6) Reserved ■ 0x7 (DLL_4): DLL Clock/4 ■ 0x8 (DLL_8): DLL Clock/8 ■ 0x9 (LPBK_8): Lpbk Clock/8 ■ 0xa (LPBK_16): Lpbk Clock/16 ■ 0xb (CAL): Cal comp clock **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
5:2	pattern_sel	R/W	This field is programmed as follows. Not listed below are reserved. These bits also determine the kind of incoming data pattern for the pattern matcher. Note that the pattern matcher does not support the JP03B and Transmitter linearity patterns. **Values:** ■ 0x0 (PRBS7): PRBS7 ■ 0x4 (PRBS15): PRBS15 ■ 0x6 (PRBS31): PRBS31 ■ 0x8 (PRQS10): PRQS10 ■ 0xb (CUSTM): Custom pattern defined by (APB::(RXS)::PATGEN::custom_pattern) **Value After Reset:** 0x0
1	gen_err	R/W	A write to 1b1 inserts 1 bit error into the ongoing pattern generation. Write 0b0 to 0b1 to create new errors. **Value After Reset:** 0x0
0	patgen_en	R/W	Write to 1b1 to enable the pattern generator. (RXS) ignores input parallel data at its input. **Value After Reset:** 0x0

11.1.21 PATGEN_1

- **Name:** Pattern Generator Configuration Settings Register 1
- **Description:** Configures Pattern Generator Settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x050



Table 11-25 Fields for Register: PATGEN_1

Bits	Name	Memory Access	Description
31:0	custom_pattern	R/W	When (pattern_sel) selects a custom pattern, this register defines the transmit pattern. Field 1. **Value After Reset:** 0xff00ff00

11.1.22 PATGEN_2

- **Name:** Pattern Generator Configuration Settings Register 2
- **Description:** Configures Pattern Generator Settings. Register 2.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x054



Table 11-26 Fields for Register: PATGEN_2

Bits	Name	Memory Access	Description
31:0	custom_pattern	R/W	When (pattern_sel) selects a custom pattern, this register defines the transmit pattern. Field 2. **Value After Reset:** 0xff00ff00

11.1.23 PATGEN_3

- **Name:** Pattern Generator Configuration Settings Register 3
- **Description:** Configures Pattern Generator Settings. Register 3.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x058



Table 11-27 Fields for Register: PATGEN_3

Bits	Name	Memory Access	Description
31:0	custom_pattern	R/W	When (pattern_sel) selects a custom pattern, this register defines the transmit pattern. Field 3. **Value After Reset:** 0xff00ff00

11.1.24 PATGEN_4

- **Name:** Pattern Generator Configuration Settings Register 4
- **Description:** Configures Pattern Generator Settings. Register 4.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x05C



Table 11-28 Fields for Register: PATGEN_4

Bits	Name	Memory Access	Description
31:0	custom_pattern	R/W	When (pattern_sel) selects a custom pattern, this register defines the transmit pattern. Field 4. **Value After Reset:** 0xff00ff00

11.1.25 PATGEN_5

- **Name:** Pattern Generator Configuration Settings Register 5
- **Description:** Configures Pattern Generator Settings. Register 5.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x060

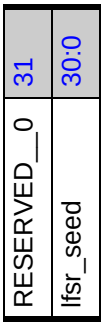


Table 11-29 Fields for Register: PATGEN_5

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:0	lfsr_seed	R/W	Defines the initial LFSR seed for the PRBS patterns as defined in (pattern_sel). **Value After Reset:** 0x1

11.1.26 PATGEN_6

- **Name:** Pattern Generator Configuration Settings Register 6
- **Description:** Configures Pattern Generator Settings. Register 6.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x064

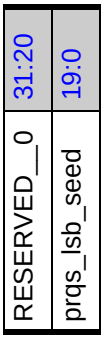


Table 11-30 Fields for Register: PATGEN_6

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19:0	prqs_lsb_seed	R/W	NOTUSED2 on the current implementation of RX PMA **Value After Reset:** 0x1

11.1.27 PATGEN_7

- **Name:** Pattern Generator Configuration Settings Register 7
- **Description:** Configures Pattern Generator Settings. Register 7.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x068

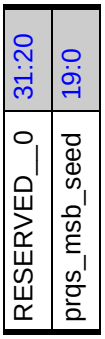


Table 11-31 Fields for Register: PATGEN_7

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19:0	prqs_msb_seed	R/W	NOTUSED1 on the current implementation of RX PMA **Value After Reset:** 0x400

11.1.28 PATGEN_8

- **Name:** Pattern Generator Configuration Settings Register 8
- **Description:** Configures Pattern Generator Settings. Register 8.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x06C

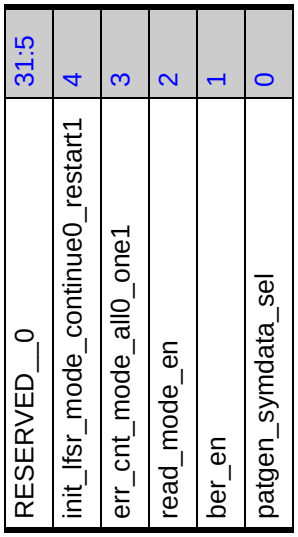


Table 11-32 Fields for Register: PATGEN_8

Bits	Name	Memory Access	Description
31:5	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	init_lfsr_mode_continue0_restart1	R/W	Internal reference pattern generation lfsr operation after error detection. Available options: **Values:** ■ 0x0 (CNT): Internal reference pattern generation lfsr continues on detection of error ■ 0x1 (RST): Internal reference pattern generation lfsr re-initializes itself on detection of error **Value After Reset:** 0x0
3	err_cnt_mode_all0_one1	R/W	Error count mode. Available options: **Values:** ■ 0x0 (ALL): Every single bit error is counted as +1 in the ber_counter ■ 0x1 (GRP): Multiple errors in a 8bit/10bit word are counted together as +1 in the ber_counter **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
2	read_mode_en	R/W	Read more. **Values:** ■ 0x1 (ENB): Enables internally taking current snapshot of ber_counter/ber_error_pattern/ber_error_location bits into corresponding fields of APB::RXS::BER_STAT so that they could be read in following APB read cycles. After read is complete, set this bit to 0b0. Even for consecutive reads, this bit should be set 1 to 0 to1. ■ 0x0 (NOTRD): ber_counter/ber_error_pattern/ber_error_location bits are not in read snapshot mode and are continuously changing internally. **Value After Reset:** 0x0
1	ber_en	R/W	When 0b1 the bit error rate computation logic is active. This can be set 0b1 after required settings for ber computation are done and CDR is locked. **Values:** ■ 0x1 (ACT): Bit error rate computation logic is active ■ 0x0 (DIS): Bit error rate computation logic is disabled **Value After Reset:** 0x0
0	patgen_symdata_sel	R/W	UNUSED **Value After Reset:** 0x0

11.1.29 PATGEN_9

- **Name:** Pattern Generator Configuration Settings Register 9
- **Description:** Configures Pattern Generator Settings. Register 9.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x070



Table 11-33 Fields for Register: PATGEN_9

Bits	Name	Memory Access	Description
31:0	ber_ref_timer	R/W	BER counting. **Values:** ■ 0x0 (CONT): BER counting is continuous. ■ 0x1 (STP): (and all non-zero) BER counting stops after (programmed_value+1) reference clocks. The internal counter starts fresh counting when ber_en is pulsed as 1-0-1. **Value After Reset:** 0x0

11.1.30 GENERIC_CONF_1

- **Name:** Generic Settings 1
- **Description:** Generic Configuration Settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x074

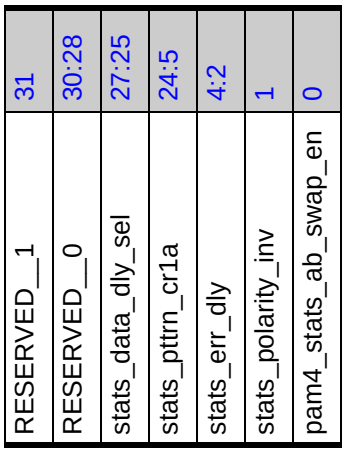


Table 11-34 Fields for Register: GENERIC_CONF_1

Bits	Name	Memory Access	Description
31	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27:25	stats_data_dly_sel	R/W	This field is the number of clock cycle delays on rx_data[63:0] **Value After Reset:** 0x0
24:5	stats_pttrn_cr1a	R/W	Value of pattern A for 1st correlator. **Value After Reset:** 0x0
4:2	stats_err_dly	R/W	This field is the number of clock cycle delays on err_rxclk. **Value After Reset:** 0x0
1	stats_polarity_inv	R/W	Polarity inversion for the PAM4/NRZ Adapt Stats Data. **Values:** ■ 0x0 (NOINV): no inversion ■ 0x1 (INVERT): invert all symbols **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
0	pam4_stats_ab_swap_en	R/W	When in PAM4 mode, enables swapping of most significant bit and least significant bit (used in adapt stats logic): ■ sym 3 -> 3 ■ sym 1 -> -1 ■ sym -1 -> 1 ■ sym -3 -> -3 **Value After Reset:** 0x1

11.1.31 GENERIC_CONF_2

- **Name:** Generic Settings 2
- **Description:** Generic Configuration Settings. Register 2.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x078

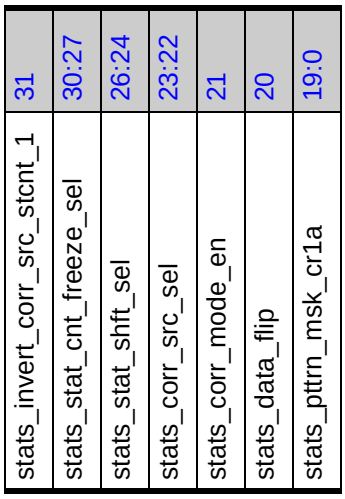


Table 11-35 Fields for Register: GENERIC_CONF_2

Bits	Name	Memory Access	Description
31	stats_invert_corr_src_stcnt_1	R/W	Select whether inverse or no inverse of corr_src_shft_s01 going to Stcnt#1: **Values:** ■ 0x0 (NINV): select non inverted corr_src_shft_s01 ■ 0x1 (IVN): select inverted corr_src_shft_s01 **Value After Reset:** 0x1
30:27	stats_stat_cnt_freeze_sel	R/W	Selects which statcnt to read as shadow register. **Value After Reset:** 0x0
26:24	stats_stat_shft_sel	R/W	Selects stat source shift value. **Values:** ■ 0x0 (NM1): Correlate N-1 > N+3 (use N for offset calibration) ■ 0x1 (NP1): Correlate N+1 > N+5 ■ 0x2 (NP6): Correlate N+6 > N+10 ■ 0x3 (NP11): Correlate N+11 > N+15 ■ 0x4 (NP16): Correlate N+16 > N+20 **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
23:22	stats_corr_src_sel	R/W	Selects correlation input source. Available options: **Values:** ■ 0x0 (RX4): error_x4 ■ 0x1 (PHX4): phase_x4 ■ 0x2 (RSVD): Reserved ■ 0x3 (_64): 64'd0 **Value After Reset:** 0x0
21	stats_corr_mode_en	R/W	Enables processing in correlation data chain. For both Error counting and Correlation of data(msb/lsb) with corresponding error/phase, this bit should be 1. **Value After Reset:** 0x0
20	stats_data_flip	R/W	Flip order of stats data (between OLD and NEW). **Values:** ■ 0x0 (OTN): set data order from old to new ■ 0x1 (NTO): set data order from new to old **Value After Reset:** 0x0
19:0	stats_pttrn_msk_cr1a	R/W	Value of pattern A mask for 1st correlator. **Value After Reset:** 0x0

11.1.32 GENERIC_CONF_3

- **Name:** Generic Settings 3
- **Description:** Generic Configuration Settings. Register 3.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x07C

31	30	29	28	27	26	25	24:22	21	20	19:0
stats_err_flip	lpbk_polarity_inv	stats_sc_timer_mode	stats_disable_sample_count	stats_sc1_stop	stats_sc_pause	stats_sc1_start	stats_stat_src_sel	stats_err_corr_polarity_inv	stats_pttrn_cr1b_en	stats_pttrn_cr1b

Table 11-36 Fields for Register: GENERIC_CONF_3

Bits	Name	Memory Access	Description
31	stats_err_flip	R/W	Flip order of error stats data (between OLD and NEW). **Values:** ■ 0x0 (OTN): set data order from old to new ■ 0x1 (NTO): set data order from new to old **Value After Reset:** 0x0
30	lpbk_polarity_inv	R/W	Inverts the Polarity for Loopback Data from RX Patgen to RX LPBK Serializer and eventually to ANA Loopback path. **Value After Reset:** 0x0
29	stats_sc_timer_mode	R/W	Sample counter operation mode. Available options: **Values:** ■ 0x0 (SMPS): counts number of matched samples. ■ 0x1 (CLKC): counts clock cycles (that is a timer). **Value After Reset:** 0x1
28	stats_disable_sample_count	R/W	Disables the sample counter. This will keep the sample counter at the current value. It is used to determine the error rate over long periods of time. This must be set back to zero to allow the sample count to complete its countdown and finish the stats call. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
27	stats_sc1_stop	R/W	Stops sample counters 1 and associated stat counters. **Value After Reset:** 0x0
26	stats_sc_pause	R/W	Pauses the sample counter and stat counters. **Value After Reset:** 0x0
25	stats_sc1_start	R/W	Starts Adaptation sample counter 1. **Value After Reset:** 0x0
24:22	stats_stat_src_sel	R/W	Selects stat source input. **Values:** ■ 0x0 (MSB): rx_data_msb ■ 0x1 (LSB): rx_data_lsb ■ 0x2 (ERX4): rx_error_x4 ■ 0x3 (PHX4): rx_phase_x4 ■ 0x4 (_641S): 64 1's ■ 0x5 (_640S): 64 0's **Value After Reset:** 0x0
21	stats_err_corr_polarity_inv	R/W	Invert option for inverting error_x4 going to corr_src branch. **Value After Reset:** 0x0
20	stats_pttrn_cr1b_en	R/W	Enables pattern B matching for 1st correlator. **Value After Reset:** 0x0
19:0	stats_pttrn_cr1b	R/W	Value of pattern B for 1st correlator. **Value After Reset:** 0x0

11.1.33 GENERIC_CONF_4

- **Name:** Generic Settings 4
- **Description:** Generic Configuration Settings. Register 4.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x080

31	30	29	28	27:25	24	23	22	21	20	19	18	17	16	15	14:12	11:10	9:0
interlane_clk_align_start	interlane_clk_align_clkgate_en	data_err_clk_align_start	data_err_clk_align_clkgate_en	err_des_dly	cg_bypass_en_bermon	RESERVED__1	stats_phase_flip	stats_stcnt_6_en	stats_stcnt_5_en	stats_stcnt_4_en	stats_stcnt_3_en	stats_stcnt_2_en	stats_stcnt_1_en	stats_freeze_stat_cnt	stats_phase_dly_sel	rx_loopback_sel_slicer	RESERVED__0

Table 11-37 Fields for Register: GENERIC_CONF_4

Bits	Name	Memory Access	Description
31	interlane_clk_align_start	R/W	Starts Signal to initiate Interlane Data Clock Alignment. **Value After Reset:** 0x0
30	interlane_clk_align_clkgate_en	R/W	Clock Gating Enable for Interlane Data Clock Alignment. Set to 1'b1 to start the Alignment. **Value After Reset:** 0x0
29	data_err_clk_align_start	R/W	Start Signal to initiate Data and Error Clock Alignment. **Value After Reset:** 0x0
28	data_err_clk_align_clkgate_en	R/W	Clock Gating Enable for Data and Error Clock Alignment. Set to 1'b1 to start the Alignment. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
27:25	err_des_dly	R/W	Options to add delay on Error Path from ANA to DIG w.r.t error clock (Will be used in Adaptation) **Values:** ■ 0x0 (_0CD): 0 cycle delay (default) ■ 0x1 (_1CD): 1 Cycle Delay ■ 0x2 (_2CD): 2 Cycle delay ■ 0x3 (_3CD): 3 Cycle Delay ■ 0x4 (_4CD): 4 Cycle Delay **Value After Reset:** 0x0
24	cg_bypass_en_bermon	R/W	Bypasses Clock Gating for bermon block. **Values:** ■ 0x1 (BYPASS): Bypass Clock Gating **Value After Reset:** 0x0
23	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x1
22	stats_phase_flip	R/W	Flip order of phase stats data (between OLD and NEW). **Values:** ■ 0x0 (OTN): set data order from old to new ■ 0x1 (NTO): set data order from new to old **Value After Reset:** 0x0
21	stats_stcnt_6_en	R/W	Enables for stat counter 6. **Value After Reset:** 0x0
20	stats_stcnt_5_en	R/W	Enables for stat counter 5. **Value After Reset:** 0x0
19	stats_stcnt_4_en	R/W	Enables for stat counter 4. **Value After Reset:** 0x0
18	stats_stcnt_3_en	R/W	Enables for stat counter 3. **Value After Reset:** 0x0
17	stats_stcnt_2_en	R/W	Enables for stat counter 2. **Value After Reset:** 0x0
16	stats_stcnt_1_en	R/W	Enables for stat counter 1. **Value After Reset:** 0x0
15	stats_freeze_stat_cnt	R/W	When this bit is enabled, it freezes the value of stat_cnt_n (enable). **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
14:12	stats_phase_dly_sel	R/W	Number of clock cycle delays on rx_phase[2:0]. **Value After Reset:** 0x0
11:10	rx_loopback_sel_slicer	R/W	Selects the origin of NRZ data from a specific slicer when in loopback mode. **Values:** ■ 0x0 (_0SS): mid slicer is selected (default). ■ 0x1 (_1SS): low slicer is selected. ■ 0x2 (_2SS): high slicer is selected. **Value After Reset:** 0x0
9:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.1.34 GENERIC_CONF_5

- **Name:** Generic Settings 5
- **Description:** Generic Configuration Settings. Register 5.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x084

31:30	29	28	27	26	25	24	23	22	21:20	19:0
stats_quad_sel	cal_stat_descal_polarity_inv	cal_stat_err_corr_polarity_inv	stats_clkgate_en	cal_stat_clkgate_dll_en	cal_stat_clkgate_cal_comp_en	cal_stat_clkgate_err_en	cal_stat_clkgate_nrz_en	cal_stat_clkgate_pam4_en	stats_corr_shft_sel	stats_pttrn_msk_cr1b

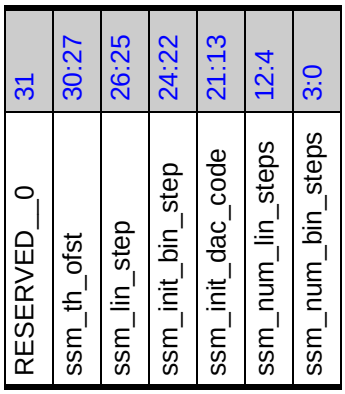
Table 11-38 Fields for Register: GENERIC_CONF_5

Bits	Name	Memory Access	Description
31:30	stats_quad_sel	R/W	Quadrant selection for the adapt stats input data. **Values:** ■ 0x0 (Q0): data from quadrant 0 ■ 0x1 (Q90): data from quadrant 90 ■ 0x2 (Q180): data from quadrant 180 ■ 0x3 (Q270): data from quadrant 270 **Value After Reset:** 0x0
29	cal_stat_descal_polarity_inv	R/W	Inverts polarity for the DES clock phase detector measure. **Value After Reset:** 0x0
28	cal_stat_err_corr_polarity_inv	R/W	Inverts polarity for the error data correlation result. **Value After Reset:** 0x0
27	stats_clkgate_en	R/W	Clock Gate Enable for Adapt Stats Block. **Value After Reset:** 0x1
26	cal_stat_clkgate_dll_en	R/W	Clock Gate Enable for DLL Clock in Cal Stats Block. **Value After Reset:** 0x1
25	cal_stat_clkgate_cal_comp_en	R/W	Clock Gate Enable for Cal Comp Clock in Cal Stats Block. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
24	cal_stat_clkgate_err_en	R/W	Clock Gate Enable for Err Clock in Cal Stats Block. **Value After Reset:** 0x1
23	cal_stat_clkgate_nrz_en	R/W	Clock Gate Enable for NRZ Clock in Cal Stats Block. **Value After Reset:** 0x1
22	cal_stat_clkgate_pam4_en	R/W	Clock Gate Enable for PAM4 Clock in Cal Stats Block. **Value After Reset:** 0x1
21:20	stats_corr_shft_sel	R/W	Selects shift for Error. Available options: **Values:** ■ 0x0 (_7_0): corr_src[7:0] ■ 0x1 (_8_1): corr_src[8:1] ■ 0x2 (_9_2): corr_src[9:2] ■ 0x3 (_10_3): corr_src[10:3] **Value After Reset:** 0x0
19:0	stats_pttrn_msk_cr1b	R/W	Value of pattern B mask for 1st correlator. **Value After Reset:** 0x0

11.1.35 GENERIC_CONF_6

- **Name:** Generic Settings 6
- **Description:** Generic Configuration Settings. Register 6.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x088



31	30:27	26:25	24:22	21:13	12:4	3:0
RESERVED__0	ssm_th_ofst	ssm_lin_step	ssm_init_bin_step	ssm_init_dac_code	ssm_num_lin_steps	ssm_num_bin_steps

Table 11-39 Fields for Register: GENERIC_CONF_6

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:27	ssm_th_ofst	R/W	SSM correlation threshold offset. **Value After Reset:** 0x0
26:25	ssm_lin_step	R/W	SSM linear serach step size (2^N). **Value After Reset:** 0x0
24:22	ssm_init_bin_step	R/W	SSM initial binary search step size (2^N). **Value After Reset:** 0x7
21:13	ssm_init_dac_code	R/W	SSM initial code for the DAC. **Value After Reset:** 0x100
12:4	ssm_num_lin_steps	R/W	SSM number of linear search steps. **Value After Reset:** 0x4
3:0	ssm_num_bin_steps	R/W	SSM number of binary search steps. **Value After Reset:** 0x8

11.1.36 GENERIC_CONF_18

- **Name:** Generic Settings 18
- **Description:** Generic Configuration Settings. Register 18.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x08C

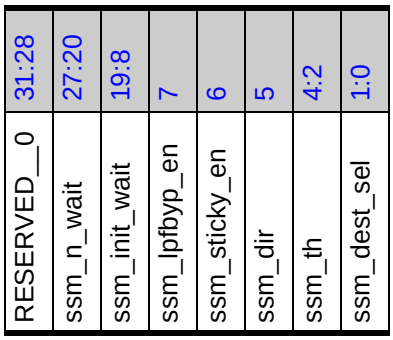


Table 11-40 Fields for Register: GENERIC_CONF_18

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27:20	ssm_n_wait	R/W	SSM wait cycles following DAC write. **Value After Reset:** 0x32
19:8	ssm_init_wait	R/W	SSM wait cycles before starting. **Value After Reset:** 0x1f4
7	ssm_lpfbyp_en	R/W	SSM enable LPF bypass. **Value After Reset:** 0x0
6	ssm_sticky_en	R/W	SSM enable sticky linear search. **Value After Reset:** 0x1
5	ssm_dir	R/W	SSM select update direction: **Values:** ■ 0x0 (POS): positive ■ 0x1 (NEG): negative **Value After Reset:** 0x0
4:2	ssm_th	R/W	SSM correlation threshold. **Value After Reset:** 0x0
1:0	ssm_dest_sel	R/W	SSM selected DAC destination. **Value After Reset:** 0x0

11.1.37 GENERIC_CONF_19

- **Name:** Generic Settings 19
- **Description:** Generic Configuration Settings. Register 19.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x090

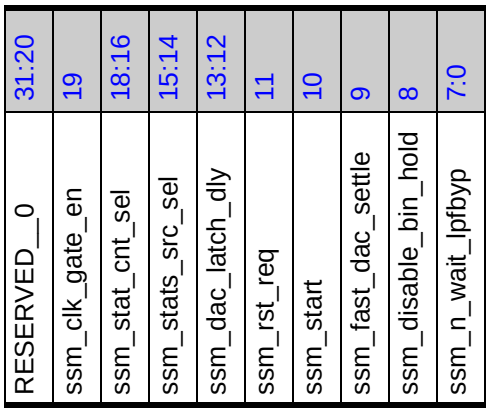


Table 11-41 Fields for Register: GENERIC_CONF_19

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19	ssm_clk_gate_en	R/W	SSM clk gating. Set 1 to ungate. **Value After Reset:** 0x0
18:16	ssm_stat_cnt_sel	R/W	SSM selection for witch stat counter to use. **Value After Reset:** 0x2
15:14	ssm_stats_src_sel	R/W	SSM input stats source selection. **Value After Reset:** 0x0
13:12	ssm_dac_latch_dly	R/W	SSM defines the latch pulse size. **Value After Reset:** 0x0
11	ssm_rst_req	R/W	SSM aborts running and resets. **Value After Reset:** 0x0
10	ssm_start	R/W	SSM starts running. **Value After Reset:** 0x0
9	ssm_fast_dac_settle	R/W	SSM bypasses the analog wait time. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
8	ssm_disable_bin_hold	R/W	SSM disables DAC hold during binary search. **Value After Reset:** 0x0
7:0	ssm_n_wait_lpfbyp	R/W	SSM wait cycles following DAC write while LPF bypass is enabled. **Value After Reset:** 0x5

11.1.38 GENERIC_CONF_7

- **Name:** Generic Settings 7
- **Description:** Generic Configuration Settings. Register 7.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x094



Table 11-42 Fields for Register: GENERIC_CONF_7

Bits	Name	Memory Access	Description
31:0	stats_sc1_ld_val	R/W	Sample counter 1 load value field 1. **Value After Reset:** 0x0

11.1.39 GENERIC_CONF_17

- **Name:** Generic Settings 17
- **Description:** Generic Configuration Settings. Register 17.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x098

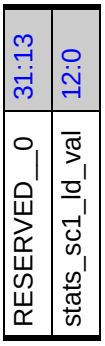


Table 11-43 Fields for Register: GENERIC_CONF_17

Bits	Name	Memory Access	Description
31:13	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
12:0	stats_sc1_ld_val	R/W	Sample counter 1 load value field 2. **Value After Reset:** 0x0

11.1.40 GENERIC_CONF_8

- **Name:** Generic Settings 8
- **Description:** Generic Configuration Settings. Register 8.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x09C

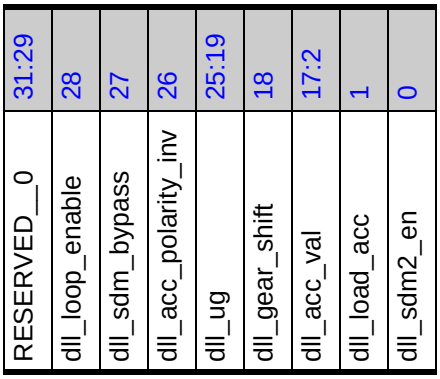


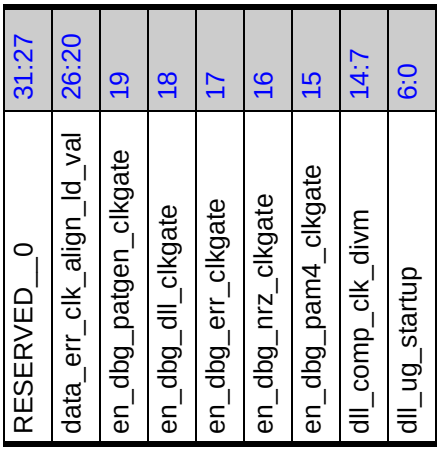
Table 11-44 Fields for Register: GENERIC_CONF_8

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	dll_loop_enable	R/W	If set to 1, enables the dll loop. **Value After Reset:** 0x0
27	dll_sdm_bypass	R/W	Bypass of SDM datapath for VREG DAC **Value After Reset:** 0x0
26	dll_acc_polarity_inv	R/W	Inverts the polarity of dll early_late data for acc. **Value After Reset:** 0x0
25:19	dll_ug	R/W	Gain for DLL loop accumulator. **Value After Reset:** 0x1
18	dll_gear_shift	R/W	Enables the dll gear shift. **Value After Reset:** 0x0
17:2	dll_acc_val	R/W	Desired value to load into DLL early_late accumulator. **Value After Reset:** 0x0
1	dll_load_acc	R/W	If set to 1, loads dll_acc_value into the DLL early_late accumulator. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
0	dll_sdm2_en	R/W	If set to 1, enables the 2nd sdm loop. **Value After Reset:** 0x0

11.1.41 GENERIC_CONF_9

- **Name:** Generic Settings 9
- **Description:** Generic Configuration Settings. Register 9.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0A0



31:27	26:20	19	18	17	16	15	14:7	6:0
RESERVED__0	data_err_clk_align_ld_val	en_dbg_patgen_clkgate	en_dbg_dll_clkgate	en_dbg_err_clkgate	en_dbg_nrz_clkgate	en_dbg_pam4_clkgate	dll_comp_clk_divm	dll_ug_startup

Table 11-45 Fields for Register: GENERIC_CONF_9

Bits	Name	Memory Access	Description
31:27	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
26:20	data_err_clk_align_ld_val	R/W	Number of Cycles of Stats Data Clock (32UI Clock) to run the Data-Error Clock Alignment. **Value After Reset:** 0x8
19	en_dbg_patgen_clkgate	R/W	Clock Gating Enable for Patgen Debug Clocks. **Value After Reset:** 0x0
18	en_dbg_dll_clkgate	R/W	Clock Gating Enable for DLL Debug Clocks. **Value After Reset:** 0x0
17	en_dbg_err_clkgate	R/W	Clock Gating Enable for Error Debug Clocks. **Value After Reset:** 0x0
16	en_dbg_nrz_clkgate	R/W	Clock Gating Enable for NRZ Debug Clocks. **Value After Reset:** 0x0
15	en_dbg_pam4_clkgate	R/W	Clock Gating Enable for PAM4 Debug Clocks. **Value After Reset:** 0x0
14:7	dll_comp_clk_divm	R/W	Configure division factor M for DLL SDM clock. **Value After Reset:** 0x8

Bits	Name	Memory Access	Description
6:0	dll_ug_startup	R/W	Gain for DLL loop accumulator during gear shift. **Value After Reset:** 0x10

11.1.42 GENERIC_CONF_10

- **Name:** Generic Settings 10
- **Description:** Generic Configuration Settings. Register 10.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0A4

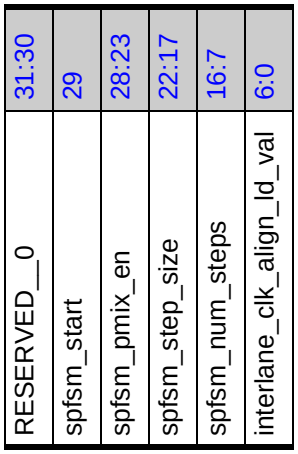


Table 11-46 Fields for Register: GENERIC_CONF_10

Bits	Name	Memory Access	Description
31:30	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	spfsm_start	R/W	Gives the order to start spinning the fsm. **Value After Reset:** 0x0
28:23	spfsm_pmix_en	R/W	Configures the desired pmix to spin (one hot): **Values:** ■ 0x0 (DIS): disabled ■ 0x1 (_0CODE): rx_ana_pmix_data0_code ■ 0x2 (_90CODE): rx_ana_pmix_data90_code ■ 0x4 (_180CODE): rx_ana_pmix_data180_code ■ 0x8 (_270CODE): rx_ana_pmix_data270_code ■ 0x10 (ERRCODE): rx_ana_pmix_error_code ■ 0x20 (PHSCODE): rx_ana_pmix_phase_code **Value After Reset:** 0x0
22:17	spfsm_step_size	R/W	Configures the number of steps per iteration, signal magnitude (-31: +31) **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
16:7	spfsm_num_steps	R/W	Configures the number of pmix steps to walk (0 - 1023). **Value After Reset:** 0x1
6:0	interlane_clk_align_ld_val	R/W	Number of Cycles of Data Clock (32UI Clock) to run the Interlane Clock Alignment. **Value After Reset:** 0x8

11.1.43 GENERIC_CONF_11

- **Name:** Generic Settings 11
- **Description:** Generic Configuration Settings. Register 11.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0A8

31	30	29	28:26	25	24	23:20	19:16	15:12	11:8	7	6:5	4	3:2	1:0
RESERVED__0	en_cdr_clkgate	en_pmix_clkgate	pmix_clkdiv	phdet_data_phase_mux_sel_update	cdr_gear_shift	cdr_phug	cdr_phug_startup	cdr_frug	cdr_frug_startup	cdr_en	phdet_data_phase_mux_sel	phdet_flip_pol	phdet_en_edge	phdet_en_slice

Table 11-47 Fields for Register: GENERIC_CONF_11

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	en_cdr_clkgate	R/W	Clock Gating for CDR Clock Going to ANA. **Values:** ■ 0x0 (GTD): Gates the Clock ■ 0x1 (PSS): Ungates the Clock **Value After Reset:** 0x0
29	en_pmix_clkgate	R/W	Clock Gating for PMIX Clock Going to ANA. **Values:** ■ 0x0 (GTD): Gates the Clock ■ 0x1 (PSS): Passes the Clock to Ana **Value After Reset:** 0x0
28:26	pmix_clkdiv	R/W	Divide by n options of Data Clock (32ui clock) to generate PMIX Clock for Analog. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
25	phdet_data_phase_mux_sel_update	R/W	Updates the signal when changing Register phdet_data_phase_mux_sel. After Changing the New Phase, assert this signal to '1' and then de-assert after few cycles so that New Phase is taken internally by the design. This way the Phase Transition is CDC proof. **Value After Reset:** 0x0
24	cdr_gear_shift	R/W	Selects the CDR Gain values based upon Startup or non-startup mode. **Values:** ■ 0x1 (STP): CDR Startup Mode ■ 0x0 (STB): CDR stable mode **Value After Reset:** 0x1
23:20	cdr_phug	R/W	Selects the Gain for proportional branch at after Startup is done. **Values:** ■ 0x0 (_x1): x1 ■ 0x1 (_x2): x2 ■ 0x2 (_x4): x4 ■ 0x3 (_x8): x8 ■ 0x4 (_x16): x16 ■ 0x5 (_x43): x32 ■ 0x6 (_x64): x64 ■ 0x7 (_x128): x128 ■ 0x8 (_x256): x256 ■ 0x9 (_x0): x0 **Value After Reset:** 0x3
19:16	cdr_phug_startup	R/W	Selects the Gain for proportional branch at Startup. **Values:** ■ 0x0 (_x1): x1 ■ 0x1 (_x2): x2 ■ 0x2 (_x4): x4 ■ 0x3 (_x8): x8 ■ 0x4 (_x16): x16 ■ 0x5 (_x43): x32 ■ 0x6 (_x64): x64 ■ 0x7 (_x128): x128 ■ 0x8 (_x256): x256 ■ 0x9 (_x0): x0 **Value After Reset:** 0x3
15:12	cdr_frug	R/W	Gain for DLL loop accumulator during normal operation. **Value After Reset:** 0x2

Bits	Name	Memory Access	Description
11:8	cdr_frug_startup	R/W	Gain for DLL loop accumulator during gear shift. **Value After Reset:** 0x8
7	cdr_en	R/W	Enables the CDR. **Value After Reset:** 0x0
6:5	phdet_data_phase_mux_sel	R/W	Selects the Data Inputs for the Phase Detector based upon the "phase" of the Phase Clock. Signal should be updated everytime "Phase" of phase Clock is moved **Values:** ■ 0x0 (_45): 45 degree Phase Clock (default) ■ 0x1 (_135): 135 degree Phase Clock ■ 0x2 (_225): 225 degree Phase Clock **Value After Reset:** 0x0
4	phdet_flip_pol	R/W	Flips the Polarity of Early and Late Decisions from DIG. **Value After Reset:** 0x0
3:2	phdet_en_edge	R/W	Edges to use for Phase Detection for CDR: **Values:** ■ 0x0 (IGN): Ignore all edges ■ 0x1 (FLNG): Use the Falling edges only ■ 0x2 (RSN): Use the Rising Edge only ■ 0x3 (BTH): Use both the edges **Value After Reset:** 0x3
1:0	phdet_en_slice	R/W	Enables the Phase Detector for CDR. Only least significant bit is used. **Value After Reset:** 0x3

11.1.44 GENERIC_CONF_12

- **Name:** Generic Settings 12
- **Description:** Generic Configuration Settings. Register 12.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0AC

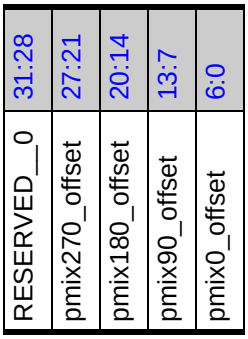


Table 11-48 Fields for Register: GENERIC_CONF_12

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27:21	pmix270_offset	R/W	PMIX270 Offset Value. Once PMIX270 Offset Calibration is externally performed, that value is provided to this Register to proceed with the CDR. **Value After Reset:** 0x0
20:14	pmix180_offset	R/W	PMIX180 Offset Value. Once PMIX180 Offset Calibration is externally performed, that value is provided to this Register to proceed with the CDR. **Value After Reset:** 0x0
13:7	pmix90_offset	R/W	PMIX90 Offset Value. Once PMIX90 Offset Calibration is externally performed, that value is provided to this Register to proceed with the CDR. **Value After Reset:** 0x0
6:0	pmix0_offset	R/W	PMIX0 Offset Value. Once PMIX0 Offset Calibration is externally performed, that value is provided to this Register to proceed with the CDR. **Value After Reset:** 0x0

11.1.45 GENERIC_CONF_13

- **Name:** Generic Settings 13
- **Description:** Generic Configuration Settings. Register 13.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0B0

31:28	27	26	25	24	23	22	21:15	14	13:7	6:0
RESERVED__0	dll_vreg_dac_sel_ovrd_sel	dll_vreg_dac_sel_ovrd_val	dll_comp_clk_ovrd_sel	dll_comp_clk_ovrd_val	pmix_la_clk_ovrd_sel	pmix_la_clk_ovrd_val	spfsm_init_val_spin	pmix_offset_code_en	pmix_err_offset	pmix_ph_offset

Table 11-49 Fields for Register: GENERIC_CONF_13

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27	dll_vreg_dac_sel_ovrd_sel	R/W	DLL Clock Select to Analog pin rx_ana_dll_vreg_dac_sel. **Values:** ■ 0x0 (INT): Selects the internally generated clock ■ 0x1 (REG): Selects (RXS) register dll_vreg_-dac_sel_ovrd_val **Value After Reset:** 0x0
26	dll_vreg_dac_sel_ovrd_val	R/W	Drives the Analog pin rx_ana_dll_vreg_dac_sel when (RXS) Register dll_vreg_dac_sel_ovrd_sel is 1. **Value After Reset:** 0x0
25	dll_comp_clk_ovrd_sel	R/W	DLL Clock Select to Analog pin rx_ana_dll_clk_comp_phd. **Values:** ■ 0x0 (INT): Select the internally generated clock ■ 0x1 (REG): Selects (RXS) register dll_comp_clk_ovrd_val **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
24	dll_comp_clk_ovrd_val	R/W	Drives the Analog pin rx_ana_dll_comp_dac_clk when (RXS) Register dll_comp_clk_ovrd_sel is 1. **Value After Reset:** 0x0
23	pmix_la_clk_ovrd_sel	R/W	PMIX Clock Select to Analog pin rx_ana_pmix_cklatch. **Values:** ■ 0x0 (INT): Selects the internally generated clock ■ 0x1 (REG): Selects (RXS) register pmix_la_clk_ovrd_val **Value After Reset:** 0x0
22	pmix_la_clk_ovrd_val	R/W	Drives the Analog pin rx_ana_pmix_cklatch when (RXS) Register pmix_la_clk_ovrd_sel is 1. **Value After Reset:** 0x0
21:15	spfsm_init_val_spin	R/W	Initialization value for the spin FSM. **Value After Reset:** 0x0
14	pmix_offset_code_en	R/W	PMIX Offset Code Enable inside Digital. When set to '1' , Digital latches the value for all PMIX* Offset and different values goes to the different PMIX codes to ANA. Digital always preserve the Offset Codes and add it to the CDR PMIX code for Analog. **Value After Reset:** 0x0
13:7	pmix_err_offset	R/W	PMIX Error Offset Value. Once PMIX Offset Calibration is externally performed, that value is provided to this Register to proceed with the CDR. **Value After Reset:** 0x40
6:0	pmix_ph_offset	R/W	PMIX Phase Offset Value. Once PMIX Offset Calibration is externally performed, that value is provided to this Register to proceed with the CDR. **Value After Reset:** 0x10

11.1.46 GENERIC_CONF_14

- **Name:** Generic Settings 14
- **Description:** Generic Configuration Settings. Register 14.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0B4

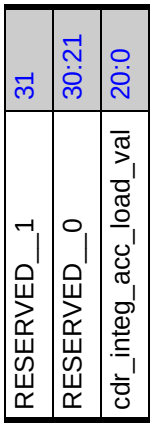


Table 11-50 Fields for Register: GENERIC_CONF_14

Bits	Name	Memory Access	Description
31	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:21	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
20:0	cdr_integ_acc_load_val	R/W	CDR Integral Accumulator Load Value. Updated when (RXS) Register cdr_load_integ_acc_en is 1. **Value After Reset:** 0x0

11.1.47 GENERIC_CONF_15

- **Name:** Generic Settings 15
- **Description:** Generic Configuration Settings. Register 15.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0B8

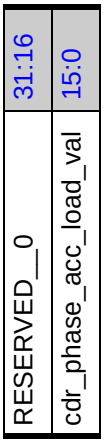


Table 11-51 Fields for Register: GENERIC_CONF_15

Bits	Name	Memory Access	Description
31:16	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
15:0	cdr_phase_acc_load_val	R/W	CDR Phase Accumulator Load Value. Updated when (RXS) Register cdr_load_phase_acc_en is 1. **Value After Reset:** 0x0

11.1.48 GENERIC_CONF_16

- **Name:** Generic Settings 16
- **Description:** Generic Configuration Settings. Register 16.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0BC

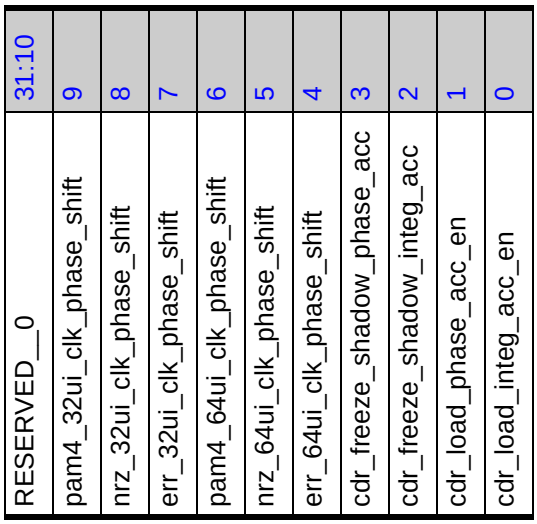


Table 11-52 Fields for Register: GENERIC_CONF_16

Bits	Name	Memory Access	Description
31:10	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
9	pam4_32ui_clk_phase_shift	R/W	pam4_32ui_clk_phase_shift is unused. **Value After Reset:** 0x0
8	nrz_32ui_clk_phase_shift	R/W	nrz_32ui_clk_phase_shift is unused. **Value After Reset:** 0x0
7	err_32ui_clk_phase_shift	R/W	err_32ui_clk_phase_shift is unused. **Value After Reset:** 0x0
6	pam4_64ui_clk_phase_shift	R/W	pam4_64ui_clk_phase_shift is unused. **Value After Reset:** 0x0
5	nrz_64ui_clk_phase_shift	R/W	nrz_64ui_clk_phase_shift is unused. **Value After Reset:** 0x0
4	err_64ui_clk_phase_shift	R/W	err_64ui_clk_phase_shift is unused. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
3	cdr_freeze_shadow_phase_acc	R/W	CDR Accumulator snapshot on (RXS) Register cdr_shadow_phase_acc when cdr_freeze_shadow_phase_acc is set to 1. **Value After Reset:** 0x0
2	cdr_freeze_shadow_integ_acc	R/W	CDR Accumulator snapshot on (RXS) Register cdr_shadow_integ_acc when cdr_freeze_shadow_integ_acc is set to 1. **Value After Reset:** 0x0
1	cdr_load_phase_acc_en	R/W	Enable to load Offset value on CDR Phase Accumulator. **Value After Reset:** 0x0
0	cdr_load_integ_acc_en	R/W	Enable to load Offset value on CDR Integral Accumulator. **Value After Reset:** 0x0

11.1.49 GENERIC_CONF_20

- **Name:** Generic Settings 20
- **Description:** Generic Configuration Settings. Register 20.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0C0

31:29	28:26	25:22	21:18	17:14	13:10	9	8:2	1	0
RESERVED__0	asm_ctle_mu_r	asm_th_ofst_r	asm_vga_th_r	asm_ctle_th_r	asm_n_wait_dac_r	asm_eyeh_en_r	asm_n_wait_asm1_r	asm_ssm_sel_r	asm_start_asm1_r

Table 11-53 Fields for Register: GENERIC_CONF_20

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28:26	asm_ctle_mu_r	R/W	This field is the CTLE boost code update gain (2^N). **Value After Reset:** 0x3
25:22	asm_th_ofst_r	R/W	This field applies an offset to the decision threshold. **Value After Reset:** 0x0
21:18	asm_vga_th_r	R/W	This field is VGA correlation decision threshold (2^N-1). During eye height measurement, the VGA_TH is repurposed for error slicer updates. **Value After Reset:** 0x0
17:14	asm_ctle_th_r	R/W	This field is CTLE correlation decision threshold (2^N-1). **Value After Reset:** 0x0
13:10	asm_n_wait_dac_r	R/W	This field selects setup time and pulse width time in terms of qword_clk for RX DFE tapes and AFE update. **Value After Reset:** 0x7
9	asm_eyeh_en_r	R/W	This bit enables eye height measurement. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
8:2	asm_n_wait_asm1_r	R/W	This field is the number of wait cycles for adaptation SM 1. **Value After Reset:** 0x9
1	asm_ssm_sel_r	R/W	This field controls whether the target DAC should be controlled from search state machine or adaptation state machine: **Values:** ■ 0x0 (SSM): SSM has priority control over target DAC ■ 0x1 (ASM): ASM has priority control over target DAC **Value After Reset:** 0x0
0	asm_start_asm1_r	R/W	This bit starts adaptation state machine 1. **Value After Reset:** 0x0

11.1.50 GENERIC_CONF_21

- **Name:** Generic Settings 21
- **Description:** Generic Configuration Settings. Register 21.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0C4

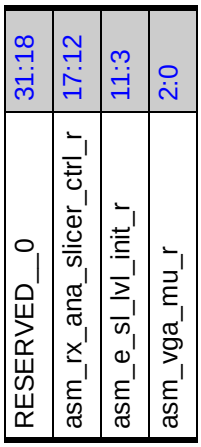


Table 11-54 Fields for Register: GENERIC_CONF_21

Bits	Name	Memory Access	Description
31:18	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
17:12	asm_rx_ana_slicer_ctrl_r	R/W	This field is the value for rx_ana_slicer_ctrl[5:0] **Value After Reset:** 0x38
11:3	asm_e_sl_lvl_init_r	R/W	The error slicer is initialized to this value at the start of a new adaptation request. **Value After Reset:** 0x0
2:0	asm_vga_mu_r	R/W	This field is the VGA gain code update gain (2^N). During eye height measurement, the VGA_MU is repurposed for error slicer updates. **Value After Reset:** 0x5

11.1.51 GENERIC_CONF_22

- **Name:** Generic Settings 22
- **Description:** Generic Configuration Settings. Register 22.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0C8

31:25	24:19	18:16	15	14:12	11:9	8:6	5:3	2:0
RESERVED__0	asm_dac_ctrl_sel_dfe_dh0_vdac_r	asm_ctle_t1_wt_r	asm_reset_asm1_r	asm_d1tod_map_val_r	asm_frc_l0_map_val_r	asm_frc_h0_map_val_r	asm_frc_l1_map_val_r	asm_frc_h1_map_val_r

Table 11-55 Fields for Register: GENERIC_CONF_22

Bits	Name	Memory Access	Description
31:25	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
24:19	asm_dac_ctrl_sel_dfe_dh0_vdac_r	R/W	This field is data0 high vDAC address. **Value After Reset:** 0x17
18:16	asm_ctle_t1_wt_r	R/W	When calculating CTLE correlation, TAP1 correlation. **Value After Reset:** 0x0
15	asm_reset_asm1_r	R/W	This bit resets adaptation state machine (ASM1), binary search as well as the stats capture block. **Value After Reset:** 0x0
14:12	asm_d1tod_map_val_r	R/W	This field is the mapping value for d1tod_o and d1tod_e signals. **Value After Reset:** 0x6
11:9	asm_frc_l0_map_val_r	R/W	This field is the mapping value for frc_l0_o and frc_l0_e signals. **Value After Reset:** 0x5

Bits	Name	Memory Access	Description
8:6	asm_frc_h0_map_val_r	R/W	This field is the mapping value for frc_h0_o and frc_h0_e signals. **Value After Reset:** 0x4
5:3	asm_frc_l1_map_val_r	R/W	This field is the mapping value for frc_l1_o and frc_l1_e signals. **Value After Reset:** 0x6
2:0	asm_frc_h1_map_val_r	R/W	This field is the mapping value for frc_h1_o and frc_h1_e signals. **Value After Reset:** 0x7

11.1.52 GENERIC_CONF_23

- **Name:** Generic Settings 23
- **Description:** Generic Configuration Settings. Register 23.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0CC

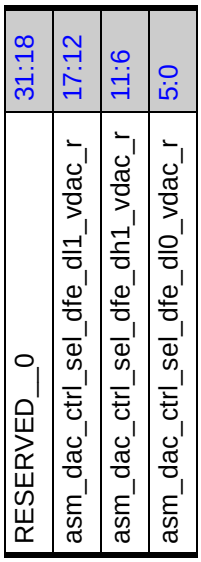


Table 11-56 Fields for Register: GENERIC_CONF_23

Bits	Name	Memory Access	Description
31:18	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
17:12	asm_dac_ctrl_sel_dfe_dl1_vdac_r	R/W	This field is data1 low DAC address. **Value After Reset:** 0x1a
11:6	asm_dac_ctrl_sel_dfe_dh1_vdac_r	R/W	This field is data1 high DAC address. **Value After Reset:** 0x19
5:0	asm_dac_ctrl_sel_dfe_dl0_vdac_r	R/W	This field is data0 low vDAC address. **Value After Reset:** 0x18

11.1.53 GENERIC_CONF_24

- **Name:** Generic Settings 24
- **Description:** Generic Configuration Settings. Register 24.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0D0

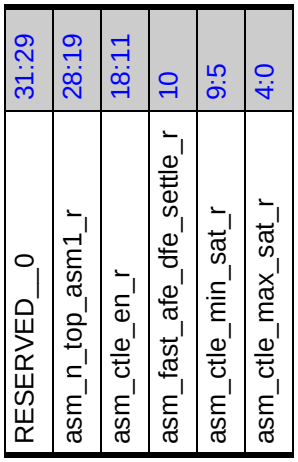


Table 11-57 Fields for Register: GENERIC_CONF_24

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28:19	asm_n_top_asm1_r	R/W	Number of iterations for ASM1 fsm. **Value After Reset:** 0x2
18:11	asm_ctle_en_r	R/W	This field enables CTLE boost adaptation. These bits determine which correlators are used to adapt the CTLE. **Value After Reset:** 0x0
10	asm_fast_afe_dfe_settle_r	R/W	This bit enables fast AFE and DFE settling time (simulation only). **Value After Reset:** 0x0
9:5	asm_ctle_min_sat_r	R/W	This field is CTLE lower saturation level. **Value After Reset:** 0x0
4:0	asm_ctle_max_sat_r	R/W	This field is CTLE upper saturation level. **Value After Reset:** 0x1f

11.1.54 STATUS_0

- **Name:** (RXS) Digital Status Register 0
- **Description:** Provides (RXS) Digital Status Register 0.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0D4

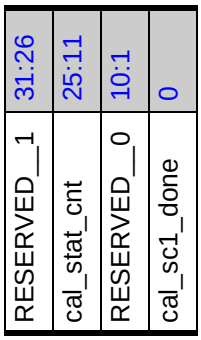


Table 11-58 Fields for Register: STATUS_0

Bits	Name	Memory Access	Description
31:26	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
25:11	cal_stat_cnt	R	Value of the Stat Counter. **Value After Reset:** 0x0
10:1	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
0	cal_sc1_done	R	Status of the Sample Counter. **Value After Reset:** 0x1

11.1.55 STATUS_1

- **Name:** (RXS) Digital Status Register 1
- **Description:** Provides (RXS) Digital Status Register 1.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0D8

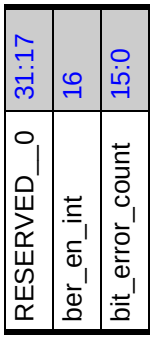


Table 11-59 Fields for Register: STATUS_1

Bits	Name	Memory Access	Description
31:17	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
16	ber_en_int	R	Internal BER enable status. When using the ber_ref_timer, this bit will be 0 when ber_ref_timer counter has expired. **Value After Reset:** 0x0
15:0	bit_error_count	R	Bit error rate (BER) counter status. **Value After Reset:** 0x0

11.1.56 STATUS_7

- **Name:** (RXS) Digital Status Register 2
- **Description:** Provides (RXS) Digital Status Register 2.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0DC



Table 11-60 Fields for Register: STATUS_7

Bits	Name	Memory Access	Description
31:0	stats_smpl_cnt1	R	Override value set on (RXS) analog port ana_rx_bmux0. This value is set when the override enable register for this (RXS) port is set to 0b1. **Value After Reset:** 0x0

11.1.57 STATUS_8

- **Name:** (RXS) Digital Status Register 3
- **Description:** Provides (RXS) Digital Status Register 3.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0E0

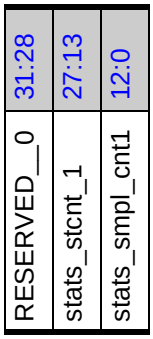


Table 11-61 Fields for Register: STATUS_8

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27:13	stats_stcnt_1	R	Adapt Stats Stat Counter 1 Current Value. **Value After Reset:** 0x0
12:0	stats_smpl_cnt1	R	Adapt Stats Sample Counter Current Value. **Value After Reset:** 0x0

11.1.58 STATUS_9

- **Name:** (RXS) Digital Status Register 4
- **Description:** Provides (RXS) Digital Status Register 4.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0E4

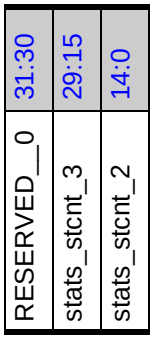


Table 11-62 Fields for Register: STATUS_9

Bits	Name	Memory Access	Description
31:30	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29:15	stats_stcnt_3	R	Adapt Stats Stat Counter 3 Current Value. **Value After Reset:** 0x0
14:0	stats_stcnt_2	R	Adapt Stats Stat Counter 2 Current Value. **Value After Reset:** 0x0

11.1.59 STATUS_10

- **Name:** (RXS) Digital Status Register 5
- **Description:** Provides (RXS) Digital Status Register 5.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0E8

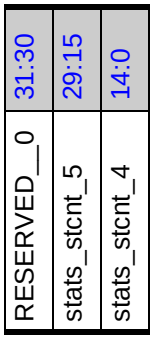


Table 11-63 Fields for Register: STATUS_10

Bits	Name	Memory Access	Description
31:30	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29:15	stats_stcnt_5	R	Adapt Stats Stat Counter 5 Current Value. **Value After Reset:** 0x0
14:0	stats_stcnt_4	R	Adapt Stats Stat Counter 4 Current Value. **Value After Reset:** 0x0

11.1.60 STATUS_11

- **Name:** (RXS) Digital Status Register 6
- **Description:** Provides (RXS) Digital Status Register 6.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0EC

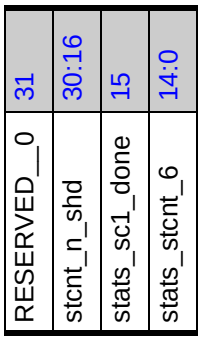


Table 11-64 Fields for Register: STATUS_11

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:16	stcnt_n_shd	R	Current value of stat counter N with the freeze bit enabled. **Value After Reset:** 0x0
15	stats_sc1_done	R	Adapt Stats Sample Counter 1 Status. **Value After Reset:** 0x1
14:0	stats_stcnt_6	R	Adapt Stats Stat Counter 6 Current Value. **Value After Reset:** 0x0

11.1.61 STATUS_12

- **Name:** (RXS) Digital Status Register 7
- **Description:** Provides (RXS) Digital Status Register 7.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0F0

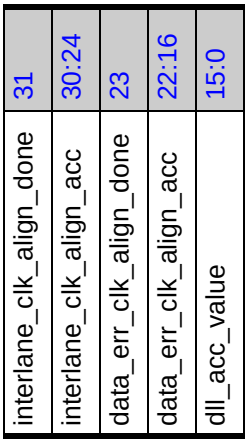


Table 11-65 Fields for Register: STATUS_12

Bits	Name	Memory Access	Description
31	interlane_clk_align_done	R	Done Status for Interlane Clock Alignment. **Value After Reset:** 0x0
30:24	interlane_clk_align_acc	R	Status of Internal Accumulator for Interlane "Data" Clock Alignment. **Value After Reset:** 0x0
23	data_err_clk_align_done	R	Done Status for Data Error Clock Alignment. **Value After Reset:** 0x0
22:16	data_err_clk_align_acc	R	Status of Internal Accumulator for Data Error Clock Alignment. **Value After Reset:** 0x0
15:0	dll_acc_value	R	Status of internal design dll_acc. **Value After Reset:** 0x0

11.1.62 STATUS_13

- **Name:** (RXS) Digital Status Register 8
- **Description:** Provides (RXS) Digital Status Register 8.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0F4



Table 11-66 Fields for Register: STATUS_13

Bits	Name	Memory Access	Description
31	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:21	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
20:0	cdr_shadow_integ_acc	R	Status of Integral Accumulator value when (RXS) cdr_freeze_shadow_integ_acc is set to '1'. **Value After Reset:** 0x0

11.1.63 STATUS_14

- **Name:** (RXS) Digital Status Register 9
- **Description:** Provides (RXS) Digital Status Register 9.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0F8

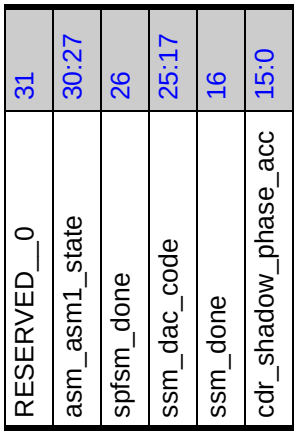


Table 11-67 Fields for Register: STATUS_14

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:27	asm_asm1_state	R	This field is the current value of the ADAPT FSM state register. **Value After Reset:** 0x0
26	spfsm_done	R	Indicates that the FSM operation ended (switches to 0 during operation and returns back to 1 after finished). **Value After Reset:** 0x1
25:17	ssm_dac_code	R	Searched DAC code from the SSM. **Value After Reset:** 0x100
16	ssm_done	R	Flag for SSM operation finish. **Value After Reset:** 0x1
15:0	cdr_shadow_phase_acc	R	Status of Phase Accumulator value when (RXS) cdr_freeze_shadow_phase_acc is set to '1'. **Value After Reset:** 0x0

11.1.64 STATUS_15

- **Name:** (RXS) Digital Status Register 10
- **Description:** Provides (RXS) Digital Status Register 10.
- **Size:** 32 bits
- **Offset:** 0x0 + j*0x100 + 0x0FC

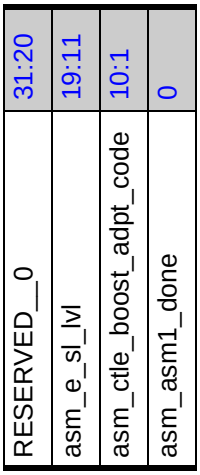


Table 11-68 Fields for Register: STATUS_15

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19:11	asm_e_sl_lvl	R	This field is the error slicer level. **Value After Reset:** 0x0
10:1	asm_ctle_boost_adpt_code	R	This field is the value of CTLE boost adaptation code. **Value After Reset:** 0x0
0	asm_asm1_done	R	This bit asserts when adaptation state machine 1 is done. **Value After Reset:** 0x0

11.2 SUP_RX_SUP Registers

11.2.1 SUP_CFG_0

- **Name:** (SUP) Configuration Register
- **Description:** Configures (SUP) settings.
- **Size:** 32 bits
- **Offset:** 0x1000

31	30	29	28:26	25:20	19:17	16:5	4	3	2	1	0
RESERVED__4	RESERVED__3	ref_ana_clk_sel	RESERVED__2	sup_dbg_clkout_sel	sup_dbg_ana_clk_sel	RESERVED__1	RESERVED__0	polarity_inv	graycode_en	pam4_ab_swap_en	signal_type

Table 11-69 Fields for Register: SUP_CFG_0

Bits	Name	Memory Access	Description
31	RESERVED__4	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	RESERVED__3	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	ref_ana_clk_sel	R/W	Selects between refa_ana_clk or ref_ro_ana_clk to be used to generate ref_dig_clk. **Values:** ■ 0x0 (REFANA): uses refa_ana_clk ■ 0x1 (ROANA): uses ref_ro_ana_clk **Value After Reset:** 0x0
28:26	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
25:20	sup_dbg_clkout_sel	R/W	Control for top debug clock selection. Choose between lanes/sup. **Values:** ■ 0x0 (_0): ZERO ■ 0x1 (L0): lane 0 ■ 0x2 (L1): lane 1 ■ 0x3 (L2): lane 2 ■ 0x4 (L3): lane 3 ■ 0x5 (L4): lane 4 ■ 0x6 (L5): lane 5 ■ 0x7 (L6): lane 6 ■ 0x8 (L7): lane 7 ■ 0x9 (L8): lane 8 ■ 0xa (L9): lane 9 ■ 0xb (L10): lane 10 ■ 0xc (L11): lane 11 ■ 0xd (L12): lane 12 ■ 0xe (L13): lane 13 ■ 0xf (L14): lane 14 ■ 0x10 (L15): lane 15 ■ 0x11 (SUP): SUP **Value After Reset:** 0x0
19:17	sup_dbg_ana_clk_sel	R/W	Selects between clks to observe on sup_dbg_clk_o. **Values:** ■ 0x0 (_0): ZERO ■ 0x1 (ANA): ref_ana_clk ■ 0x2 (ROANA): ref_ro_ana_clk ■ 0x3 (DIG): ref_dig_clk ■ 0x4 (ANAFB): mpll_ana_fb_clk ■ 0x5 (ANAWRD/8): mpll_ana_word_clk / 8 ■ 0x6 (ANAWRD/16): ropll_ana_word_clk / 16 ■ 0x7 (ANADIV): ropll_ana_div_clk **Value After Reset:** 0x0
16:5	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
3	polarity_inv	R/W	RESERVED **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
2	graycode_en	R/W	RESERVED **Value After Reset:** 0x1
1	pam4_ab_swap_en	R/W	RESERVED **Value After Reset:** 0x0
0	signal_type	R/W	RESERVED **Value After Reset:** 0x0

11.2.2 SUP_OVRDEN

- **Name:** (SUP) Internal Logic Override Enable Register
- **Description:** Register bits select signal of the mux which overrides value on (SUP) internal logic signals.
- **Size:** 32 bits
- **Offset:** 0x1004



Table 11-70 Fields for Register: SUP_OVRDEN

Bits	Name	Memory Access	Description
31:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.2.3 SUP_OVRDVAL

- **Name:** (SUP) Internal Logic Override Value Register
- **Description:** Register bits act as override value at the mux which can override values on (SUP) internal logic signals. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1008



Table 11-71 Fields for Register: SUP_OVRDVAL

Bits	Name	Memory Access	Description
31:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.2.4 SUP_STAT

- **Name:** (SUP) Status Register
- **Description:** RESERVED
- **Size:** 32 bits
- **Offset:** 0x100c

31:14	13	12	11:0
RESERVED__3	RESERVED__2	RESERVED__1	RESERVED__0

Table 11-72 Fields for Register: SUP_STAT

Bits	Name	Memory Access	Description
31:14	RESERVED__3	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
13	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
12	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
11:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.2.5 ANA_OVRDVAL_0

- **Name:** OSC Configuration Register 0
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 0. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1010

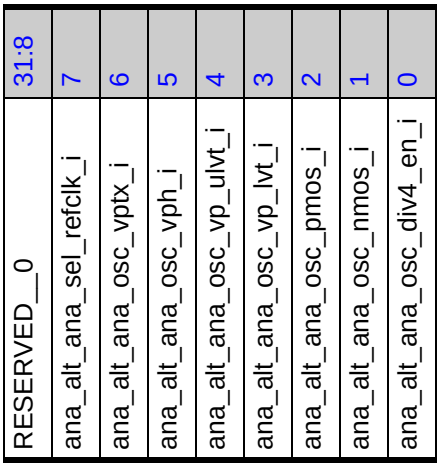


Table 11-73 Fields for Register: ANA_OVRDVAL_0

Bits	Name	Memory Access	Description
31:8	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
7	ana_alt_ana_sel_refclk_i	R/W	Override value set on (SUP) analog port ana_alt_ana_sel_refclk_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
6	ana_alt_ana_osc_vptx_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_vptx_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5	ana_alt_ana_osc_vph_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_vph_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
4	ana_alt_ana_osc_vp_ulvt_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_vp_ulvt_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3	ana_alt_ana_osc_vp_lvt_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_vp_lvt_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_alt_ana_osc_pmos_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_pmos_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1	ana_alt_ana_osc_nmos_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_nmos_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_alt_ana_osc_div4_en_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_div4_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.6 ANA_OVRDVAL_1

- **Name:** ANA Configuration Register 1
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 1. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1014

31:23	22	21:18	17:10	9:8	7:6	5	4	3	2	1	0
RESERVED__0	ana_vref_pre_i	ana_sup_ana_term_ctrl_i	ana_sup_ana_misc_i	ana_nominal_vph_sel_i	ana_nominal_vp_sel_i	ana_bg_ana_vref_fast_start_i	ana_bg_ana_kick_start_en_i	ana_bg_ana_en_i	ana_ana_pwrdn_i	ana_ana_flyover_en_i	ana_ana_burnin_i

Table 11-74 Fields for Register: ANA_OVRDVAL_1

Bits	Name	Memory Access	Description
31:23	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
22	ana_vref_pre_i	R/W	Override value set on (SUP) analog port ana_vref_pre_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
21:18	ana_sup_ana_term_ctrl_i	R/W	Override value set on (SUP) analog port ana_sup_ana_term_ctrl_i[3:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
17:10	ana_sup_ana_misc_i	R/W	Override value set on (SUP) analog port ana_sup_ana_misc_i[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9:8	ana_nominal_vph_sel_i	R/W	Override value set on (SUP) analog port ana_nominal_vph_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
7:6	ana_nominal_vp_sel_i	R/W	Override value set on (SUP) analog port ana_nominal_vp_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5	ana_bg_ana_vref_fast_start_i	R/W	Override value set on (SUP) analog port ana_bg_ana_vref_fast_start_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4	ana_bg_ana_kick_start_en_i	R/W	Override value set on (SUP) analog port ana_bg_ana_kick_start_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3	ana_bg_ana_en_i	R/W	Override value set on (SUP) analog port ana_bg_ana_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_ana_pwrdn_i	R/W	Override value set on (SUP) analog port ana_ana_pwrdn_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
1	ana_ana_flyover_en_i	R/W	Override value set on (SUP) analog port ana_ana_flyover_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
0	ana_ana_burnin_i	R/W	Override value set on (SUP) analog port ana_ana_burnin_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.7 ANA_OVRDVAL_2

- **Name:** CRU Access Register 2
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 2. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1018

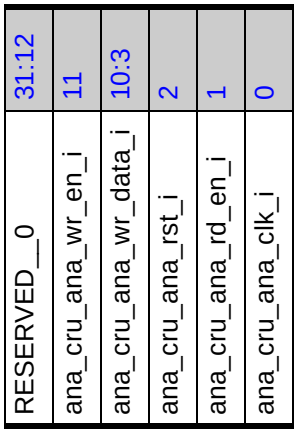


Table 11-75 Fields for Register: ANA_OVRDVAL_2

Bits	Name	Memory Access	Description
31:12	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
11	ana_cru_ana_wr_en_i	R/W	Override value set on (SUP) analog port ana_cru_ana_wr_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:3	ana_cru_ana_wr_data_i	R/W	Override value set on (SUP) analog port ana_cru_ana_wr_data_i[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_cru_ana_rst_i	R/W	Override value set on (SUP) analog port ana_cru_ana_rst_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
1	ana_cru_ana_rd_en_i	R/W	Override value set on (SUP) analog port ana_cru_ana_rd_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_cru_ana_clk_i	R/W	Override value set on (SUP) analog port ana_cru_ana_clk_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.8 ANA_OVRDVAL_3

- **Name:** CRU Access Register 3
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 3. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x101c

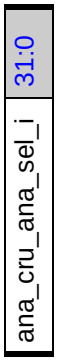


Table 11-76 Fields for Register: ANA_OVRDVAL_3

Bits	Name	Memory Access	Description
31:0	ana_cru_ana_sel_i	R/W	Override value set on (SUP) analog port ana_cru_ana_sel_i[27:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.9 ANA_OVRDVAL_4

- **Name:** DIFFBUF Configuration Register 4
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 4. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1020

31:16	15	14	13	12	11	10:6	5	4:2	1:0
RESERVED__0	ana_atb_ana_sup_en_i	ana_atb_ana_lane_right_en_i	ana_atb_ana_lane_left_en_i	ana_diffbuf_tx_en_i	ana_diffbuf_pulldn_i	ana_diffbuf_imp_ctrl_i	ana_diffbuf_de_emp_en_i	ana_diffbuf_de_emp_ctrl_i	ana_diffbuf_clk_sel_i

Table 11-77 Fields for Register: ANA_OVRDVAL_4

Bits	Name	Memory Access	Description
31:16	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
15	ana_atb_ana_sup_en_i	R/W	Override value set on (SUP) analog port ana_atb_ana_sup_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14	ana_atb_ana_lane_right_en_i	R/W	Override value set on (SUP) analog port ana_atb_ana_lane_right_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
13	ana_atb_ana_lane_left_en_i	R/W	Override value set on (SUP) analog port ana_atb_ana_lane_left_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
12	ana_diffbuf_tx_en_i	R/W	Override value set on (SUP) analog port ana_diffbuf_tx_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11	ana_diffbuf_pulldn_i	R/W	Override value set on (SUP) analog port ana_diffbuf_pulldn_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:6	ana_diffbuf_imp_ctrl_i	R/W	Override value set on (SUP) analog port ana_diffbuf_imp_ctrl_i[4:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5	ana_diffbuf_de_emp_en_i	R/W	Override value set on (SUP) analog port ana_diffbuf_de_emp_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4:2	ana_diffbuf_de_emp_ctrl_i	R/W	Override value set on (SUP) analog port ana_diffbuf_de_emp_ctrl_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1:0	ana_diffbuf_clk_sel_i	R/W	Override value set on (SUP) analog port ana_diffbuf_clk_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.10 ANA_OVRDVAL_5

- **Name:** MPLL Analog Configuration Register 5
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 5. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1024

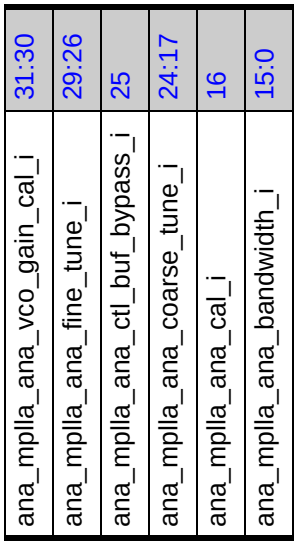


Table 11-78 Fields for Register: ANA_OVRDVAL_5

Bits	Name	Memory Access	Description
31:30	ana_mplla_ana_vco_gain_cal_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_vco_gain_cal_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
29:26	ana_mplla_ana_fine_tune_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_fine_tune_i[3:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_mplla_ana_ctl_buf_bypass_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_ctl_buf_bypass_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
24:17	ana_mplla_ana_coarse_tune_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_coarse_tune_i[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_mplla_ana_cal_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_cal_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:0	ana_mplla_ana_bandwidth_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_bandwidth_i[15:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x161f **Volatile:** true

11.2.11 ANA_OVRDVAL_6

- **Name:** MPLL Startup Configuration Register 6
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 6. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1028

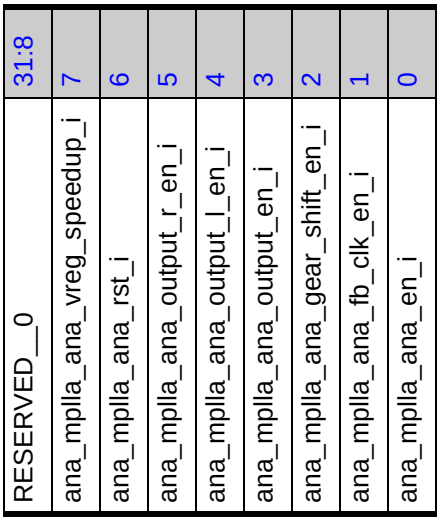


Table 11-79 Fields for Register: ANA_OVRDVAL_6

Bits	Name	Memory Access	Description
31:8	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
7	ana_mplla_ana_vreg_speedup_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_vreg_speedup_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
6	ana_mplla_ana_rst_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_rst_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true

Bits	Name	Memory Access	Description
5	ana_mplla_ana_output_r_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_output_r_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4	ana_mplla_ana_output_l_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_output_l_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3	ana_mplla_ana_output_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_output_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_mplla_ana_gear_shift_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_gear_shift_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
1	ana_mplla_ana_fb_clk_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_fb_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_mplla_ana_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.12 ANA_OVRDVAL_7

- **Name:** MPLL Dividers Configuration Register 7
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 7. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x102c

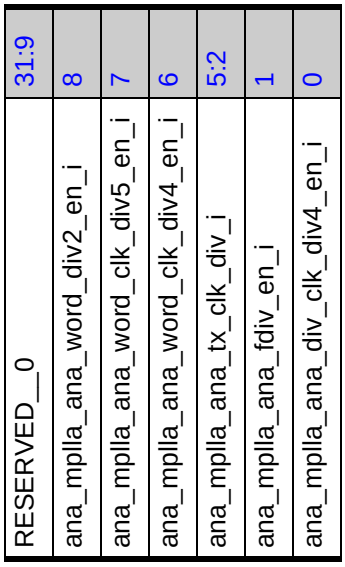


Table 11-80 Fields for Register: ANA_OVRDVAL_7

Bits	Name	Memory Access	Description
31:9	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
8	ana_mplla_ana_word_div2_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_word_div2_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7	ana_mplla_ana_word_clk_div5_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_word_clk_div5_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true

Bits	Name	Memory Access	Description
6	ana_mplla_ana_word_clk_div4_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_word_clk_div4_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5:2	ana_mplla_ana_tx_clk_div_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_tx_clk_div_i[3:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
1	ana_mplla_ana_fdiv_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_fdiv_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_mplla_ana_div_clk_div4_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_div_clk_div4_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.13 ANA_OVRDVAL_8

- **Name:** MPLL PMIX Configuration Register 8
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 8. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1030

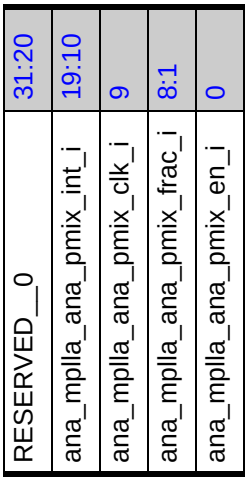


Table 11-81 Fields for Register: ANA_OVRDVAL_8

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19:10	ana_mplla_ana_pmix_int_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_pmix_int_i[9:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9	ana_mplla_ana_pmix_clk_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_pmix_clk_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8:1	ana_mplla_ana_pmix_frac_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_pmix_frac_i[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
0	ana_mplla_ana_pmix_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_pmix_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.14 ANA_OVRDVAL_9

- **Name:** REF Configuration Register 9
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 9. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1034

31	30	29	28:27	26	25	24	23:22	21	20:19	18	17:15	14:13	12	11:9	8	7:6	5:0
RESERVED__0	ana_refa_ana_lane_clk_en_i	ana_refa_ana_clkdet_en_i	ana_refa_ana_clk_sel_i	ana_refa_ana_clk_en_i	ana_ref_ro_ana_lane_clk_en_i	ana_ref_ro_ana_clkdet_en_i	ana_ref_ro_ana_clk_sel_i	ana_ref_ro_ana_clk_en_i	ana_ref_ana_vreg_ref_sel_i	ana_ref_ana_vreg_fast_start_i	ana_ref_ana_ropll_div_i	ana_ref_ana_repeat_clk_sel_i	ana_ref_ana_repeat_clk_en_i	ana_ref_ana_mplla_div_i	ana_ref_ana_diffbuf_clk_en_i	ana_ref_ana_dco_range_i	ana_ref_ana_dco_finetune_i

Table 11-82 Fields for Register: ANA_OVRDVAL_9

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	ana_refa_ana_lane_clk_en_i	R/W	Override value set on (SUP) analog port ana_refa_ana_lane_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
29	ana_refa_ana_clkdet_en_i	R/W	Override value set on (SUP) analog port ana_refa_ana_clkdet_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
28:27	ana_refa_ana_clk_sel_i	R/W	Override value set on (SUP) analog port ana_refa_ana_clk_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
26	ana_refa_ana_clk_en_i	R/W	Override value set on (SUP) analog port ana_refa_ana_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_ref_ro_ana_lane_clk_en_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_lane_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	ana_ref_ro_ana_clkdet_en_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clkdet_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
23:22	ana_ref_ro_ana_clk_sel_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clk_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
21	ana_ref_ro_ana_clk_en_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
20:19	ana_ref_ana_vreg_ref_sel_i	R/W	Override value set on (SUP) analog port ana_ref_ana_vreg_ref_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x3 **Volatile:** true
18	ana_ref_ana_vreg_fast_start_i	R/W	Override value set on (SUP) analog port ana_ref_ana_vreg_fast_start_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
17:15	ana_ref_ana_ropll_div_i	R/W	Override value set on (SUP) analog port ana_ref_ana_ropll_div_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14:13	ana_ref_ana_repeat_clk_sel_i	R/W	Override value set on (SUP) analog port ana_ref_ana_repeat_clk_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
12	ana_ref_ana_repeat_clk_en_i	R/W	Override value set on (SUP) analog port ana_ref_ana_repeat_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11:9	ana_ref_ana_mplla_div_i	R/W	Override value set on (SUP) analog port ana_ref_ana_mplla_div_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8	ana_ref_ana_diffbuf_clk_en_i	R/W	Override value set on (SUP) analog port ana_ref_ana_diffbuf_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7:6	ana_ref_ana_dco_range_i	R/W	Override value set on (SUP) analog port ana_ref_ana_dco_range_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5:0	ana_ref_ana_dco_finetune_i	R/W	Override value set on (SUP) analog port ana_ref_ana_dco_finetune_i[5:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0xd **Volatile:** true

11.2.15 ANA_OVRDVAL_10

- **Name:** ROPLL Startup Configuration Register 10
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 10. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1038

31:28	27	26:25	24:23	22:21	20:18	17	16	15	14:8	7:1	0
RESERVED__0	ana_ropll_ana_vreg_fast_start_i	ana_ropll_ana_vco_low_freq_i	ana_ropll_ana_v2i_mode_i	ana_ropll_ana_refsel_i	ana_ropll_ana_rc_filter_i	ana_ropll_ana_preset_i	ana_ropll_ana_gearshift_i	ana_ropll_ana_en_i	ana_ropll_ana_cp_ctrl_prop_i	ana_ropll_ana_cp_ctrl_int_i	ana_ropll_ana_bypass_i

Table 11-83 Fields for Register: ANA_OVRDVAL_10

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27	ana_ropll_ana_vreg_fast_start_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_vreg_fast_start_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
26:25	ana_ropll_ana_vco_low_freq_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_vco_low_freq_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
24:23	ana_ropll_ana_v2i_mode_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_v2i_mode_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x3 **Volatile:** true
22:21	ana_ropll_ana_refsel_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_refsel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
20:18	ana_ropll_ana_rc_filter_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_rc_filter_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x4 **Volatile:** true
17	ana_ropll_ana_preset_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_preset_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_ropll_ana_gearshift_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_gearshift_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
15	ana_ropll_ana_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14:8	ana_ropll_ana_cp_ctrl_prop_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_cp_ctrl_prop_i[6:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7:1	ana_ropll_ana_cp_ctrl_int_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_cp_ctrl_int_i[6:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
0	ana_ropll_ana_bypass_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_bypass_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.16 ANA_OVRDVAL_11

- **Name:** ROPLL Dividers Configuration Register 11
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 11. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x103c

31:29	28	27	26	25	24	23:20	19:17	16:11	10:9	8	7:1	0
RESERVED__0	ana_ref_ro_ana_clk_div8_en_i	ana_ropll_ana_word_div2_en_i	ana_ropll_ana_word_clk_en_i	ana_ropll_ana_word_clk_div5_en_i	ana_ropll_ana_word_clk_div4_en_i	ana_ropll_ana_refdiv_i	ana_ropll_ana_postdiv_i	ana_ropll_ana_out_div_msb_i	ana_ropll_ana_out_div_lsb_i	ana_ropll_ana_fbdiv_rst_i	ana_ropll_ana_fbdiv_i	ana_ropll_ana_div_clk_en_i

Table 11-84 Fields for Register: ANA_OVRDVAL_11

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	ana_ref_ro_ana_clk_div8_en_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clk_div8_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
27	ana_ropll_ana_word_div2_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_word_div2_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
26	ana_ropll_ana_word_clk_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_word_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_ropll_ana_word_clk_div5_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_word_clk_div5_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	ana_ropll_ana_word_clk_div4_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_word_clk_div4_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
23:20	ana_ropll_ana_refdiv_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_refdiv_i[3:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
19:17	ana_ropll_ana_postdiv_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_postdiv_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
16:11	ana_ropll_ana_out_div_msb_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_out_div_msb_i[5:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:9	ana_ropll_ana_out_div_lsb_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_out_div_lsb_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8	ana_ropll_ana_fbdiv_rst_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_fbdiv_rst_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
7:1	ana_ropll_ana_fbdiv_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_fbdiv_i[6:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0xe **Volatile:** true
0	ana_ropll_ana_div_clk_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_div_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.17 ANA_OVRDVAL_12

- **Name:** RTRX Configuration Register 12
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 12. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1040

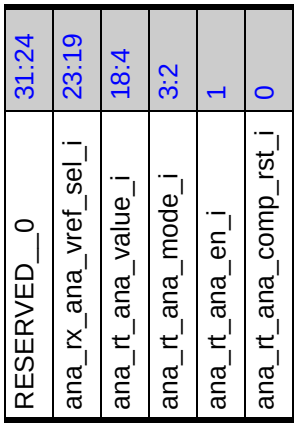


Table 11-85 Fields for Register: ANA_OVRDVAL_12

Bits	Name	Memory Access	Description
31:24	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
23:19	ana_rx_ana_vref_sel_i	R/W	Override value set on (SUP) analog port ana_rx_ana_vref_sel_i[4:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
18:4	ana_rt_ana_value_i	R/W	Override value set on (SUP) analog port ana_rt_ana_value_i[9:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3:2	ana_rt_ana_mode_i	R/W	Override value set on (SUP) analog port ana_rt_ana_mode_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
1	ana_rt_ana_en_i	R/W	Override value set on (SUP) analog port ana_rt_ana_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_rt_ana_comp_rst_i	R/W	Override value set on (SUP) analog port ana_rt_ana_comp_rst_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.18 ANA_OVRDEN_0

- **Name:** (SUP) Internal Override Enable Register 0
- **Description:** Bits that act as select signal of the mux which overrides value on (SUP) input and output ports Register 0
- **Size:** 32 bits
- **Offset:** 0x1044

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED__0	ovrd_en_ana_diffbuf_tx_en_i	ovrd_en_ana_diffbuf_pulldn_i	ovrd_en_ana_diffbuf_imp_ctrl_i	ovrd_en_ana_diffbuf_de_emp_en_i	ovrd_en_ana_diffbuf_de_emp_ctrl_i	ovrd_en_ana_diffbuf_clk_sel_i	ovrd_en_ana_cru_ana_sel_i	ovrd_en_ana_cru_ana_wr_en_i	ovrd_en_ana_cru_ana_wr_data_i	ovrd_en_ana_cru_ana_rst_i	ovrd_en_ana_cru_ana_rd_en_i	ovrd_en_ana_cru_ana_clk_i	ovrd_en_ana_vref_pre_i	ovrd_en_ana_sup_ana_term_ctrl_i	ovrd_en_ana_sup_ana_misc_i	ovrd_en_ana_nominal_vph_sel_i	ovrd_en_ana_nominal_vp_sel_i	ovrd_en_ana_bg_ana_vref_fast_start_i	ovrd_en_ana_bg_ana_kick_start_en_i	ovrd_en_ana_bg_ana_en_i	ovrd_en_ana_ana_pwrdn_i	ovrd_en_ana_ana_flyover_en_i	ovrd_en_ana_ana_burnin_i	ovrd_en_ana_alt_ana_sel_refclk_i	ovrd_en_ana_alt_ana_osc_vptx_i	ovrd_en_ana_alt_ana_osc_vph_i	ovrd_en_ana_alt_ana_osc_vp_ulvt_i	ovrd_en_ana_alt_ana_osc_vp_lvt_i	ovrd_en_ana_alt_ana_osc_pmos_i	ovrd_en_ana_alt_ana_osc_nmos_i	ovrd_en_ana_alt_ana_osc_div4_en_i

Table 11-86 Fields for Register: ANA_OVRDEN_0

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x1
30	ovrd_en_ana_diffbuf_tx_en_i	R/W	Override enable for (SUP) port ana_diffbuf_tx_en_i. **Value After Reset:** 0x1
29	ovrd_en_ana_diffbuf_pulldn_i	R/W	Override enable for (SUP) port ana_diffbuf_pulldn_i. **Value After Reset:** 0x1
28	ovrd_en_ana_diffbuf_imp_ctrl_i	R/W	Override enable for (SUP) port ana_diffbuf_imp_ctrl_i. **Value After Reset:** 0x1
27	ovrd_en_ana_diffbuf_de_emp_en_i	R/W	Override enable for (SUP) port ana_diffbuf_de_emp_en_i. **Value After Reset:** 0x1
26	ovrd_en_ana_diffbuf_de_emp_ctrl_i	R/W	Override enable for (SUP) port ana_diffbuf_de_emp_ctrl_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
25	ovrd_en_ana_diffbuf_clk_sel_i	R/W	Override enable for (SUP) port ana_diffbuf_clk_sel_i. **Value After Reset:** 0x1
24	ovrd_en_ana_cru_ana_sel_i	R/W	Override enable for (SUP) port ana_cru_ana_sel_i. **Value After Reset:** 0x1
23	ovrd_en_ana_cru_ana_wr_en_i	R/W	Override enable for (SUP) port ana_cru_ana_wr_en_i. **Value After Reset:** 0x1
22	ovrd_en_ana_cru_ana_wr_data_i	R/W	Override enable for (SUP) port ana_cru_ana_wr_data_i. **Value After Reset:** 0x1
21	ovrd_en_ana_cru_ana_rst_i	R/W	Override enable for (SUP) port ana_cru_ana_rst_i. **Value After Reset:** 0x1
20	ovrd_en_ana_cru_ana_rd_en_i	R/W	Override enable for (SUP) port ana_cru_ana_rd_en_i. **Value After Reset:** 0x1
19	ovrd_en_ana_cru_ana_clk_i	R/W	Override enable for (SUP) port ana_cru_ana_clk_i. **Value After Reset:** 0x1
18	ovrd_en_ana_vref_pre_i	R/W	Override enable for (SUP) port ana_vref_pre_i. **Value After Reset:** 0x1
17	ovrd_en_ana_sup_ana_term_ctrl_i	R/W	Override enable for (SUP) port ana_sup_ana_term_ctrl_i. **Value After Reset:** 0x1
16	ovrd_en_ana_sup_ana_misc_i	R/W	Override enable for (SUP) port ana_sup_ana_misc_i. **Value After Reset:** 0x1
15	ovrd_en_ana_nominal_vph_sel_i	R/W	Override enable for (SUP) port ana_nominal_vph_sel_i. **Value After Reset:** 0x1
14	ovrd_en_ana_nominal_vp_sel_i	R/W	Override enable for (SUP) port ana_nominal_vp_sel_i. **Value After Reset:** 0x1
13	ovrd_en_ana_bg_ana_vref_fast_start_i	R/W	Override enable for (SUP) port ana_bg_ana_vref_fast_start_i. **Value After Reset:** 0x1
12	ovrd_en_ana_bg_ana_kick_start_en_i	R/W	Override enable for (SUP) port ana_bg_ana_kick_start_en_i. **Value After Reset:** 0x1
11	ovrd_en_ana_bg_ana_en_i	R/W	Override enable for (SUP) port ana_bg_ana_en_i. **Value After Reset:** 0x1
10	ovrd_en_ana_ana_pwrdn_i	R/W	Override enable for (SUP) port ana_ana_pwrdn_i. **Value After Reset:** 0x1
9	ovrd_en_ana_ana_flyover_en_i	R/W	Override enable for (SUP) port ana_ana_flyover_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
8	ovrd_en_ana_ana_burnin_i	R/W	Override enable for (SUP) port ana_ana_burnin_i. **Value After Reset:** 0x1
7	ovrd_en_ana_alt_ana_sel_refclk_i	R/W	Override enable for (SUP) port ana_alt_ana_sel_refclk_i. **Value After Reset:** 0x1
6	ovrd_en_ana_alt_ana_osc_vptx_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_vptx_i. **Value After Reset:** 0x1
5	ovrd_en_ana_alt_ana_osc_vph_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_vph_i. **Value After Reset:** 0x1
4	ovrd_en_ana_alt_ana_osc_vp_ulvt_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_vp_ulvt_i. **Value After Reset:** 0x1
3	ovrd_en_ana_alt_ana_osc_vp_lvt_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_vp_lvt_i. **Value After Reset:** 0x1
2	ovrd_en_ana_alt_ana_osc_pmos_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_pmos_i. **Value After Reset:** 0x1
1	ovrd_en_ana_alt_ana_osc_nmos_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_nmos_i. **Value After Reset:** 0x1
0	ovrd_en_ana_alt_ana_osc_div4_en_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_div4_en_i. **Value After Reset:** 0x1

11.2.19 ANA_OVRDEN_1

- **Name:** (SUP) Internal Override Enable Register 1
- **Description:** Bits that act as select signal of the mux which overrides value on (SUP) input and output ports Register 1
- **Size:** 32 bits
- **Offset:** 0x1048

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ovrd_en_ana_ref_ana_vreg_fast_start_i	ovrd_en_ana_ref_ana_ropll_div_i	ovrd_en_ana_ref_ana_repeat_clk_sel_i	ovrd_en_ana_ref_ana_repeat_clk_en_i	ovrd_en_ana_ref_ana_mplla_div_i	ovrd_en_ana_ref_ana_diffbuf_clk_en_i	ovrd_en_ana_ref_ana_dco_range_i	ovrd_en_ana_ref_ana_dco_finetune_i	ovrd_en_ana_mplla_ana_pmix_int_i	ovrd_en_ana_mplla_ana_pmix_frac_i	ovrd_en_ana_mplla_ana_pmix_en_i	ovrd_en_ana_mplla_ana_pmix_clk_i	ovrd_en_ana_mplla_ana_word_div2_en_i	ovrd_en_ana_mplla_ana_word_clk_div5_en_i	ovrd_en_ana_mplla_ana_word_clk_div4_en_i	ovrd_en_ana_mplla_ana_tx_clk_div_i	ovrd_en_ana_mplla_ana_fdiv_en_i	ovrd_en_ana_mplla_ana_div_clk_div4_en_i	ovrd_en_ana_mplla_ana_vreg_speedup_i	ovrd_en_ana_mplla_ana_rst_i	ovrd_en_ana_mplla_ana_output_r_en_i	ovrd_en_ana_mplla_ana_output_l_en_i	ovrd_en_ana_mplla_ana_output_en_i	ovrd_en_ana_mplla_ana_gear_shift_en_i	ovrd_en_ana_mplla_ana_fb_clk_en_i	ovrd_en_ana_mplla_ana_en_i	ovrd_en_ana_mplla_ana_vco_gain_cal_i	ovrd_en_ana_mplla_ana_fine_tune_i	ovrd_en_ana_mplla_ana_ctl_buf_bypass_i	ovrd_en_ana_mplla_ana_coarse_tune_i	ovrd_en_ana_mplla_ana_cal_i	ovrd_en_ana_mplla_ana_bandwidth_i

Table 11-87 Fields for Register: ANA_OVRDEN_1

Bits	Name	Memory Access	Description
31	ovrd_en_ana_ref_ana_vreg_fast_start_i	R/W	Override enable for (SUP) port ana_ref_ana_vreg_fast_start_i. **Value After Reset:** 0x1
30	ovrd_en_ana_ref_ana_ropll_div_i	R/W	Override enable for (SUP) port ana_ref_ana_ropll_div_i. **Value After Reset:** 0x1
29	ovrd_en_ana_ref_ana_repeat_clk_sel_i	R/W	Override enable for (SUP) port ana_ref_ana_repeat_clk_sel_i. **Value After Reset:** 0x1
28	ovrd_en_ana_ref_ana_repeat_clk_en_i	R/W	Override enable for (SUP) port ana_ref_ana_repeat_clk_en_i. **Value After Reset:** 0x1
27	ovrd_en_ana_ref_ana_mplla_div_i	R/W	Override enable for (SUP) port ana_ref_ana_mplla_div_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
26	ovrd_en_ana_ref_ana_diffbuf_clk_en_i	R/W	Override enable for (SUP) port ana_ref_ana_diffbuf_clk_en_i. **Value After Reset:** 0x1
25	ovrd_en_ana_ref_ana_dco_range_i	R/W	Override enable for (SUP) port ana_ref_ana_dco_range_i. **Value After Reset:** 0x1
24	ovrd_en_ana_ref_ana_dco_finetune_i	R/W	Override enable for (SUP) port ana_ref_ana_dco_finetune_i. **Value After Reset:** 0x1
23	ovrd_en_ana_mplla_ana_pmix_int_i	R/W	Override enable for (SUP) port ana_mplla_ana_pmix_int_i. **Value After Reset:** 0x1
22	ovrd_en_ana_mplla_ana_pmix_frac_i	R/W	Override enable for (SUP) port ana_mplla_ana_pmix_frac_i. **Value After Reset:** 0x1
21	ovrd_en_ana_mplla_ana_pmix_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_pmix_en_i. **Value After Reset:** 0x1
20	ovrd_en_ana_mplla_ana_pmix_clk_i	R/W	Override enable for (SUP) port ana_mplla_ana_pmix_clk_i. **Value After Reset:** 0x1
19	ovrd_en_ana_mplla_ana_word_div2_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_word_div2_en_i. **Value After Reset:** 0x1
18	ovrd_en_ana_mplla_ana_word_clk_div5_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_word_clk_div5_en_i. **Value After Reset:** 0x1
17	ovrd_en_ana_mplla_ana_word_clk_div4_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_word_clk_div4_en_i. **Value After Reset:** 0x1
16	ovrd_en_ana_mplla_ana_tx_clk_div_i	R/W	Override enable for (SUP) port ana_mplla_ana_tx_clk_div_i. **Value After Reset:** 0x1
15	ovrd_en_ana_mplla_ana_fdiv_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_fdiv_en_i. **Value After Reset:** 0x1
14	ovrd_en_ana_mplla_ana_div_clk_div4_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_div_clk_div4_en_i. **Value After Reset:** 0x1
13	ovrd_en_ana_mplla_ana_vreg_speedup_i	R/W	Override enable for (SUP) port ana_mplla_ana_vreg_speedup_i. **Value After Reset:** 0x1
12	ovrd_en_ana_mplla_ana_rst_i	R/W	Override enable for (SUP) port ana_mplla_ana_rst_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
11	ovrd_en_ana_mplla_ana_output_r_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_output_r_en_i. **Value After Reset:** 0x1
10	ovrd_en_ana_mplla_ana_output_l_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_output_l_en_i. **Value After Reset:** 0x1
9	ovrd_en_ana_mplla_ana_output_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_output_en_i. **Value After Reset:** 0x1
8	ovrd_en_ana_mplla_ana_gear_shift_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_gear_shift_en_i. **Value After Reset:** 0x1
7	ovrd_en_ana_mplla_ana_fb_clk_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_fb_clk_en_i. **Value After Reset:** 0x1
6	ovrd_en_ana_mplla_ana_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_en_i. **Value After Reset:** 0x1
5	ovrd_en_ana_mplla_ana_vco_gain_cal_i	R/W	Override enable for (SUP) port ana_mplla_ana_vco_gain_cal_i. **Value After Reset:** 0x1
4	ovrd_en_ana_mplla_ana_fine_tune_i	R/W	Override enable for (SUP) port ana_mplla_ana_fine_tune_i. **Value After Reset:** 0x1
3	ovrd_en_ana_mplla_ana_ctl_buf_bypass_i	R/W	Override enable for (SUP) port ana_mplla_ana_ctl_buf_bypass_i. **Value After Reset:** 0x1
2	ovrd_en_ana_mplla_ana_coarse_tune_i	R/W	Override enable for (SUP) port ana_mplla_ana_coarse_tune_i. **Value After Reset:** 0x1
1	ovrd_en_ana_mplla_ana_cal_i	R/W	Override enable for (SUP) port ana_mplla_ana_cal_i. **Value After Reset:** 0x1
0	ovrd_en_ana_mplla_ana_bandwidth_i	R/W	Override enable for (SUP) port ana_mplla_ana_bandwidth_i. **Value After Reset:** 0x1

11.2.20 ANA_OVRDEN_2

- **Name:** (SUP) Internal Override Enable Register 2
- **Description:** Bits that act as select signal of the mux which overrides value on (SUP) input and output ports Register 2
- **Size:** 32 bits
- **Offset:** 0x104c

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ovrd_en_ana_rt_ana_comp_rst_i	ovrd_en_ana_ropll_ana_word_div2_en_i	ovrd_en_ana_ropll_ana_word_clk_en_i	ovrd_en_ana_ropll_ana_word_clk_div5_en_i	ovrd_en_ana_ropll_ana_word_clk_div4_en_i	ovrd_en_ana_ropll_ana_refdiv_i	ovrd_en_ana_ropll_ana_postdiv_i	ovrd_en_ana_ropll_ana_out_div_msb_i	ovrd_en_ana_ropll_ana_out_div_lsb_i	ovrd_en_ana_ropll_ana_fbdiv_rst_i	ovrd_en_ana_ropll_ana_fbdiv_i	ovrd_en_ana_ropll_ana_div_clk_en_i	ovrd_en_ana_ropll_ana_vreg_fast_start_i	ovrd_en_ana_ropll_ana_vco_low_freq_i	ovrd_en_ana_ropll_ana_v2i_mode_i	ovrd_en_ana_ropll_ana_refsel_i	ovrd_en_ana_ropll_ana_rc_filter_i	ovrd_en_ana_ropll_ana_preset_i	ovrd_en_ana_ropll_ana_gearshift_i	ovrd_en_ana_ropll_ana_en_i	ovrd_en_ana_ropll_ana_cp_ctrl_prop_i	ovrd_en_ana_ropll_ana_cp_ctrl_int_i	ovrd_en_ana_ropll_ana_bypass_i	ovrd_en_ana_refa_ana_lane_clk_en_i	ovrd_en_ana_refa_ana_clkdet_en_i	ovrd_en_ana_refa_ana_clk_sel_i	ovrd_en_ana_refa_ana_clk_en_i	ovrd_en_ana_ref_ro_ana_lane_clk_en_i	ovrd_en_ana_ref_ro_ana_clkdet_en_i	ovrd_en_ana_ref_ro_ana_clk_sel_i	ovrd_en_ana_ref_ro_ana_clk_en_i	ovrd_en_ana_ref_ana_vreg_ref_sel_i

Table 11-88 Fields for Register: ANA_OVRDEN_2

Bits	Name	Memory Access	Description
31	ovrd_en_ana_rt_ana_comp_rst_i	R/W	Override enable for (SUP) port ana_rt_ana_comp_rst_i. **Value After Reset:** 0x1
30	ovrd_en_ana_ropll_ana_word_div2_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_word_div2_en_i. **Value After Reset:** 0x1
29	ovrd_en_ana_ropll_ana_word_clk_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_word_clk_en_i. **Value After Reset:** 0x1
28	ovrd_en_ana_ropll_ana_word_clk_div5_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_word_clk_div5_en_i. **Value After Reset:** 0x1
27	ovrd_en_ana_ropll_ana_word_clk_div4_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_word_clk_div4_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
26	ovrd_en_ana_ropll_ana_refdiv_i	R/W	Override enable for (SUP) port ana_ropll_ana_refdiv_i. **Value After Reset:** 0x1
25	ovrd_en_ana_ropll_ana_postdiv_i	R/W	Override enable for (SUP) port ana_ropll_ana_postdiv_i. **Value After Reset:** 0x1
24	ovrd_en_ana_ropll_ana_out_div_msb_i	R/W	Override enable for (SUP) port ana_ropll_ana_out_div_msb_i. **Value After Reset:** 0x1
23	ovrd_en_ana_ropll_ana_out_div_lsb_i	R/W	Override enable for (SUP) port ana_ropll_ana_out_div_lsb_i. **Value After Reset:** 0x1
22	ovrd_en_ana_ropll_ana_fbdiv_rst_i	R/W	Override enable for (SUP) port ana_ropll_ana_fbdiv_rst_i. **Value After Reset:** 0x1
21	ovrd_en_ana_ropll_ana_fbdiv_i	R/W	Override enable for (SUP) port ana_ropll_ana_fbdiv_i. **Value After Reset:** 0x1
20	ovrd_en_ana_ropll_ana_div_clk_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_div_clk_en_i. **Value After Reset:** 0x1
19	ovrd_en_ana_ropll_ana_vreg_fast_start_i	R/W	Override enable for (SUP) port ana_ropll_ana_vreg_fast_start_i. **Value After Reset:** 0x1
18	ovrd_en_ana_ropll_ana_vco_low_freq_i	R/W	Override enable for (SUP) port ana_ropll_ana_vco_low_freq_i. **Value After Reset:** 0x1
17	ovrd_en_ana_ropll_ana_v2i_mode_i	R/W	Override enable for (SUP) port ana_ropll_ana_v2i_mode_i. **Value After Reset:** 0x1
16	ovrd_en_ana_ropll_ana_refsel_i	R/W	Override enable for (SUP) port ana_ropll_ana_refsel_i. **Value After Reset:** 0x1
15	ovrd_en_ana_ropll_ana_rc_filter_i	R/W	Override enable for (SUP) port ana_ropll_ana_rc_filter_i. **Value After Reset:** 0x1
14	ovrd_en_ana_ropll_ana_preset_i	R/W	Override enable for (SUP) port ana_ropll_ana_preset_i. **Value After Reset:** 0x1
13	ovrd_en_ana_ropll_ana_gearshift_i	R/W	Override enable for (SUP) port ana_ropll_ana_gearshift_i. **Value After Reset:** 0x1
12	ovrd_en_ana_ropll_ana_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_en_i. **Value After Reset:** 0x1
11	ovrd_en_ana_ropll_ana_cp_ctrl_prop_i	R/W	Override enable for (SUP) port ana_ropll_ana_cp_ctrl_prop_i. **Value After Reset:** 0x1
10	ovrd_en_ana_ropll_ana_cp_ctrl_int_i	R/W	Override enable for (SUP) port ana_ropll_ana_cp_ctrl_int_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
9	ovrd_en_ana_ropll_ana_bypass_i	R/W	Override enable for (SUP) port ana_ropll_ana_bypass_i. **Value After Reset:** 0x1
8	ovrd_en_ana_refa_ana_lane_clk_en_i	R/W	Override enable for (SUP) port ana_refa_ana_lane_clk_en_i. **Value After Reset:** 0x1
7	ovrd_en_ana_refa_ana_clkdet_en_i	R/W	Override enable for (SUP) port ana_refa_ana_clkdet_en_i. **Value After Reset:** 0x1
6	ovrd_en_ana_refa_ana_clk_sel_i	R/W	Override enable for (SUP) port ana_refa_ana_clk_sel_i. **Value After Reset:** 0x1
5	ovrd_en_ana_refa_ana_clk_en_i	R/W	Override enable for (SUP) port ana_refa_ana_clk_en_i. **Value After Reset:** 0x1
4	ovrd_en_ana_ref_ro_ana_lane_clk_en_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_lane_clk_en_i. **Value After Reset:** 0x1
3	ovrd_en_ana_ref_ro_ana_clkdet_en_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_clkdet_en_i. **Value After Reset:** 0x1
2	ovrd_en_ana_ref_ro_ana_clk_sel_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_clk_sel_i. **Value After Reset:** 0x1
1	ovrd_en_ana_ref_ro_ana_clk_en_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_clk_en_i. **Value After Reset:** 0x1
0	ovrd_en_ana_ref_ana_vreg_ref_sel_i	R/W	Override enable for (SUP) port ana_ref_ana_vreg_ref_sel_i. **Value After Reset:** 0x1

11.2.21 ANA_OVRDEN_3

- **Name:** (SUP) Internal Override Enable Register 3
- **Description:** Bits that act as select signal of the mux which overrides value on (SUP) input and output ports Register 3
- **Size:** 32 bits
- **Offset:** 0x1050

31:19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED__2	RESERVED__1	RESERVED__0	ovrd_en_ana_sup_ana_spare_in	ovrd_en_ana_atb_ana_sup_en_i	ovrd_en_ana_atb_ana_lane_right_en_i	ovrd_en_ana_atb_ana_lane_left_en_i	ovrd_en_ana_rt_ana_comp_result_o	ovrd_en_ana_ropll_ana_cru_rd_data_o	ovrd_en_ana_refa_ana_clkdet_result_o	ovrd_en_ana_refa_ana_clk_ack_o	ovrd_en_ana_ref_ro_ana_clkdet_result_o	ovrd_en_ana_ref_ro_ana_clk_ack_o	ovrd_en_ana_mplla_ana_cru_rd_data_o	ovrd_en_ana_sup_ana_cru_rd_data_o	ovrd_en_ana_ref_ro_ana_clk_div8_en_i	ovrd_en_ana_rx_ana_vref_sel_i	ovrd_en_ana_rt_ana_value_i	ovrd_en_ana_rt_ana_mode_i	ovrd_en_ana_rt_ana_en_i

Table 11-89 Fields for Register: ANA_OVRDEN_3

Bits	Name	Memory Access	Description
31:19	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
18	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
17	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
16	ovrd_en_ana_sup_ana_spare_in	R/W	Override enable for (SUP) port ana_spare_i **Value After Reset:** 0x0
15	ovrd_en_ana_atb_ana_sup_en_i	R/W	Override enable for (SUP) port ana_atb_ana_sup_en_i. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
14	ovrd_en_ana_atb_ana_lane_right_en_i	R/W	Override enable for (SUP) port ana_atb_ana_lane_right_en_i. **Value After Reset:** 0x0
13	ovrd_en_ana_atb_ana_lane_left_en_i	R/W	Override enable for (SUP) port ana_atb_ana_lane_left_en_i. **Value After Reset:** 0x0
12	ovrd_en_ana_rt_ana_comp_result_o	R/W	Override enable for (SUP) port ana_rt_ana_comp_result_o. **Value After Reset:** 0x0
11	ovrd_en_ana_ropll_ana_cru_rd_data_o	R/W	Override enable for (SUP) port ana_ropll_ana_cru_rd_data_o. **Value After Reset:** 0x0
10	ovrd_en_ana_refa_ana_clkdet_result_o	R/W	Override enable for (SUP) port ana_refa_ana_clkdet_result_o. **Value After Reset:** 0x0
9	ovrd_en_ana_refa_ana_clk_ack_o	R/W	Override enable for (SUP) port ana_refa_ana_clk_ack_o. **Value After Reset:** 0x0
8	ovrd_en_ana_ref_ro_ana_clkdet_result_o	R/W	Override enable for (SUP) port ana_ref_ro_ana_clkdet_result_o. **Value After Reset:** 0x0
7	ovrd_en_ana_ref_ro_ana_clk_ack_o	R/W	Override enable for (SUP) port ana_ref_ro_ana_clk_ack_o. **Value After Reset:** 0x0
6	ovrd_en_ana_mplla_ana_cru_rd_data_o	R/W	Override enable for (SUP) port ana_mplla_ana_cru_rd_data_o. **Value After Reset:** 0x0
5	ovrd_en_ana_sup_ana_cru_rd_data_o	R/W	Override enable for (SUP) port ana_sup_ana_cru_rd_data_o. **Value After Reset:** 0x0
4	ovrd_en_ana_ref_ro_ana_clk_div8_en_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_clk_div8_en_i. **Value After Reset:** 0x1
3	ovrd_en_ana_rx_ana_vref_sel_i	R/W	Override enable for (SUP) port ana_rx_ana_vref_sel_i. **Value After Reset:** 0x1
2	ovrd_en_ana_rt_ana_value_i	R/W	Override enable for (SUP) port ana_rt_ana_value_i. **Value After Reset:** 0x1
1	ovrd_en_ana_rt_ana_mode_i	R/W	Override enable for (SUP) port ana_rt_ana_mode_i. **Value After Reset:** 0x1
0	ovrd_en_ana_rt_ana_en_i	R/W	Override enable for (SUP) port ana_rt_ana_en_i. **Value After Reset:** 0x1

11.2.22 ANA_OVRDVAL_13

- **Name:** SUP ANA Output Register 13
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 13. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1054

31	30	29	28	27:20	19	18	17	16	15:8	7:0
RESERVED__2	RESERVED__1	RESERVED__0	ana_rt_ana_comp_result_o	ana_ropll_ana_cru_rd_data_o	ana_refa_ana_clkdet_result_o	ana_refa_ana_clk_ack_o	ana_ref_ro_ana_clkdet_result_o	ana_ref_ro_ana_clk_ack_o	ana_mplla_ana_cru_rd_data_o	ana_sup_ana_cru_rd_data_o

Table 11-90 Fields for Register: ANA_OVRDVAL_13

Bits	Name	Memory Access	Description
31	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	ana_rt_ana_comp_result_o	R/W	Override value set on (SUP) analog port ana_rt_ana_comp_result_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
27:20	ana_ropll_ana_cru_rd_data_o	R/W	Override value set on (SUP) analog port ana_ropll_ana_cru_rd_data_o[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
19	ana_refa_ana_clkdet_result_o	R/W	Override value set on (SUP) analog port ana_refa_ana_clkdet_result_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
18	ana_refa_ana_clk_ack_o	R/W	Override value set on (SUP) analog port ana_refa_ana_clk_ack_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
17	ana_ref_ro_ana_clkdet_result_o	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clkdet_result_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_ref_ro_ana_clk_ack_o	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clk_ack_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:8	ana_mplla_ana_cru_rd_data_o	R/W	Override value set on (SUP) analog port ana_mplla_ana_cru_rd_data_o[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7:0	ana_sup_ana_cru_rd_data_o	R/W	Override value set on (SUP) analog port ana_sup_ana_cru_rd_data_o[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.23 ANA_OVRDVAL_14

- **Name:** SUP ANA Output Register 14
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 14. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x1058

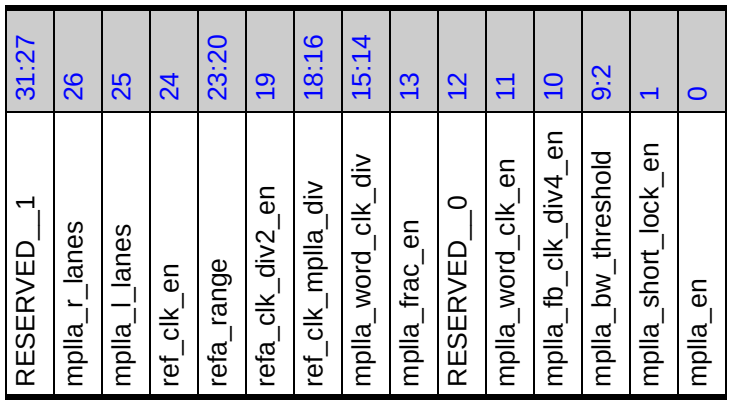


Table 11-91 Fields for Register: ANA_OVRDVAL_14

Bits	Name	Memory Access	Description
31:0	ana_sup_ana_spare_in	R/W	Override value set on (SUP) analog port sup_spare_in. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.2.24 SUP_CFG_1

- **Name:** (SUP) Configuration Register 1
- **Description:** Configures (SUP) settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x105c



31:27	26	25	24	23:20	19	18:16	15:14	13	12	11	10	9:2	1	0
RESERVED__1	mplla_r_lanes	mplla_l_lanes	ref_clk_en	refa_range	refa_clk_div2_en	ref_clk_mplla_div	mplla_word_clk_div	mplla_frac_en	RESERVED__0	mplla_word_clk_en	mplla_fb_clk_div4_en	mplla_bw_threshold	mplla_short_lock_en	mplla_en

Table 11-92 Fields for Register: SUP_CFG_1

Bits	Name	Memory Access	Description
31:27	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
26	mplla_r_lanes	R/W	Enables clock distribution to the right lanes of the PHY, for both TX and RX. **Value After Reset:** 0x0
25	mplla_l_lanes	R/W	Enables clock distribution to the left lanes of the PHY, for both TX and RX. **Value After Reset:** 0x0
24	ref_clk_en	R/W	Enables ref_clk distribution. **Value After Reset:** 0x0
23:20	refa_range	R/W	Configurability to divide ref_dig_clk to generate ref_range_clk. **Value After Reset:** 0x0
19	refa_clk_div2_en	R/W	Multiplies ref_dig_clk by 2 to generate ref_dig_clk_div. **Value After Reset:** 0x0
18:16	ref_clk_mplla_div	R/W	Divides ref_dig_clk by 2 power ref_clk_mplla_div to generate ref_dig_clk_div. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
15:14	mplla_word_clk_div	R/W	Sets MPLL clock divider settings. **Value After Reset:** 0x0
13	mplla_frac_en	R/W	Enables the fractional divider sub-block **Value After Reset:** 0x0
12	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
11	mplla_word_clk_en	R/W	Enables output clock divider settings. **Value After Reset:** 0x0
10	mplla_fb_clk_div4_en	R/W	When enabled it divides fb clock by 4. **Value After Reset:** 0x0
9:2	mplla_bw_threshold	R/W	Threshold for vco_fb_cnt_diff to drive the output bandwidth. **Value After Reset:** 0x0
1	mplla_short_lock_en	R/W	Reduces MPLL startup time. **Value After Reset:** 0x0
0	mplla_en	R/W	Enables the analog mpll. **Value After Reset:** 0x0

11.2.25 SUP_CFG_2

- **Name:** (SUP) Configuration Register 2
- **Description:** Configures (SUP) settings. Register 2.
- **Size:** 32 bits
- **Offset:** 0x1060

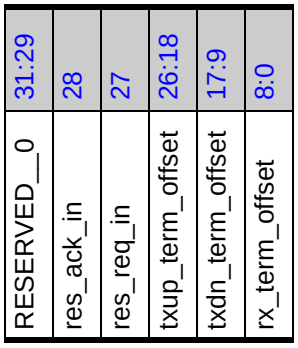


Table 11-93 Fields for Register: SUP_CFG_2

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	res_ack_in	R/W	Input acknowledgment used in resistance sharing. ■ Several PHY instances may be daisy chained in this manner to allow sharing a single reference resistor. ■ The first PHY its res_req_in tied low, while the last PHY must have its res_ack_in tied to its res_req_out. ■ The res_ack lines must be connected in reverse order with respect to the connections of the res_req lines. **Value After Reset:** 0x0
27	res_req_in	R/W	Input request used in resistance sharing. **Value After Reset:** 0x0
26:18	txup_term_offset	R/W	txup rtune offset. **Value After Reset:** 0x0
17:9	txdn_term_offset	R/W	txdn rtune offset. **Value After Reset:** 0x0
8:0	rx_term_offset	R/W	RX rtune offset. **Value After Reset:** 0x0

11.2.26 SUP_CFG_3

- **Name:** (SUP) Configuration Register 3
- **Description:** Configures (SUP) settings. Register 3.
- **Size:** 32 bits
- **Offset:** 0x1064

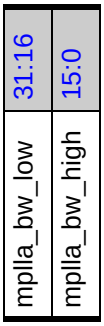


Table 11-94 Fields for Register: SUP_CFG_3

Bits	Name	Memory Access	Description
31:16	mplla_bw_low	R/W	Low bandwidth. **Value After Reset:** 0x0
15:0	mplla_bw_high	R/W	High bandwitdh. **Value After Reset:** 0x0

11.2.27 SUP_CFG_4

- **Name:** (SUP) Configuration Register 4
- **Description:** Configures (SUP) settings. Register 4.
- **Size:** 32 bits
- **Offset:** 0x1068

31	30	29	28	27	26:22	21	20:17	16	15:12	11	10	9	8:1	0
ovrd_sel	mpll_fb_dig_clk_en	mpll_pclk_en	fast_mpll_pwrup	fast_mpll_lock	dtb_sel	mpll_extbw_sel	mpll_extfine_tune	mpll_refclk_gating_dis	chkfrq_ld_cnt	mpll_skipcal	mpll_extcal	mpll_extchkfrq_en	mpll_extcoarse_tune	mpll_extcal_done

Table 11-95 Fields for Register: SUP_CFG_4

Bits	Name	Memory Access	Description
31	ovrd_sel	R/W	Override enable bit. **Value After Reset:** 0x0
30	mpll_fb_dig_clk_en	R/W	Overrides the PWR FSM mpll_fb_dig_clk_en signal. **Value After Reset:** 0x0
29	mpll_pclk_en	R/W	Overrides the PWR FSM mpll_pclk_en signal. **Value After Reset:** 0x0
28	fast_mpll_pwrup	R/W	Enables fast MPLL powerup. **Value After Reset:** 0x0
27	fast_mpll_lock	R/W	Enables fast MPLL locking. **Value After Reset:** 0x0
26:22	dtb_sel	R/W	DTB select for MPLL dtb signals. **Value After Reset:** 0x0
21	mpll_extbw_sel	R/W	Value of mpll_ana_fine_tune_i in external calibration mode. **Value After Reset:** 0x0
20:17	mpll_extfine_tune	R/W	Value of mpll_bw_sel in external calibration mode. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
16	mpll_refclk_gating_dis	R/W	Disables waiting for the IDLE state to deassert ref_ana_clk_en . **Value After Reset:** 0x0
15:12	chkfrq_ld_cnt	R/W	MSBs for the CHKFRQ FSM ld_val load value. **Value After Reset:** 0x0
11	mpll_skipcal	R/W	This bit skips automatic (internal) calibration of MPLL then MPLL_SKIPCAL_COARSE_TUNE is used. **Value After Reset:** 0x0
10	mpll_extcal	R/W	Enables external calibration of MPLL. **Value After Reset:** 0x0
9	mpll_extchkfrq_en	R/W	Checks the frequency of the MPLL. **Value After Reset:** 0x0
8:1	mpll_extcoarse_tune	R/W	Value of mpll_ana_coarse_tune_i in external calibration mode. **Value After Reset:** 0x0
0	mpll_extcal_done	R/W	Sets the external calibration status to done. **Value After Reset:** 0x0

11.2.28 SUP_CFG_5

- **Name:** (SUP) Configuration Register 5
- **Description:** Configures (SUP) settings. Register 5.
- **Size:** 32 bits
- **Offset:** 0x106c

31:28	27:16	15:14	13:12	11	10	9:8	7:2	1	0
RESERVED__0	chkfrq_ld_val	mpll_gain_cal_val0	mpll_gain_cal_val1	mpll_skip_gain_cal	mpll_bw_sel_invert	mpll_fine_start	mpll_vco_drift_thresh	mpll_lock_det_en	mpll_state_disable

Table 11-96 Fields for Register: SUP_CFG_5

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27:16	chkfrq_ld_val	R/W	This field is the MSBs for the CHKFRQ FSM ld_val 10:0 load value. **Value After Reset:** 0x0
15:14	mpll_gain_cal_val0	R/W	This is the value of vco_gain_cal = 2'b01. **Value After Reset:** 0x0
13:12	mpll_gain_cal_val1	R/W	This is the value of vco_gain_cal = 2'b10. **Value After Reset:** 0x0
11	mpll_skip_gain_cal	R/W	This bit skips gain calibration of MPLL. **Value After Reset:** 0x0
10	mpll_bw_sel_invert	R/W	Selects low bandwidth inversion. **Value After Reset:** 0x0
9:8	mpll_fine_start	R/W	This field is the value for the starting point of the MPLL fine bank. **Value After Reset:** 0x0
7:2	mpll_vco_drift_thresh	R/W	FB clock drift threshold. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
1	mpll_lock_det_en	R/W	Enables continuous MPLL lock detection. **Value After Reset:** 0x0
0	mpll_state_disable	R/W	Enables de-asserting mpll_state when MPLL VCO drift away. **Value After Reset:** 0x0

11.2.29 SUP_CFG_6

- **Name:** (SUP) Configuration Register 6
- **Description:** Configures (SUP) settings. Register 6.
- **Size:** 32 bits
- **Offset:** 0x1070

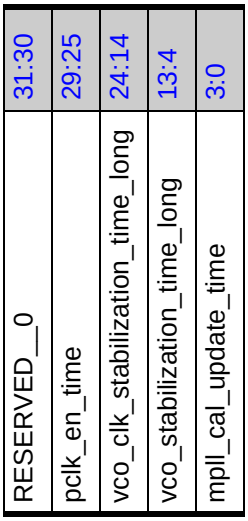


Table 11-97 Fields for Register: SUP_CFG_6

Bits	Name	Memory Access	Description
31:30	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29:25	pclk_en_time	R/W	Threshold for the PCLK enable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
24:14	vco_clk_stabilization_time_long	R/W	Threshold for the VCO clock stabilization timer in terms of number of ref_range_clk cycles. Default value is 3us. **Value After Reset:** 0x0
13:4	vco_stabilization_time_long	R/W	Threshold for the VCO stabilization timer in terms of number of ref_range_clk cycles. Default value is 12us. **Value After Reset:** 0x0
3:0	mpll_cal_update_time	R/W	Threshold for the MPLL calibration control word update timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0

11.2.30 SUP_CFG_7

- **Name:** (SUP) Configuration Register 7
- **Description:** Configures (SUP) settings. Register 7.
- **Size:** 32 bits
- **Offset:** 0x1074

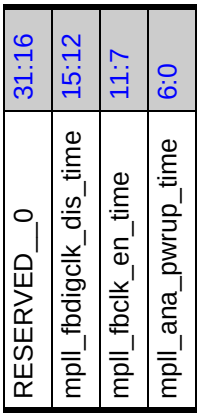


Table 11-98 Fields for Register: SUP_CFG_7

Bits	Name	Memory Access	Description
31:16	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
15:12	mpll_fbdigclk_dis_time	R/W	Threshold for the MPLL feedback digital clock disable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
11:7	mpll_fbclk_en_time	R/W	Threshold for the MPLL feedback digital clock enable timer in terms of number of mpll_fb_clk cycles. **Value After Reset:** 0x0
6:0	mpll_ana_pwrup_time	R/W	Threshold for the MPLL analog power up timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0

11.2.31 SUP_CFG_8

- **Name:** (SUP) Configuration Register 8
- **Description:** Configures (SUP) settings. Register 8.
- **Size:** 32 bits
- **Offset:** 0x1078

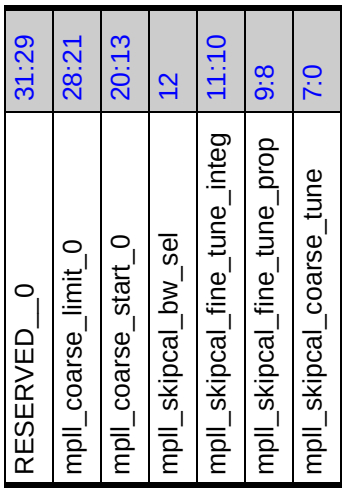


Table 11-99 Fields for Register: SUP_CFG_8

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28:21	mpll_coarse_limit_0	R/W	Value for the ending point of the MPLL coarse range 0. **Value After Reset:** 0x0
20:13	mpll_coarse_start_0	R/W	Value for the starting point of the MPLL coarse range 0. **Value After Reset:** 0x0
12	mpll_skipcal_bw_sel	R/W	This bit is the value for MPLL bw_sel when calibration or gain calibration is skipped. **Value After Reset:** 0x0
11:10	mpll_skipcal_fine_tune_integ	R/W	Value for MPLL fine_tune (integral portion) when calibration is skipped. **Value After Reset:** 0x0
9:8	mpll_skipcal_fine_tune_prop	R/W	Value for MPLL fine_tune (proportional portion) when calibration is skipped. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
7:0	mpll_skipcal_coarse_tune	R/W	Value for MPLL coarse_tune when calibration is skipped. **Value After Reset:** 0x0

11.2.32 SUP_CFG_9

- **Name:** (SUP) Configuration Register 9
- **Description:** Configures (SUP) settings. Register 9.
- **Size:** 32 bits
- **Offset:** 0x107c

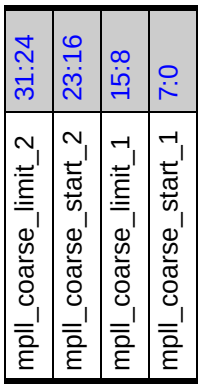


Table 11-100 Fields for Register: SUP_CFG_9

Bits	Name	Memory Access	Description
31:24	mpll_coarse_limit_2	R/W	Value for the ending point of the MPLL coarse range 2. **Value After Reset:** 0x0
23:16	mpll_coarse_start_2	R/W	Value for the starting point of the MPLL coarse range 2. **Value After Reset:** 0x0
15:8	mpll_coarse_limit_1	R/W	Value for the ending point of the MPLL coarse range 1. **Value After Reset:** 0x0
7:0	mpll_coarse_start_1	R/W	Value for the starting point of the MPLL coarse range 1. **Value After Reset:** 0x0

11.2.33 SUP_CFG_10

- **Name:** (SUP) Configuration Register 10
- **Description:** Configures (SUP) settings. Register 10.
- **Size:** 32 bits
- **Offset:** 0x1080

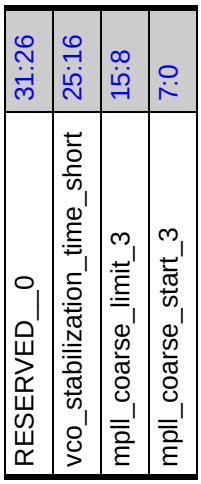


Table 11-101 Fields for Register: SUP_CFG_10

Bits	Name	Memory Access	Description
31:26	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
25:16	vco_stabilization_time_short	R/W	Threshold for the VCO clock stabilization short timer in terms of number of ref_range_clk cycles. Default value is 35us. **Value After Reset:** 0x0
15:8	mpll_coarse_limit_3	R/W	Value for the ending point of the MPLL coarse range 3. **Value After Reset:** 0x0
7:0	mpll_coarse_start_3	R/W	Value for the starting point of the MPLL coarse range 3. **Value After Reset:** 0x0

11.2.34 SUP_CFG_11

- **Name:** (SUP) Configuration Register 11
- **Description:** Configures (SUP) settings. Register 11.
- **Size:** 32 bits
- **Offset:** 0x1084

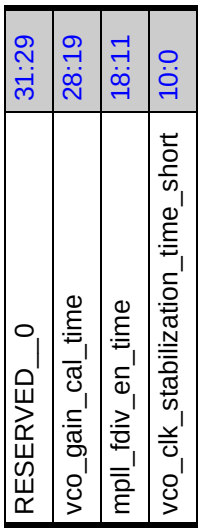


Table 11-102 Fields for Register: SUP_CFG_11

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28:19	vco_gain_cal_time	R/W	Threshold for the MPLL VCO control voltage settling timer in terms of number of ref_range_clk cycles. Default value is 10us. **Value After Reset:** 0x0
18:11	mpll_fdiv_en_time	R/W	Threshold for the MPLL fdiv enable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
10:0	vco_clk_stabilization_time_short	R/W	Threshold for the VCO stabilization timer in terms of number of ref_range_clk cycles. Default value is 4us. **Value After Reset:** 0x0

11.2.35 MPLL_STAT_0

- **Name:** MPLL Configuration Register 0
- **Description:** Configures MPLL settings. Register 0.
- **Size:** 32 bits
- **Offset:** 0x1088

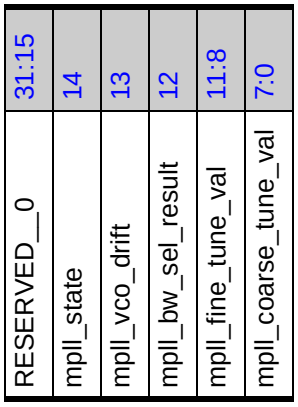


Table 11-103 Fields for Register: MPLL_STAT_0

Bits	Name	Memory Access	Description
31:15	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
14	mpll_state	R	Current value of mpll_state. **Value After Reset:** 0x0
13	mpll_vco_drift	R	This bit indicates if the MPLL lost LOCK while in the ON state. **Value After Reset:** 0x0
12	mpll_bw_sel_result	R	Current value of mpll_bw_sel_result. **Value After Reset:** 0x0
11:8	mpll_fine_tune_val	R	Current value of mpll_ana_fine_tune_i. **Value After Reset:** 0x0
7:0	mpll_coarse_tune_val	R	Current value of mpll_ana_coarse_tune_i. **Value After Reset:** 0x0

11.2.36 MPLL_STAT_1

- **Name:** MPLL Configuration Register 1
- **Description:** Configures MPLL settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x108c

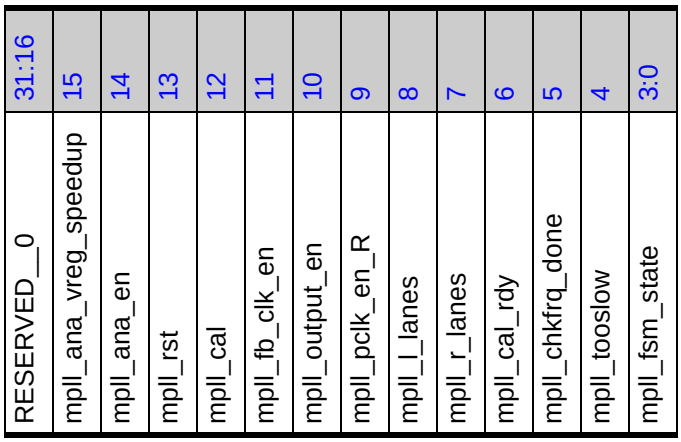


Table 11-104 Fields for Register: MPLL_STAT_1

Bits	Name	Memory Access	Description
31:16	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
15	mpll_ana_vreg_speedup	R	Current value of mpll_ana_vreg_speedup_i. **Value After Reset:** 0x0
14	mpll_ana_en	R	Current value of mpll_ana_en_i. **Value After Reset:** 0x0
13	mpll_rst	R	Current value of mpll_ana_rst_i. **Value After Reset:** 0x0
12	mpll_cal	R	Current value of mpll_ana_cal_i. **Value After Reset:** 0x0
11	mpll_fb_clk_en	R	Current value of mpll_ana_fb_clk_en_i. **Value After Reset:** 0x0
10	mpll_output_en	R	Current value of mpll_ana_output_en_i. **Value After Reset:** 0x0
9	mpll_pclk_en_R	R	Current value of mpll_pclk_en. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
8	mpll_l_lanes	R	Current value of lane_mpll_l. **Value After Reset:** 0x0
7	mpll_r_lanes	R	Current value of lane_mpll_r. **Value After Reset:** 0x0
6	mpll_cal_rdy	R	Current value of mpll_cal_rdy. **Value After Reset:** 0x0
5	mpll_chkfrq_done	R	Current value of mpll_chkfrq_done. **Value After Reset:** 0x0
4	mpll_tooslow	R	Current value of mpll_tooslow. **Value After Reset:** 0x0
3:0	mpll_fsm_state	R	Current value of the PWR FSM state register. **Value After Reset:** 0x0

11.2.37 MPLL_STAT_2

- **Name:** MPLL Configuration Register 2
- **Description:** Configures MPLL settings. Register 2.
- **Size:** 32 bits
- **Offset:** 0x1090



Table 11-105 Fields for Register: MPLL_STAT_2

Bits	Name	Memory Access	Description
31:26	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
25:13	mpll_ana_vco_fb_cnt2	R	Current value of mpll_ana_vco_fb_cnt2. **Value After Reset:** 0x0
12:0	mpll_ana_vco_fb_cnt1	R	Current value of mpll_ana_vco_fb_cnt1. **Value After Reset:** 0x0

11.2.38 MPLL_STAT_3

- **Name:** MPLL Configuration Register 3
- **Description:** Configures MPLL settings. Register 3.
- **Size:** 32 bits
- **Offset:** 0x1094

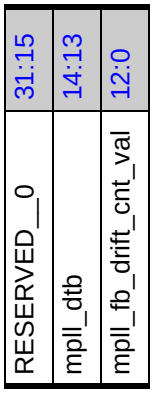


Table 11-106 Fields for Register: MPLL_STAT_3

Bits	Name	Memory Access	Description
31:15	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
14:13	mpll_dtb	R	Config of dtb mpll bus. **Value After Reset:** 0x0
12:0	mpll_fb_drift_cnt_val	R	Current value of mpll_fb_drift_cnt. **Value After Reset:** 0x0

11.2.39 SUP_CFG_12

- **Name:** (SUP) Configuration Register 12
- **Description:** Configures (SUP) settings. Register 12.
- **Size:** 32 bits
- **Offset:** 0x1098

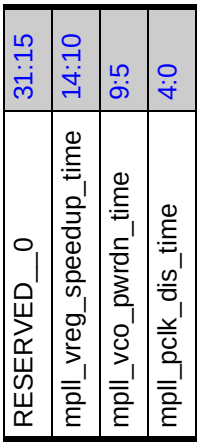


Table 11-107 Fields for Register: SUP_CFG_12

Bits	Name	Memory Access	Description
31:15	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
14:10	mpll_vreg_speedup_time	R/W	Threshold for the MPLL vreg speedup timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
9:5	mpll_vco_pwrdn_time	R/W	Threshold for the MPLL VCO power down timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
4:0	mpll_pclk_dis_time	R/W	Threshold for the PCLK disable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0

11.2.40 SUP_CFG_13

- **Name:** (SUP) Configuration Register 13
- **Description:** Configures (SUP) settings. Register 13.
- **Size:** 32 bits
- **Offset:** 0x109c

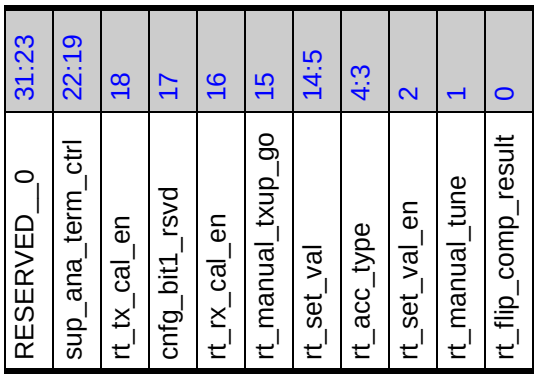


Table 11-108 Fields for Register: SUP_CFG_13

Bits	Name	Memory Access	Description
31:23	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
22:19	sup_ana_term_ctrl	R/W	This field sets the reference resistor in the analog. Value Impedance (ohm). **Values:** ■ 0x0 (_54): 54 ■ 0x1 (_52): 52 ■ 0x2 (_50): 50 ■ 0x3 (_48): 48 ■ 0x4 (_46): 46 ■ 0x5 (_44): 44 ■ 0x6 (_42): 42 ■ 0x7 (_40): (and 1xxx) 40 **Value After Reset:** 0x0
18	rt_tx_cal_en	R/W	Enables calibration of TX resistor. **Value After Reset:** 0x0
17	cnfg_bit1_rsvd	R/W	RESERVED **Value After Reset:** 0x0
16	rt_rx_cal_en	R/W	Enables calibration of RX resistor. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
15	rt_manual_txup_go	R/W	Enables TxUP tune to continue in manual tune mode, when TYPE is TxUP tune. When in non TxUP manual tune mode, this bit must be 0. When in TxUP manual tune mode and after TxUP manual tune is triggered, if the read only register SUP.RTUNE_STAT.DTB_RTUNE equals to 2'b01, it is required to first release the rtune related asynchronous override on analog/digital interface then turn this bit TXUP_GO from low to high to continue TxUP manual tune which also indicates that this bit needs to be low when triggering TxUP manual tune. **Value After Reset:** 0x0
14:5	rt_set_val	R/W	Value to use when triggering SET_VAL field: **Values:** ■ 0x0 (ADC): ADC, or read/write rt_value. ■ 0x1 (RX_TUNE): Rx tune, or read/write rx_cal_val (only 6 bits). ■ 0x2 (TX_DN): TxDN tune, or read/write txdn_cal_val (10 bits). ■ 0x3 (TX_UP): TxUP tune, or read/write txup_cal_val (10 bits) or Resref detect (no affect when triggering SET_VAL field). **Value After Reset:** 0x0
4:3	rt_acc_type	R/W	Type of manual tuning or register read/write to execute. **Value After Reset:** 0x0
2	rt_set_val_en	R/W	This bit sets value. Write 1 to manually write the register specified by the TYPE field to the value in the VALUE field. **Value After Reset:** 0x0
1	rt_manual_tune	R/W	Write 1 to perform manual tuning specified by TYPE field. Starting a manual tune while a tune is currently running can cause unpredictable results. For use only when you know what the part is doing (w.r.t. resistor tuning). **Value After Reset:** 0x0
0	rt_flip_comp_result	R/W	Invert analog comparator output. **Value After Reset:** 0x0

11.2.41 SUP_CFG_14

- **Name:** (SUP) Configuration Register 14
- **Description:** Configures (SUP) settings. Register 14.
- **Size:** 32 bits
- **Offset:** 0x10a0

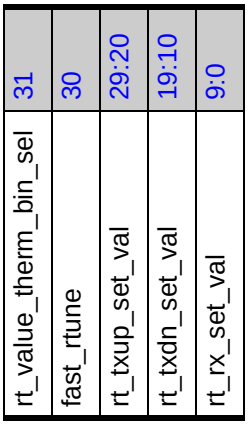


Table 11-109 Fields for Register: SUP_CFG_14

Bits	Name	Memory Access	Description
31	rt_value_therm_bin_sel	R/W	Selects between binary or themometer decoder rt_ana_value **Values:** ■ 0x0 (THM): Thermometer rt value ■ 0x1 (BIN): Binary rt value **Value After Reset:** 0x0
30	fast_rtune	R/W	Enables fast resistor tuning (simulation only). **Value After Reset:** 0x0
29:20	rt_txup_set_val	R/W	This field sets value of tx,up resistor. Writing a value to this register sets the tx,up resistor value. **Value After Reset:** 0x0
19:10	rt_txdn_set_val	R/W	This field sets value of tx,dn resistor. Writing a value to this register sets the tx,dn resistor value. **Value After Reset:** 0x0
9:0	rt_rx_set_val	R/W	This field sets value of rx resistor. Writing a value to this register sets the rx resistor value. **Value After Reset:** 0x0

11.2.42 rt_stat_0

- **Name:** rt_stat Configuration register 0
- **Description:** Configures rt_stat settings. Register 0.
- **Size:** 32 bits
- **Offset:** 0x10a4

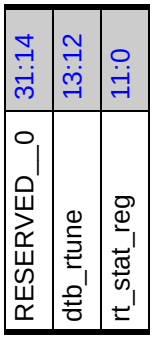


Table 11-110 Fields for Register: rt_stat_0

Bits	Name	Memory Access	Description
31:14	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
13:12	dtb_rtune	R	dtb rtune bus **Value After Reset:** 0x0
11:0	rt_stat_reg	R	Current value of the register specified by the DEBUG.TYPE field. **Value After Reset:** 0x0

11.2.43 rt_stat_1

- **Name:** rt_stat Configuration register 1
- **Description:** Configures rt_stat settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x10a8

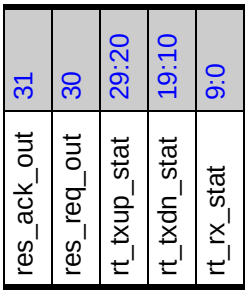


Table 11-111 Fields for Register: rt_stat_1

Bits	Name	Memory Access	Description
31	res_ack_out	R	Output acknowlegment used in resistance sharing. **Value After Reset:** 0x0
30	res_req_out	R	Output request used in resistance sharing. **Value After Reset:** 0x0
29:20	rt_txup_stat	R	Current value of the tx, up resistor tuning register. **Value After Reset:** 0x0
19:10	rt_txdn_stat	R	Current value of the tx, dn resistor tuning register. **Value After Reset:** 0x0
9:0	rt_rx_stat	R	Current value of the rx resistor tuning register. **Value After Reset:** 0x0

11.2.44 SUP_CFG_15

- **Name:** (SUP) Configuration Register 15
- **Description:** Configures (SUP) settings. Register 15.
- **Size:** 32 bits
- **Offset:** 0x10ac

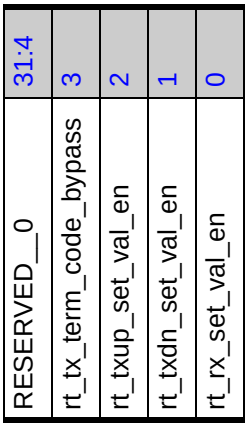


Table 11-112 Fields for Register: SUP_CFG_15

Bits	Name	Memory Access	Description
31:4	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
3	rt_tx_term_code_bypass	R/W	Bypass txup term code **Value After Reset:** 0x1
2	rt_txup_set_val_en	R/W	Input to pulse catcher to generate txup_set_val pulse. **Value After Reset:** 0x0
1	rt_txdn_set_val_en	R/W	Input to pulse catcher to generate txdn_set_val pulse. **Value After Reset:** 0x0
0	rt_rx_set_val_en	R/W	Input to pulse catcher to generate rx_set_val pulse. **Value After Reset:** 0x0

11.2.45 SUP_CFG_16

- **Name:** (SUP) Configuration Register 16
- **Description:** Configures (SUP) settings. Register 16.
- **Size:** 32 bits
- **Offset:** 0x10b0

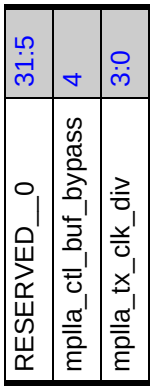


Table 11-113 Fields for Register: SUP_CFG_16

Bits	Name	Memory Access	Description
31:5	RESERVED__0	R	RESREVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	mplla_ctl_buf_bypass	R/W	Controler buffer bypass signal is connected directly to analog. **Value After Reset:** 0x0
3:0	mplla_tx_clk_div	R/W	Configurability to divide mplla clock is connected directly to analog. **Value After Reset:** 0x0

11.2.46 SUP_CFG_17

- **Name:** (SUP) Configuration Register 17
- **Description:** Configures (SUP) settings. Register 17.
- **Size:** 32 bits
- **Offset:** 0x10b4

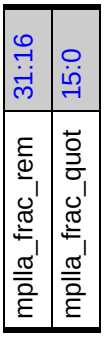


Table 11-114 Fields for Register: SUP_CFG_17

Bits	Name	Memory Access	Description
31:16	mplla_frac_rem	R/W	Control the remainder of the fractional ratio wanted **Value After Reset:** 0x0
15:0	mplla_frac_quot	R/W	Control the quotient of the fractional ratio wanted **Value After Reset:** 0x0

11.2.47 SUP_CFG_18

- **Name:** (SUP) Configuration Register 18
- **Description:** Configures (SUP) settings. Register 18.
- **Size:** 32 bits
- **Offset:** 0x10b8

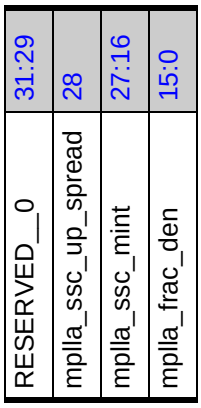


Table 11-115 Fields for Register: SUP_CFG_18

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESREVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	mplla_ssc_up_spread	R/W	Controls if the SSC profile is up (1) or down (0) spread **Value After Reset:** 0x0
27:16	mplla_ssc_mint	R/W	Control the feedback divider ratio that will be modulated by the SSC/FracN machine resulting in the Mout_int <11:0> **Value After Reset:** 0x0
15:0	mplla_frac_den	R/W	Control the denominator of the fractional ratio wanted **Value After Reset:** 0x0

11.2.48 SUP_CFG_19

- **Name:** (SUP) Configuration Register 19
- **Description:** Configures (SUP) settings. Register 19.
- **Size:** 32 bits
- **Offset:** 0x10bc

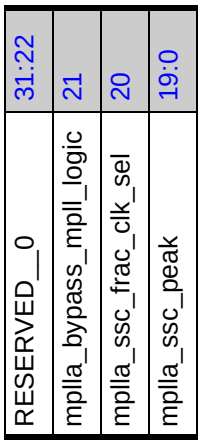


Table 11-116 Fields for Register: SUP_CFG_19

Bits	Name	Memory Access	Description
31:22	RESERVED__0	R	RESREVED **Reserved Field:** Yes **Value After Reset:** 0x0
21	mplla_bypass_mpll_logic	R/W	Bypass mint_i mapping logic **Value After Reset:** 0x0
20	mplla_ssc_frac_clk_sel	R/W	Selects between reference and feedback clock to be the ss_gen module operating clock **Value After Reset:** 0x0
19:0	mplla_ssc_peak	R/W	Controls the ppm shift amplitude of the SSC profile **Value After Reset:** 0x0

11.2.49 SUP_CFG_20

- **Name:** (SUP) Configuration Register 20
- **Description:** Configures (SUP) settings. Register 20.
- **Size:** 32 bits
- **Offset:** 0x10c0



Table 11-117 Fields for Register: SUP_CFG_20

Bits	Name	Memory Access	Description
31:22	RESERVED__0	R	RESREVED **Reserved Field:** Yes **Value After Reset:** 0x0
21	mplla_ssc_en	R/W	Enables the ssc sub block **Value After Reset:** 0x0
20:0	mplla_ssc_step_size	R/W	Controls the SSC profile step **Value After Reset:** 0x0

11.2.50 SUP_CFG_21

- **Name:** (SUP) Configuration Register 21
- **Description:** Configures (SUP) settings. Register 21.
- **Size:** 32 bits
- **Offset:** 0x10c4

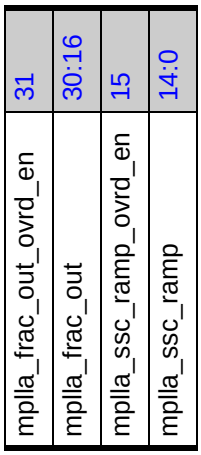


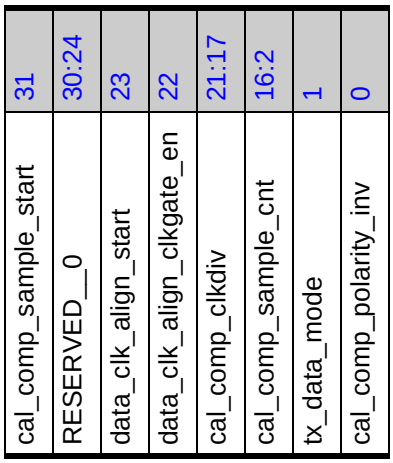
Table 11-118 Fields for Register: SUP_CFG_21

Bits	Name	Memory Access	Description
31	mplla_frac_out_ovrd_en	R/W	The frac_out override enable **Value After Reset:** 0x0
30:16	mplla_frac_out	R/W	This field is the frac_out <19:5> override value **Value After Reset:** 0x0
15	mplla_ssc_ramp_ovrd_en	R/W	The ssc_ramp override enable **Value After Reset:** 0x0
14:0	mplla_ssc_ramp	R/W	This file is the ssc_ramp <20:6> override value **Value After Reset:** 0x0

11.3 TX_TX_j Registers

11.3.1 TXS_CFG_0

- **Name:** (TXS) Configuration Register 0
- **Description:** Configures (TXS) functional settings. Register 0.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x00



31	30:24	23	22	21:17	16:2	1	0
cal_comp_sample_start	RESERVED__0	data_clk_align_start	data_clk_align_clkgate_en	cal_comp_clkdiv	cal_comp_sample_cnt	tx_data_mode	cal_comp_polarity_inv

Table 11-119 Fields for Register: TXS_CFG_0

Bits	Name	Memory Access	Description
31	cal_comp_sample_start	R/W	Write 1'b1 to this field to signal the start of sampling of the cal_result_o signal for calibration. **Value After Reset:** 0x0
30:24	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
23	data_clk_align_start	R/W	Start Signal to initiate Data Clock Alignment. **Value After Reset:** 0x0
22	data_clk_align_clkgate_en	R/W	Clock Gating Enable for Data Clock Alignment. Set to 1'b1 to start the Alignment. **Value After Reset:** 0x0
21:17	cal_comp_clkdiv	R/W	Clock divider value used to produce the comparator clock for calibration comparator logic. The default value is established for a 100 MHz master clock. **Value After Reset:** 0x9
16:2	cal_comp_sample_cnt	R/W	Number of samples of cal_result_o to capture for calibration. Default is 16K samples. **Value After Reset:** 0x4000

Bits	Name	Memory Access	Description
1	tx_data_mode	R/W	UNUSED **Value After Reset:** 0x0
0	cal_comp_polarity_inv	R/W	Inverts the polarity of cal_result signal. **Value After Reset:** 0x0

11.3.2 TXS_CFG_1

- **Name:** (TXS) Configuration Register 1
- **Description:** Configures (TXS) functional settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x04

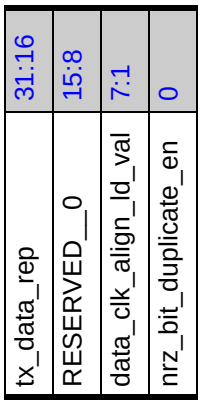


Table 11-120 Fields for Register: TXS_CFG_1

Bits	Name	Memory Access	Description
31:16	tx_data_rep	R/W	Static data pattern for replica datapath used for TX serial clock alignment and TX loopback. **Value After Reset:** 0xf0f
15:8	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
7:1	data_clk_align_ld_val	R/W	Number of Cycles of Data Clock (32UI Clock) to run the Data Clock Alignment. **Value After Reset:** 0x8
0	nrz_bit_duplicate_en	R/W	If set to 1, when in NRZ mode, duplicates the bits going to ANA. **Value After Reset:** 0x0

11.3.3 TXS_CFG_2

- **Name:** (TXS) Configuration Register 2
- **Description:** Configures (TXS) functional settings. Register 2.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x08

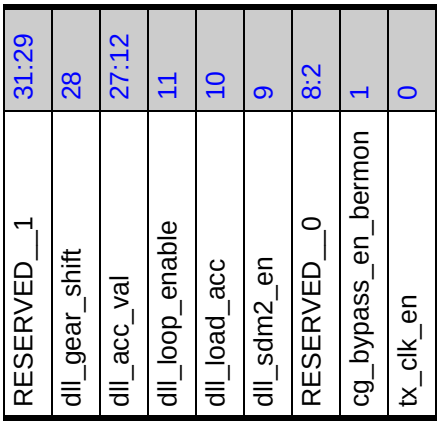


Table 11-121 Fields for Register: TXS_CFG_2

Bits	Name	Memory Access	Description
31:29	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	dll_gear_shift	R/W	Enables dll gear shift. **Value After Reset:** 0x0
27:12	dll_acc_val	R/W	Desired value to load into DLL early_late accumulator. **Value After Reset:** 0x0
11	dll_loop_enable	R/W	if set to 1, enables the dll loop. **Value After Reset:** 0x0
10	dll_load_acc	R/W	If set to 1, loads dll_acc_value into the DLL early_late accumulator. **Value After Reset:** 0x0
9	dll_sdm2_en	R/W	If set to 1, enables the sdm second order loop. **Value After Reset:** 0x0
8:2	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
1	cg_bypass_en_bermon	R/W	Bypass Clock Gating for bermon block. **Values:** ■ 0x1 (BPS): Bypass clock gating **Value After Reset:** 0x0
0	tx_clk_en	R/W	UNUSED **Value After Reset:** 0x1

11.3.4 TXS_CFG_3

- **Name:** (TXS) Configuration Register 3
- **Description:** Configures (TXS) functional settings. Register 3.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x0C

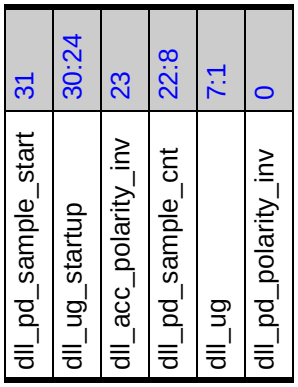


Table 11-122 Fields for Register: TXS_CFG_3

Bits	Name	Memory Access	Description
31	dll_pd_sample_start	R/W	Write 1'b1 to this field to signal the start of sampling of the dll_pd_out_o signal for DLL calibration. **Value After Reset:** 0x0
30:24	dll_ug_startup	R/W	Gain to be applied during gear_shift phase. **Value After Reset:** 0x10
23	dll_acc_polarity_inv	R/W	Inverts the polarity of dll early_late data for acc. **Value After Reset:** 0x0
22:8	dll_pd_sample_cnt	R/W	Number of samples of dll_pd_out to capture for calibration. Default is 16K samples. **Value After Reset:** 0x4000
7:1	dll_ug	R/W	Gain for DLL loop accumulator. **Value After Reset:** 0x1
0	dll_pd_polarity_inv	R/W	Inverts the polarity of dll_pd_out signal. **Value After Reset:** 0x0

11.3.5 TXS_CFG_4

- **Name:** (TXS) Configuration Register 4
- **Description:** Configures (TXS) functional settings. Register 4.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x10

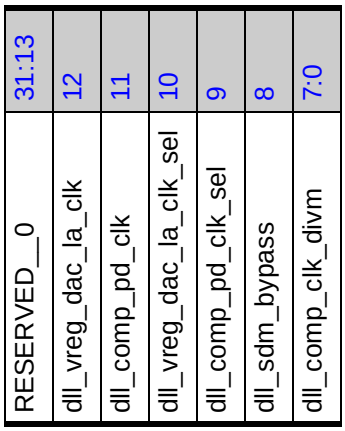


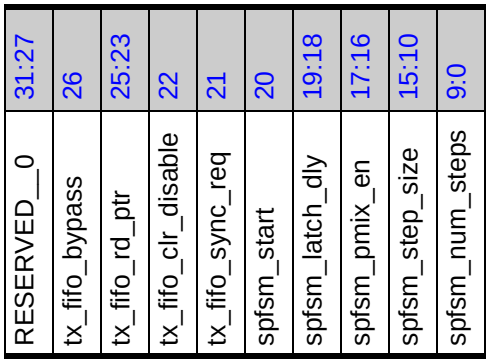
Table 11-123 Fields for Register: TXS_CFG_4

Bits	Name	Memory Access	Description
31:13	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
12	dll_vreg_dac_la_clk	R/W	If dll_vreg_dac_la_clk_sel is 1, this register connects directly to analog pin tx_ana_dll_vreg_dac_sel. **Value After Reset:** 0x0
11	dll_comp_pd_clk	R/W	If dll_comp_pd_clk_sel is 1, this register connects directly to analog pin tx_ana_dll_clk_comp_pd. **Value After Reset:** 0x0
10	dll_vreg_dac_la_clk_sel	R/W	DLL phase detector clock select. **Values:** ■ 0x0 (INT): select the internally generated clock ■ 0x1 (REG): select (TXS) register dll_vreg_dac_la_clk **Value After Reset:** 0x0
9	dll_comp_pd_clk_sel	R/W	DLL phase detector clock select. **Values:** ■ 0x0 (INT): select the internally generated clock ■ 0x1 (REG): select (TXS) register dll_comp_pd_clk **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
8	dll_sdm_bypass	R/W	Bypass of SDM datapath for VREG DAC. **Value After Reset:** 0x0
7:0	dll_comp_clk_divm	R/W	Configures division factor M for DLL comparator clock. **Value After Reset:** 0x8

11.3.6 TXS_CFG_5

- **Name:** (TXS) Configuration Register 5
- **Description:** Configures (TXS) functional settings. Register 5.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x14



31:27	26	25:23	22	21	20	19:18	17:16	15:10	9:0
RESERVED__0	tx_fifo_bypass	tx_fifo_rd_ptr	tx_fifo_clr_disable	tx_fifo_sync_req	spfsm_start	spfsm_latch_dly	spfsm_pmix_en	spfsm_step_size	spfsm_num_steps

Table 11-124 Fields for Register: TXS_CFG_5

Bits	Name	Memory Access	Description
31:27	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
26	tx_fifo_bypass	R/W	Enables FIFO bypass mode. **Value After Reset:** 0x1
25:23	tx_fifo_rd_ptr	R/W	Read pointer initial FIFO state. **Value After Reset:** 0x2
22	tx_fifo_clr_disable	R/W	This bit disables clearing TX datapath FIFO **Value After Reset:** 0x0
21	tx_fifo_sync_req	R/W	Reset trigger which asserts after completion of clocks alignment **Value After Reset:** 0x0
20	spfsm_start	R/W	Gives the order to start spinning the fsm. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
19:18	spfsm_latch_dly	R/W	Configures the numbers of 32UI cycles to use between pmix_code and pmix_dec_clk to ensure successful capture: **Values:** ■ 0x0 (_2C): 2 cycles ■ 0x1 (_4C): 4 cycles ■ 0x2 (_8C): 8 cycles ■ 0x3 (_16C): 16 cycles **Value After Reset:** 0x0
17:16	spfsm_pmix_en	R/W	Configures the desired pmix to spin: **Values:** ■ 0x0 (DIS): disabled ■ 0x1 (SER): tx_ana_pmix_ser_code ■ 0x2 (LPBK): tx_ana_pmix_lpbk_code **Value After Reset:** 0x0
15:10	spfsm_step_size	R/W	Configures the number of steps per iteration, signal magnitude (-31 : +31). **Value After Reset:** 0x1
9:0	spfsm_num_steps	R/W	Configures the number of pmix steps to walk (0 - 1023). **Value After Reset:** 0x1

11.3.7 PIN_OVRDEN_0

- **Name:** (TXS) Port Override Enable Register 0
- **Description:** Register bits act as select signal of the mux which overrides value on PMA (TXS) input and output ports. Register 0.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x18

31:7	6	5	4	3	2	1	0
RESERVED__0	ovrd_en_tx_rate_i	ovrd_en_tx_polarity_inv_i	ovrd_en_tx_graycode_en_i	ovrd_en_tx_pam4_ab_swap_en_i	ovrd_en_tx_signal_type_i	ovrd_en_tx_data_i	ovrd_en_tx_rstn_i

Table 11-125 Fields for Register: PIN_OVRDEN_0

Bits	Name	Memory Access	Description
31:7	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
6	ovrd_en_tx_rate_i	R/W	Override enable for (TXS) port tx_clock_mode_i. **Value After Reset:** 0x0
5	ovrd_en_tx_polarity_inv_i	R/W	Override enable for (TXS) port tx_polarity_inv_i. **Value After Reset:** 0x0
4	ovrd_en_tx_graycode_en_i	R/W	Override enable for (TXS) port tx_graycode_en_i. **Value After Reset:** 0x0
3	ovrd_en_tx_pam4_ab_swap_en_i	R/W	Override enable for (TXS) port tx_pam4_ab_swap_en_i. **Value After Reset:** 0x0
2	ovrd_en_tx_signal_type_i	R/W	Override enable for (TXS) port tx_signal_type_i. **Value After Reset:** 0x0
1	ovrd_en_tx_data_i	R/W	Override enable for (TXS) port tx_data_i. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
0	ovrd_en_tx_rstn_i	R/W	Override enable for (TXS) port tx_rstn_i. **Value After Reset:** 0x0

11.3.8 PIN_OVRDVAL_0

- **Name:** (TXS) Port Override Value Register 0
- **Description:** Register bits that act as override value at the mux which can override values on (TXS) input and output ports. Register 0. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x1C

31:8	7:5	4	3	2	1	0
RESERVED__0	int_tx_rate_i	int_tx_polarity_inv_i	int_tx_graycode_en_i	int_tx_pam4_ab_swap_en_i	int_tx_signal_type_i	int_tx_rstn_i

Table 11-126 Fields for Register: PIN_OVRDVAL_0

Bits	Name	Memory Access	Description
31:8	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
7:5	int_tx_rate_i	R/W	Setup PMA TX data rate modes. Available options: **Values:** ■ 0x0 (R/4): Quarter rate ■ 0x1 (R/2): Half rate ■ 0x2 (R/1): Full rate ■ 0x3 (_2XOS): 2x oversampling ■ 0x4 (_4XOS): 4x oversampling **Value After Reset:** 0x0 **Volatile:** true
4	int_tx_polarity_inv_i	R/W	Override value set on (TXS) port tx_polarity_inv_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
3	int_tx_graycode_en_i	R/W	Override value set on (TXS) port tx_graycode_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	int_tx_pam4_ab_swap_en_i	R/W	Override value set on (TXS) port tx_pam4_ab_swap_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1	int_tx_signal_type_i	R/W	Override value set on (TXS) port tx_signal_type_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	int_tx_rstn_i	R/W	Override value set on (TXS) port tx_rstn_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.9 PIN_OVRDVAL_1

- **Name:** (TXS) Port Override Value Register 1
- **Description:** Register bits act as override value at the mux which can override values on (TXS) input and output ports. Register 1. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x20

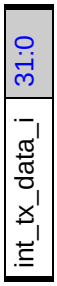


Table 11-127 Fields for Register: PIN_OVRDVAL_1

Bits	Name	Memory Access	Description
31:0	int_tx_data_i	R/W	Override value set on (TXS) port tx0_symdata_i[31:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.10 PIN_OVRDVAL_2

- **Name:** (TXS) Port Override Value Register 2
- **Description:** Register bits act as override value at the mux which can override values on (TXS) input and output ports. Register 2. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x24

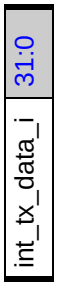


Table 11-128 Fields for Register: PIN_OVRDVAL_2

Bits	Name	Memory Access	Description
31:0	int_tx_data_i	R/W	Override value set on (TXS) port tx0_symdata_i[63:32]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.11 TST_CFG_0

- **Name:** (TXS) Test Mode Configuration Register 0
- **Description:** Configures the (TXS) internal test. Register 0.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x28

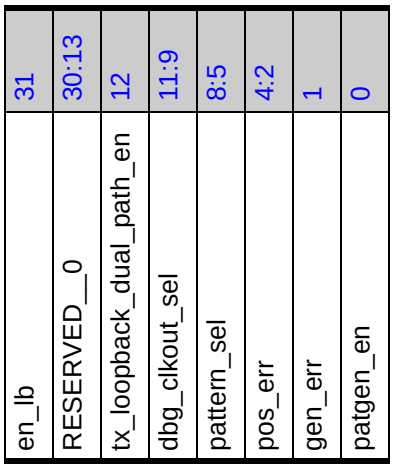


Table 11-129 Fields for Register: TST_CFG_0

Bits	Name	Memory Access	Description
31	en_lb	R/W	Loopback enable. The loopback path uses the tx_ana_rep path. **Value After Reset:** 0x0
30:13	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
12	tx_loopback_dual_path_en	R/W	Selects dual path data (2*LSB) to use in loopback mode. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
11:9	dbg_clkout_sel	R/W	Select signal to bring-out different internal clocks for observation on (PIN::(TXS)::tx_dbg_clk_o). Available options: **Values:** ■ 0x0 (ZERO): 0 ■ 0x1 (SYMB_INT_8): SYMCLK_INT/8 ■ 0x2 (SYMB_INT_16): SYMCLK_INT/16 ■ 0x3 (DLL_4): DLL_CLK/4 ■ 0x4 (DLL_8): DLL_CLK/8 ■ 0x5 (CAL_COMP): CAL_COMP_CLK ■ 0x6 (SYMB_8): SYMCLK/8 ■ 0x7 (SYMB_16): SYMCLK/16 **Value After Reset:** 0x0
8:5	pattern_sel	R/W	Pattern selection. These bits also determine the kind of incoming data pattern for the pattern matcher. Note that the pattern matcher does not support the JP03B and Transmitter linearity patterns. This field is programmed as follows. Values not listed below are reserved. **Values:** ■ 0x0 (PRBS7): PRBS7 ■ 0x4 (PRBS15): PRBS15 ■ 0x6 (PRBS31): PRBS31 ■ 0x8 (PRQS10): PRQS10 ■ 0xb (CUSTM): Custom pattern defined by (APB::(TXS)::TST_CFG::custom_pattern) **Value After Reset:** 0x0
4:2	pos_err	R/W	Controls the position, in the word to serialize, of the error inserted by gen_error: **Values:** ■ 0x0 (ERRF): 4 bits of error inserted in the 4 LSBs ■ 0x1 (ERR1): 1 bit of error inserted in the LSB ■ 0x2 (ERR2): 1 bit of error inserted in the 2nd LSB ■ 0x3 (ERR3): 1 bit of error inserted in the 3rd LSB ■ 0x4 (ERR4): 1 bit of error inserted in the 4th LSB **Value After Reset:** 0x0
1	gen_err	R/W	A write to 0b1 inserts 1 bit error into the ongoing pattern generation. Write 0b0 to 0b1 to create new errors. **Value After Reset:** 0x0
0	patgen_en	R/W	Write to 0b1 to enable the pattern generator. (TXS) ignores input parallel data at its input. **Value After Reset:** 0x0

11.3.12 TST_CFG_1

- **Name:** (TXS) Test Mode Configuration Register 1
- **Description:** Configures the (TXS) internal test. Register 1.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x2C



Table 11-130 Fields for Register: TST_CFG_1

Bits	Name	Memory Access	Description
31:0	custom_pattern	R/W	When (pattern_sel) selects a custom pattern, this register defines the transmit pattern. Field 1. **Value After Reset:** 0xff00ff00

11.3.13 TST_CFG_2

- **Name:** (TXS) Test Mode Configuration Register 2
- **Description:** Configures the (TXS) internal test. Register 2.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x30



Table 11-131 Fields for Register: TST_CFG_2

Bits	Name	Memory Access	Description
31:0	custom_pattern	R/W	When (pattern_sel) selects a custom pattern, this register defines the transmit pattern. Field 2. **Value After Reset:** 0xff00ff00

11.3.14 TST_CFG_3

- **Name:** (TXS) Test Mode Configuration Register 3
- **Description:** Configures the (TXS) internal test. Register 3.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x34



Table 11-132 Fields for Register: TST_CFG_3

Bits	Name	Memory Access	Description
31:0	custom_pattern	R/W	When (pattern_sel) selects a custom pattern, this register defines the transmit pattern. Field 3. **Value After Reset:** 0xff00ff00

11.3.15 TST_CFG_4

- **Name:** (TXS) Test Mode Configuration Register 4
- **Description:** Configures the (TXS) internal test. Register 4.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x38



Table 11-133 Fields for Register: TST_CFG_4

Bits	Name	Memory Access	Description
31:0	custom_pattern	R/W	When (pattern_sel) selects a custom pattern, this register defines the transmit pattern. Field 4. **Value After Reset:** 0xff00ff00

11.3.16 TST_CFG_5

- **Name:** (TXS) Test Mode Configuration Register 5
- **Description:** Configures the (TXS) internal test. Register 5.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x3C

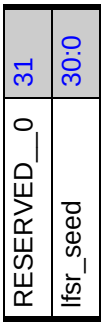


Table 11-134 Fields for Register: TST_CFG_5

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:0	lfsr_seed	R/W	Defines the initial LFSR seed for the PRBS patterns as defined in (pattern_sel). **Value After Reset:** 0x1

11.3.17 TST_CFG_6

- **Name:** (TXS) Test Mode Configuration Register 6
- **Description:** Configures the (TXS) internal test. Register 6.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x40

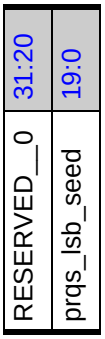


Table 11-135 Fields for Register: TST_CFG_6

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19:0	prqs_lsb_seed	R/W	Initializes PRBS20 for generation of least significant bit of PRQS10. (Not Available on this implementation) **Value After Reset:** 0x1

11.3.18 TST_CFG_7

- **Name:** (TXS) Test Mode Configuration Register 7
- **Description:** Configures the (TXS) internal test. Register 7.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x44

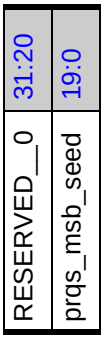


Table 11-136 Fields for Register: TST_CFG_7

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19:0	prqs_msb_seed	R/W	Based on value of (prbs_lsb_seed), it is used to initialize PRBS20 for generation of the most significant bit of PRQS10. Default value corresponds to shift10 of default least significant bit seed value. (not available on this implementation) **Value After Reset:** 0x400

11.3.19 TST_CFG_8

- **Name:** (TXS) Test Mode Configuration Register 8
- **Description:** Configures the (TXS) internal test. Register 8.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x48

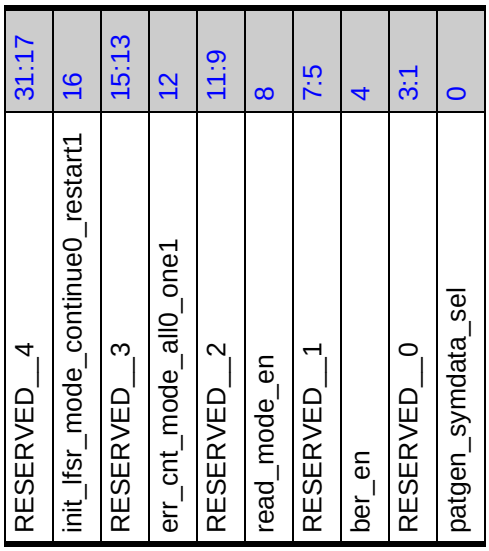


Table 11-137 Fields for Register: TST_CFG_8

Bits	Name	Memory Access	Description
31:17	RESERVED__4	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
16	init_lfsr_mode_continue0_restart1	R/W	Internal reference pattern generation lfsr operation after error detection. Available options: **Values:** ■ 0x0 (CNT): Internal reference pattern generation lfsr continues on detection of error ■ 0x1 (RST): Internal reference pattern generation lfsr re-initializes itself on detection of error **Value After Reset:** 0x0
15:13	RESERVED__3	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
12	err_cnt_mode_all0_one1	R/W	Error count mode. **Values:** ■ 0x0 (ALL): Every single bit error is counted as +1 in the ber_counter ■ 0x1 (GRP): Multiple errors in a 8bit/10bit word are counted together as +1 in the ber_counter **Value After Reset:** 0x0
11:9	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
8	read_mode_en	R/W	Read more. **Values:** ■ 0x1 (ENB): Enables internally taking current snapshot of ber_counter/ber_error_pattern/ber_error_location bits into corresponding fields of APB::RXS::BER_STAT so that they could be read in following APB read cycles. After read is complete, set this bit to 0b0. Even for consecutive reads, this bit should be set 1 to 0 to1. ■ 0x0 (NOTRD): ber_counter/ber_error_pattern/ber_error_location bits are not in read snapshot mode and are continuously changing internally. **Value After Reset:** 0x0
7:5	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	ber_en	R/W	When 0b1 the bit error rate computation logic is active. This can be set 0b1 after required settings for ber computation are done and CDR is locked. **Values:** ■ 0x1 (ACT): Bit error rate computation logic is active ■ 0x0 (DIS): Bit error rate computation logic is disabled **Value After Reset:** 0x0
3:1	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
0	patgen_symdata_sel	R/W	UNUSED **Value After Reset:** 0x0

11.3.20 TST_CFG_9

- **Name:** (TXS) Test Mode Configuration Register 9
- **Description:** Configures the (TXS) internal test. Register 9.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x4C



Table 11-138 Fields for Register: TST_CFG_9

Bits	Name	Memory Access	Description
31:0	ber_ref_timer	R/W	BER counting. **Values:** ■ 0x0 (CONT): BER counting is continuous. ■ 0x1 (STP): (and all non-zero) BER counting stops after (programmed_value+1) reference clocks. The internal counter starts fresh counting when ber_en is pulsed as 1-0-1. **Value After Reset:** 0x0

11.3.21 TST_CFG_10

- **Name:** (TXS) Test Mode Configuration Register 10
- **Description:** Configures the (TXS) internal test. Register 10.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x50

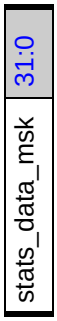


Table 11-139 Fields for Register: TST_CFG_10

Bits	Name	Memory Access	Description
31:0	stats_data_msk	R/W	Scope data mask configuration. **Value After Reset:** 0x0

11.3.22 TST_CFG_11

- **Name:** (TXS) Test Mode Configuration Register 11
- **Description:** Configures the (TXS) internal test. Register 11.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x54

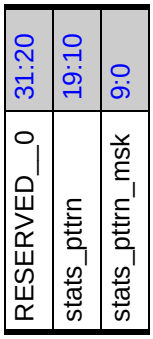


Table 11-140 Fields for Register: TST_CFG_11

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19:10	stats_pttrn	R/W	Scope pattern configuration. **Value After Reset:** 0x0
9:0	stats_pttrn_msk	R/W	Scope pattern mask configuration. **Value After Reset:** 0x0

11.3.23 TST_CFG_12

- **Name:** (TXS) Test Mode Configuration Register 12
- **Description:** Configures the (TXS) internal test. Register 12.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x58

31:30	29	28:26	25	24	23	22	21	20:6	5	4:2	1:0
RESERVED__0	stats_ab_swap_en	stats_data_dly_4ui_sel	stats_disable_smpl_cnt	stats_sc_stop	stats_sc_pause	stats_sc_start	stats_sc_timer_mode	stats_sc_ld_val	stats_invert_sel	stats_data_dly_sel	stats_quad_sel

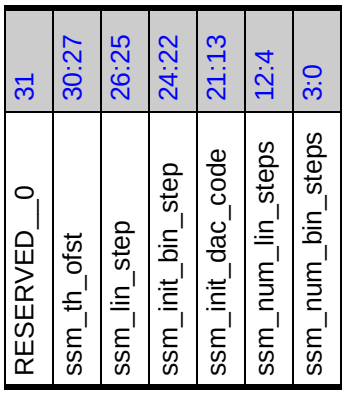
Table 11-141 Fields for Register: TST_CFG_12

Bits	Name	Memory Access	Description
31:30	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	stats_ab_swap_en	R/W	Inverts the symbol position in the scope data. **Value After Reset:** 0x0
28:26	stats_data_dly_4ui_sel	R/W	Configures number of 1ui delays on scope data. **Value After Reset:** 0x0
25	stats_disable_smpl_cnt	R/W	Disable timer counter of scope sample counter. **Value After Reset:** 0x0
24	stats_sc_stop	R/W	Scope abort. **Value After Reset:** 0x0
23	stats_sc_pause	R/W	Scope pause. **Value After Reset:** 0x0
22	stats_sc_start	R/W	Scope start flag. **Value After Reset:** 0x0
21	stats_sc_timer_mode	R/W	Selects between timer or event mode on the scope sample counter. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
20:6	stats_sc_ld_val	R/W	Scope sample counter target. **Value After Reset:** 0x0
5	stats_invert_sel	R/W	Invert polarity of the loopback scope data. **Value After Reset:** 0x0
4:2	stats_data_dly_sel	R/W	Select number of clock cycles delay in the data to be latched in analog: **Values:** ■ 0x0 (_0CLKDELAY): 0 clocks of delay ■ 0x1 (_1CLKDELAY): 1 clocks of delay ■ 0x2 (_2CLKDELAY): 2 clocks of delay ■ 0x3 (_3CLKDELAY): 3 clocks of delay ■ 0x4 (_4CLKDELAY): 4 clocks of delay ■ 0x5 (_5CLKDELAY): 5 clocks of delay ■ 0x6 (_6CLKDELAY): 6 clocks of delay ■ 0x7 (_7CLKDELAY): 7 clocks of delay **Value After Reset:** 0x0
1:0	stats_quad_sel	R/W	Filter quadrant data collected by the scope: **Values:** ■ 0x0 (_0): phase 0 ■ 0x1 (_180): phase 180 ■ 0x2 (_90): - phase 90 ■ 0x3 (_270): - phase 270 **Value After Reset:** 0x0

11.3.24 TST_CFG_13

- **Name:** (TXS) Test Mode Configuration Register 13
- **Description:** Configures the (TXS) internal test. Register 13.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x5C



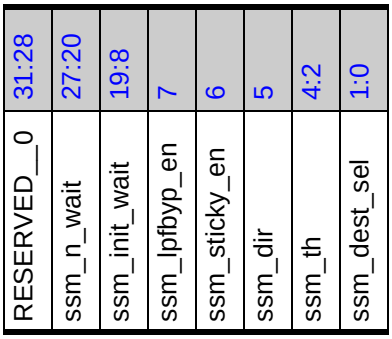
31	30:27	26:25	24:22	21:13	12:4	3:0
RESERVED__0	ssm_th_ofst	ssm_lin_step	ssm_init_bin_step	ssm_init_dac_code	ssm_num_lin_steps	ssm_num_bin_steps

Table 11-142 Fields for Register: TST_CFG_13

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:27	ssm_th_ofst	R/W	SSM correlation threshold offset. **Value After Reset:** 0x0
26:25	ssm_lin_step	R/W	SSM linear serach step size (2^N). **Value After Reset:** 0x0
24:22	ssm_init_bin_step	R/W	SSM initial binary search step size (2^N). **Value After Reset:** 0x7
21:13	ssm_init_dac_code	R/W	SSM initial code for the DAC. **Value After Reset:** 0x100
12:4	ssm_num_lin_steps	R/W	SSM number of linear search steps. **Value After Reset:** 0x4
3:0	ssm_num_bin_steps	R/W	SSM number of binary search steps. **Value After Reset:** 0x8

11.3.25 TST_CFG_14

- **Name:** (TXS) Test Mode Configuration Register 14
- **Description:** Configures the (TXS) internal test. Register 14.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x60



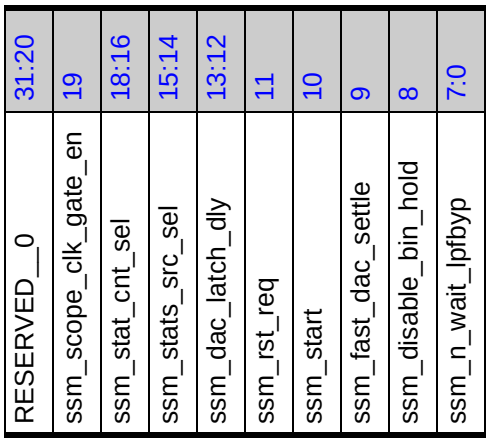
31:28	27:20	19:8	7	6	5	4:2	1:0
RESERVED__0	ssm_n_wait	ssm_init_wait	ssm_lpfbyp_en	ssm_sticky_en	ssm_dir	ssm_th	ssm_dest_sel

Table 11-143 Fields for Register: TST_CFG_14

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27:20	ssm_n_wait	R/W	SSM wait cycles following DAC write. **Value After Reset:** 0x32
19:8	ssm_init_wait	R/W	SSM wait cycles before starting. **Value After Reset:** 0x1f4
7	ssm_lpfbyp_en	R/W	SSM enable LPF bypass. **Value After Reset:** 0x0
6	ssm_sticky_en	R/W	SSM enable sticky linear search. **Value After Reset:** 0x1
5	ssm_dir	R/W	SSM select update direction: **Values:** ■ 0x0 (POS): positive ■ 0x1 (NEG): negative **Value After Reset:** 0x0
4:2	ssm_th	R/W	SSM correlation threshold. **Value After Reset:** 0x0
1:0	ssm_dest_sel	R/W	SSM selected DAC destination. **Value After Reset:** 0x0

11.3.26 TST_CFG_15

- **Name:** (TXS) Test Mode Configuration Register 15
- **Description:** Configures the (TXS) internal test. Register 15.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x64



31:20	19	18:16	15:14	13:12	11	10	9	8	7:0
RESERVED__0	ssm_scope_clk_gate_en	ssm_stat_cnt_sel	ssm_stats_src_sel	ssm_dac_latch_dly	ssm_rst_req	ssm_start	ssm_fast_dac_settle	ssm_disable_bin_hold	ssm_n_wait_lpfbyp

Table 11-144 Fields for Register: TST_CFG_15

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19	ssm_scope_clk_gate_en	R/W	SSM and Scope modules clk gating. Set 1 to ungate. **Value After Reset:** 0x0
18:16	ssm_stat_cnt_sel	R/W	SSM selection for witch stat counter to use. **Value After Reset:** 0x2
15:14	ssm_stats_src_sel	R/W	SSM input stats source selection. **Value After Reset:** 0x0
13:12	ssm_dac_latch_dly	R/W	SSM defines the latch pulse size. **Value After Reset:** 0x0
11	ssm_rst_req	R/W	SSM aborts running and resets. **Value After Reset:** 0x0
10	ssm_start	R/W	SSM starts running. **Value After Reset:** 0x0
9	ssm_fast_dac_settle	R/W	SSM bypasses the analog wait time. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
8	ssm_disable_bin_hold	R/W	SSM disables DAC hold during binary search. **Value After Reset:** 0x0
7:0	ssm_n_wait_lpfbyp	R/W	SSM wait cycles following DAC write while LPF bypass is enabled. **Value After Reset:** 0x5

11.3.27 ANA_OVRDEN_0

- **Name:** (TXS) Internal Override Enable Register 0
- **Description:** Bits that act as select signal of the mux which overrides value on (TXS) input and output ports. Register 0.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x68

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ovrd_en_ana_tx_ana_drv_pre_en_i	ovrd_en_ana_tx_ana_drv_post_en_i	ovrd_en_ana_tx_ana_drv_main_en_i	ovrd_en_ana_tx_ana_drv_lsb_en_i	ovrd_en_ana_tx_ana_dll_vreg_dac_en_i	ovrd_en_ana_tx_ana_dll_vreg_en_i	ovrd_en_ana_tx_ana_dll_vreg_dac_coarse_i	ovrd_en_ana_tx_ana_dll_vreg_dac_range_i	ovrd_en_ana_tx_ana_dll_vreg_dac_code_i	ovrd_en_ana_tx_ana_dll_dac_dec_rst_n_i	ovrd_en_ana_tx_ana_dll_phd_en_i	ovrd_en_ana_tx_ana_dll_phd_dig_sel_i	ovrd_en_ana_tx_ana_dll_phd_ana_sel_i	ovrd_en_ana_tx_ana_dll_ac_en_i	ovrd_en_ana_tx_ana_dll_dcc_en_i	ovrd_en_ana_tx_ana_dll_dcc_dac_range_i	ovrd_en_ana_tx_ana_dll_comp_dcc_dac_addr_i	ovrd_en_ana_tx_ana_dll_comp_dac_range_i	ovrd_en_ana_tx_ana_dll_comp_dcc_dac_code_i	ovrd_en_ana_tx_ana_dll_delay_fine_en_i	ovrd_en_ana_tx_ana_dll_comp_dcc_dac_en_i	ovrd_en_ana_tx_ana_dll_vreg_byp_i	ovrd_en_ana_tx_ana_dll_delay_leg_en_i	ovrd_en_ana_tx_ana_dll_delay_addr_i	ovrd_en_ana_tx_ana_dll_phd_early_late_o	RESERVED__1	RESERVED__0	ovrd_en_ana_tx_ana_cal_muxb_sel_i	ovrd_en_ana_tx_ana_cal_muxa_sel_i	ovrd_en_ana_tx_ana_cal_comp_en_i	ovrd_en_ana_tx_ana_cal_comp_clk_i	ovrd_en_ana_tx_ana_atb_s_en_i

Table 11-145 Fields for Register: ANA_OVRDEN_0

Bits	Name	Memory Access	Description
31	ovrd_en_ana_tx_ana_drv_pre_en_i	R/W	Override enable for (TXS) port tx_ana_drv_pre_en_i. **Value After Reset:** 0x1
30	ovrd_en_ana_tx_ana_drv_post_en_i	R/W	Override enable for (TXS) port tx_ana_drv_post_en_i. **Value After Reset:** 0x1
29	ovrd_en_ana_tx_ana_drv_main_en_i	R/W	Override enable for (TXS) port tx_ana_drv_main_en_i. **Value After Reset:** 0x1
28	ovrd_en_ana_tx_ana_drv_lsb_en_i	R/W	Override enable for (TXS) port tx_ana_drv_lsb_en_i. **Value After Reset:** 0x1
27	ovrd_en_ana_tx_ana_dll_vreg_dac_en_i	R/W	Override enable for (TXS) port tx_ana_dll_vreg_dac_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
26	ovrd_en_ana_tx_ana_dll_vreg_en_i	R/W	Override enable for (TXS) port tx_ana_dll_vreg_en_i. **Value After Reset:** 0x1
25	ovrd_en_ana_tx_ana_dll_vreg_dac_coarse_i	R/W	Override enable for (TXS) port tx_ana_dll_vreg_dac_coarse_i. **Value After Reset:** 0x1
24	ovrd_en_ana_tx_ana_dll_vreg_dac_range_i	R/W	Override enable for (TXS) port tx_ana_dll_vreg_dac_range_i. **Value After Reset:** 0x1
23	ovrd_en_ana_tx_ana_dll_vreg_dac_code_i	R/W	Override enable for (TXS) port tx_ana_dll_vreg_dac_code_i. **Value After Reset:** 0x0
22	ovrd_en_ana_tx_ana_dll_dac_dec_rst_n_i	R/W	Override enable for (TXS) port tx_ana_dll_dac_dec_rst_n_i. **Value After Reset:** 0x1
21	ovrd_en_ana_tx_ana_dll_phd_en_i	R/W	Override enable for (TXS) port tx_ana_dll_phd_en_i. **Value After Reset:** 0x1
20	ovrd_en_ana_tx_ana_dll_phd_dig_sel_i	R/W	Override enable for (TXS) port tx_ana_dll_phd_dig_sel_i. **Value After Reset:** 0x1
19	ovrd_en_ana_tx_ana_dll_phd_ana_sel_i	R/W	Override enable for (TXS) port tx_ana_dll_phd_ana_sel_i. **Value After Reset:** 0x1
18	ovrd_en_ana_tx_ana_dll_ac_en_i	R/W	Override enable for (TXS) port tx_ana_dll_ac_en_i. **Value After Reset:** 0x1
17	ovrd_en_ana_tx_ana_dll_dcc_en_i	R/W	Override enable for (TXS) port tx_ana_dll_dcc_en_i. **Value After Reset:** 0x1
16	ovrd_en_ana_tx_ana_dll_dcc_dac_range_i	R/W	Override enable for (TXS) port tx_ana_dll_dcc_dac_range_i. **Value After Reset:** 0x1
15	ovrd_en_ana_tx_ana_dll_comp_dcc_dac_addr_i	R/W	Override enable for (TXS) port tx_ana_dll_comp_dcc_dac_addr_i. **Value After Reset:** 0x1
14	ovrd_en_ana_tx_ana_dll_comp_dac_range_i	R/W	Override enable for (TXS) port tx_ana_dll_comp_dac_range_i. **Value After Reset:** 0x1
13	ovrd_en_ana_tx_ana_dll_comp_dcc_dac_code_i	R/W	Override enable for (TXS) port tx_ana_dll_comp_dcc_dac_code_i. **Value After Reset:** 0x1
12	ovrd_en_ana_tx_ana_dll_delay_fine_en_i	R/W	Override enable for (TXS) port tx_ana_dll_delay_fine_en_i. **Value After Reset:** 0x1
11	ovrd_en_ana_tx_ana_dll_comp_dcc_dac_en_i	R/W	Override enable for (TXS) port tx_ana_dll_comp_dcc_dac_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
10	ovrd_en_ana_tx_ana_dll_vreg_byp_i	R/W	Override enable for (TXS) port tx_ana_dll_vreg_byp_i. **Value After Reset:** 0x1
9	ovrd_en_ana_tx_ana_dll_delay_leg_en_i	R/W	Override enable for (TXS) port tx_ana_dll_delay_leg_en_i. **Value After Reset:** 0x1
8	ovrd_en_ana_tx_ana_dll_delay_addr_i	R/W	Override enable for (TXS) port tx_ana_dll_delay_addr_i. **Value After Reset:** 0x1
7	ovrd_en_ana_tx_ana_dll_phd_early_late_o	R/W	Override enable for (TXS) port tx_ana_dll_phd_early_late_o. **Value After Reset:** 0x0
6	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
5	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	ovrd_en_ana_tx_ana_cal_muxb_sel_i	R/W	Override enable for (TXS) port tx_ana_cal_muxb_sel_i. **Value After Reset:** 0x1
3	ovrd_en_ana_tx_ana_cal_muxa_sel_i	R/W	Override enable for (TXS) port tx_ana_cal_muxa_sel_i. **Value After Reset:** 0x1
2	ovrd_en_ana_tx_ana_cal_comp_en_i	R/W	Override enable for (TXS) port tx_ana_cal_comp_en_i. **Value After Reset:** 0x1
1	ovrd_en_ana_tx_ana_cal_comp_clk_i	R/W	Override enable for (TXS) port tx_ana_cal_comp_clk_i. **Value After Reset:** 0x0
0	ovrd_en_ana_tx_ana_atb_s_en_i	R/W	Override enable for (TXS) port tx_ana_atb_s_en_i. **Value After Reset:** 0x1

11.3.28 ANA_OVRDEN_1

- **Name:** (TXS) Internal Override Enable Register 1
- **Description:** Bits that act as select signal of the mux which overrides value on (TXS) input and output ports. Register 1.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x6C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ovrd_en_ana_tx_ana_pmix_lpbk_en_i	ovrd_en_ana_tx_ana_pmix_dec_rst_n_i	ovrd_en_ana_tx_ana_ser_msb_en_i	ovrd_en_ana_tx_ana_atb_meas_ibias_ref_i	ovrd_en_ana_tx_ana_atb_meas_enable_i	ovrd_en_ana_tx_ana_atb_meas_dll_vreg_vctrl_i	ovrd_en_ana_tx_ana_atb_meas_dll_vreg_nbias_i	ovrd_en_ana_tx_ana_atb_meas_dll_vp_dll_i	ovrd_en_ana_tx_ana_atb_meas_dll_phd_mux_i	ovrd_en_ana_tx_ana_atb_meas_dcc_ibias_idac_i	ovrd_en_ana_tx_ana_atb_meas_cal_mux_i	ovrd_en_ana_tx_ana_atb_meas_cal_comp_i	ovrd_en_ana_tx_ana_ser_main_en_i	ovrd_en_ana_tx_ana_ser_lsb_en_i	ovrd_en_ana_tx_ana_lpbk_en_i	ovrd_en_ana_tx_ana_ibias_xor_en_i	ovrd_en_ana_tx_ana_ibias_xor_cont_i	ovrd_en_ana_tx_ana_drv_xor_iq_sense_i	ovrd_en_ana_tx_ana_drv_msb_en_i	ovrd_en_ana_tx_ana_ibias_vdac_dll_vreg_en_i	ovrd_en_ana_tx_ana_ibias_vdac_dll_vreg_cont_i	ovrd_en_ana_tx_ana_ibias_lpbk_vdac_en_i	ovrd_en_ana_tx_ana_ibias_lpbk_vdac_cont_i	ovrd_en_ana_tx_ana_ibias_ref_en_i	ovrd_en_ana_tx_ana_ibias_ref_cont_i	ovrd_en_ana_tx_ana_ibias_dcc_idac_en_i	ovrd_en_ana_tx_ana_ibias_dcc_idac_cont_i	ovrd_en_ana_tx_ana_cal_res_div_en_i	ovrd_en_ana_tx_ana_drv_bias_sel_i	ovrd_en_ana_tx_ana_ibias_cal_comp_en_i	ovrd_en_ana_tx_ana_ibias_cal_comp_cont_i	ovrd_en_ana_tx_ana_drv_rep_en_i

Table 11-146 Fields for Register: ANA_OVRDEN_1

Bits	Name	Memory Access	Description
31	ovrd_en_ana_tx_ana_pmix_lpbk_en_i	R/W	Override enable for (TXS) port tx_ana_pmix_lpbk_en_i. **Value After Reset:** 0x1
30	ovrd_en_ana_tx_ana_pmix_dec_rst_n_i	R/W	Override enable for (TXS) port tx_ana_pmix_dec_rst_n_i. **Value After Reset:** 0x1
29	ovrd_en_ana_tx_ana_ser_msb_en_i	R/W	Override enable for (TXS) port tx_ana_ser_msb_en_i. **Value After Reset:** 0x1
28	ovrd_en_ana_tx_ana_atb_meas_ibias_ref_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_ibias_ref_i. **Value After Reset:** 0x1
27	ovrd_en_ana_tx_ana_atb_meas_enable_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_enable_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
26	ovrd_en_ana_tx_ana_atb_meas_dll_vreg_vctrl_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_dll_vreg_vctrl_i. **Value After Reset:** 0x1
25	ovrd_en_ana_tx_ana_atb_meas_dll_vreg_nbias_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_dll_vreg_nbias_i. **Value After Reset:** 0x1
24	ovrd_en_ana_tx_ana_atb_meas_dll_vp_dll_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_dll_vp_dll_i. **Value After Reset:** 0x1
23	ovrd_en_ana_tx_ana_atb_meas_dll_phd_mux_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_dll_phd_mux_i. **Value After Reset:** 0x1
22	ovrd_en_ana_tx_ana_atb_meas_dcc_ibias_idac_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_dcc_ibias_idac_i. **Value After Reset:** 0x1
21	ovrd_en_ana_tx_ana_atb_meas_cal_mux_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_cal_mux_i. **Value After Reset:** 0x1
20	ovrd_en_ana_tx_ana_atb_meas_cal_comp_i	R/W	Override enable for (TXS) port tx_ana_atb_meas_cal_comp_i. **Value After Reset:** 0x1
19	ovrd_en_ana_tx_ana_ser_main_en_i	R/W	Override enable for (TXS) port tx_ana_ser_main_en_i. **Value After Reset:** 0x1
18	ovrd_en_ana_tx_ana_ser_lsb_en_i	R/W	Override enable for (TXS) port tx_ana_ser_lsb_en_i. **Value After Reset:** 0x0
17	ovrd_en_ana_tx_ana_lpbk_en_i	R/W	Override enable for (TXS) port tx_ana_lpbk_en_i. **Value After Reset:** 0x1
16	ovrd_en_ana_tx_ana_ibias_xor_en_i	R/W	Override enable for (TXS) port tx_ana_ibias_xor_en_i. **Value After Reset:** 0x1
15	ovrd_en_ana_tx_ana_ibias_xor_cont_i	R/W	Override enable for (TXS) port tx_ana_ibias_xor_cont_i. **Value After Reset:** 0x1
14	ovrd_en_ana_tx_ana_drv_xor_iq_sense_i	R/W	Override enable for (TXS) port tx_ana_drv_xor_iq_sense_i. **Value After Reset:** 0x1
13	ovrd_en_ana_tx_ana_drv_msb_en_i	R/W	Override enable for (TXS) port tx_ana_drv_msb_en_i. **Value After Reset:** 0x1
12	ovrd_en_ana_tx_ana_ibias_vdac_dll_vreg_en_i	R/W	Override enable for (TXS) port tx_ana_ibias_vdac_dll_vreg_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
11	ovrd_en_ana_tx_ana_ibias_vdac_dll_vreg_cont_i	R/W	Override enable for (TXS) port tx_ana_ibias_vdac_dll_vreg_cont_i. **Value After Reset:** 0x1
10	ovrd_en_ana_tx_ana_ibias_lpbk_vdac_en_i	R/W	Override enable for (TXS) port tx_ana_ibias_lpbk_vdac_en_i. **Value After Reset:** 0x1
9	ovrd_en_ana_tx_ana_ibias_lpbk_vdac_cont_i	R/W	Override enable for (TXS) port tx_ana_ibias_lpbk_vdac_cont_i. **Value After Reset:** 0x1
8	ovrd_en_ana_tx_ana_ibias_ref_en_i	R/W	Override enable for (TXS) port tx_ana_ibias_ref_en_i. **Value After Reset:** 0x1
7	ovrd_en_ana_tx_ana_ibias_ref_cont_i	R/W	Override enable for (TXS) port tx_ana_ibias_ref_cont_i. **Value After Reset:** 0x1
6	ovrd_en_ana_tx_ana_ibias_dcc_idac_en_i	R/W	Override enable for (TXS) port tx_ana_ibias_dcc_idac_en_i. **Value After Reset:** 0x1
5	ovrd_en_ana_tx_ana_ibias_dcc_idac_cont_i	R/W	Override enable for (TXS) port tx_ana_ibias_dcc_idac_cont_i. **Value After Reset:** 0x1
4	ovrd_en_ana_tx_ana_cal_res_div_en_i	R/W	Override enable for (TXS) port tx_ana_cal_res_div_en_i. **Value After Reset:** 0x1
3	ovrd_en_ana_tx_ana_drv_bias_sel_i	R/W	Override enable for (TXS) port tx_ana_drv_bias_sel_i. **Value After Reset:** 0x1
2	ovrd_en_ana_tx_ana_ibias_cal_comp_en_i	R/W	Override enable for (TXS) port tx_ana_ibias_cal_comp_en_i. **Value After Reset:** 0x1
1	ovrd_en_ana_tx_ana_ibias_cal_comp_cont_i	R/W	Override enable for (TXS) port tx_ana_ibias_cal_comp_cont_i. **Value After Reset:** 0x1
0	ovrd_en_ana_tx_ana_drv_rep_en_i	R/W	Override enable for (TXS) port tx_ana_drv_rep_en_i. **Value After Reset:** 0x1

11.3.29 ANA_OVRDEN_2

- **Name:** (TXS) Internal Override Enable Register 2
- **Description:** Bits that act as select signal of the mux which overrides value on (TXS) input and output ports. Register 2.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x70

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ovrd_en_ana_tx_ana_dll_delay_rst_n_i	ovrd_en_ana_tx_ana_dll_comp_dac_vcm_i	ovrd_en_ana_tx_ana_lpbk_des_rst_n_i	ovrd_en_ana_tx_ana_lpbk_data_32ui_o	ovrd_en_ana_tx_ana_lpbk_dac_en_i	ovrd_en_ana_tx_ana_lpbk_dac_dec_rst_n_i	ovrd_en_ana_tx_ana_lpbk_dac_range_i	ovrd_en_ana_tx_ana_cal_result_o	ovrd_en_ana_tx_ana_lpbk_dac_dec_clk_i	ovrd_en_ana_tx_ana_lpbk_dac_code_i	ovrd_en_ana_tx_ana_idrv_dac_addr_i	ovrd_en_ana_tx_ana_lpbk_dac_extra_range_i	RESERVED__0	ovrd_en_ana_tx_ana_idrv_dac_code_i	ovrd_en_ana_tx_ana_cmldrv_bias_rst_n_i	ovrd_en_ana_tx_ana_idrv_dac_en_i	ovrd_en_ana_tx_ana_term_mos_bias_en_i	ovrd_en_ana_tx_ana_idrv_bias_clk_i	ovrd_en_ana_tx_ana_ser_pre_config_i	ovrd_en_ana_tx_ana_ibias_idac_drv_en_i	ovrd_en_ana_tx_ana_ibias_idac_drv_cont_i	ovrd_en_ana_tx_ana_term_ctrl_vp_th_i	ovrd_en_ana_tx_ana_spare_i	ovrd_en_ana_tx_ana_ser_rst_n_i	ovrd_en_ana_tx_ana_ser_rep_en_i	ovrd_en_ana_tx_ana_ser_pre_en_i	ovrd_en_ana_tx_ana_ser_post_en_i	ovrd_en_ana_tx_ana_pmix_ser_en_i	ovrd_en_ana_tx_ana_drv_xor_rep_en_i	ovrd_en_ana_tx_ana_pmix_ser_dec_clk_i	ovrd_en_ana_tx_ana_pmix_lpbk_code_i	ovrd_en_ana_tx_ana_pmix_ser_code_i

Table 11-147 Fields for Register: ANA_OVRDEN_2

Bits	Name	Memory Access	Description
31	ovrd_en_ana_tx_ana_dll_delay_rst_n_i	R/W	Override enable for (TXS) port tx_ana_dll_delay_rst_n_i. **Value After Reset:** 0x1
30	ovrd_en_ana_tx_ana_dll_comp_dac_vcm_i	R/W	Override enable for (TXS) port tx_ana_dll_comp_dac_vcm_i. **Value After Reset:** 0x1
29	ovrd_en_ana_tx_ana_lpbk_des_rst_n_i	R/W	Override enable for (TXS) port tx_ana_lpbk_des_rst_n_i. **Value After Reset:** 0x1
28	ovrd_en_ana_tx_ana_lpbk_data_32ui_o	R/W	Override enable for (TXS) port tx_ana_lpbk_data_32ui_o. **Value After Reset:** 0x0
27	ovrd_en_ana_tx_ana_lpbk_dac_en_i	R/W	Override enable for (TXS) port ana_tx_ana_lpbk_dac_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
26	ovrd_en_ana_tx_ana_lpbk_dac_dec_rst_n_i	R/W	Override enable for (TXS) port ana_tx_ana_lpbk_dac_dec_rst_n_i. **Value After Reset:** 0x1
25	ovrd_en_ana_tx_ana_lpbk_dac_range_i	R/W	Override enable for (TXS) port tx_ana_lpbk_dac_range_i. **Value After Reset:** 0x1
24	ovrd_en_ana_tx_ana_cal_result_o	R/W	Override enable for (TXS) port tx_ana_cal_result_o. **Value After Reset:** 0x0
23	ovrd_en_ana_tx_ana_lpbk_dac_dec_clk_i	R/W	Override enable for (TXS) port tx_ana_lpbk_dac_dec_clk_i. **Value After Reset:** 0x1
22	ovrd_en_ana_tx_ana_lpbk_dac_code_i	R/W	Override enable for (TXS) port tx_ana_lpbk_dac_code_i. **Value After Reset:** 0x1
21	ovrd_en_ana_tx_ana_idrv_dac_addr_i	R/W	Override enable for (TXS) port tx_ana_idrv_dac_addr_i. **Value After Reset:** 0x1
20	ovrd_en_ana_tx_ana_lpbk_dac_extra_range_i	R/W	Override enable for (TXS) port tx_ana_lpbk_dac_extra_range_i. **Value After Reset:** 0x1
19	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x1
18	ovrd_en_ana_tx_ana_idrv_dac_code_i	R/W	Override enable for (TXS) port tx_ana_idrv_dac_code_i. **Value After Reset:** 0x1
17	ovrd_en_ana_tx_ana_cmldrv_bias_rst_n_i	R/W	Override enable for (TXS) port tx_ana_cmldrv_bias_rst_n_i. **Value After Reset:** 0x1
16	ovrd_en_ana_tx_ana_idrv_dac_en_i	R/W	Override enable for (TXS) port tx_ana_idrv_dac_en_i. **Value After Reset:** 0x1
15	ovrd_en_ana_tx_ana_term_mos_bias_en_i	R/W	Override enable for (TXS) port tx_ana_term_mos_bias_en_i. **Value After Reset:** 0x1
14	ovrd_en_ana_tx_ana_idrv_bias_clk_i	R/W	Override enable for (TXS) port tx_ana_idrv_bias_clk_i. **Value After Reset:** 0x1
13	ovrd_en_ana_tx_ana_ser_pre_config_i	R/W	Override enable for (TXS) port tx_ana_ser_pre_config_i. **Value After Reset:** 0x1
12	ovrd_en_ana_tx_ana_ibias_idac_drv_en_i	R/W	Override enable for (TXS) port tx_ana_ibias_idac_drv_en_i. **Value After Reset:** 0x1
11	ovrd_en_ana_tx_ana_ibias_idac_drv_cont_i	R/W	Override enable for (TXS) port tx_ana_ibias_idac_drv_cont_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
10	ovrd_en_ana_tx_ana_term_ctrl_vp_th_i	R/W	Override enable for (TXS) port tx_ana_term_ctrl_vp_th_i. **Value After Reset:** 0x1
9	ovrd_en_ana_tx_ana_spare_i	R/W	Override enable for (TXS) port tx_ana_spare_i. **Value After Reset:** 0x1
8	ovrd_en_ana_tx_ana_ser_rst_n_i	R/W	Override enable for (TXS) port tx_ana_ser_rst_n_i. **Value After Reset:** 0x1
7	ovrd_en_ana_tx_ana_ser_rep_en_i	R/W	Override enable for (TXS) port tx_ana_ser_rep_en_i. **Value After Reset:** 0x1
6	ovrd_en_ana_tx_ana_ser_pre_en_i	R/W	Override enable for (TXS) port tx_ana_ser_pre_en_i. **Value After Reset:** 0x1
5	ovrd_en_ana_tx_ana_ser_post_en_i	R/W	Override enable for (TXS) port tx_ana_ser_post_en_i. **Value After Reset:** 0x1
4	ovrd_en_ana_tx_ana_pmix_ser_en_i	R/W	Override enable for (TXS) port tx_ana_pmix_ser_en_i. **Value After Reset:** 0x1
3	ovrd_en_ana_tx_ana_drv_xor_rep_en_i	R/W	Override enable for (TXS) port tx_ana_drv_xor_rep_en_i. **Value After Reset:** 0x1
2	ovrd_en_ana_tx_ana_pmix_ser_dec_clk_i	R/W	Override enable for (TXS) port tx_ana_pmix_ser_dec_clk_i. **Value After Reset:** 0x1
1	ovrd_en_ana_tx_ana_pmix_lpbk_code_i	R/W	Override enable for (TXS) port tx_ana_pmix_lpbk_code_i. **Value After Reset:** 0x1
0	ovrd_en_ana_tx_ana_pmix_ser_code_i	R/W	Override enable for (TXS) port tx_ana_pmix_ser_code_i. **Value After Reset:** 0x1

11.3.30 ANA_OVRDEN_3

- **Name:** (TXS) Internal Override Enable Register 3
- **Description:** Bits that act as select signal of the mux which overrides value on (TXS) input and output ports. Register 3.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x74

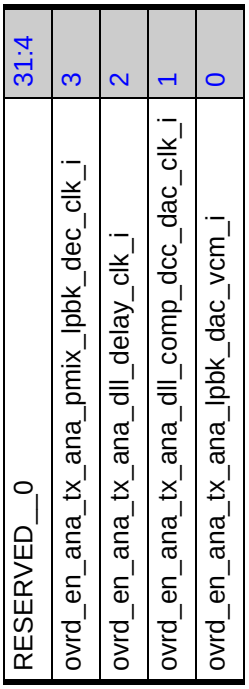


Table 11-148 Fields for Register: ANA_OVRDEN_3

Bits	Name	Memory Access	Description
31:4	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
3	ovrd_en_ana_tx_ana_pmix_lpbk_dec_clk_i	R/W	Override enable for (TXS) port tx_ana_pmix_lpbk_dec_clk_i. **Value After Reset:** 0x1
2	ovrd_en_ana_tx_ana_dll_delay_clk_i	R/W	Override enable for (TXS) port tx_ana_dll_delay_clk_i. **Value After Reset:** 0x1
1	ovrd_en_ana_tx_ana_dll_comp_dcc_dac_clk_i	R/W	Override enable for (TXS) port tx_ana_dll_comp_dcc_dac_clk_i. **Value After Reset:** 0x1
0	ovrd_en_ana_tx_ana_lpbk_dac_vcm_i	R/W	Override enable for (TXS) port tx_ana_lpbk_dac_vcm_i. **Value After Reset:** 0x1

11.3.31 ANA_OVRDVAL_0

- **Name:** (TXS) Internal Override Value Register 0
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 0. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x78

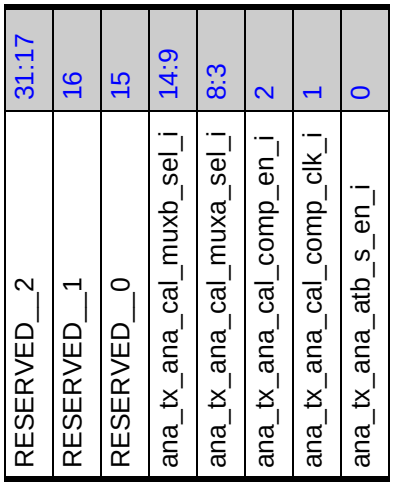


Table 11-149 Fields for Register: ANA_OVRDVAL_0

Bits	Name	Memory Access	Description
31:17	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
16	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
15	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
14:9	ana_tx_ana_cal_muxb_sel_i	R/W	Override value set on (TXS) analog port tx_ana_cal_muxb_sel_i[5:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
8:3	ana_tx_ana_cal_muxa_sel_i	R/W	Override value set on (TXS) analog port tx_ana_cal_muxa_sel_i[5:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_tx_ana_cal_comp_en_i	R/W	Override value set on (TXS) analog port tx_ana_cal_comp_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1	ana_tx_ana_cal_comp_clk_i	R/W	Override value set on (TXS) analog port tx_ana_cal_comp_clk_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_tx_ana_atb_s_en_i	R/W	Override value set on (TXS) analog port tx_ana_atb_s_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.32 ANA_OVRDVAL_1

- **Name:** (TXS) Internal Override Value Register 1
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 1. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x7C

31:29	28	27:26	25	24	23:21	20:11	10	9:0
ana_tx_ana_dll_delay_fine_en_i	ana_tx_ana_dll_delay_rst_n_i	ana_tx_ana_dll_comp_dac_vcm_i	ana_tx_ana_dll_comp_dcc_dac_en_i	ana_tx_ana_dll_comp_dcc_dac_clk_i	ana_tx_ana_dll_delay_leg_en_i	ana_tx_ana_dll_delay_addr_i	ana_tx_ana_dll_dac_dec_rst_n_i	RESERVED__0

Table 11-150 Fields for Register: ANA_OVRDVAL_1

Bits	Name	Memory Access	Description
31:29	ana_tx_ana_dll_delay_fine_en_i	R/W	Override value set on (TXS) analog port tx_ana_dll_delay_fine_en_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
28	ana_tx_ana_dll_delay_rst_n_i	R/W	Override value set on (TXS) analog port tx_ana_dll_delay_rst_n_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
27:26	ana_tx_ana_dll_comp_dac_vcm_i	R/W	Override value set on (TXS) analog port tx_ana_dll_comp_dac_vcm[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true

Bits	Name	Memory Access	Description
25	ana_tx_ana_dll_comp_dcc_dac_en_i	R/W	Override value set on (TXS) analog port tx_ana_dll_comp_dcc_dac_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	ana_tx_ana_dll_comp_dcc_dac_clk_i	R/W	Override value set on (TXS) analog port tx_ana_dll_comp_dcc_dac_clk_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
23:21	ana_tx_ana_dll_delay_leg_en_i	R/W	Override value set on (TXS) analog port tx_ana_dll_delay_leg_en_i[2:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
20:11	ana_tx_ana_dll_delay_addr_i	R/W	Override value set on (TXS) analog port tx_ana_dll_delay_addr_i[8:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x3ff **Volatile:** true
10	ana_tx_ana_dll_dac_dec_rst_n_i	R/W	Override value set on (TXS) analog port tx_ana_dll_dac_dec_rst_n_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.3.33 ANA_OVRDVAL_2

- **Name:** (TXS) Internal Override Value Register 2
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 2. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x80

31:26	25	24	23:20	19:16	15:11	10:9	8:0
RESERVED__0	ana_tx_ana_dll_phd_early_late_o	ana_tx_ana_dll_ac_en_i	ana_tx_ana_dll_dcc_en_i	ana_tx_ana_dll_dcc_dac_range_i	ana_tx_ana_dll_comp_dcc_dac_addr_i	ana_tx_ana_dll_comp_dac_range_i	ana_tx_ana_dll_comp_dcc_dac_code_i

Table 11-151 Fields for Register: ANA_OVRDVAL_2

Bits	Name	Memory Access	Description
31:26	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
25	ana_tx_ana_dll_phd_early_late_o	R/W	Override value set on (TXS) analog port tx_ana_dll_phd_early_late_o. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	ana_tx_ana_dll_ac_en_i	R/W	Override value set on (TXS) analog port tx_ana_dll_ac_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
23:20	ana_tx_ana_dll_dcc_en_i	R/W	Override value set on (TXS) analog port tx_ana_dll_dcc_en_i[7:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
19:16	ana_tx_ana_dll_dcc_dac_range_i	R/W	Override value set on (TXS) analog port tx_ana_dll_dcc_dac_range_i[3:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x8 **Volatile:** true
15:11	ana_tx_ana_dll_comp_dcc_dac_addr_i	R/W	Override value set on (TXS) analog port tx_ana_dll_comp_dcc_dac_addr_i[4:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:9	ana_tx_ana_dll_comp_dac_range_i	R/W	Override value set on (TXS) analog port tx_ana_dll_comp_dac_range_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
8:0	ana_tx_ana_dll_comp_dcc_dac_code_i	R/W	Override value set on (TXS) analog port tx_ana_dll_comp_dcc_dac_code_i[8:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.34 ANA_OVRDVAL_3

- **Name:** (TXS) Internal Override Value Register 3
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 3. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x84

31:29	28:26	25	24	23	22:21	20:12	11	10	9:5	4:0
RESERVED__0	ana_tx_ana_dll_vreg_dac_coarse_i	ana_tx_ana_dll_vreg_byp_i	ana_tx_ana_dll_vreg_en_i	ana_tx_ana_dll_vreg_dac_en_i	ana_tx_ana_dll_vreg_dac_range_i	ana_tx_ana_dll_vreg_dac_code_i	ana_tx_ana_dll_delay_clk_i	ana_tx_ana_dll_phd_en_i	ana_tx_ana_dll_phd_dig_sel_i	ana_tx_ana_dll_phd_ana_sel_i

Table 11-152 Fields for Register: ANA_OVRDVAL_3

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28:26	ana_tx_ana_dll_vreg_dac_coarse_i	R/W	Override value set on (TXS) analog port tx_ana_dll_vreg_dac_coarse_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_tx_ana_dll_vreg_byp_i	R/W	Override value set on (TXS) analog port tx_ana_dll_vreg_byp_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
24	ana_tx_ana_dll_vreg_en_i	R/W	Override value set on (TXS) analog port tx_ana_dll_vreg_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
23	ana_tx_ana_dll_vreg_dac_en_i	R/W	Override value set on (TXS) analog port tx_ana_dll_vreg_dac_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
22:21	ana_tx_ana_dll_vreg_dac_range_i	R/W	Override value set on (TXS) analog port tx_ana_dll_vreg_dac_range_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
20:12	ana_tx_ana_dll_vreg_dac_code_i	R/W	Override value set on (TXS) analog port tx_ana_dll_vreg_dac_code_i[8:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11	ana_tx_ana_dll_delay_clk_i	R/W	Override value set on (TXS) analog port tx_ana_dll_delay_clk_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10	ana_tx_ana_dll_phd_en_i	R/W	Override value set on (TXS) analog port tx_ana_dll_phd_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9:5	ana_tx_ana_dll_phd_dig_sel_i	R/W	Override value set on (TXS) analog port tx_ana_dll_phd_dig_sel_i[4:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4:0	ana_tx_ana_dll_phd_ana_sel_i	R/W	Override value set on (TXS) analog port tx_ana_dll_phd_ana_sel_i[4:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.35 ANA_OVRDVAL_4

- **Name:** (TXS) Internal Override Value Register 4
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 4. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x88

31	30:29	28	27:26	25	24:23	22	21:20	19	18:17	16	15:14	13	12	11	10:9	8	7	6	5:4	3:0
ana_tx_ana_ibias_vdac_dll_vreg_en_i	ana_tx_ana_ibias_vdac_dll_vreg_cont_i	ana_tx_ana_ibias_lpbk_vdac_en_i	ana_tx_ana_ibias_lpbk_vdac_cont_i	ana_tx_ana_ibias_ref_en_i	ana_tx_ana_ibias_ref_cont_i	ana_tx_ana_ibias_dcc_idac_en_i	ana_tx_ana_ibias_dcc_idac_cont_i	ana_tx_ana_cal_res_div_en_i	ana_tx_ana_drv_bias_sel_i	ana_tx_ana_ibias_cal_comp_en_i	ana_tx_ana_ibias_cal_comp_cont_i	ana_tx_ana_drv_rep_en_i	ana_tx_ana_drv_pre_en_i	ana_tx_ana_drv_post_en_i	ana_tx_ana_drv_main_en_i	ana_tx_ana_drv_lsb_en_i	ana_tx_ana_drv_msb_en_i	ana_tx_ana_drv_xor_iq_sense_i	ana_tx_ana_drv_xor_rep_en_i	RESERVED__0

Table 11-153 Fields for Register: ANA_OVRDVAL_4

Bits	Name	Memory Access	Description
31	ana_tx_ana_ibias_vdac_dll_vreg_en_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_vdac_dll_vreg_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
30:29	ana_tx_ana_ibias_vdac_dll_vreg_cont_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_vdac_dll_vreg_cont_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true

Bits	Name	Memory Access	Description
28	ana_tx_ana_ibias_lpbk_vdac_en_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_lpbk_vdac_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
27:26	ana_tx_ana_ibias_lpbk_vdac_cont_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_lpbk_vdac_cont_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
25	ana_tx_ana_ibias_ref_en_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_ref_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24:23	ana_tx_ana_ibias_ref_cont_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_ref_cont_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
22	ana_tx_ana_ibias_dcc_idac_en_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_dcc_idac_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
21:20	ana_tx_ana_ibias_dcc_idac_cont_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_dcc_idac_cont_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
19	ana_tx_ana_cal_res_div_en_i	R/W	Override value set on (TXS) analog port tx_ana_cal_res_div_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
18:17	ana_tx_ana_drv_bias_sel_i	R/W	Override value set on (TXS) analog port tx_ana_drv_bias_sel_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
16	ana_tx_ana_ibias_cal_comp_en_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_cal_comp_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:14	ana_tx_ana_ibias_cal_comp_cont_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_cal_comp_cont_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
13	ana_tx_ana_drv_rep_en_i	R/W	Override value set on (TXS) analog port tx_ana_drv_rep_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
12	ana_tx_ana_drv_pre_en_i	R/W	Override value set on (TXS) analog port tx_ana_drv_pre_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11	ana_tx_ana_drv_post_en_i	R/W	Override value set on (TXS) analog port tx_ana_drv_post_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:9	ana_tx_ana_drv_main_en_i	R/W	Override value set on (TXS) analog port tx_ana_drv_main_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8	ana_tx_ana_drv_lsb_en_i	R/W	Override value set on (TXS) analog port tx_ana_drv_lsb_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7	ana_tx_ana_drv_msb_en_i	R/W	Override value set on (TXS) analog port tx_ana_drv_msb_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
6	ana_tx_ana_drv_xor_iq_sense_i	R/W	Override value set on (TXS) analog port tx_ana_drv_xor_iq_sense_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5:4	ana_tx_ana_drv_xor_rep_en_i	R/W	Override value set on (TXS) analog port tx_ana_drv_xor_rep_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
3:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.3.36 ANA_OVRDVAL_5

- **Name:** (TXS) Internal Override Value Register 5
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 5. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x8C

31:25	24	23	22	21	20:16	15	14	13	12:11	10	9	8	7	6	5	4	3:2	1:0
ana_tx_ana_pmix_ser_code_i	ana_tx_ana_pmix_lpbk_en_i	ana_tx_ana_pmix_dec_rst_n_i	ana_tx_ana_pmix_ser_en_i	ana_tx_ana_ser_msb_en_i	ana_tx_ana_atb_meas_ibias_ref_i	ana_tx_ana_atb_meas_dll_vreg_vctrl_i	ana_tx_ana_atb_meas_dll_vreg_nbias_i	ana_tx_ana_atb_meas_dll_vp_dll_i	ana_tx_ana_atb_meas_dll_phd_mux_i	ana_tx_ana_atb_meas_dcc_ibias_idac_i	ana_tx_ana_atb_meas_cal_mux_i	ana_tx_ana_atb_meas_cal_comp_i	ana_tx_ana_ser_main_en_i	ana_tx_ana_ser_lsb_en_i	ana_tx_ana_lpbk_en_i	ana_tx_ana_ibias_xor_en_i	ana_tx_ana_ibias_xor_cont_i	RESERVED__0

Table 11-154 Fields for Register: ANA_OVRDVAL_5

Bits	Name	Memory Access	Description
31:25	ana_tx_ana_pmix_ser_code_i	R/W	Override value set on (TXS) analog port tx_ana_pmix_ser_code_i[2:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	ana_tx_ana_pmix_lpbk_en_i	R/W	Override value set on (TXS) analog port tx_ana_pmix_lpbk_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
23	ana_tx_ana_pmix_dec_rst_n_i	R/W	Override value set on (TXS) analog port tx_ana_pmix_dec_rst_n_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
22	ana_tx_ana_pmix_ser_en_i	R/W	Override value set on (TXS) analog port tx_ana_pmix_ser_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
21	ana_tx_ana_ser_msb_en_i	R/W	Override value set on (TXS) analog port tx_ana_ser_msb_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
20:16	ana_tx_ana_atb_meas_ibias_ref_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_ibias_ref_i[4:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15	ana_tx_ana_atb_meas_dll_vreg_vctrl_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_dll_vreg_vctrl_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14	ana_tx_ana_atb_meas_dll_vreg_nbias_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_dll_vreg_nbias_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
13	ana_tx_ana_atb_meas_dll_vp_dll_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_dll_vp_dll_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
12:11	ana_tx_ana_atb_meas_dll_phd_mux_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_dll_phd_mux_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
10	ana_tx_ana_atb_meas_dcc_ibias_idac_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_dcc_ibias_idac_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9	ana_tx_ana_atb_meas_cal_mux_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_cal_mux_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8	ana_tx_ana_atb_meas_cal_comp_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_cal_comp_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7	ana_tx_ana_ser_main_en_i	R/W	Override value set on (TXS) analog port tx_ana_ser_main_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
6	ana_tx_ana_ser_lsb_en_i	R/W	Override value set on (TXS) analog port tx_ana_ser_lsb_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5	ana_tx_ana_lpbk_en_i	R/W	Override value set on (TXS) analog port tx_ana_lpbk_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4	ana_tx_ana_ibias_xor_en_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_xor_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3:2	ana_tx_ana_ibias_xor_cont_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_xor_cont_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true

Bits	Name	Memory Access	Description
1:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.3.37 ANA_OVRDVAL_6

- **Name:** (TXS) Internal Override Value Register 6
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 6. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x90

31:30	29	28	27	26	25	24	23	22:16	15:0
RESERVED__1	ana_tx_ana_ser_rst_n_i	ana_tx_ana_ser_rep_en_i	ana_tx_ana_ser_pre_en_i	ana_tx_ana_ser_post_en_i	RESERVED__0	ana_tx_ana_pmix_lpbk_dec_clk_i	ana_tx_ana_pmix_ser_dec_clk_i	ana_tx_ana_pmix_lpbk_code_i	ana_tx_ana_atb_meas_enable_i

Table 11-155 Fields for Register: ANA_OVRDVAL_6

Bits	Name	Memory Access	Description
31:30	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	ana_tx_ana_ser_rst_n_i	R/W	Override value set on (TXS) analog port tx_ana_ser_rst_n_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
28	ana_tx_ana_ser_rep_en_i	R/W	Override value set on (TXS) analog port tx_ana_ser_rep_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
27	ana_tx_ana_ser_pre_en_i	R/W	Override value set on (TXS) analog port tx_ana_ser_pre_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
26	ana_tx_ana_ser_post_en_i	R/W	Override value set on (TXS) analog port tx_ana_ser_post_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
24	ana_tx_ana_pmix_lpbk_dec_clk_i	R/W	Override value set on (TXS) analog port tx_ana_pmix_lpbk_dec_clk_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
23	ana_tx_ana_pmix_ser_dec_clk_i	R/W	Override value set on (TXS) analog port tx_ana_pmix_ser_dec_clk_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
22:16	ana_tx_ana_pmix_lpbk_code_i	R/W	Override value set on (TXS) analog port tx_ana_pmix_lpbk_code_i[6:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:0	ana_tx_ana_atb_meas_enable_i	R/W	Override value set on (TXS) analog port tx_ana_atb_meas_enable_i[15:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.38 ANA_OVRDVAL_7

- **Name:** (TXS) Internal Override Value Register 7
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 7. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x94

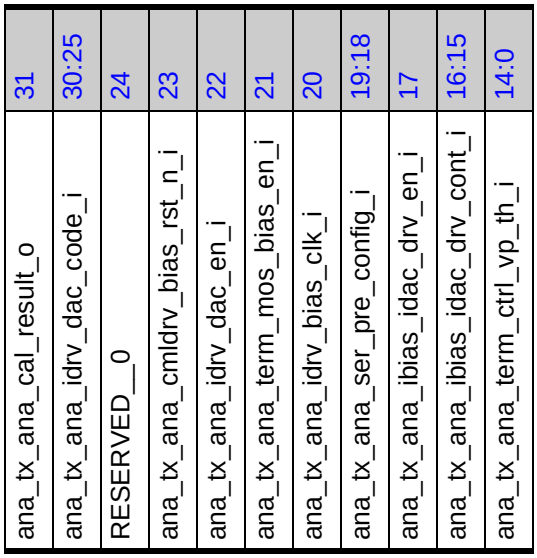


Table 11-156 Fields for Register: ANA_OVRDVAL_7

Bits	Name	Memory Access	Description
31:0	ana_tx_ana_spare_i	R/W	Override value set on (TXS) analog port tx_ana_spare_i[31:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x80002 **Volatile:** true

11.3.39 ANA_OVRDVAL_8

- **Name:** (TXS) Internal Override Value Register 8
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 8. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x98



31	30:25	24	23	22	21	20	19:18	17	16:15	14:0
ana_tx_ana_cal_result_o	ana_tx_ana_idrv_dac_code_i	RESERVED__0	ana_tx_ana_cmldrv_bias_rst_n_i	ana_tx_ana_idrv_dac_en_i	ana_tx_ana_term_mos_bias_en_i	ana_tx_ana_idrv_bias_clk_i	ana_tx_ana_ser_pre_config_i	ana_tx_ana_ibias_idac_drv_en_i	ana_tx_ana_ibias_idac_drv_cont_i	ana_tx_ana_term_ctrl_vp_th_i

Table 11-157 Fields for Register: ANA_OVRDVAL_8

Bits	Name	Memory Access	Description
31	ana_tx_ana_cal_result_o	R/W	Override value set on (TXS) analog port tx_ana_cal_result_o. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
30:25	ana_tx_ana_idrv_dac_code_i	R/W	Override value set on (TXS) analog port tx_ana_idrv_dac_code_i[5:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
23	ana_tx_ana_cmldrv_bias_rst_n_i	R/W	Override value set on (TXS) analog port tx_ana_cmldrv_bias_rst_n_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
22	ana_tx_ana_idrv_dac_en_i	R/W	Override value set on (TXS) analog port tx_ana_idrv_dac_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
21	ana_tx_ana_term_mos_bias_en_i	R/W	Override value set on (TXS) analog port tx_ana_term_mos_bias_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
20	ana_tx_ana_idrv_bias_clk_i	R/W	Override value set on (TXS) analog port tx_ana_idrv_bias_clk_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
19:18	ana_tx_ana_ser_pre_config_i	R/W	Override value set on (TXS) analog port tx_ana_ser_pre_config_i[1:0]i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
17	ana_tx_ana_ibias_idac_drv_en_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_idac_drv_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16:15	ana_tx_ana_ibias_idac_drv_cont_i	R/W	Override value set on (TXS) analog port tx_ana_ibias_idac_drv_cont_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14:0	ana_tx_ana_term_ctrl_vp_th_i	R/W	Override value set on (TXS) analog port tx_ana_term_ctrl_vp_th_i[14:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.40 ANA_OVRDVAL_9

- **Name:** (TXS) Internal Override Value Register 9
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 9. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0x9C

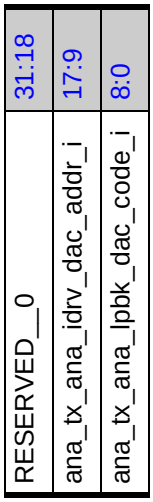


Table 11-158 Fields for Register: ANA_OVRDVAL_9

Bits	Name	Memory Access	Description
31:18	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
17:9	ana_tx_ana_idrv_dac_addr_i	R/W	Override value set on (TXS) analog port tx_ana_idrv_dac_addr_i[8:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8:0	ana_tx_ana_lpbk_dac_code_i	R/W	Override value set on (TXS) analog port tx_ana_lpbk_dac_code_i[8:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.41 ANA_OVRDVAL_10

- **Name:** (TXS) Internal Override Value Register 10
- **Description:** Register bits act as override value at the mux which can override values on (TXS) analog-digital interface ports. Register 11. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xA0

31:18	17:16	15:14	13	12:5	4:3	2	1	0
RESERVED__0	ana_tx_ana_lpbk_dac_extra_range_i	ana_tx_ana_lpbk_dac_range_i	ana_tx_ana_lpbk_des_rst_n_i	ana_tx_ana_lpbk_data_32ui_o	ana_tx_ana_lpbk_dac_vcm_i	ana_tx_ana_lpbk_dac_en_i	ana_tx_ana_lpbk_dac_dec_rst_n_i	ana_tx_ana_lpbk_dac_dec_clk_i

Table 11-159 Fields for Register: ANA_OVRDVAL_10

Bits	Name	Memory Access	Description
31:18	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
17:16	ana_tx_ana_lpbk_dac_extra_range_i	R/W	Override value set on (TXS) analog port tx_ana_lpbk_dac_extra_range_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
15:14	ana_tx_ana_lpbk_dac_range_i	R/W	Override value set on (TXS) analog port tx_ana_lpbk_dac_range_i[2:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true

Bits	Name	Memory Access	Description
13	ana_tx_ana_lpbk_des_rst_n_i	R/W	Override value set on (TXS) analog port tx_ana_lpbk_des_rst_n_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
12:5	ana_tx_ana_lpbk_data_32ui_o	R/W	Override value set on (TXS) analog port tx_ana_lpbk_data_32ui_o[7:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4:3	ana_tx_ana_lpbk_dac_vcm_i	R/W	Override value set on (TXS) analog port tx_ana_lpbk_dac_vcm_i[1:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
2	ana_tx_ana_lpbk_dac_en_i	R/W	Override value set on (TXS) analog port ana_tx_ana_lpbk_dac_en_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1	ana_tx_ana_lpbk_dac_dec_rst_n_i	R/W	Override value set on (TXS) analog port ana_tx_ana_lpbk_dac_dec_rst_n_i[3:0]. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_tx_ana_lpbk_dac_dec_clk_i	R/W	Override value set on (TXS) analog port tx_ana_lpbk_dac_dec_clk_i. This value is set when the override enable register for this (TXS) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.3.42 TXS_STAT_0

- **Name:** (TXS) Status Register 0
- **Description:** Returns status of various internal signals in (TXS). Register 0.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xA4

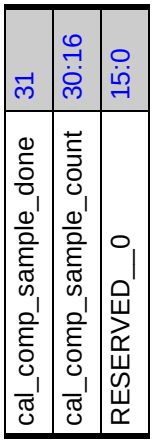


Table 11-160 Fields for Register: TXS_STAT_0

Bits	Name	Memory Access	Description
31	cal_comp_sample_done	R	Status of internal design signal cal_sample_done. **Value After Reset:** 0x0
30:16	cal_comp_sample_count	R	Status of internal design signal cal_comp_sample_count. **Value After Reset:** 0x0
15:0	RESERVED__0	R	Reserved **Reserved Field:** Yes **Value After Reset:** 0x0

11.3.43 TXS_STAT_1

- **Name:** (TXS) Status Register 1
- **Description:** Returns status of various internal signals in (TXS). Register 1.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xA8

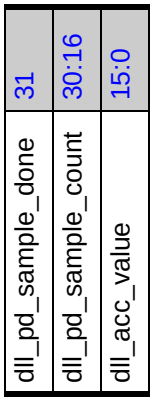


Table 11-161 Fields for Register: TXS_STAT_1

Bits	Name	Memory Access	Description
31	dll_pd_sample_done	R	Status of internal design signal dll_pd_sample_done. **Value After Reset:** 0x0
30:16	dll_pd_sample_count	R	Status of internal design signal dll_pd_sample_count. **Value After Reset:** 0x0
15:0	dll_acc_value	R	Status of internal design dll_acc. **Value After Reset:** 0x0

11.3.44 TXS_STAT_2

- **Name:** (TXS) Status Register 2
- **Description:** Returns status of various internal signals in (TXS). Register 2.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xAC

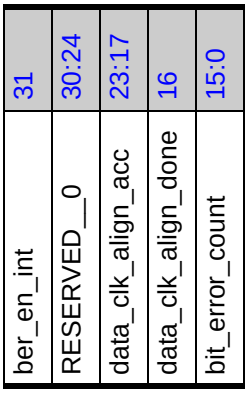


Table 11-162 Fields for Register: TXS_STAT_2

Bits	Name	Memory Access	Description
31	ber_en_int	R	Internal BER enable status. When using the ber_ref_timer, this bit will be 0 when ber_ref_timer counter has expired. **Value After Reset:** 0x0
30:24	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
23:17	data_clk_align_acc	R	Status of Internal Accumulator for Data"Clock Alignment. **Value After Reset:** 0x0
16	data_clk_align_done	R	Done Status for Data Clock Alignment. **Value After Reset:** 0x0
15:0	bit_error_count	R	Bit error rate (BER) counter status. **Value After Reset:** 0x0

11.3.45 TXS_STAT_3

- **Name:** (TXS) Status Register 3
- **Description:** Returns status of various internal signals in (TXS). Register 3.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xB0



Table 11-163 Fields for Register: TXS_STAT_3

Bits	Name	Memory Access	Description
31:16	odd_bit_error_count	R	Odd position of received parallel data is defined as bit locations 1, 3, 5, 7, etc. Count of bit errors, in odd location, during BER test. **Value After Reset:** 0x0
15:0	even_bit_error_count	R	Even position of received parallel data is defined as bit locations 0, 2, 4, 8, etc. Count of bit errors, in even location, during BER test. **Value After Reset:** 0x0

11.3.46 TXS_STAT_4

- **Name:** (TXS) Status Register 4
- **Description:** Returns status of various internal signals in (TXS). Register 4.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xB4

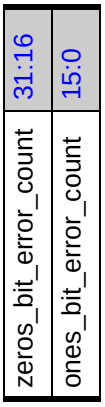


Table 11-164 Fields for Register: TXS_STAT_4

Bits	Name	Memory Access	Description
31:16	zeros_bit_error_count	R	Counter of bit errors in locations where expected value is 0b0 during BER test. **Value After Reset:** 0x0
15:0	ones_bit_error_count	R	Counter of bit errors in locations where expected value is 0b1 during BER test. **Value After Reset:** 0x0

11.3.47 TXS_STAT_5

- **Name:** (TXS) Status Register 5
- **Description:** Returns status of various internal signals in (TXS). Register 5.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xB8



Table 11-165 Fields for Register: TXS_STAT_5

Bits	Name	Memory Access	Description
31:0	bit_error_location	R	The latest erroneous received data pattern which caused an increment in BER counter. Field 1. **Value After Reset:** 0x0

11.3.48 TXS_STAT_6

- **Name:** (TXS) Status Register 6
- **Description:** Returns status of various internal signals in (TXS). Register 6.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xBC



Table 11-166 Fields for Register: TXS_STAT_6

Bits	Name	Memory Access	Description
31:0	bit_error_location	R	The latest erroneous received data pattern which caused an increment in BER counter. Field 2. **Value After Reset:** 0x0

11.3.49 TXS_STAT_7

- **Name:** (TXS) Status Register 7
- **Description:** Returns status of various internal signals in (TXS). Register 7.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xC0



Table 11-167 Fields for Register: TXS_STAT_7

Bits	Name	Memory Access	Description
31:0	bit_error_pattern	R	If any bit is 0b1, bit error was detected for that bit in the reference pattern as available on (APB::RXS::BER_STAT::ber_error_pattern). Field 1. **Value After Reset:** 0x0

11.3.50 TXS_STAT_8

- **Name:** (TXS) Status Register 8
- **Description:** Returns status of various internal signals in (TXS). Register 8.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xC4



Table 11-168 Fields for Register: TXS_STAT_8

Bits	Name	Memory Access	Description
31:0	bit_error_pattern	R	If any bit is 0b1, bit error was detected for that bit in the reference pattern as available on (APB::RXS::BER_STAT::ber_error_pattern). Field 2. **Value After Reset:** 0x0

11.3.51 TXS_STAT_9

- **Name:** (TXS) Status Register 9
- **Description:** Returns status of various internal signals in (TXS). Register 9.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xC8

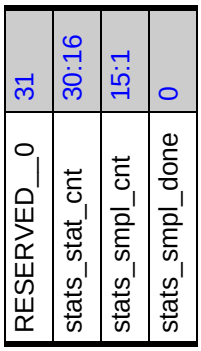


Table 11-169 Fields for Register: TXS_STAT_9

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30:16	stats_stat_cnt	R	Value for capture scope level. **Value After Reset:** 0x0
15:1	stats_smpl_cnt	R	Number of samples counted in the scope. **Value After Reset:** 0x0
0	stats_smpl_done	R	Indication of stats scope sample counter done. **Value After Reset:** 0x1

11.3.52 TXS_STAT_10

- **Name:** (TXS) Status Register 10
- **Description:** Returns status of various internal signals in (TXS). Register 10.
- **Size:** 32 bits
- **Offset:** 0x2000 + j*0x100 + 0xCC

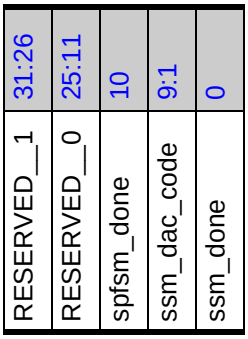


Table 11-170 Fields for Register: TXS_STAT_10

Bits	Name	Memory Access	Description
31:26	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
25:11	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
10	spfsm_done	R	Flag for SPFSM operation finish. **Value After Reset:** 0x1
9:1	ssm_dac_code	R	Searched DAC code from the SSM. **Value After Reset:** 0x100
0	ssm_done	R	Flag for SSM operation finish. **Value After Reset:** 0x1

11.4 SUP_TX_SUP Registers

11.4.1 SUP_CFG_0

- **Name:** (SUP) Configuration Register
- **Description:** Configures (SUP) settings.
- **Size:** 32 bits
- **Offset:** 0x3000

31	30	29	28:26	25:20	19:17	16:5	4	3	2	1	0
RESERVED__4	RESERVED__3	ref_ana_clk_sel	RESERVED__2	sup_dbg_clkout_sel	sup_dbg_ana_clk_sel	RESERVED__1	RESERVED__0	polarity_inv	graycode_en	pam4_ab_swap_en	signal_type

Table 11-171 Fields for Register: SUP_CFG_0

Bits	Name	Memory Access	Description
31	RESERVED__4	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	RESERVED__3	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	ref_ana_clk_sel	R/W	Selects between refa_ana_clk or ref_ro_ana_clk to be used to generate ref_dig_clk. **Values:** ■ 0x0 (REFANA): uses refa_ana_clk ■ 0x1 (ROANA): uses ref_ro_ana_clk **Value After Reset:** 0x0
28:26	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
25:20	sup_dbg_clkout_sel	R/W	Control for top debug clock selection. Choose between lanes/sup. **Values:** ■ 0x0 (_0): ZERO ■ 0x1 (L0): lane 0 ■ 0x2 (L1): lane 1 ■ 0x3 (L2): lane 2 ■ 0x4 (L3): lane 3 ■ 0x5 (L4): lane 4 ■ 0x6 (L5): lane 5 ■ 0x7 (L6): lane 6 ■ 0x8 (L7): lane 7 ■ 0x9 (L8): lane 8 ■ 0xa (L9): lane 9 ■ 0xb (L10): lane 10 ■ 0xc (L11): lane 11 ■ 0xd (L12): lane 12 ■ 0xe (L13): lane 13 ■ 0xf (L14): lane 14 ■ 0x10 (L15): lane 15 ■ 0x11 (SUP): SUP **Value After Reset:** 0x0
19:17	sup_dbg_ana_clk_sel	R/W	Selects between clks to observe on sup_dbg_clk_o. **Values:** ■ 0x0 (_0): ZERO ■ 0x1 (ANA): ref_ana_clk ■ 0x2 (ROANA): ref_ro_ana_clk ■ 0x3 (DIG): ref_dig_clk ■ 0x4 (ANAFB): mpll_ana_fb_clk ■ 0x5 (ANAWRD/8): mpll_ana_word_clk / 8 ■ 0x6 (ANAWRD/16): ropll_ana_word_clk / 16 ■ 0x7 (ANADIV): ropll_ana_div_clk **Value After Reset:** 0x0
16:5	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
3	polarity_inv	R/W	RESERVED **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
2	graycode_en	R/W	RESERVED **Value After Reset:** 0x1
1	pam4_ab_swap_en	R/W	RESERVED **Value After Reset:** 0x0
0	signal_type	R/W	RESERVED **Value After Reset:** 0x0

11.4.2 SUP_OVRDEN

- **Name:** (SUP) Internal Logic Override Enable Register
- **Description:** Register bits select signal of the mux which overrides value on (SUP) internal logic signals.
- **Size:** 32 bits
- **Offset:** 0x3004



Table 11-172 Fields for Register: SUP_OVRDEN

Bits	Name	Memory Access	Description
31:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.4.3 SUP_OVRDVAL

- **Name:** (SUP) Internal Logic Override Value Register
- **Description:** Register bits act as override value at the mux which can override values on (SUP) internal logic signals. Note that when the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3008



Table 11-173 Fields for Register: SUP_OVRDVAL

Bits	Name	Memory Access	Description
31:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.4.4 SUP_STAT

- **Name:** (SUP) Status Register
- **Description:** RESERVED
- **Size:** 32 bits
- **Offset:** 0x300c

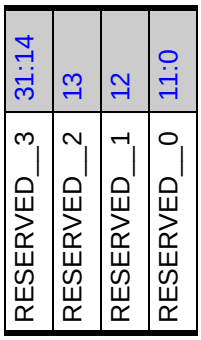


Table 11-174 Fields for Register: SUP_STAT

Bits	Name	Memory Access	Description
31:14	RESERVED__3	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
13	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
12	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
11:0	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0

11.4.5 ANA_OVRDVAL_0

- **Name:** OSC Configuration Register 0
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 0. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3010

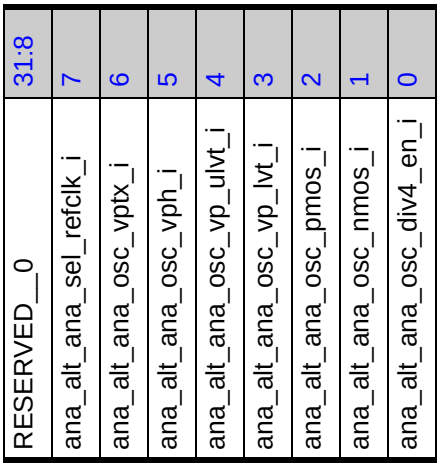


Table 11-175 Fields for Register: ANA_OVRDVAL_0

Bits	Name	Memory Access	Description
31:8	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
7	ana_alt_ana_sel_refclk_i	R/W	Override value set on (SUP) analog port ana_alt_ana_sel_refclk_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
6	ana_alt_ana_osc_vptx_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_vptx_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5	ana_alt_ana_osc_vph_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_vph_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
4	ana_alt_ana_osc_vp_ulvt_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_vp_ulvt_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3	ana_alt_ana_osc_vp_lvt_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_vp_lvt_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_alt_ana_osc_pmos_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_pmos_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1	ana_alt_ana_osc_nmos_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_nmos_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_alt_ana_osc_div4_en_i	R/W	Override value set on (SUP) analog port ana_alt_ana_osc_div4_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.6 ANA_OVRDVAL_1

- **Name:** ANA Configuration Register 1
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 1. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3014

31:23	22	21:18	17:10	9:8	7:6	5	4	3	2	1	0
RESERVED__0	ana_vref_pre_i	ana_sup_ana_term_ctrl_i	ana_sup_ana_misc_i	ana_nominal_vph_sel_i	ana_nominal_vp_sel_i	ana_bg_ana_vref_fast_start_i	ana_bg_ana_kick_start_en_i	ana_bg_ana_en_i	ana_ana_pwrdn_i	ana_ana_flyover_en_i	ana_ana_burnin_i

Table 11-176 Fields for Register: ANA_OVRDVAL_1

Bits	Name	Memory Access	Description
31:23	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
22	ana_vref_pre_i	R/W	Override value set on (SUP) analog port ana_vref_pre_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
21:18	ana_sup_ana_term_ctrl_i	R/W	Override value set on (SUP) analog port ana_sup_ana_term_ctrl_i[3:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
17:10	ana_sup_ana_misc_i	R/W	Override value set on (SUP) analog port ana_sup_ana_misc_i[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9:8	ana_nominal_vph_sel_i	R/W	Override value set on (SUP) analog port ana_nominal_vph_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x2 **Volatile:** true
7:6	ana_nominal_vp_sel_i	R/W	Override value set on (SUP) analog port ana_nominal_vp_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5	ana_bg_ana_vref_fast_start_i	R/W	Override value set on (SUP) analog port ana_bg_ana_vref_fast_start_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4	ana_bg_ana_kick_start_en_i	R/W	Override value set on (SUP) analog port ana_bg_ana_kick_start_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3	ana_bg_ana_en_i	R/W	Override value set on (SUP) analog port ana_bg_ana_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_ana_pwrdn_i	R/W	Override value set on (SUP) analog port ana_ana_pwrdn_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
1	ana_ana_flyover_en_i	R/W	Override value set on (SUP) analog port ana_ana_flyover_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
0	ana_ana_burnin_i	R/W	Override value set on (SUP) analog port ana_ana_burnin_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.7 ANA_OVRDVAL_2

- **Name:** CRU Access Register 2
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 2. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3018

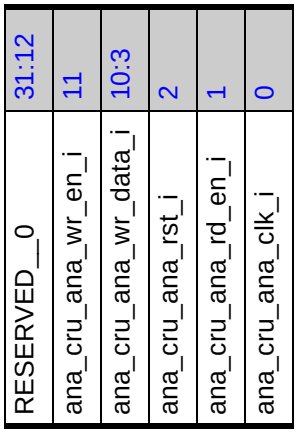


Table 11-177 Fields for Register: ANA_OVRDVAL_2

Bits	Name	Memory Access	Description
31:12	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
11	ana_cru_ana_wr_en_i	R/W	Override value set on (SUP) analog port ana_cru_ana_wr_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:3	ana_cru_ana_wr_data_i	R/W	Override value set on (SUP) analog port ana_cru_ana_wr_data_i[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_cru_ana_rst_i	R/W	Override value set on (SUP) analog port ana_cru_ana_rst_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
1	ana_cru_ana_rd_en_i	R/W	Override value set on (SUP) analog port ana_cru_ana_rd_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_cru_ana_clk_i	R/W	Override value set on (SUP) analog port ana_cru_ana_clk_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.8 ANA_OVRDVAL_3

- **Name:** CRU Access Register 3
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 3. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x301c

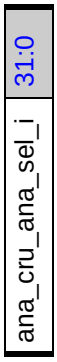


Table 11-178 Fields for Register: ANA_OVRDVAL_3

Bits	Name	Memory Access	Description
31:0	ana_cru_ana_sel_i	R/W	Override value set on (SUP) analog port ana_cru_ana_sel_i[27:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.9 ANA_OVRDVAL_4

- **Name:** DIFFBUF Configuration Register 4
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 4. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3020

31:16	15	14	13	12	11	10:6	5	4:2	1:0
RESERVED__0	ana_atb_ana_sup_en_i	ana_atb_ana_lane_right_en_i	ana_atb_ana_lane_left_en_i	ana_diffbuf_tx_en_i	ana_diffbuf_pulldn_i	ana_diffbuf_imp_ctrl_i	ana_diffbuf_de_emp_en_i	ana_diffbuf_de_emp_ctrl_i	ana_diffbuf_clk_sel_i

Table 11-179 Fields for Register: ANA_OVRDVAL_4

Bits	Name	Memory Access	Description
31:16	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
15	ana_atb_ana_sup_en_i	R/W	Override value set on (SUP) analog port ana_atb_ana_sup_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14	ana_atb_ana_lane_right_en_i	R/W	Override value set on (SUP) analog port ana_atb_ana_lane_right_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
13	ana_atb_ana_lane_left_en_i	R/W	Override value set on (SUP) analog port ana_atb_ana_lane_left_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
12	ana_diffbuf_tx_en_i	R/W	Override value set on (SUP) analog port ana_diffbuf_tx_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11	ana_diffbuf_pulldn_i	R/W	Override value set on (SUP) analog port ana_diffbuf_pulldn_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:6	ana_diffbuf_imp_ctrl_i	R/W	Override value set on (SUP) analog port ana_diffbuf_imp_ctrl_i[4:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5	ana_diffbuf_de_emp_en_i	R/W	Override value set on (SUP) analog port ana_diffbuf_de_emp_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4:2	ana_diffbuf_de_emp_ctrl_i	R/W	Override value set on (SUP) analog port ana_diffbuf_de_emp_ctrl_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
1:0	ana_diffbuf_clk_sel_i	R/W	Override value set on (SUP) analog port ana_diffbuf_clk_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.10 ANA_OVRDVAL_5

- **Name:** MPLL Analog Configuration Register 5
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 5. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3024

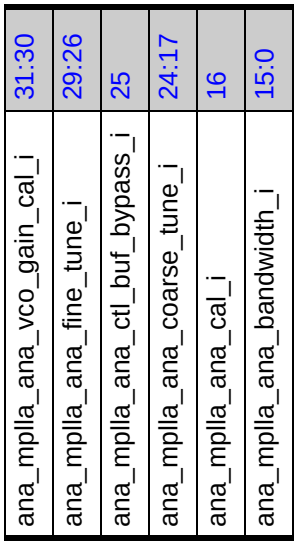


Table 11-180 Fields for Register: ANA_OVRDVAL_5

Bits	Name	Memory Access	Description
31:30	ana_mplla_ana_vco_gain_cal_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_vco_gain_cal_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
29:26	ana_mplla_ana_fine_tune_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_fine_tune_i[3:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_mplla_ana_ctl_buf_bypass_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_ctl_buf_bypass_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
24:17	ana_mplla_ana_coarse_tune_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_coarse_tune_i[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_mplla_ana_cal_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_cal_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:0	ana_mplla_ana_bandwidth_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_bandwidth_i[15:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x161f **Volatile:** true

11.4.11 ANA_OVRDVAL_6

- **Name:** MPLL Startup Configuration Register 6
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 6. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3028

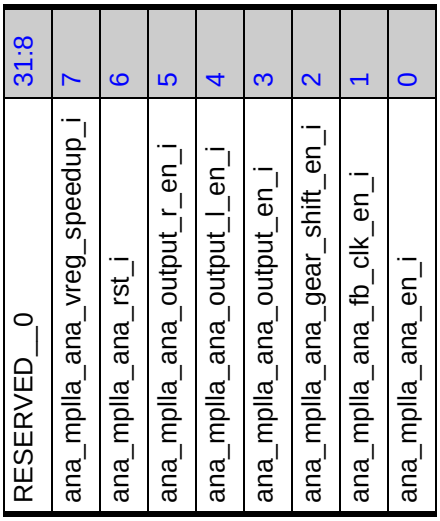


Table 11-181 Fields for Register: ANA_OVRDVAL_6

Bits	Name	Memory Access	Description
31:8	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
7	ana_mplla_ana_vreg_speedup_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_vreg_speedup_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
6	ana_mplla_ana_rst_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_rst_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true

Bits	Name	Memory Access	Description
5	ana_mplla_ana_output_r_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_output_r_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
4	ana_mplla_ana_output_l_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_output_l_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3	ana_mplla_ana_output_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_output_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
2	ana_mplla_ana_gear_shift_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_gear_shift_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
1	ana_mplla_ana_fb_clk_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_fb_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_mplla_ana_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.12 ANA_OVRDVAL_7

- **Name:** MPLL Dividers Configuration Register 7
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 7. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x302c

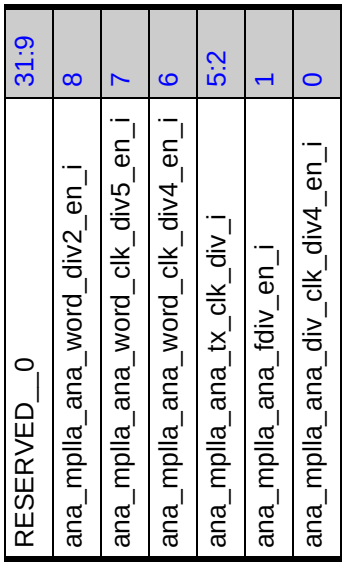


Table 11-182 Fields for Register: ANA_OVRDVAL_7

Bits	Name	Memory Access	Description
31:9	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
8	ana_mplla_ana_word_div2_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_word_div2_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7	ana_mplla_ana_word_clk_div5_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_word_clk_div5_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true

Bits	Name	Memory Access	Description
6	ana_mplla_ana_word_clk_div4_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_word_clk_div4_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5:2	ana_mplla_ana_tx_clk_div_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_tx_clk_div_i[3:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
1	ana_mplla_ana_fdiv_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_fdiv_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_mplla_ana_div_clk_div4_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_div_clk_div4_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.13 ANA_OVRDVAL_8

- **Name:** MPLL PMIX Configuration Register 8
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 8. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3030

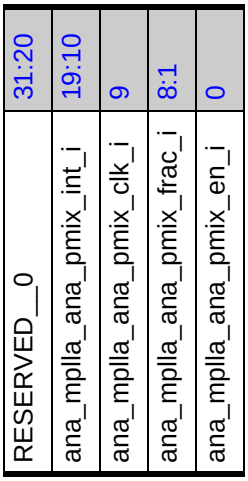


Table 11-183 Fields for Register: ANA_OVRDVAL_8

Bits	Name	Memory Access	Description
31:20	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
19:10	ana_mplla_ana_pmix_int_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_pmix_int_i[9:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
9	ana_mplla_ana_pmix_clk_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_pmix_clk_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8:1	ana_mplla_ana_pmix_frac_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_pmix_frac_i[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
0	ana_mplla_ana_pmix_en_i	R/W	Override value set on (SUP) analog port ana_mplla_ana_pmix_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.14 ANA_OVRDVAL_9

- **Name:** REF Configuration Register 9
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 9. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3034

31	30	29	28:27	26	25	24	23:22	21	20:19	18	17:15	14:13	12	11:9	8	7:6	5:0
RESERVED__0	ana_refa_ana_lane_clk_en_i	ana_refa_ana_clkdet_en_i	ana_refa_ana_clk_sel_i	ana_refa_ana_clk_en_i	ana_ref_ro_ana_lane_clk_en_i	ana_ref_ro_ana_clkdet_en_i	ana_ref_ro_ana_clk_sel_i	ana_ref_ro_ana_clk_en_i	ana_ref_ana_vreg_ref_sel_i	ana_ref_ana_vreg_fast_start_i	ana_ref_ana_ropll_div_i	ana_ref_ana_repeat_clk_sel_i	ana_ref_ana_repeat_clk_en_i	ana_ref_ana_mplla_div_i	ana_ref_ana_diffbuf_clk_en_i	ana_ref_ana_dco_range_i	ana_ref_ana_dco_finetune_i

Table 11-184 Fields for Register: ANA_OVRDVAL_9

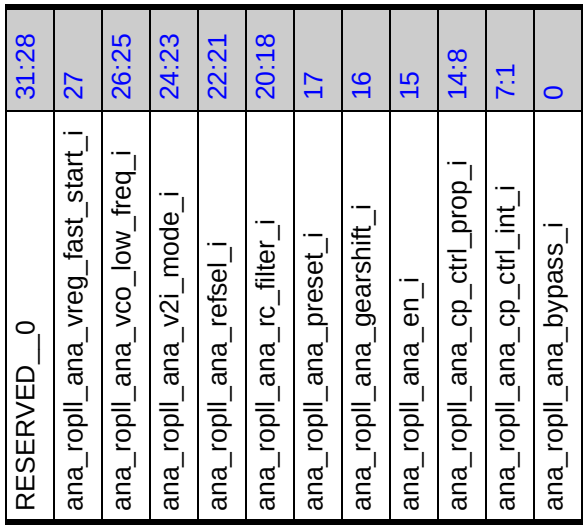
Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	ana_refa_ana_lane_clk_en_i	R/W	Override value set on (SUP) analog port ana_refa_ana_lane_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
29	ana_refa_ana_clkdet_en_i	R/W	Override value set on (SUP) analog port ana_refa_ana_clkdet_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
28:27	ana_refa_ana_clk_sel_i	R/W	Override value set on (SUP) analog port ana_refa_ana_clk_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
26	ana_refa_ana_clk_en_i	R/W	Override value set on (SUP) analog port ana_refa_ana_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_ref_ro_ana_lane_clk_en_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_lane_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	ana_ref_ro_ana_clkdet_en_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clkdet_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
23:22	ana_ref_ro_ana_clk_sel_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clk_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
21	ana_ref_ro_ana_clk_en_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
20:19	ana_ref_ana_vreg_ref_sel_i	R/W	Override value set on (SUP) analog port ana_ref_ana_vreg_ref_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x3 **Volatile:** true
18	ana_ref_ana_vreg_fast_start_i	R/W	Override value set on (SUP) analog port ana_ref_ana_vreg_fast_start_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
17:15	ana_ref_ana_ropll_div_i	R/W	Override value set on (SUP) analog port ana_ref_ana_ropll_div_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14:13	ana_ref_ana_repeat_clk_sel_i	R/W	Override value set on (SUP) analog port ana_ref_ana_repeat_clk_sel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
12	ana_ref_ana_repeat_clk_en_i	R/W	Override value set on (SUP) analog port ana_ref_ana_repeat_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
11:9	ana_ref_ana_mplla_div_i	R/W	Override value set on (SUP) analog port ana_ref_ana_mplla_div_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8	ana_ref_ana_diffbuf_clk_en_i	R/W	Override value set on (SUP) analog port ana_ref_ana_diffbuf_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7:6	ana_ref_ana_dco_range_i	R/W	Override value set on (SUP) analog port ana_ref_ana_dco_range_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
5:0	ana_ref_ana_dco_finetune_i	R/W	Override value set on (SUP) analog port ana_ref_ana_dco_finetune_i[5:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0xd **Volatile:** true

11.4.15 ANA_OVRDVAL_10

- **Name:** ROPLL Startup Configuration Register 10
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 10. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3038



31:28	27	26:25	24:23	22:21	20:18	17	16	15	14:8	7:1	0
RESERVED__0	ana_ropll_ana_vreg_fast_start_i	ana_ropll_ana_vco_low_freq_i	ana_ropll_ana_v2i_mode_i	ana_ropll_ana_refsel_i	ana_ropll_ana_rc_filter_i	ana_ropll_ana_preset_i	ana_ropll_ana_gearshift_i	ana_ropll_ana_en_i	ana_ropll_ana_cp_ctrl_prop_i	ana_ropll_ana_cp_ctrl_int_i	ana_ropll_ana_bypass_i

Table 11-185 Fields for Register: ANA_OVRDVAL_10

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27	ana_ropll_ana_vreg_fast_start_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_vreg_fast_start_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
26:25	ana_ropll_ana_vco_low_freq_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_vco_low_freq_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
24:23	ana_ropll_ana_v2i_mode_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_v2i_mode_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x3 **Volatile:** true
22:21	ana_ropll_ana_refsel_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_refsel_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
20:18	ana_ropll_ana_rc_filter_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_rc_filter_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x4 **Volatile:** true
17	ana_ropll_ana_preset_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_preset_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_ropll_ana_gearshift_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_gearshift_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
15	ana_ropll_ana_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
14:8	ana_ropll_ana_cp_ctrl_prop_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_cp_ctrl_prop_i[6:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7:1	ana_ropll_ana_cp_ctrl_int_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_cp_ctrl_int_i[6:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
0	ana_ropll_ana_bypass_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_bypass_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.16 ANA_OVRDVAL_11

- **Name:** ROPLL Dividers Configuration Register 11
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 11. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x303c

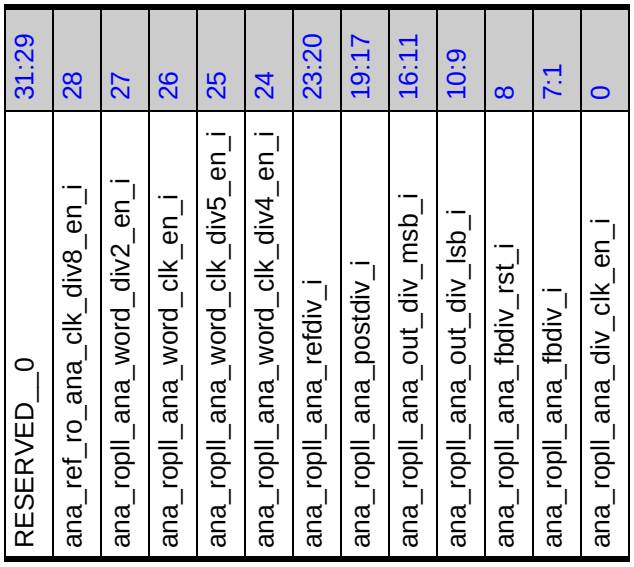


Table 11-186 Fields for Register: ANA_OVRDVAL_11

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	ana_ref_ro_ana_clk_div8_en_i	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clk_div8_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
27	ana_ropll_ana_word_div2_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_word_div2_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
26	ana_ropll_ana_word_clk_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_word_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
25	ana_ropll_ana_word_clk_div5_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_word_clk_div5_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
24	ana_ropll_ana_word_clk_div4_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_word_clk_div4_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
23:20	ana_ropll_ana_refdiv_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_refdiv_i[3:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
19:17	ana_ropll_ana_postdiv_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_postdiv_i[2:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x1 **Volatile:** true
16:11	ana_ropll_ana_out_div_msb_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_out_div_msb_i[5:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
10:9	ana_ropll_ana_out_div_lsb_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_out_div_lsb_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
8	ana_ropll_ana_fbdiv_rst_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_fbdiv_rst_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
7:1	ana_ropll_ana_fbdiv_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_fbdiv_i[6:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0xe **Volatile:** true
0	ana_ropll_ana_div_clk_en_i	R/W	Override value set on (SUP) analog port ana_ropll_ana_div_clk_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.17 ANA_OVRDVAL_12

- **Name:** RTRX Configuration Register 12
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 12. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3040

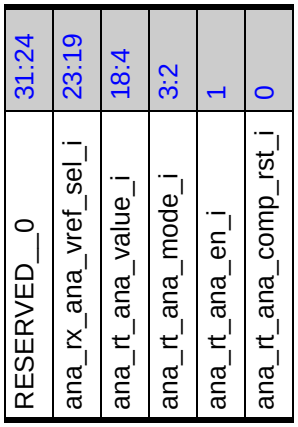


Table 11-187 Fields for Register: ANA_OVRDVAL_12

Bits	Name	Memory Access	Description
31:24	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
23:19	ana_rx_ana_vref_sel_i	R/W	Override value set on (SUP) analog port ana_rx_ana_vref_sel_i[4:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
18:4	ana_rt_ana_value_i	R/W	Override value set on (SUP) analog port ana_rt_ana_value_i[9:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
3:2	ana_rt_ana_mode_i	R/W	Override value set on (SUP) analog port ana_rt_ana_mode_i[1:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
1	ana_rt_ana_en_i	R/W	Override value set on (SUP) analog port ana_rt_ana_en_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
0	ana_rt_ana_comp_rst_i	R/W	Override value set on (SUP) analog port ana_rt_ana_comp_rst_i. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.18 ANA_OVRDEN_0

- **Name:** (SUP) Internal Override Enable Register 0
- **Description:** Bits that act as select signal of the mux which overrides value on (SUP) input and output ports Register 0
- **Size:** 32 bits
- **Offset:** 0x3044

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED__0	ovrd_en_ana_diffbuf_tx_en_i	ovrd_en_ana_diffbuf_pulldn_i	ovrd_en_ana_diffbuf_imp_ctrl_i	ovrd_en_ana_diffbuf_de_emp_en_i	ovrd_en_ana_diffbuf_de_emp_ctrl_i	ovrd_en_ana_diffbuf_clk_sel_i	ovrd_en_ana_cru_ana_sel_i	ovrd_en_ana_cru_ana_wr_en_i	ovrd_en_ana_cru_ana_wr_data_i	ovrd_en_ana_cru_ana_rst_i	ovrd_en_ana_cru_ana_rd_en_i	ovrd_en_ana_cru_ana_clk_i	ovrd_en_ana_vref_pre_i	ovrd_en_ana_sup_ana_term_ctrl_i	ovrd_en_ana_sup_ana_misc_i	ovrd_en_ana_nominal_vph_sel_i	ovrd_en_ana_nominal_vp_sel_i	ovrd_en_ana_bg_ana_vref_fast_start_i	ovrd_en_ana_bg_ana_kick_start_en_i	ovrd_en_ana_bg_ana_en_i	ovrd_en_ana_ana_pwrdn_i	ovrd_en_ana_ana_flyover_en_i	ovrd_en_ana_ana_burnin_i	ovrd_en_ana_alt_ana_sel_refclk_i	ovrd_en_ana_alt_ana_osc_vptx_i	ovrd_en_ana_alt_ana_osc_vph_i	ovrd_en_ana_alt_ana_osc_vp_ulvt_i	ovrd_en_ana_alt_ana_osc_vp_lvt_i	ovrd_en_ana_alt_ana_osc_pmos_i	ovrd_en_ana_alt_ana_osc_nmos_i	ovrd_en_ana_alt_ana_osc_div4_en_i

Table 11-188 Fields for Register: ANA_OVRDEN_0

Bits	Name	Memory Access	Description
31	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x1
30	ovrd_en_ana_diffbuf_tx_en_i	R/W	Override enable for (SUP) port ana_diffbuf_tx_en_i. **Value After Reset:** 0x1
29	ovrd_en_ana_diffbuf_pulldn_i	R/W	Override enable for (SUP) port ana_diffbuf_pulldn_i. **Value After Reset:** 0x1
28	ovrd_en_ana_diffbuf_imp_ctrl_i	R/W	Override enable for (SUP) port ana_diffbuf_imp_ctrl_i. **Value After Reset:** 0x1
27	ovrd_en_ana_diffbuf_de_emp_en_i	R/W	Override enable for (SUP) port ana_diffbuf_de_emp_en_i. **Value After Reset:** 0x1
26	ovrd_en_ana_diffbuf_de_emp_ctrl_i	R/W	Override enable for (SUP) port ana_diffbuf_de_emp_ctrl_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
25	ovrd_en_ana_diffbuf_clk_sel_i	R/W	Override enable for (SUP) port ana_diffbuf_clk_sel_i. **Value After Reset:** 0x1
24	ovrd_en_ana_cru_ana_sel_i	R/W	Override enable for (SUP) port ana_cru_ana_sel_i. **Value After Reset:** 0x1
23	ovrd_en_ana_cru_ana_wr_en_i	R/W	Override enable for (SUP) port ana_cru_ana_wr_en_i. **Value After Reset:** 0x1
22	ovrd_en_ana_cru_ana_wr_data_i	R/W	Override enable for (SUP) port ana_cru_ana_wr_data_i. **Value After Reset:** 0x1
21	ovrd_en_ana_cru_ana_rst_i	R/W	Override enable for (SUP) port ana_cru_ana_rst_i. **Value After Reset:** 0x1
20	ovrd_en_ana_cru_ana_rd_en_i	R/W	Override enable for (SUP) port ana_cru_ana_rd_en_i. **Value After Reset:** 0x1
19	ovrd_en_ana_cru_ana_clk_i	R/W	Override enable for (SUP) port ana_cru_ana_clk_i. **Value After Reset:** 0x1
18	ovrd_en_ana_vref_pre_i	R/W	Override enable for (SUP) port ana_vref_pre_i. **Value After Reset:** 0x1
17	ovrd_en_ana_sup_ana_term_ctrl_i	R/W	Override enable for (SUP) port ana_sup_ana_term_ctrl_i. **Value After Reset:** 0x1
16	ovrd_en_ana_sup_ana_misc_i	R/W	Override enable for (SUP) port ana_sup_ana_misc_i. **Value After Reset:** 0x1
15	ovrd_en_ana_nominal_vph_sel_i	R/W	Override enable for (SUP) port ana_nominal_vph_sel_i. **Value After Reset:** 0x1
14	ovrd_en_ana_nominal_vp_sel_i	R/W	Override enable for (SUP) port ana_nominal_vp_sel_i. **Value After Reset:** 0x1
13	ovrd_en_ana_bg_ana_vref_fast_start_i	R/W	Override enable for (SUP) port ana_bg_ana_vref_fast_start_i. **Value After Reset:** 0x1
12	ovrd_en_ana_bg_ana_kick_start_en_i	R/W	Override enable for (SUP) port ana_bg_ana_kick_start_en_i. **Value After Reset:** 0x1
11	ovrd_en_ana_bg_ana_en_i	R/W	Override enable for (SUP) port ana_bg_ana_en_i. **Value After Reset:** 0x1
10	ovrd_en_ana_ana_pwrdn_i	R/W	Override enable for (SUP) port ana_ana_pwrdn_i. **Value After Reset:** 0x1
9	ovrd_en_ana_ana_flyover_en_i	R/W	Override enable for (SUP) port ana_ana_flyover_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
8	ovrd_en_ana_ana_burnin_i	R/W	Override enable for (SUP) port ana_ana_burnin_i. **Value After Reset:** 0x1
7	ovrd_en_ana_alt_ana_sel_refclk_i	R/W	Override enable for (SUP) port ana_alt_ana_sel_refclk_i. **Value After Reset:** 0x1
6	ovrd_en_ana_alt_ana_osc_vptx_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_vptx_i. **Value After Reset:** 0x1
5	ovrd_en_ana_alt_ana_osc_vph_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_vph_i. **Value After Reset:** 0x1
4	ovrd_en_ana_alt_ana_osc_vp_ulvt_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_vp_ulvt_i. **Value After Reset:** 0x1
3	ovrd_en_ana_alt_ana_osc_vp_lvt_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_vp_lvt_i. **Value After Reset:** 0x1
2	ovrd_en_ana_alt_ana_osc_pmos_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_pmos_i. **Value After Reset:** 0x1
1	ovrd_en_ana_alt_ana_osc_nmos_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_nmos_i. **Value After Reset:** 0x1
0	ovrd_en_ana_alt_ana_osc_div4_en_i	R/W	Override enable for (SUP) port ana_alt_ana_osc_div4_en_i. **Value After Reset:** 0x1

11.4.19 ANA_OVRDEN_1

- **Name:** (SUP) Internal Override Enable Register 1
- **Description:** Bits that act as select signal of the mux which overrides value on (SUP) input and output ports Register 1
- **Size:** 32 bits
- **Offset:** 0x3048

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ovrd_en_ana_ref_ana_vreg_fast_start_i	ovrd_en_ana_ref_ana_ropll_div_i	ovrd_en_ana_ref_ana_repeat_clk_sel_i	ovrd_en_ana_ref_ana_repeat_clk_en_i	ovrd_en_ana_ref_ana_mplla_div_i	ovrd_en_ana_ref_ana_diffbuf_clk_en_i	ovrd_en_ana_ref_ana_dco_range_i	ovrd_en_ana_ref_ana_dco_finetune_i	ovrd_en_ana_mplla_ana_pmix_int_i	ovrd_en_ana_mplla_ana_pmix_frac_i	ovrd_en_ana_mplla_ana_pmix_en_i	ovrd_en_ana_mplla_ana_pmix_clk_i	ovrd_en_ana_mplla_ana_word_div2_en_i	ovrd_en_ana_mplla_ana_word_clk_div5_en_i	ovrd_en_ana_mplla_ana_word_clk_div4_en_i	ovrd_en_ana_mplla_ana_tx_clk_div_i	ovrd_en_ana_mplla_ana_fdiv_en_i	ovrd_en_ana_mplla_ana_div_clk_div4_en_i	ovrd_en_ana_mplla_ana_vreg_speedup_i	ovrd_en_ana_mplla_ana_rst_i	ovrd_en_ana_mplla_ana_output_r_en_i	ovrd_en_ana_mplla_ana_output_l_en_i	ovrd_en_ana_mplla_ana_output_en_i	ovrd_en_ana_mplla_ana_gear_shift_en_i	ovrd_en_ana_mplla_ana_fb_clk_en_i	ovrd_en_ana_mplla_ana_en_i	ovrd_en_ana_mplla_ana_vco_gain_cal_i	ovrd_en_ana_mplla_ana_fine_tune_i	ovrd_en_ana_mplla_ana_ctl_buf_bypass_i	ovrd_en_ana_mplla_ana_coarse_tune_i	ovrd_en_ana_mplla_ana_cal_i	ovrd_en_ana_mplla_ana_bandwidth_i

Table 11-189 Fields for Register: ANA_OVRDEN_1

Bits	Name	Memory Access	Description
31	ovrd_en_ana_ref_ana_vreg_fast_start_i	R/W	Override enable for (SUP) port ana_ref_ana_vreg_fast_start_i. **Value After Reset:** 0x1
30	ovrd_en_ana_ref_ana_ropll_div_i	R/W	Override enable for (SUP) port ana_ref_ana_ropll_div_i. **Value After Reset:** 0x1
29	ovrd_en_ana_ref_ana_repeat_clk_sel_i	R/W	Override enable for (SUP) port ana_ref_ana_repeat_clk_sel_i. **Value After Reset:** 0x1
28	ovrd_en_ana_ref_ana_repeat_clk_en_i	R/W	Override enable for (SUP) port ana_ref_ana_repeat_clk_en_i. **Value After Reset:** 0x1
27	ovrd_en_ana_ref_ana_mplla_div_i	R/W	Override enable for (SUP) port ana_ref_ana_mplla_div_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
26	ovrd_en_ana_ref_ana_diffbuf_clk_en_i	R/W	Override enable for (SUP) port ana_ref_ana_diffbuf_clk_en_i. **Value After Reset:** 0x1
25	ovrd_en_ana_ref_ana_dco_range_i	R/W	Override enable for (SUP) port ana_ref_ana_dco_range_i. **Value After Reset:** 0x1
24	ovrd_en_ana_ref_ana_dco_finetune_i	R/W	Override enable for (SUP) port ana_ref_ana_dco_finetune_i. **Value After Reset:** 0x1
23	ovrd_en_ana_mplla_ana_pmix_int_i	R/W	Override enable for (SUP) port ana_mplla_ana_pmix_int_i. **Value After Reset:** 0x1
22	ovrd_en_ana_mplla_ana_pmix_frac_i	R/W	Override enable for (SUP) port ana_mplla_ana_pmix_frac_i. **Value After Reset:** 0x1
21	ovrd_en_ana_mplla_ana_pmix_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_pmix_en_i. **Value After Reset:** 0x1
20	ovrd_en_ana_mplla_ana_pmix_clk_i	R/W	Override enable for (SUP) port ana_mplla_ana_pmix_clk_i. **Value After Reset:** 0x1
19	ovrd_en_ana_mplla_ana_word_div2_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_word_div2_en_i. **Value After Reset:** 0x1
18	ovrd_en_ana_mplla_ana_word_clk_div5_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_word_clk_div5_en_i. **Value After Reset:** 0x1
17	ovrd_en_ana_mplla_ana_word_clk_div4_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_word_clk_div4_en_i. **Value After Reset:** 0x1
16	ovrd_en_ana_mplla_ana_tx_clk_div_i	R/W	Override enable for (SUP) port ana_mplla_ana_tx_clk_div_i. **Value After Reset:** 0x1
15	ovrd_en_ana_mplla_ana_fdiv_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_fdiv_en_i. **Value After Reset:** 0x1
14	ovrd_en_ana_mplla_ana_div_clk_div4_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_div_clk_div4_en_i. **Value After Reset:** 0x1
13	ovrd_en_ana_mplla_ana_vreg_speedup_i	R/W	Override enable for (SUP) port ana_mplla_ana_vreg_speedup_i. **Value After Reset:** 0x1
12	ovrd_en_ana_mplla_ana_rst_i	R/W	Override enable for (SUP) port ana_mplla_ana_rst_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
11	ovrd_en_ana_mplla_ana_output_r_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_output_r_en_i. **Value After Reset:** 0x1
10	ovrd_en_ana_mplla_ana_output_l_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_output_l_en_i. **Value After Reset:** 0x1
9	ovrd_en_ana_mplla_ana_output_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_output_en_i. **Value After Reset:** 0x1
8	ovrd_en_ana_mplla_ana_gear_shift_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_gear_shift_en_i. **Value After Reset:** 0x1
7	ovrd_en_ana_mplla_ana_fb_clk_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_fb_clk_en_i. **Value After Reset:** 0x1
6	ovrd_en_ana_mplla_ana_en_i	R/W	Override enable for (SUP) port ana_mplla_ana_en_i. **Value After Reset:** 0x1
5	ovrd_en_ana_mplla_ana_vco_gain_cal_i	R/W	Override enable for (SUP) port ana_mplla_ana_vco_gain_cal_i. **Value After Reset:** 0x1
4	ovrd_en_ana_mplla_ana_fine_tune_i	R/W	Override enable for (SUP) port ana_mplla_ana_fine_tune_i. **Value After Reset:** 0x1
3	ovrd_en_ana_mplla_ana_ctl_buf_bypass_i	R/W	Override enable for (SUP) port ana_mplla_ana_ctl_buf_bypass_i. **Value After Reset:** 0x1
2	ovrd_en_ana_mplla_ana_coarse_tune_i	R/W	Override enable for (SUP) port ana_mplla_ana_coarse_tune_i. **Value After Reset:** 0x1
1	ovrd_en_ana_mplla_ana_cal_i	R/W	Override enable for (SUP) port ana_mplla_ana_cal_i. **Value After Reset:** 0x1
0	ovrd_en_ana_mplla_ana_bandwidth_i	R/W	Override enable for (SUP) port ana_mplla_ana_bandwidth_i. **Value After Reset:** 0x1

11.4.20 ANA_OVRDEN_2

- **Name:** (SUP) Internal Override Enable Register 2
- **Description:** Bits that act as select signal of the mux which overrides value on (SUP) input and output ports Register 2
- **Size:** 32 bits
- **Offset:** 0x304c

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ovrd_en_ana_rt_ana_comp_rst_i	ovrd_en_ana_ropll_ana_word_div2_en_i	ovrd_en_ana_ropll_ana_word_clk_en_i	ovrd_en_ana_ropll_ana_word_clk_div5_en_i	ovrd_en_ana_ropll_ana_word_clk_div4_en_i	ovrd_en_ana_ropll_ana_refdiv_i	ovrd_en_ana_ropll_ana_postdiv_i	ovrd_en_ana_ropll_ana_out_div_msb_i	ovrd_en_ana_ropll_ana_out_div_lsb_i	ovrd_en_ana_ropll_ana_fbdiv_rst_i	ovrd_en_ana_ropll_ana_fbdiv_i	ovrd_en_ana_ropll_ana_div_clk_en_i	ovrd_en_ana_ropll_ana_vreg_fast_start_i	ovrd_en_ana_ropll_ana_vco_low_freq_i	ovrd_en_ana_ropll_ana_v2i_mode_i	ovrd_en_ana_ropll_ana_refsel_i	ovrd_en_ana_ropll_ana_rc_filter_i	ovrd_en_ana_ropll_ana_preset_i	ovrd_en_ana_ropll_ana_gearshift_i	ovrd_en_ana_ropll_ana_en_i	ovrd_en_ana_ropll_ana_cp_ctrl_prop_i	ovrd_en_ana_ropll_ana_cp_ctrl_int_i	ovrd_en_ana_ropll_ana_bypass_i	ovrd_en_ana_refa_ana_lane_clk_en_i	ovrd_en_ana_refa_ana_clkdet_en_i	ovrd_en_ana_refa_ana_clk_sel_i	ovrd_en_ana_refa_ana_clk_en_i	ovrd_en_ana_ref_ro_ana_lane_clk_en_i	ovrd_en_ana_ref_ro_ana_clkdet_en_i	ovrd_en_ana_ref_ro_ana_clk_sel_i	ovrd_en_ana_ref_ro_ana_clk_en_i	ovrd_en_ana_ref_ana_vreg_ref_sel_i

Table 11-190 Fields for Register: ANA_OVRDEN_2

Bits	Name	Memory Access	Description
31	ovrd_en_ana_rt_ana_comp_rst_i	R/W	Override enable for (SUP) port ana_rt_ana_comp_rst_i. **Value After Reset:** 0x1
30	ovrd_en_ana_ropll_ana_word_div2_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_word_div2_en_i. **Value After Reset:** 0x1
29	ovrd_en_ana_ropll_ana_word_clk_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_word_clk_en_i. **Value After Reset:** 0x1
28	ovrd_en_ana_ropll_ana_word_clk_div5_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_word_clk_div5_en_i. **Value After Reset:** 0x1
27	ovrd_en_ana_ropll_ana_word_clk_div4_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_word_clk_div4_en_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
26	ovrd_en_ana_ropll_ana_refdiv_i	R/W	Override enable for (SUP) port ana_ropll_ana_refdiv_i. **Value After Reset:** 0x1
25	ovrd_en_ana_ropll_ana_postdiv_i	R/W	Override enable for (SUP) port ana_ropll_ana_postdiv_i. **Value After Reset:** 0x1
24	ovrd_en_ana_ropll_ana_out_div_msb_i	R/W	Override enable for (SUP) port ana_ropll_ana_out_div_msb_i. **Value After Reset:** 0x1
23	ovrd_en_ana_ropll_ana_out_div_lsb_i	R/W	Override enable for (SUP) port ana_ropll_ana_out_div_lsb_i. **Value After Reset:** 0x1
22	ovrd_en_ana_ropll_ana_fbdiv_rst_i	R/W	Override enable for (SUP) port ana_ropll_ana_fbdiv_rst_i. **Value After Reset:** 0x1
21	ovrd_en_ana_ropll_ana_fbdiv_i	R/W	Override enable for (SUP) port ana_ropll_ana_fbdiv_i. **Value After Reset:** 0x1
20	ovrd_en_ana_ropll_ana_div_clk_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_div_clk_en_i. **Value After Reset:** 0x1
19	ovrd_en_ana_ropll_ana_vreg_fast_start_i	R/W	Override enable for (SUP) port ana_ropll_ana_vreg_fast_start_i. **Value After Reset:** 0x1
18	ovrd_en_ana_ropll_ana_vco_low_freq_i	R/W	Override enable for (SUP) port ana_ropll_ana_vco_low_freq_i. **Value After Reset:** 0x1
17	ovrd_en_ana_ropll_ana_v2i_mode_i	R/W	Override enable for (SUP) port ana_ropll_ana_v2i_mode_i. **Value After Reset:** 0x1
16	ovrd_en_ana_ropll_ana_refsel_i	R/W	Override enable for (SUP) port ana_ropll_ana_refsel_i. **Value After Reset:** 0x1
15	ovrd_en_ana_ropll_ana_rc_filter_i	R/W	Override enable for (SUP) port ana_ropll_ana_rc_filter_i. **Value After Reset:** 0x1
14	ovrd_en_ana_ropll_ana_preset_i	R/W	Override enable for (SUP) port ana_ropll_ana_preset_i. **Value After Reset:** 0x1
13	ovrd_en_ana_ropll_ana_gearshift_i	R/W	Override enable for (SUP) port ana_ropll_ana_gearshift_i. **Value After Reset:** 0x1
12	ovrd_en_ana_ropll_ana_en_i	R/W	Override enable for (SUP) port ana_ropll_ana_en_i. **Value After Reset:** 0x1
11	ovrd_en_ana_ropll_ana_cp_ctrl_prop_i	R/W	Override enable for (SUP) port ana_ropll_ana_cp_ctrl_prop_i. **Value After Reset:** 0x1
10	ovrd_en_ana_ropll_ana_cp_ctrl_int_i	R/W	Override enable for (SUP) port ana_ropll_ana_cp_ctrl_int_i. **Value After Reset:** 0x1

Bits	Name	Memory Access	Description
9	ovrd_en_ana_ropll_ana_bypass_i	R/W	Override enable for (SUP) port ana_ropll_ana_bypass_i. **Value After Reset:** 0x1
8	ovrd_en_ana_refa_ana_lane_clk_en_i	R/W	Override enable for (SUP) port ana_refa_ana_lane_clk_en_i. **Value After Reset:** 0x1
7	ovrd_en_ana_refa_ana_clkdet_en_i	R/W	Override enable for (SUP) port ana_refa_ana_clkdet_en_i. **Value After Reset:** 0x1
6	ovrd_en_ana_refa_ana_clk_sel_i	R/W	Override enable for (SUP) port ana_refa_ana_clk_sel_i. **Value After Reset:** 0x1
5	ovrd_en_ana_refa_ana_clk_en_i	R/W	Override enable for (SUP) port ana_refa_ana_clk_en_i. **Value After Reset:** 0x1
4	ovrd_en_ana_ref_ro_ana_lane_clk_en_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_lane_clk_en_i. **Value After Reset:** 0x1
3	ovrd_en_ana_ref_ro_ana_clkdet_en_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_clkdet_en_i. **Value After Reset:** 0x1
2	ovrd_en_ana_ref_ro_ana_clk_sel_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_clk_sel_i. **Value After Reset:** 0x1
1	ovrd_en_ana_ref_ro_ana_clk_en_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_clk_en_i. **Value After Reset:** 0x1
0	ovrd_en_ana_ref_ana_vreg_ref_sel_i	R/W	Override enable for (SUP) port ana_ref_ana_vreg_ref_sel_i. **Value After Reset:** 0x1

11.4.21 ANA_OVRDEN_3

- **Name:** (SUP) Internal Override Enable Register 3
- **Description:** Bits that act as select signal of the mux which overrides value on (SUP) input and output ports Register 3
- **Size:** 32 bits
- **Offset:** 0x3050

31:19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED__2	RESERVED__1	RESERVED__0	ovrd_en_ana_sup_ana_spare_in	ovrd_en_ana_atb_ana_sup_en_i	ovrd_en_ana_atb_ana_lane_right_en_i	ovrd_en_ana_atb_ana_lane_left_en_i	ovrd_en_ana_rt_ana_comp_result_o	ovrd_en_ana_ropll_ana_cru_rd_data_o	ovrd_en_ana_refa_ana_clkdet_result_o	ovrd_en_ana_refa_ana_clk_ack_o	ovrd_en_ana_ref_ro_ana_clkdet_result_o	ovrd_en_ana_ref_ro_ana_clk_ack_o	ovrd_en_ana_mplla_ana_cru_rd_data_o	ovrd_en_ana_sup_ana_cru_rd_data_o	ovrd_en_ana_ref_ro_ana_clk_div8_en_i	ovrd_en_ana_rx_ana_vref_sel_i	ovrd_en_ana_rt_ana_value_i	ovrd_en_ana_rt_ana_mode_i	ovrd_en_ana_rt_ana_en_i

Table 11-191 Fields for Register: ANA_OVRDEN_3

Bits	Name	Memory Access	Description
31:19	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
18	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
17	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
16	ovrd_en_ana_sup_ana_spare_in	R/W	Override enable for (SUP) port ana_spare_i **Value After Reset:** 0x0
15	ovrd_en_ana_atb_ana_sup_en_i	R/W	Override enable for (SUP) port ana_atb_ana_sup_en_i. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
14	ovrd_en_ana_atb_ana_lane_right_en_i	R/W	Override enable for (SUP) port ana_atb_ana_lane_right_en_i. **Value After Reset:** 0x0
13	ovrd_en_ana_atb_ana_lane_left_en_i	R/W	Override enable for (SUP) port ana_atb_ana_lane_left_en_i. **Value After Reset:** 0x0
12	ovrd_en_ana_rt_ana_comp_result_o	R/W	Override enable for (SUP) port ana_rt_ana_comp_result_o. **Value After Reset:** 0x0
11	ovrd_en_ana_ropll_ana_cru_rd_data_o	R/W	Override enable for (SUP) port ana_ropll_ana_cru_rd_data_o. **Value After Reset:** 0x0
10	ovrd_en_ana_refa_ana_clkdet_result_o	R/W	Override enable for (SUP) port ana_refa_ana_clkdet_result_o. **Value After Reset:** 0x0
9	ovrd_en_ana_refa_ana_clk_ack_o	R/W	Override enable for (SUP) port ana_refa_ana_clk_ack_o. **Value After Reset:** 0x0
8	ovrd_en_ana_ref_ro_ana_clkdet_result_o	R/W	Override enable for (SUP) port ana_ref_ro_ana_clkdet_result_o. **Value After Reset:** 0x0
7	ovrd_en_ana_ref_ro_ana_clk_ack_o	R/W	Override enable for (SUP) port ana_ref_ro_ana_clk_ack_o. **Value After Reset:** 0x0
6	ovrd_en_ana_mplla_ana_cru_rd_data_o	R/W	Override enable for (SUP) port ana_mplla_ana_cru_rd_data_o. **Value After Reset:** 0x0
5	ovrd_en_ana_sup_ana_cru_rd_data_o	R/W	Override enable for (SUP) port ana_sup_ana_cru_rd_data_o. **Value After Reset:** 0x0
4	ovrd_en_ana_ref_ro_ana_clk_div8_en_i	R/W	Override enable for (SUP) port ana_ref_ro_ana_clk_div8_en_i. **Value After Reset:** 0x1
3	ovrd_en_ana_rx_ana_vref_sel_i	R/W	Override enable for (SUP) port ana_rx_ana_vref_sel_i. **Value After Reset:** 0x1
2	ovrd_en_ana_rt_ana_value_i	R/W	Override enable for (SUP) port ana_rt_ana_value_i. **Value After Reset:** 0x1
1	ovrd_en_ana_rt_ana_mode_i	R/W	Override enable for (SUP) port ana_rt_ana_mode_i. **Value After Reset:** 0x1
0	ovrd_en_ana_rt_ana_en_i	R/W	Override enable for (SUP) port ana_rt_ana_en_i. **Value After Reset:** 0x1

11.4.22 ANA_OVRDVAL_13

- **Name:** SUP ANA Output Register 13
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 13. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3054

31	30	29	28	27:20	19	18	17	16	15:8	7:0
RESERVED__2	RESERVED__1	RESERVED__0	ana_rt_ana_comp_result_o	ana_ropll_ana_cru_rd_data_o	ana_refa_ana_clkdet_result_o	ana_refa_ana_clk_ack_o	ana_ref_ro_ana_clkdet_result_o	ana_ref_ro_ana_clk_ack_o	ana_mplla_ana_cru_rd_data_o	ana_sup_ana_cru_rd_data_o

Table 11-192 Fields for Register: ANA_OVRDVAL_13

Bits	Name	Memory Access	Description
31	RESERVED__2	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
30	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	ana_rt_ana_comp_result_o	R/W	Override value set on (SUP) analog port ana_rt_ana_comp_result_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

Bits	Name	Memory Access	Description
27:20	ana_ropll_ana_cru_rd_data_o	R/W	Override value set on (SUP) analog port ana_ropll_ana_cru_rd_data_o[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
19	ana_refa_ana_clkdet_result_o	R/W	Override value set on (SUP) analog port ana_refa_ana_clkdet_result_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
18	ana_refa_ana_clk_ack_o	R/W	Override value set on (SUP) analog port ana_refa_ana_clk_ack_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
17	ana_ref_ro_ana_clkdet_result_o	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clkdet_result_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
16	ana_ref_ro_ana_clk_ack_o	R/W	Override value set on (SUP) analog port ana_ref_ro_ana_clk_ack_o. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
15:8	ana_mplla_ana_cru_rd_data_o	R/W	Override value set on (SUP) analog port ana_mplla_ana_cru_rd_data_o[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true
7:0	ana_sup_ana_cru_rd_data_o	R/W	Override value set on (SUP) analog port ana_sup_ana_cru_rd_data_o[7:0]. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.23 ANA_OVRDVAL_14

- **Name:** SUP ANA Output Register 14
- **Description:** Register bits act as override value at the mux which can override values on (SUP) analog-digital interface ports. Register 14. When the respective override enable bit is 0b0, the readback of this register returns the original value of the signal.
- **Size:** 32 bits
- **Offset:** 0x3058

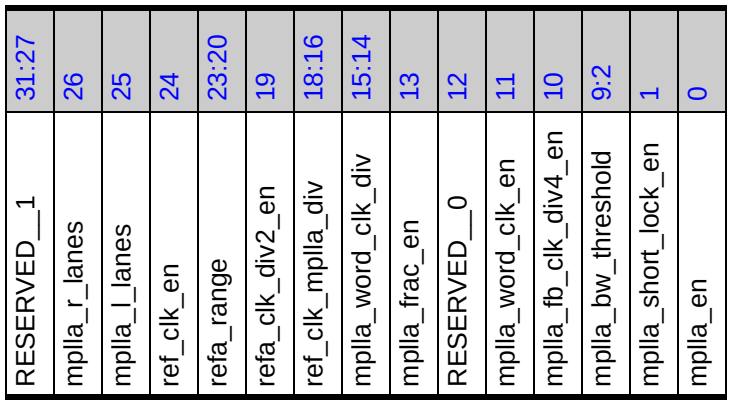


Table 11-193 Fields for Register: ANA_OVRDVAL_14

Bits	Name	Memory Access	Description
31:0	ana_sup_ana_spare_in	R/W	Override value set on (SUP) analog port sup_spare_in. This value is set when the override enable register for this (SUP) port is set to 0b1. **Value After Reset:** 0x0 **Volatile:** true

11.4.24 SUP_CFG_1

- **Name:** (SUP) Configuration Register 1
- **Description:** Configures (SUP) settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x305c



31:27	26	25	24	23:20	19	18:16	15:14	13	12	11	10	9:2	1	0
RESERVED__1	mplla_r_lanes	mplla_l_lanes	ref_clk_en	refa_range	refa_clk_div2_en	ref_clk_mplla_div	mplla_word_clk_div	mplla_frac_en	RESERVED__0	mplla_word_clk_en	mplla_fb_clk_div4_en	mplla_bw_threshold	mplla_short_lock_en	mplla_en

Table 11-194 Fields for Register: SUP_CFG_1

Bits	Name	Memory Access	Description
31:27	RESERVED__1	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
26	mplla_r_lanes	R/W	Enables clock distribution to the right lanes of the PHY, for both TX and RX. **Value After Reset:** 0x0
25	mplla_l_lanes	R/W	Enables clock distribution to the left lanes of the PHY, for both TX and RX. **Value After Reset:** 0x0
24	ref_clk_en	R/W	Enables ref_clk distribution. **Value After Reset:** 0x0
23:20	refa_range	R/W	Configurability to divide ref_dig_clk to generate ref_range_clk. **Value After Reset:** 0x0
19	refa_clk_div2_en	R/W	Multiplies ref_dig_clk by 2 to generate ref_dig_clk_div. **Value After Reset:** 0x0
18:16	ref_clk_mplla_div	R/W	Divides ref_dig_clk by 2 power ref_clk_mplla_div to generate ref_dig_clk_div. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
15:14	mplla_word_clk_div	R/W	Sets MPLL clock divider settings. **Value After Reset:** 0x0
13	mplla_frac_en	R/W	Enables the fractional divider sub-block **Value After Reset:** 0x0
12	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
11	mplla_word_clk_en	R/W	Enables output clock divider settings. **Value After Reset:** 0x0
10	mplla_fb_clk_div4_en	R/W	When enabled it divides fb clock by 4. **Value After Reset:** 0x0
9:2	mplla_bw_threshold	R/W	Threshold for vco_fb_cnt_diff to drive the output bandwidth. **Value After Reset:** 0x0
1	mplla_short_lock_en	R/W	Reduces MPLL startup time. **Value After Reset:** 0x0
0	mplla_en	R/W	Enables the analog mpll. **Value After Reset:** 0x0

11.4.25 SUP_CFG_2

- **Name:** (SUP) Configuration Register 2
- **Description:** Configures (SUP) settings. Register 2.
- **Size:** 32 bits
- **Offset:** 0x3060

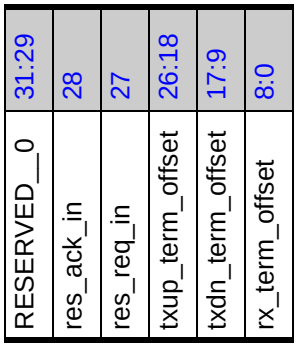


Table 11-195 Fields for Register: SUP_CFG_2

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	res_ack_in	R/W	Input acknowledgment used in resistance sharing. ■ Several PHY instances may be daisy chained in this manner to allow sharing a single reference resistor. ■ The first PHY its res_req_in tied low, while the last PHY must have its res_ack_in tied to its res_req_out. ■ The res_ack lines must be connected in reverse order with respect to the connections of the res_req lines. **Value After Reset:** 0x0
27	res_req_in	R/W	Input request used in resistance sharing. **Value After Reset:** 0x0
26:18	txup_term_offset	R/W	txup rtune offset. **Value After Reset:** 0x0
17:9	txdn_term_offset	R/W	txdn rtune offset. **Value After Reset:** 0x0
8:0	rx_term_offset	R/W	RX rtune offset. **Value After Reset:** 0x0

11.4.26 SUP_CFG_3

- **Name:** (SUP) Configuration Register 3
- **Description:** Configures (SUP) settings. Register 3.
- **Size:** 32 bits
- **Offset:** 0x3064

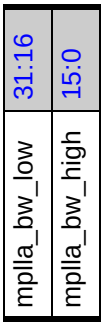


Table 11-196 Fields for Register: SUP_CFG_3

Bits	Name	Memory Access	Description
31:16	mplla_bw_low	R/W	Low bandwidth. **Value After Reset:** 0x0
15:0	mplla_bw_high	R/W	High bandwitdh. **Value After Reset:** 0x0

11.4.27 SUP_CFG_4

- **Name:** (SUP) Configuration Register 4
- **Description:** Configures (SUP) settings. Register 4.
- **Size:** 32 bits
- **Offset:** 0x3068

31	30	29	28	27	26:22	21	20:17	16	15:12	11	10	9	8:1	0
ovrd_sel	mpll_fb_dig_clk_en	mpll_pclk_en	fast_mpll_pwrup	fast_mpll_lock	dtb_sel	mpll_extbw_sel	mpll_extfine_tune	mpll_refclk_gating_dis	chkfrq_ld_cnt	mpll_skipcal	mpll_extcal	mpll_extchkfrq_en	mpll_extcoarse_tune	mpll_extcal_done

Table 11-197 Fields for Register: SUP_CFG_4

Bits	Name	Memory Access	Description
31	ovrd_sel	R/W	Override enable bit. **Value After Reset:** 0x0
30	mpll_fb_dig_clk_en	R/W	Overrides the PWR FSM mpll_fb_dig_clk_en signal. **Value After Reset:** 0x0
29	mpll_pclk_en	R/W	Overrides the PWR FSM mpll_pclk_en signal. **Value After Reset:** 0x0
28	fast_mpll_pwrup	R/W	Enables fast MPLL powerup. **Value After Reset:** 0x0
27	fast_mpll_lock	R/W	Enables fast MPLL locking. **Value After Reset:** 0x0
26:22	dtb_sel	R/W	DTB select for MPLL dtb signals. **Value After Reset:** 0x0
21	mpll_extbw_sel	R/W	Value of mpll_ana_fine_tune_i in external calibration mode. **Value After Reset:** 0x0
20:17	mpll_extfine_tune	R/W	Value of mpll_bw_sel in external calibration mode. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
16	mpll_refclk_gating_dis	R/W	Disables waiting for the IDLE state to deassert ref_ana_clk_en . **Value After Reset:** 0x0
15:12	chkfrq_ld_cnt	R/W	MSBs for the CHKFRQ FSM ld_val load value. **Value After Reset:** 0x0
11	mpll_skipcal	R/W	This bit skips automatic (internal) calibration of MPLL then MPLL_SKIPCAL_COARSE_TUNE is used. **Value After Reset:** 0x0
10	mpll_extcal	R/W	Enables external calibration of MPLL. **Value After Reset:** 0x0
9	mpll_extchkfrq_en	R/W	Checks the frequency of the MPLL. **Value After Reset:** 0x0
8:1	mpll_extcoarse_tune	R/W	Value of mpll_ana_coarse_tune_i in external calibration mode. **Value After Reset:** 0x0
0	mpll_extcal_done	R/W	Sets the external calibration status to done. **Value After Reset:** 0x0

11.4.28 SUP_CFG_5

- **Name:** (SUP) Configuration Register 5
- **Description:** Configures (SUP) settings. Register 5.
- **Size:** 32 bits
- **Offset:** 0x306c

31:28	27:16	15:14	13:12	11	10	9:8	7:2	1	0
RESERVED__0	chkfrq_ld_val	mpll_gain_cal_val0	mpll_gain_cal_val1	mpll_skip_gain_cal	mpll_bw_sel_invert	mpll_fine_start	mpll_vco_drift_thresh	mpll_lock_det_en	mpll_state_disable

Table 11-198 Fields for Register: SUP_CFG_5

Bits	Name	Memory Access	Description
31:28	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
27:16	chkfrq_ld_val	R/W	This field is the MSBs for the CHKFRQ FSM ld_val 10:0 load value. **Value After Reset:** 0x0
15:14	mpll_gain_cal_val0	R/W	This is the value of vco_gain_cal = 2'b01. **Value After Reset:** 0x0
13:12	mpll_gain_cal_val1	R/W	This is the value of vco_gain_cal = 2'b10. **Value After Reset:** 0x0
11	mpll_skip_gain_cal	R/W	This bit skips gain calibration of MPLL. **Value After Reset:** 0x0
10	mpll_bw_sel_invert	R/W	Selects low bandwidth inversion. **Value After Reset:** 0x0
9:8	mpll_fine_start	R/W	This field is the value for the starting point of the MPLL fine bank. **Value After Reset:** 0x0
7:2	mpll_vco_drift_thresh	R/W	FB clock drift threshold. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
1	mpll_lock_det_en	R/W	Enables continuous MPLL lock detection. **Value After Reset:** 0x0
0	mpll_state_disable	R/W	Enables de-asserting mpll_state when MPLL VCO drift away. **Value After Reset:** 0x0

11.4.29 SUP_CFG_6

- **Name:** (SUP) Configuration Register 6
- **Description:** Configures (SUP) settings. Register 6.
- **Size:** 32 bits
- **Offset:** 0x3070

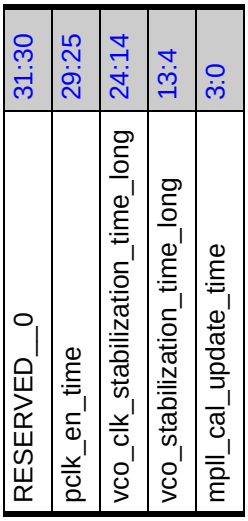


Table 11-199 Fields for Register: SUP_CFG_6

Bits	Name	Memory Access	Description
31:30	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
29:25	pclk_en_time	R/W	Threshold for the PCLK enable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
24:14	vco_clk_stabilization_time_long	R/W	Threshold for the VCO clock stabilization timer in terms of number of ref_range_clk cycles. Default value is 3us. **Value After Reset:** 0x0
13:4	vco_stabilization_time_long	R/W	Threshold for the VCO stabilization timer in terms of number of ref_range_clk cycles. Default value is 12us. **Value After Reset:** 0x0
3:0	mpll_cal_update_time	R/W	Threshold for the MPLL calibration control word update timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0

11.4.30 SUP_CFG_7

- **Name:** (SUP) Configuration Register 7
- **Description:** Configures (SUP) settings. Register 7.
- **Size:** 32 bits
- **Offset:** 0x3074

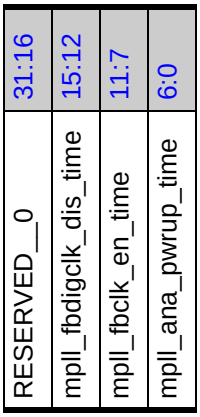


Table 11-200 Fields for Register: SUP_CFG_7

Bits	Name	Memory Access	Description
31:16	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
15:12	mpll_fbdigclk_dis_time	R/W	Threshold for the MPLL feedback digital clock disable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
11:7	mpll_fbclk_en_time	R/W	Threshold for the MPLL feedback digital clock enable timer in terms of number of mpll_fb_clk cycles. **Value After Reset:** 0x0
6:0	mpll_ana_pwrup_time	R/W	Threshold for the MPLL analog power up timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0

11.4.31 SUP_CFG_8

- **Name:** (SUP) Configuration Register 8
- **Description:** Configures (SUP) settings. Register 8.
- **Size:** 32 bits
- **Offset:** 0x3078

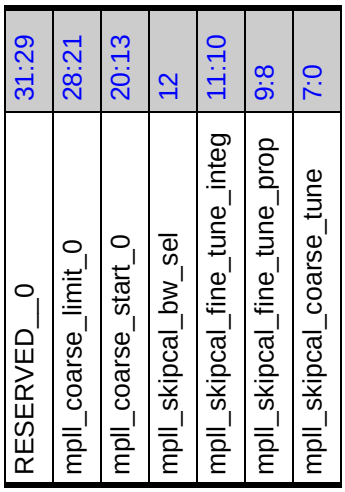


Table 11-201 Fields for Register: SUP_CFG_8

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28:21	mpll_coarse_limit_0	R/W	Value for the ending point of the MPLL coarse range 0. **Value After Reset:** 0x0
20:13	mpll_coarse_start_0	R/W	Value for the starting point of the MPLL coarse range 0. **Value After Reset:** 0x0
12	mpll_skipcal_bw_sel	R/W	This bit is the value for MPLL bw_sel when calibration or gain calibration is skipped. **Value After Reset:** 0x0
11:10	mpll_skipcal_fine_tune_integ	R/W	Value for MPLL fine_tune (integral portion) when calibration is skipped. **Value After Reset:** 0x0
9:8	mpll_skipcal_fine_tune_prop	R/W	Value for MPLL fine_tune (proportional portion) when calibration is skipped. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
7:0	mpll_skipcal_coarse_tune	R/W	Value for MPLL coarse_tune when calibration is skipped. **Value After Reset:** 0x0

11.4.32 SUP_CFG_9

- **Name:** (SUP) Configuration Register 9
- **Description:** Configures (SUP) settings. Register 9.
- **Size:** 32 bits
- **Offset:** 0x307c

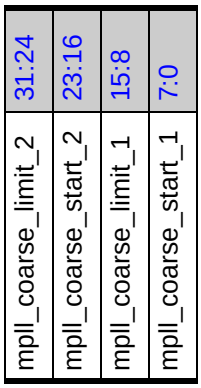


Table 11-202 Fields for Register: SUP_CFG_9

Bits	Name	Memory Access	Description
31:24	mpll_coarse_limit_2	R/W	Value for the ending point of the MPLL coarse range 2. **Value After Reset:** 0x0
23:16	mpll_coarse_start_2	R/W	Value for the starting point of the MPLL coarse range 2. **Value After Reset:** 0x0
15:8	mpll_coarse_limit_1	R/W	Value for the ending point of the MPLL coarse range 1. **Value After Reset:** 0x0
7:0	mpll_coarse_start_1	R/W	Value for the starting point of the MPLL coarse range 1. **Value After Reset:** 0x0

11.4.33 SUP_CFG_10

- **Name:** (SUP) Configuration Register 10
- **Description:** Configures (SUP) settings. Register 10.
- **Size:** 32 bits
- **Offset:** 0x3080

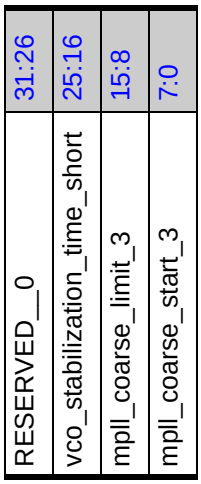


Table 11-203 Fields for Register: SUP_CFG_10

Bits	Name	Memory Access	Description
31:26	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
25:16	vco_stabilization_time_short	R/W	Threshold for the VCO clock stabilization short timer in terms of number of ref_range_clk cycles. Default value is 35us. **Value After Reset:** 0x0
15:8	mpll_coarse_limit_3	R/W	Value for the ending point of the MPLL coarse range 3. **Value After Reset:** 0x0
7:0	mpll_coarse_start_3	R/W	Value for the starting point of the MPLL coarse range 3. **Value After Reset:** 0x0

11.4.34 SUP_CFG_11

- **Name:** (SUP) Configuration Register 11
- **Description:** Configures (SUP) settings. Register 11.
- **Size:** 32 bits
- **Offset:** 0x3084

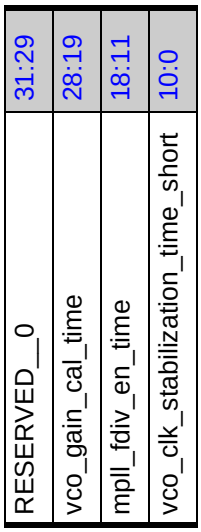


Table 11-204 Fields for Register: SUP_CFG_11

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
28:19	vco_gain_cal_time	R/W	Threshold for the MPLL VCO control voltage settling timer in terms of number of ref_range_clk cycles. Default value is 10us. **Value After Reset:** 0x0
18:11	mpll_fdiv_en_time	R/W	Threshold for the MPLL fdiv enable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
10:0	vco_clk_stabilization_time_short	R/W	Threshold for the VCO stabilization timer in terms of number of ref_range_clk cycles. Default value is 4us. **Value After Reset:** 0x0

11.4.35 MPLL_STAT_0

- **Name:** MPLL Configuration Register 0
- **Description:** Configures MPLL settings. Register 0.
- **Size:** 32 bits
- **Offset:** 0x3088

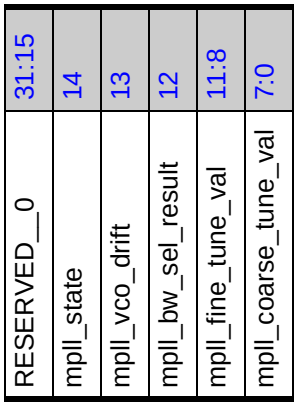


Table 11-205 Fields for Register: MPLL_STAT_0

Bits	Name	Memory Access	Description
31:15	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
14	mpll_state	R	Current value of mpll_state. **Value After Reset:** 0x0
13	mpll_vco_drift	R	This bit indicates if the MPLL lost LOCK while in the ON state. **Value After Reset:** 0x0
12	mpll_bw_sel_result	R	Current value of mpll_bw_sel_result. **Value After Reset:** 0x0
11:8	mpll_fine_tune_val	R	Current value of mpll_ana_fine_tune_i. **Value After Reset:** 0x0
7:0	mpll_coarse_tune_val	R	Current value of mpll_ana_coarse_tune_i. **Value After Reset:** 0x0

11.4.36 MPLL_STAT_1

- **Name:** MPLL Configuration Register 1
- **Description:** Configures MPLL settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x308c

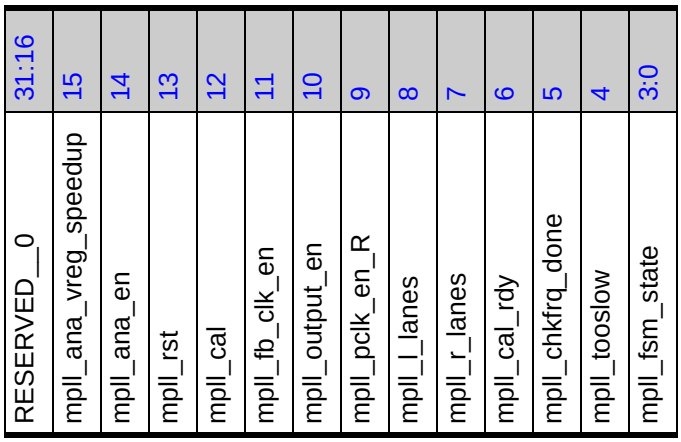


Table 11-206 Fields for Register: MPLL_STAT_1

Bits	Name	Memory Access	Description
31:16	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
15	mpll_ana_vreg_speedup	R	Current value of mpll_ana_vreg_speedup_i. **Value After Reset:** 0x0
14	mpll_ana_en	R	Current value of mpll_ana_en_i. **Value After Reset:** 0x0
13	mpll_rst	R	Current value of mpll_ana_rst_i. **Value After Reset:** 0x0
12	mpll_cal	R	Current value of mpll_ana_cal_i. **Value After Reset:** 0x0
11	mpll_fb_clk_en	R	Current value of mpll_ana_fb_clk_en_i. **Value After Reset:** 0x0
10	mpll_output_en	R	Current value of mpll_ana_output_en_i. **Value After Reset:** 0x0
9	mpll_pclk_en_R	R	Current value of mpll_pclk_en. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
8	mpll_l_lanes	R	Current value of lane_mpll_l. **Value After Reset:** 0x0
7	mpll_r_lanes	R	Current value of lane_mpll_r. **Value After Reset:** 0x0
6	mpll_cal_rdy	R	Current value of mpll_cal_rdy. **Value After Reset:** 0x0
5	mpll_chkfrq_done	R	Current value of mpll_chkfrq_done. **Value After Reset:** 0x0
4	mpll_tooslow	R	Current value of mpll_tooslow. **Value After Reset:** 0x0
3:0	mpll_fsm_state	R	Current value of the PWR FSM state register. **Value After Reset:** 0x0

11.4.37 MPLL_STAT_2

- **Name:** MPLL Configuration Register 2
- **Description:** Configures MPLL settings. Register 2.
- **Size:** 32 bits
- **Offset:** 0x3090



Table 11-207 Fields for Register: MPLL_STAT_2

Bits	Name	Memory Access	Description
31:26	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
25:13	mpll_ana_vco_fb_cnt2	R	Current value of mpll_ana_vco_fb_cnt2. **Value After Reset:** 0x0
12:0	mpll_ana_vco_fb_cnt1	R	Current value of mpll_ana_vco_fb_cnt1. **Value After Reset:** 0x0

11.4.38 MPLL_STAT_3

- **Name:** MPLL Configuration Register 3
- **Description:** Configures MPLL settings. Register 3.
- **Size:** 32 bits
- **Offset:** 0x3094

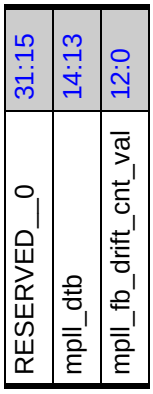


Table 11-208 Fields for Register: MPLL_STAT_3

Bits	Name	Memory Access	Description
31:15	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
14:13	mpll_dtb	R	Config of dtb mpll bus. **Value After Reset:** 0x0
12:0	mpll_fb_drift_cnt_val	R	Current value of mpll_fb_drift_cnt. **Value After Reset:** 0x0

11.4.39 SUP_CFG_12

- **Name:** (SUP) Configuration Register 12
- **Description:** Configures (SUP) settings. Register 12.
- **Size:** 32 bits
- **Offset:** 0x3098

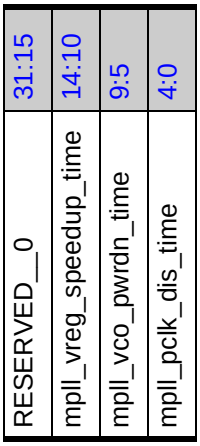
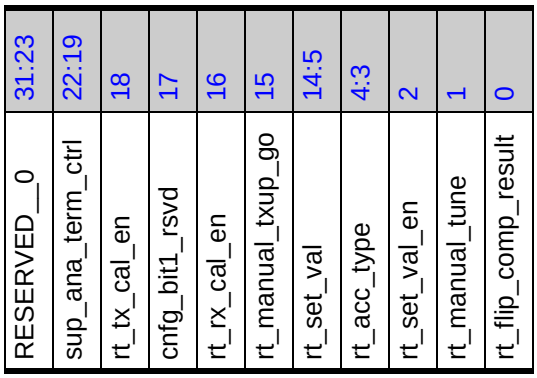


Table 11-209 Fields for Register: SUP_CFG_12

Bits	Name	Memory Access	Description
31:15	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
14:10	mpll_vreg_speedup_time	R/W	Threshold for the MPLL vreg speedup timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
9:5	mpll_vco_pwrdn_time	R/W	Threshold for the MPLL VCO power down timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0
4:0	mpll_pclk_dis_time	R/W	Threshold for the PCLK disable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x0

11.4.40 SUP_CFG_13

- **Name:** (SUP) Configuration Register 13
- **Description:** Configures (SUP) settings. Register 13.
- **Size:** 32 bits
- **Offset:** 0x309c



31:23	22:19	18	17	16	15	14:5	4:3	2	1	0
RESERVED__0	sup_ana_term_ctrl	rt_tx_cal_en	cnfg_bit1_rsvd	rt_rx_cal_en	rt_manual_txup_go	rt_set_val	rt_acc_type	rt_set_val_en	rt_manual_tune	rt_flip_comp_result

Table 11-210 Fields for Register: SUP_CFG_13

Bits	Name	Memory Access	Description
31:23	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
22:19	sup_ana_term_ctrl	R/W	This field sets the reference resistor in the analog. Value Impedance (ohm). **Values:** ■ 0x0 (_54): 54 ■ 0x1 (_52): 52 ■ 0x2 (_50): 50 ■ 0x3 (_48): 48 ■ 0x4 (_46): 46 ■ 0x5 (_44): 44 ■ 0x6 (_42): 42 ■ 0x7 (_40): (and 1xxx) 40 **Value After Reset:** 0x0
18	rt_tx_cal_en	R/W	Enables calibration of TX resistor. **Value After Reset:** 0x0
17	cnfg_bit1_rsvd	R/W	RESERVED **Value After Reset:** 0x0
16	rt_rx_cal_en	R/W	Enables calibration of RX resistor. **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
15	rt_manual_txup_go	R/W	Enables TxUP tune to continue in manual tune mode, when TYPE is TxUP tune. When in non TxUP manual tune mode, this bit must be 0. When in TxUP manual tune mode and after TxUP manual tune is triggered, if the read only register SUP.RTUNE_STAT.DTB_RTUNE equals to 2'b01, it is required to first release the rtune related asynchronous override on analog/digital interface then turn this bit TXUP_GO from low to high to continue TxUP manual tune which also indicates that this bit needs to be low when triggering TxUP manual tune. **Value After Reset:** 0x0
14:5	rt_set_val	R/W	Value to use when triggering SET_VAL field: **Values:** ■ 0x0 (ADC): ADC, or read/write rt_value. ■ 0x1 (RX_TUNE): Rx tune, or read/write rx_cal_val (only 6 bits). ■ 0x2 (TX_DN): TxDN tune, or read/write txdn_cal_val (10 bits). ■ 0x3 (TX_UP): TxUP tune, or read/write txup_cal_val (10 bits) or Resref detect (no affect when triggering SET_VAL field). **Value After Reset:** 0x0
4:3	rt_acc_type	R/W	Type of manual tuning or register read/write to execute. **Value After Reset:** 0x0
2	rt_set_val_en	R/W	This bit sets value. Write 1 to manually write the register specified by the TYPE field to the value in the VALUE field. **Value After Reset:** 0x0
1	rt_manual_tune	R/W	Write 1 to perform manual tuning specified by TYPE field. Starting a manual tune while a tune is currently running can cause unpredictable results. For use only when you know what the part is doing (w.r.t. resistor tuning). **Value After Reset:** 0x0
0	rt_flip_comp_result	R/W	Invert analog comparator output. **Value After Reset:** 0x0

11.4.41 SUP_CFG_14

- **Name:** (SUP) Configuration Register 14
- **Description:** Configures (SUP) settings. Register 14.
- **Size:** 32 bits
- **Offset:** 0x30a0

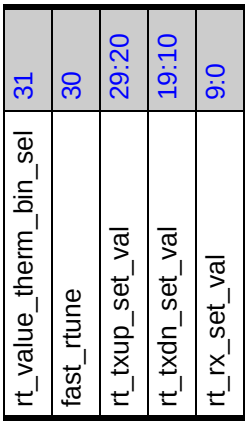


Table 11-211 Fields for Register: SUP_CFG_14

Bits	Name	Memory Access	Description
31	rt_value_therm_bin_sel	R/W	Selects between binary or themometer decoder rt_ana_value **Values:** ■ 0x0 (THM): Thermometer rt value ■ 0x1 (BIN): Binary rt value **Value After Reset:** 0x0
30	fast_rtune	R/W	Enables fast resistor tuning (simulation only). **Value After Reset:** 0x0
29:20	rt_txup_set_val	R/W	This field sets value of tx,up resistor. Writing a value to this register sets the tx,up resistor value. **Value After Reset:** 0x0
19:10	rt_txdn_set_val	R/W	This field sets value of tx,dn resistor. Writing a value to this register sets the tx,dn resistor value. **Value After Reset:** 0x0
9:0	rt_rx_set_val	R/W	This field sets value of rx resistor. Writing a value to this register sets the rx resistor value. **Value After Reset:** 0x0

11.4.42 rt_stat_0

- **Name:** rt_stat Configuration register 0
- **Description:** Configures rt_stat settings. Register 0.
- **Size:** 32 bits
- **Offset:** 0x30a4

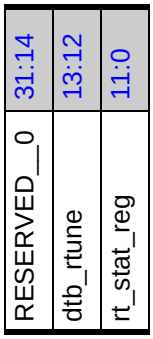


Table 11-212 Fields for Register: rt_stat_0

Bits	Name	Memory Access	Description
31:14	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
13:12	dtb_rtune	R	dtb rtune bus **Value After Reset:** 0x0
11:0	rt_stat_reg	R	Current value of the register specified by the DEBUG.TYPE field. **Value After Reset:** 0x0

11.4.43 rt_stat_1

- **Name:** rt_stat Configuration register 1
- **Description:** Configures rt_stat settings. Register 1.
- **Size:** 32 bits
- **Offset:** 0x30a8

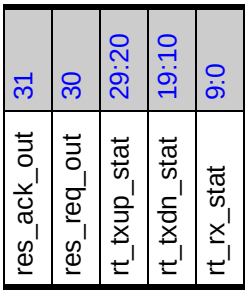


Table 11-213 Fields for Register: rt_stat_1

Bits	Name	Memory Access	Description
31	res_ack_out	R	Output acknowlegment used in resistance sharing. **Value After Reset:** 0x0
30	res_req_out	R	Output request used in resistance sharing. **Value After Reset:** 0x0
29:20	rt_txup_stat	R	Current value of the tx, up resistor tuning register. **Value After Reset:** 0x0
19:10	rt_txdn_stat	R	Current value of the tx, dn resistor tuning register. **Value After Reset:** 0x0
9:0	rt_rx_stat	R	Current value of the rx resistor tuning register. **Value After Reset:** 0x0

11.4.44 SUP_CFG_15

- **Name:** (SUP) Configuration Register 15
- **Description:** Configures (SUP) settings. Register 15.
- **Size:** 32 bits
- **Offset:** 0x30ac

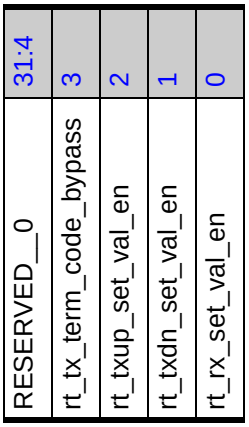


Table 11-214 Fields for Register: SUP_CFG_15

Bits	Name	Memory Access	Description
31:4	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
3	rt_tx_term_code_bypass	R/W	Bypass txup term code **Value After Reset:** 0x1
2	rt_txup_set_val_en	R/W	Input to pulse catcher to generate txup_set_val pulse. **Value After Reset:** 0x0
1	rt_txdn_set_val_en	R/W	Input to pulse catcher to generate txdn_set_val pulse. **Value After Reset:** 0x0
0	rt_rx_set_val_en	R/W	Input to pulse catcher to generate rx_set_val pulse. **Value After Reset:** 0x0

11.4.45 SUP_CFG_16

- **Name:** (SUP) Configuration Register 16
- **Description:** Configures (SUP) settings. Register 16.
- **Size:** 32 bits
- **Offset:** 0x30b0

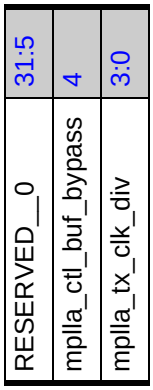


Table 11-215 Fields for Register: SUP_CFG_16

Bits	Name	Memory Access	Description
31:5	RESERVED__0	R	RESREVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	mplla_ctl_buf_bypass	R/W	Controler buffer bypass signal is connected directly to analog. **Value After Reset:** 0x0
3:0	mplla_tx_clk_div	R/W	Configurability to divide mplla clock is connected directly to analog. **Value After Reset:** 0x0

11.4.46 SUP_CFG_17

- **Name:** (SUP) Configuration Register 17
- **Description:** Configures (SUP) settings. Register 17.
- **Size:** 32 bits
- **Offset:** 0x30b4

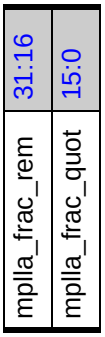


Table 11-216 Fields for Register: SUP_CFG_17

Bits	Name	Memory Access	Description
31:16	mplla_frac_rem	R/W	Control the remainder of the fractional ratio wanted **Value After Reset:** 0x0
15:0	mplla_frac_quot	R/W	Control the quotient of the fractional ratio wanted **Value After Reset:** 0x0

11.4.47 SUP_CFG_18

- **Name:** (SUP) Configuration Register 18
- **Description:** Configures (SUP) settings. Register 18.
- **Size:** 32 bits
- **Offset:** 0x30b8

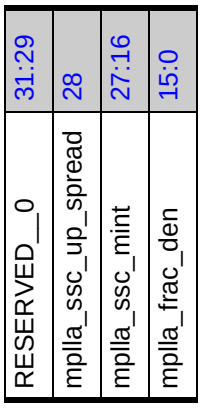


Table 11-217 Fields for Register: SUP_CFG_18

Bits	Name	Memory Access	Description
31:29	RESERVED__0	R	RESREVED **Reserved Field:** Yes **Value After Reset:** 0x0
28	mplla_ssc_up_spread	R/W	Controls if the SSC profile is up (1) or down (0) spread **Value After Reset:** 0x0
27:16	mplla_ssc_mint	R/W	Control the feedback divider ratio that will be modulated by the SSC/FracN machine resulting in the Mout_int <11:0> **Value After Reset:** 0x0
15:0	mplla_frac_den	R/W	Control the denominator of the fractional ratio wanted **Value After Reset:** 0x0

11.4.48 SUP_CFG_19

- **Name:** (SUP) Configuration Register 19
- **Description:** Configures (SUP) settings. Register 19.
- **Size:** 32 bits
- **Offset:** 0x30bc

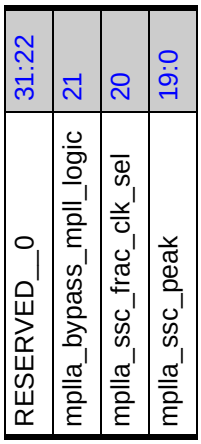


Table 11-218 Fields for Register: SUP_CFG_19

Bits	Name	Memory Access	Description
31:22	RESERVED__0	R	RESREVED **Reserved Field:** Yes **Value After Reset:** 0x0
21	mplla_bypass_mpll_logic	R/W	Bypass mint_i mapping logic **Value After Reset:** 0x0
20	mplla_ssc_frac_clk_sel	R/W	Selects between reference and feedback clock to be the ss_gen module operating clock **Value After Reset:** 0x0
19:0	mplla_ssc_peak	R/W	Controls the ppm shift amplitude of the SSC profile **Value After Reset:** 0x0

11.4.49 SUP_CFG_20

- **Name:** (SUP) Configuration Register 20
- **Description:** Configures (SUP) settings. Register 20.
- **Size:** 32 bits
- **Offset:** 0x30c0



Table 11-219 Fields for Register: SUP_CFG_20

Bits	Name	Memory Access	Description
31:22	RESERVED__0	R	RESREVED **Reserved Field:** Yes **Value After Reset:** 0x0
21	mplla_ssc_en	R/W	Enables the ssc sub block **Value After Reset:** 0x0
20:0	mplla_ssc_step_size	R/W	Controls the SSC profile step **Value After Reset:** 0x0

11.4.50 SUP_CFG_21

- **Name:** (SUP) Configuration Register 21
- **Description:** Configures (SUP) settings. Register 21.
- **Size:** 32 bits
- **Offset:** 0x30c4

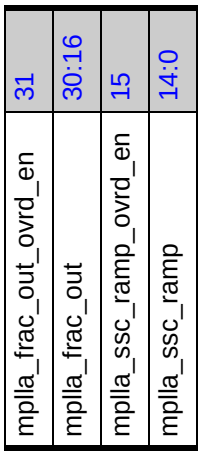


Table 11-220 Fields for Register: SUP_CFG_21

Bits	Name	Memory Access	Description
31	mplla_frac_out_ovrd_en	R/W	The frac_out override enable **Value After Reset:** 0x0
30:16	mplla_frac_out	R/W	This field is the frac_out <19:5> override value **Value After Reset:** 0x0
15	mplla_ssc_ramp_ovrd_en	R/W	The ssc_ramp override enable **Value After Reset:** 0x0
14:0	mplla_ssc_ramp	R/W	This file is the ssc_ramp <20:6> override value **Value After Reset:** 0x0

11.5 RAW_PCS_RAW_PCS Registers

11.5.1 CPU_CFG

- **Name:** RAW PCS Configuration Register 0
- **Description:** CPU configuration register.
- **Size:** 32 bits
- **Offset:** 0x4000

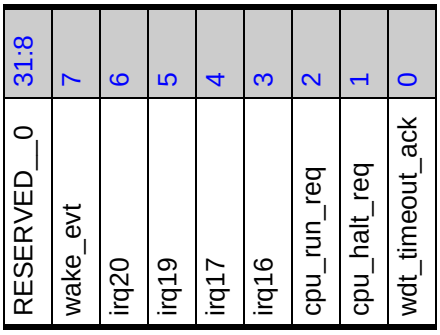


Table 11-221 Fields for Register: CPU_CFG

Bits	Name	Memory Access	Description
31:8	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
7	wake_evt	R/W	Asynchronous signal for waking up the CPU core that has been put to sleep by firmware. **Value After Reset:** 0x0
6	irq20	R/W	CPU interrupt request irq20. This is level or edge sensitive, run time configurable by firmware. **Value After Reset:** 0x0
5	irq19	R/W	CPU interrupt request irq19. This is level or edge sensitive, run time configurable by firmware. **Value After Reset:** 0x0
4	irq17	R/W	CPU interrupt request irq17. This is level or edge sensitive, run time configurable by firmware. **Value After Reset:** 0x0
3	irq16	R/W	CPU interrupt request irq16. This is level or edge sensitive, run time configurable by firmware. **Value After Reset:** 0x0
2	cpu_run_req	R/W	Active-high run request. Assert this signal to request an CPU core to run. Clear the signal when the CPU core responds with (cpu_run_ack). **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
1	cpu_halt_req	R/W	Active-high halt request. Assert this signal to request the CPU core to halt. Remove the signal when the CPU core responds with (cpu_halt_ack). **Value After Reset:** 0x0
0	wdt_timeout_ack	R/W	Watch-dog timout acknowledge. Acknowledge signal for the time out of Watchdog Timer of the internal CPU. You can use this signal to clear the wdt_timeout register without resetting the CPU. **Value After Reset:** 0x0

11.5.2 CPU_STAT

- **Name:** CPU Status Register
- **Description:** CPU status register.
- **Size:** 32 bits
- **Offset:** 0x4004

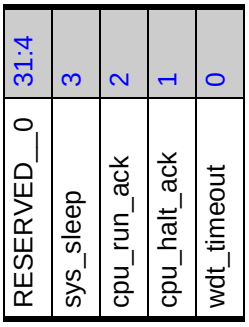


Table 11-222 Fields for Register: CPU_STAT

Bits	Name	Memory Access	Description
31:4	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
3	sys_sleep	R	Indicates that the CPU core is sleeping. **Value After Reset:** 0x0
2	cpu_run_ack	R	Active-high acknowledgement signal. The CPU core acknowledges the run request and is now running. Cleared after (cpu_run_req) is deasserted. **Value After Reset:** 0x0
1	cpu_halt_ack	R	Active-high acknowledgement signal. The CPU core acknowledges the halt request and is now halted. Cleared after (cpu_halt_req) is deasserted. **Value After Reset:** 0x0
0	wdt_timeout	R	Watch-dog timout status. Asserted high to indicate that the Watchdog Timer of the internal CPU has timed out. The signal remains high until it has been accepted by the accompanying acknowledge signal wdt_timeout_ack or until a phy reset is applied. **Value After Reset:** 0x0

11.5.3 raw_pcs_CFG

- **Name:** RAW PCS Configuration Register 1
- **Description:** Configures (\raw_pcs) settings.
- **Size:** 32 bits
- **Offset:** 0x4008

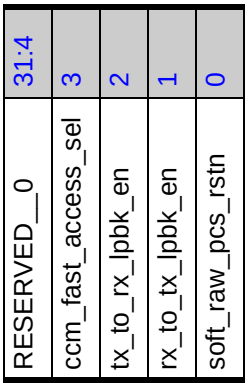


Table 11-223 Fields for Register: raw_pcs_CFG

Bits	Name	Memory Access	Description
31:4	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
3	ccm_fast_access_sel	R/W	Selects method of accessing ICCM and DCCM. **Values:** ■ 0x0 (EXTERNAL_JTAG): External apb interface or jtag (based on (jtag_apb_sel_i). This is a slow access. ■ 0x1 (CCM_AHB_INT): CCM AHB master interface. This is a fast access and supports AHB burst feature. **Value After Reset:** 0x0
2	tx_to_rx_lpbk_en	R/W	Enables a tx clock, and tx data parallel loopback from (TX) to (RX). **Value After Reset:** 0x0
1	rx_to_tx_lpbk_en	R/W	Enables a recovered clock and recovered data parallel loopback from (RX) to (TX). **Value After Reset:** 0x0
0	soft_raw_pcs_rstn	R/W	Write 0 to soft reset the raw_pcs. **Value After Reset:** 0x1

11.5.4 SUP_CFG

- **Name:** SUP Block Configuration Register
- **Description:** Configures (SUP) settings.
- **Size:** 32 bits
- **Offset:** 0x400c

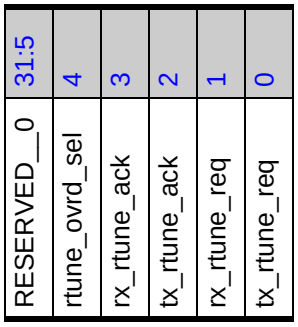


Table 11-224 Fields for Register: SUP_CFG

Bits	Name	Memory Access	Description
31:5	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
4	rtune_ovrd_sel	R/W	Rtune override register (1'b0 when overriden by regbank), (1'b1 when derived from top level). **Value After Reset:** 0x0
3	rx_rtune_ack	R	Assert this flag whenever rtune have completed tuning in receiver. **Value After Reset:** 0x0
2	tx_rtune_ack	R	Assert this flag whenever rtune have completed tuning in transmitter. **Value After Reset:** 0x0
1	rx_rtune_req	R/W	Rtune request signal for receiver. **Value After Reset:** 0x0
0	tx_rtune_req	R/W	Rtune request signal for transmitter. **Value After Reset:** 0x0

11.5.5 FW_CFG_0

- **Name:** Firmware Configuration Register 0
- **Description:** Configures (FW) settings.
- **Size:** 32 bits
- **Offset:** 0x4010

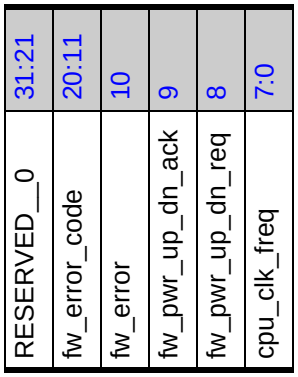


Table 11-225 Fields for Register: FW_CFG_0

Bits	Name	Memory Access	Description
31:21	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
20:11	fw_error_code	R/W	Firmware error code. Refer to firmware specifications for more details. Value After Reset: 0x0 **Value After Reset:** 0x0
10	fw_error	R/W	Firmware error flag. If set, check (fw_error_code). **Value After Reset:** 0x0
9	fw_pwr_up_dn_ack	R/W	This is a handshake with the firmware. Firmware writes to this register to indicate it received power up/down request. When signal transitions from 0b0 to 0b1, then firmware indicates that it has received power up request. When this signal transitions from 0b1 to 0b0, then firmware indicates that it has received power down request. **Value After Reset:** 0x0
8	fw_pwr_up_dn_req	R/W	This is a handshake with the firmware. Requests the firmware to power up/down the phy. Use along with (fw_pwr_up_dn_ack). **Values:** ■ 0x1 (PU_REQ): power up request ■ 0x0 (PD_REQ): power down request **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
7:0	cpu_clk_freq	R/W	CPU clock frequency in units of 10 MHz. **Value After Reset:** 0xa

11.5.6 FW_CFG_1

- **Name:** Firmware Configuration Register 1
- **Description:** Configures (FW) settings.
- **Size:** 32 bits
- **Offset:** 0x4014

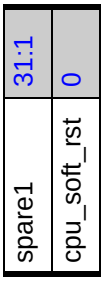


Table 11-226 Fields for Register: FW_CFG_1

Bits	Name	Memory Access	Description
31:1	spare1	R/W	Spare 1 register, to be defined for firmware use. Value After Reset: 0x0 **Value After Reset:** 0x0
0	cpu_soft_rst	R/W	For debug use only. CPU soft reset. Value After Reset: 0x0 **Value After Reset:** 0x0

11.5.7 LANES_STATUS

- **Name:** Lane Status Register
- **Description:** Status of lanes settings.
- **Size:** 32 bits
- **Offset:** 0x4018

31:25	24:23	22	21	20	19	18:17	16:14	13:7	6	5	4	3	2:0
RESERVED__0	conf_mode	rx_spareout	tx_spareout	rx_sparein	tx_sparein	tx_nrz_amp_ctrl	clk_mode	sup_cntx_sel	signal_type	polarity_inv	pam4_ab_swap_en	graycode_en	rate

Table 11-227 Fields for Register: LANES_STATUS

Bits	Name	Memory Access	Description
31:25	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
24:23	conf_mode	R	Reserved **Value After Reset:** 0x0
22	rx_spareout	R/W	RX spare out **Value After Reset:** 0x0
21	tx_spareout	R/W	TX spare out **Value After Reset:** 0x0
20	rx_sparein	R	RX spare in **Value After Reset:** 0x0
19	tx_sparein	R	Tx spare in **Value After Reset:** 0x0
18:17	tx_nrz_amp_ctrl	R	Controls the amplitude of the TX in NRZ mode, i.e. 11: = +/3, 10=+/-2, 01=+/-1, 00=reserverd **Value After Reset:** 0x0
16:14	clk_mode	R	Access value of input port clk_mode_i **Value After Reset:** 0x0

Bits	Name	Memory Access	Description
13:7	sup_cntx_sel	R	Access value of input port sup_cntx_sel_i **Value After Reset:** 0x0
6	signal_type	R	Access value of input port signal_type_i **Value After Reset:** 0x0
5	polarity_inv	R	Access value of input port polarity_inv_i **Value After Reset:** 0x0
4	pam4_ab_swap_en	R	Access value of input port pam4_ab_swap_en_i **Value After Reset:** 0x0
3	graycode_en	R	Access value of input port garycode_en_i **Value After Reset:** 0x0
2:0	rate	R	Access value of input port rate_i **Value After Reset:** 0x0

11.5.8 JTAG_ID_CODE

- **Name:** JTAG ID Code Configuration Register
- **Description:** Jtag ID code configuration.
- **Size:** 32 bits
- **Offset:** 0x401c



Table 11-228 Fields for Register: JTAG_ID_CODE

Bits	Name	Memory Access	Description
31:0	sreg_id_code	R	ID code register **Value After Reset:** 0x0

11.5.9 PIPE_CONFIG_0

- **Name:** PIPE Interface Configuration Register 0
- **Description:** Pipe interface configuration.
- **Size:** 32 bits
- **Offset:** 0x4020

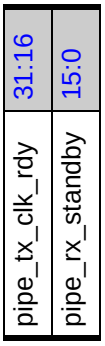


Table 11-229 Fields for Register: PIPE_CONFIG_0

Bits	Name	Memory Access	Description
31:16	pipe_tx_clk_rdy	R	Signal is asserted when the input clock of the TX is running **Value After Reset:** 0x0
15:0	pipe_rx_standby	R	This signal is asserted when phy_status is de-asserted. **Value After Reset:** 0x0

11.5.10 PIPE_CONFIG_1

- **Name:** PIPE Interface Configuration Register 1
- **Description:** Pipe interface configuration.
- **Size:** 32 bits
- **Offset:** 0x4024

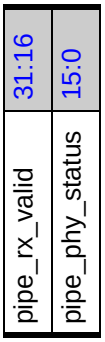


Table 11-230 Fields for Register: PIPE_CONFIG_1

Bits	Name	Memory Access	Description
31:16	pipe_rx_valid	R/W	This signal will indicate that All calibrations are done and RX output data is valid **Value After Reset:** 0x0
15:0	pipe_phy_status	R/W	This signal will be de-asserted when the TX is calibrated and ready for transmission. TX data will be ignored until this signal is de-asserted. **Value After Reset:** 0xffff

11.5.11 PIPE_CONFIG_2

- **Name:** PIPE Interface Configuration Register 2
- **Description:** Pipe interface configuration.
- **Size:** 32 bits
- **Offset:** 0x4028

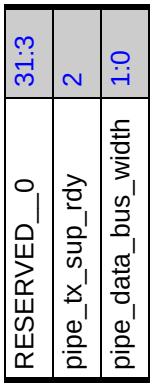


Table 11-231 Fields for Register: PIPE_CONFIG_2

Bits	Name	Memory Access	Description
31:3	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
2	pipe_tx_sup_rdy	R/W	This signal is asserted when the SUP of the tX is calibrated and clk on the pipe_max_pclk_o is running. **Value After Reset:** 0x0
1:0	pipe_data_bus_width	R	Reports the PHY data bus width. 1'b0 is 64bit per lane (only option valid at this point. **Value After Reset:** 0x0

11.5.12 EQ_CONFIG

- **Name:** Equalization Configuration Register
- **Description:** Equalization configuration.
- **Size:** 32 bits
- **Offset:** 0x402c

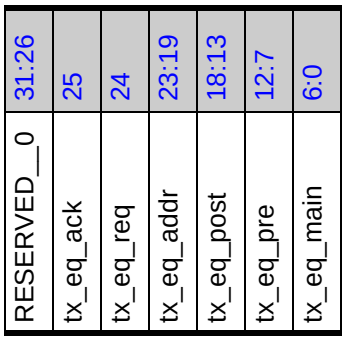


Table 11-232 Fields for Register: EQ_CONFIG

Bits	Name	Memory Access	Description
31:26	RESERVED__0	R	RESERVED **Reserved Field:** Yes **Value After Reset:** 0x0
25	tx_eq_ack	R/W	tx_eq_ack **Value After Reset:** 0x0
24	tx_eq_req	R	tx equalization req **Value After Reset:** 0x0
23:19	tx_eq_addr	R	tx equalization addr (MSB used as a broadcast to all TXs) **Value After Reset:** 0x0
18:13	tx_eq_post	R	tx equalization post **Value After Reset:** 0x0
12:7	tx_eq_pre	R	tx equalization pre **Value After Reset:** 0x0
6:0	tx_eq_main	R	tx equalization main **Value After Reset:** 0x0

11.5.13 SPARE_FW_0

- **Name:** Spare Register 0
- **Description:** Spare for firmware
- **Size:** 32 bits
- **Offset:** 0x4030

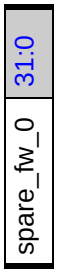


Table 11-233 Fields for Register: SPARE_FW_0

Bits	Name	Memory Access	Description
31:0	spare_fw_0	R/W	spare_0 for firmware **Value After Reset:** 0x0

11.5.14 SPARE_FW_1

- **Name:** Spare Register 1
- **Description:** Spare for firmware
- **Size:** 32 bits
- **Offset:** 0x4034

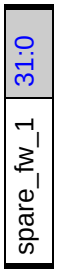


Table 11-234 Fields for Register: SPARE_FW_1

Bits	Name	Memory Access	Description
31:0	spare_fw_1	R/W	spare_1 for firmware **Value After Reset:** 0x0

A

PHY Configuration

This appendix provides configuration settings for the PHY.

- Table A-1: “sup_cntx_sel [6:0] Settings”
- Table A-2: “clk_mode[2:0] Settings”

Table A-1 sup_cntx_sel [6:0] Settings

Information/Primary Inputs		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44
Primary Inputs	ref_alt0/1_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	100	100	78.125	78.125	78.125	78.125	78.125	78.125	78.125	78.125	78.125	78.125	100	100	100	100	100	100	100	100	100	156.25	156.25	156.25	156.25	156.25	156.25	156.25	156.25	156.25	156.25	200	200	200	200	200	200	200	200	200	125	187.5	100	125	156.25
	MPLL VCO frequency (GHz)	14	14	14.0625	14.0625	14.0625	14.0625	14.0625	14.0625	14.0625	14.0625	14.0625	13.2813	14	14	14	14	14	14	12.5	13	14	14.0625	14.0625	14.0625	14.0625	14.0625	14.0625	14.0625	14.0625	14.0625	13.2813	14	14	14	14	14	14	12.5	13	14	13.875	13.875	13.2813	13.2813	14
	ROPLL input clock frequency (GHz) (sup_ana/ropll/ropll_mplla_clk)	1.75	1.75	0.87891	0.87891	0.87891	0.87891	0.87891	0.87891	0.87891	0.87891	0.87891	0.83008	0.875	0.875	0.875	0.875	0.875	0.875	0.78125	0.8125	0.875	0.87891	0.87891	0.87891	0.87891	0.87891	0.87891	0.87891	0.87891	0.87891	0.83008	0.875	0.875	0.875	0.875	0.875	0.875	0.78125	0.8125	0.875	0.86719	0.86719	0.83008	0.83008	0.875
	Clock from SUP to TX/RX (GHz) (sup_ana/ropll_clk_0_left)	14	14	7.03125	7.91016	8.78906	9.66797	10.5469	11.4258	12.3047	13.1836	14.0625	13.2813	7	7.875	8.75	9.625	10.5	11.375	12.5	13	14	7.03125	7.91016	8.78906	9.66797	10.5469	11.4258	12.3047	13.1836	14.0625	13.2813	7	7.875	8.75	9.625	10.5	11.375	12.5	13	14	13.875	13.875	13.2813	13.2813	14
	ropll wordclk (MHz) (sup_ana/ropll_div/ropll_word_clk)	1750	1750	878.906	988.77	1098.63	1208.5	1318.36	1428.22	1538.09	1647.95	1757.81	1660.16	875	984.375	1093.75	1203.13	1312.5	1421.88	1562.5	1625	1750	878.906	988.77	1098.63	1208.5	1318.36	1428.22	1538.09	1647.95	1757.81	1660.16	875	984.375	1093.75	1203.13	1312.5	1421.88	1562.5	1625	1750	1734.38	1734.38	1660.16	1660.16	1750
	Max DataRate (GSps)	56.00	56.00	28.13	31.64	35.16	38.67	42.19	45.70	49.22	52.73	56.25	53.13	28.00	31.50	35.00	38.50	42.00	45.50	50.00	52.00	56.00	28.13	31.64	35.16	38.67	42.19	45.70	49.22	52.73	56.25	53.13	28.00	31.50	35.00	38.50	42.00	45.50	50.00	52.00	56.00	55.50	55.50	53.13	53.13	56.00
	ref_ana_mplla_div[2:0]	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	2	1	0	1	1	1	1	1	1	1	1	1	2	1	1	1	1	1	1	3	2	1	2	1	0	0	0
	mplla_ana_div_clk_div4_en	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	mplla_ana_tx_clk_div[3:0]	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	mplla_ana_word_div2_en	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	mplla_ana_word_clk_div5_en	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	mplla_ana_word_clk_div4_en	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	mplla_ana_frac_quot[15:0]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26624	8192	19660
	mplla_ana_frac_den[15:0]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	5
	mplla_ana_frac_rem[15:0]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4
	ropll_ana_refdiv[3:0]	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	ropll_ana_fbdiv[6:0]	8	8	8	9	10	11	12	13	14	15	16	16	8	9	10	11	12	13	16	16	16	8	9	10	11	12	13	14	15	16	16	8	9	10	11	12	13	16	16	16	16	16	16	16	16
	ropll_ana_div_clk_en	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	ropll_ana_word_div2_en	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	ropll_ana_word_clk_div5_en	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	ropll_ana_word_clk_div4_en	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Primary Inputs [all CLK_MODE cases except Rx fwd clock]	ref_range[3:0]	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	6	6	6	6	6	6	6	6	6	6	7	7	7	7	7	7	7	7	4	7	3	4	6
	ref_clk_div2_en	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	ref_raw_clk_div2_en	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	1
Primary Inputs [CLK_MODE = Rx fwd clock]	ref_range[3:0]	8	8	4	4	4	4	4	4	4	4	4	3	4	4	4	4	4	4	3	3	4	4	4	4	4	4	4	4	4	4	3	4	4	4	4	4	4	3	3	4	4	4	3	3	4
	ref_clk_div2_en	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	ref_raw_clk_div2_en	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Info	sup_ctrl_sel<6:0>	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44
Primary Inputs	mplla_ana_bandwidth[15:0]	101	101	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663	5663
	mplla_ana_ctl_buf_bypass	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	mplla_ana_pmix_en	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
	ropll_ana_cp_ctrl_int[6:0]	26	26	8	8	12	12	14	26	26	26	26	26	8	12	14	12	14	14	26	26	26	8	8	12	12	14	26	26	26	26	26	8	12	14	12	14	14	26	26	26	26	26	26	26	26
	ropll_ana_cp_ctrl_prop[6:0]	47	47	65	57	69	77	67	75	81	75	81	81	57	69	67	69	67	67	75	81	81	65	57	69	77	67	75	81	75	81	81	57	69	67	69	67	67	75	81	81	81	81	81	81	81
	ropll_ana_rc_filter[2:0]	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
	ropll_ana_v2i_mode[1:0]	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	ropll_ana_vco_low_freq[1:0]	2	2	3	3	3	3	2	2	2	2	2	2	3	3	3	3	2	2	2	2	2	3	3	3	3	2	2	2	2	2	2	3	3	3	3	2	2	2	2	2	2	2	2	2	2
	sup_ana_misc[7:0]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	ropll_ana_postdiv[2:0]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table A-2 clk_mode[2:0] Settings

Information/Primary Inputs		clk_mode<2:0>							
		0	1	2	3	4	5	6	7
	Clock source	internal	internal	internal	internal	external	external	external	external
	Clock mode	fwd clk	common clk	CDR assist	reserved	fwd clk	common clk	CDR assist	reserved
tx sup_ana settings	mplla_ana_en	1	1	1	1	1	1	1	1
	mplla_ana_output_en	1	1	1	1	1	1	1	1
	ropll_ana_refsel[1:0]	0	0	0	0	0	0	0	0
	diffbuf_tx_en	1	0	0	0	1	0	0	0
	diffbuf_pulldn	0	1	1	1	0	1	1	1
	ref_ana_diffbuf_clk_en	0	0	0	0	0	0	0	0
	refa_ana_clk_en	1	1	1	1	1	1	1	1
	refa_ana_clk_sel[1:0]	0	0	0	0	1	1	1	1
	ref_ro_ana_clk_div8_en	0	0	0	0	0	0	0	0
	ref_ana_ropll_div[2:0]	0	0	0	0	0	0	0	0
	ref_ro_ana_clk_en	0	0	0	0	0	0	0	0
	ref_ana_clk_sel	0	0	0	0	0	0	0	0
tx sup_ana settings DIRECT	ropll_ana_en	0	0	0	0	0	0	0	0
	spare_in[9]	0	0	0	0	0	0	0	0
	spare_in[3]	0	0	0	0	0	0	0	0
tx sup_ana settings CASCADED	ropll_ana_en	1	1	1	1	1	1	1	1
	spare_in[9]	1	1	1	1	1	1	1	1
	spare_in[3]	1	1	1	1	1	1	1	1
rx sup_ana settings	mplla_ana_en	0	1	1	1	0	1	1	1
	mplla_ana_output_en	0	1	1	1	0	1	1	1
	ropll_ana_refsel[1:0]	1	0	0	0	1	0	0	0
	diffbuf_tx_en	0	0	0	0	0	0	0	0
	diffbuf_pulldn	1	1	1	1	1	1	1	1
	ref_ana_diffbuf_clk_en	1	0	0	0	1	0	0	0
	refa_ana_clk_en	0	1	1	1	0	1	1	1
	refa_ana_clk_sel[1:0]	0	0	0	0	0	1	1	1
	ref_ro_ana_clk_div8_en	1	0	0	0	1	0	0	0
	ref_ana_ropll_div[2:0]	3	0	0	0	3	0	0	0
	ref_ro_ana_clk_en	1	0	0	0	1	0	0	0
	ref_ana_clk_sel	1	0	0	0	1	0	0	0
rx sup_ana settings DIRECT	ropll_ana_en	1	0	0	0	1	0	0	0
	spare_in[9]	1	0	0	0	1	0	0	0
	spare_in[3]	1	0	0	0	1	0	0	0
rx sup_ana settings CASCADED	ropll_ana_en	1	1	1	0	1	1	1	0
	spare_in[9]	1	1	1	0	1	1	1	0
	spare_in[3]	1	1	1	1	1	1	1	1